One Number Explains Everything

How $\xi = 4/30000$ generates lepton masses, the fine structure constant, and gravity

Fundamental Fractal Geometric Field Theory (FFGFT)
A-Series · Canonical Edition · 48 Documents

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Layer markers:

[AXIOM]	Axiom — not derived, declared as starting point
[K]	Core — derived from ξ, numerically verified
[B]	Bridge — algebraically proven
[S]	Sketch — plausible, not fully worked out

The author bears full responsibility for content and errors.

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Part I

Foundations

Chapter 1

A010 · Purpose, Structure and Guide to Reading

1.1 Use of AI Tools

This document series was created with the assistance of AI tools. The notice appears at the front, not in small print, and belongs as its own section in every document of the series.

The division of labour is clear. The questions, the theoretical stipulations, the geometric ideas, and all substantive decisions come from the author. The AI tools are tools in the literal sense: they formulate, compute, check for contradiction, create verification scripts, find inconsistencies between documents, and translate between representations. They make no substantive decisions and generate no theses.

This separation is not merely an attribution but is methodologically essential. A language model generates plausible-sounding text even where there is no substance – exactly the type of error that the layer marking of the following section is designed to counter. Therefore, throughout this series: every computational statement has a verification script that can be independently checked; every number has a tolerance standard; every derivation step has a layer marker. What does not pass this test does not appear as a result in the text.

Responsibility for content and errors rests entirely with the author.

1.2 What Is Not Part of the A-Series

Two areas are explicitly excluded.

The cosmological sector. The A-series does not include it. The corresponding documents from the legacy corpus remain and retain their numbers, but are not carried into the new structure. The reason is a difference in standards: the geometric core derives results from ξ; the cosmological sector works throughout with imported scales and declared calibrations. Mixing both in one series would obscure precisely the separation that motivates the new structure.

Predictions beyond the fundamental quantities. The only quantities carried as predictions are those that follow directly from the apparatus and at which the theory can fail: the tau mass, the ξ-order of the CHSH deviation, the fractal dimension D_f, and the fundamental length L_0. Everything else that appeared as a prediction in earlier documents appears here as a structural statement, or not at all.

1.3 Why a New Series

The existing FFGFT document series grew historically. Its numbers follow the order of creation, not the subject matter. Over roughly three years, more than two hundred German-language documents accumulated – individually comprehensible, but collectively exhibiting four properties that harm the whole.

First, reference decay. A substantial portion of the early documents is rarely cited by the later ones. The content has not become wrong, but it is presented more completely and precisely elsewhere. The old number persists and continues to be found.

Second, corrections sit outside the texts. A separate correction register amends documents from the outside instead of modifying them; it records corrections, refinements, and revisions as dated entries. A reader of a single old document therefore reads the uncorrected state. The working rule of not modifying older documents was methodologically correct – it keeps the development history honest – but it shifts the burden onto the reader.

Third, multiple coverage. The mass sector appears in at least five documents, the concept of time in seven, the information question in six. Each of these presentations has its occasion, but there is no single place that says: *this* is the current state.

Fourth, layers are typographically invisible. A statement derived from ξ, an algebraically proved translation, and a plausibility sketch appear in the same typeface side by side. This is the most dangerous of the four properties, because it inflates the theory's apparent claim.

[STIPULATION] The A-series addresses all four points with the same measure: subject order instead of chronological order, corrections integrated instead of appended, one topic in one place, layer status declared per statement.

1.4 Introductory Reading Before the A-Series

A reader entering the A-series encounters stipulations, layer markers, and geometric structures from A020 onward. Two documents in the legacy corpus prepare for this – in different ways.

"The Journey" (legacy corpus, popular science) describes the T0 theory in everyday language – with analogies, without formulas, without requiring a physics background. A reader who reads it before A010 arrives with a picture in mind: why a single number ξ might suffice, what time-mass duality means, what is different about this approach from established theories.

"FFGFT in Plain Language" (legacy corpus, narrative-technical) is the narrative introduction to the Lagrangian core. It translates the central Lagrangian formulations into accessible language – using images from acoustics (vibrating string, resonator, Pythagorean comma) that appear in FFGFT itself as structural parallels, not arbitrary metaphors. Every statement is anchored in the formal documents. A reader who wants to understand how the theory computes before seeing the computations should read this document.

Both documents are not part of the A-series and are not superseded by it. They are independent and remain so.

1.5 What the A-Series Is Not

[STIPULATION] It is *not* a revision in the sense of reappraisal. Nothing is retracted that has not already been retracted in the correction register. The content is the same; the order is new.

It does not physically replace the old documents. The legacy corpus is fully preserved, unchanged, under its old numbers. It is the development history and remains citable. The A-series is the version to read when one wants to know what the theory claims – not how it got there.

And it is not an abridgment. Where an old text is more detailed than the corresponding A-section, the latter refers to it. What is to be lost is redundancy, not substance.

1.6 The Three Layers

FFGFT has always had three methodological layers. What is new is that they are marked in the A-series – per statement, not per document.

[K] Core. Derived from the geometry and the single parameter ξ. Example: the Koide relation $Q = 2/3$ as operator condition $r = \sqrt{2}$; the non-closure of the recursion; $D_f = 3 - \xi$.

[B] Bridge. Algebraically proved translation between two representations. Example: the bijection $\mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3$; the DFT diagonalisation of the Z_3 circulant. A bridge carries everything established on one side, and proves nothing that was not already there.

[S] Sketch. Plausibility, analogy, candidate. Example: the neutrino sector; $N_0 \approx 38.6$; the fine value of the refraction strength δ.

A fourth marker is added that is not a layer but a booking category:

[STIPULATION] Declared, not derived. Example: $\tilde{T} \cdot m = 1$; the choice of T^4; the numerical value $\xi = 4/30000$; the reference point m_μ/m_e.

[B] The purpose of the markers is not modesty but testability. A theory that names its stipulations can be measured by how few it needs. A theory that conceals them cannot.

1.7 The Tolerance Standard

Every numerical statement in the A-series names the standard against which it is tested. This is a consequence of an experience older than the theory: a measuring instrument is always less capable than the reality it measures. Exact agreement on the last digit is therefore neither achievable nor the goal.

[STIPULATION] Completeness in the A-series means: *no residual remains that exceeds the tolerances and can only be closed by a free parameter.* A residual within the determination tolerance is not a deficiency but the normal case. A residual that exceeds the tolerance is a statement – and is booked as such, not rounded away.

In practice: at every deviation, three quantities appear together – the value, the measurement precision of the comparison quantity, and the determination tolerance of the theoretical quantity. Only the ratio of these three decides whether a deviation is meaningful.

1.8 Citation Policy

[STIPULATION] The A-series carries no reference lists. The rule:

Standard physics without source. Textbook knowledge – constants, standard formulas, established results (Dirac equation, Casimir formula, CHSH bound, Koide relation, Landauer principle) – is used without citation. Name references (Bell, Tsirelson, Weyl) are concept labels, not literature references.

Specific external measurements with inline source. Where a concrete, dated measurement by a third party with error bars is claimed, the source appears in parentheses directly at the number.

Own measurements with reference to the measurement protocol. Hardware runs and own measurement campaigns refer to raw data and evaluation scripts in the corpus, so that the claim is independently verifiable.

1.9 Structure and Numbering

The series has four blocks.

Block 0 (A010–A090): Foundation. Stipulations, geometry, fractal correction, recursion, time, Hilbert space, field theory, units, projection chain. A reader who has worked through this block knows the apparatus completely – without a single result. Current documents: A010, A015, A020, A030, A040, A050, A060, A070, A075, A080, A085, A090.

Block 1 (A100–A190): Sectors. Leptons, Koide relation, $\theta = 2/9$, fine-structure constant, g-2, gravitation, gravitational constant, quarks and neutrinos, Bell and hardware, information, Standard Model. The results appear here, each referred back to the apparatus from Block 0. Current documents: A100, A105, A110, A120, A130, A135, A138, A140, A142, A145, A150, A160, A165, A180, A190.

Block 2 (A200–A250): Methodology and assessment. The distinction of law, norm, and stipulation; the test quantities; the tolerance standard; the open questions; the demarcation from external frameworks; the index. Current documents: A200, A210, A220, A230, A240, A250.

Block 3 (A260–): Extensions. Topics without back-reference from earlier blocks – self-contained units that supplement the core but are not embedded in the main flow. Current documents: A260 (Casimir effect), A261 (scale hierarchy), A262 (c-convention and $E = m$), A263 (Dirac complete), A264 (asymmetry and CP), A265 (spectral shift and SI anchor), A266 (unit test and SI back-computation), A267 ($\alpha = 1$ as stipulation).

Methodological status of Block 3. A260–A264 and A265 are physics result documents without back-reference from Block 0/1. A266 (unit test) and A267 ($\alpha = 1$ convention) are convention and methodology documents – they contain computational tools, not new theoretical results. This does not change their place in Block 3, but the reader should know that the claim differs.

[STIPULATION] Numbers within blocks increase in steps of ten. Intermediate numbers (A015, A075, A138, A142, etc.) are assigned when a later document refers back to an earlier one and the ordering requires it. A document without back-reference from the existing block goes to the end – in Block 3, not as an intermediate number. That is the rule guiding the decision: not the topic, but the reference.

1.10 Sources

Every A-document names in its closing section the old numbers whose content it subsumes. This is not a reference list but a replacement declaration: a reader who has read the A-document need not read the listed old documents to know the current state.

The reverse direction – from each old number to its A-successor – is collected in A250. There it is also recorded which entries of the correction register are incorporated where, and which old documents deliberately have no successor because their content is outdated or purely historical.

1.11 Language Status

The A-series is initially produced in German only. This is a deliberate departure from the previous rule of keeping every change bilingual. The reason is practical: the structure must settle first. An English version is produced when a block is complete and stable – then in one pass from the finished German state, not change by change.

Until then, the bilingual legacy series remains the reference for external correspondence.

1.12 Symbol Glossary of the A-Series

The following table explains the symbols used consistently throughout the entire A-series. Symbols that appear only in a single document are introduced there at their first occurrence.

Symbol	Meaning
Fundamental constants and parameters	
$\xi = 4/30000$	Geometric fundamental constant of FFGFT; $\xi = 4/3 \times 10^{-4}$
ℓ_P	Planck length, $\ell_P = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}$ m
$\hbar$	Reduced Planck constant
c	Speed of light (in natural units: $c = 1$)
α	Fine-structure constant, $\alpha \approx 1/137.036$
φ	Golden ratio, $\varphi = (1 + \sqrt{5})/2 \approx 1.618$
FFGFT-specific quantities	
$D_f = 3 - \xi$	Fractal dimension of the FFGFT torus
$L_0 = \xi \cdot \ell_P$	Minimal length scale (sub-Planck), $L_0 \approx 2.15 \times 10^{-39}$ m
$L_T = \ell_P/\xi$	Torus fundamental period, $L_T \approx 7500\, \ell_P$
$\tau_0 = \xi \cdot t_P$	Minimal time scale; by $c = 1$ the same quantity as L_0 in time dress (four-dress principle, A080)
$K_{\text{frak}} = 1 - 100\xi$	Fractal correction factor, $K_{\text{frak}} \approx 0.9867$
$E_0 = \sqrt{m_e m_\mu}$	Characteristic energy, $E_0 \approx 7.398$ MeV *[Note 3 August 2026: E_0 is the calibration of the unit system – by A085 there is exactly one – and the value carried is the one that provides the anchoring in SI: 7.398 MeV matches the measured α to 5×10^{-6} via $\alpha = \xi E_0^2$. The geometric mean $\sqrt{m_e m_\mu} = 7.348$ MeV (A130, route 1) names the origin of the order of magnitude but is unsuitable as an anchor: inserted, it would give $1/138.9$, i.e. $-1.35\,\%$ in α, and that deviation would propagate into every derived absolute value and accumulate in comparisons. The formula notation here is abbreviated; the anchor value governs (A130, route 2: $E_0^2 = 4\sqrt{2}\, m_\mu/\xi^p$, without m_e).]*
$\tilde{T} \cdot m = 1$	Time-mass duality (stipulation); $\tilde{T}$ is the intrinsic time field
$v = 246$ GeV	Higgs vacuum expectation value (energy unit)
Layer markers	
[STIPULATION]	Axiom of the theory; not further derived

(continued on next page)

Symbol	Meaning
[B]	Bridge; algebraically proved
[K]	Core; numerically verified derivation
[S]	Sketch; plausible but not fully worked out
Mathematical structures	
$T^4/\mathbb{Z}_3$	Compact 4-torus with $\mathbb{Z}_3$ orbifold structure
$\mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3$	Hilbert space of the theory
$\pi_1(T^4) \cong \mathbb{Z}^4$	First homotopy group of the 4-torus
$P_i = \lvert\mathbf{e}_i\rangle\langle\mathbf{e}_i\rvert$	Projector onto mode i
$\Phi = \sum_i m_i P_i$	Mode operator (field operator)
Masses and ratios	
m_e, m_μ, m_τ	Electron, muon, tau mass
$m_i = r_i \cdot \xi^{p_i} \cdot v$	FFGFT mass formula; $r_i \in \mathbb{Q}$ rational prefactor, $p_i \in \mathbb{Q}$ exponent
(n, l, j)	Quantum numbers: principal, angular momentum, spin

1.13 Independent Documents of the Legacy Corpus

[STIPULATION] The A-series subsumes 102 documents of the legacy corpus. The remainder – more than 200 numbers – stands independently. This section lists the main thematic groups of these independent documents, so the reader knows where the legacy corpus ends and where the A-series begins. Complete mapping: A250.

Cosmological sector (excluded, Doc. 036, 063, 064, 140 et al.). Hubble constant, cosmic microwave background, vacuum field at galactic scales, Λ_{cosmo}. This sector is not wrong – it is of a different ambition: it works throughout with imported scales and calibrations that the A-series cannot reproduce. Excluded, not superseded.

Gauge sector and QFT extensions (Doc. 186, 192, 281 et al.). Complete Feynman rules, renormalisation group beyond

ξ-corrections, Yang-Mills formulation. These documents treat topics declared as open points in the A-series (A190, A240). They remain independent because the projection mapping is still missing.

IPI framework comparisons (Doc. 246, 251, 252, 265, 267, 269, 271, 272, 274, 283, 285, 294, 297, 299 et al.). Structural comparisons with HLV (Helix-Light-Vortex), RA/PMT, Dot Theory, SICC, and other works from the IPI community. These documents are explicitly not FFGFT statements but comparison documents: they investigate where external frameworks and FFGFT share points of contact. The A-series takes over none of these comparisons (A240 names the contact points without carrying them).

Quantum hardware and photonic systems (Doc. 083–085, 161, 167, 173–176, 183 et al.). Photonic chips in communications technology, quantum dots, Shor's algorithm on photonic platforms, Google Willow analysis, Ising machines. These documents apply FFGFT ideas to concrete hardware systems. The application is independent and not back-anchored in the A-series core.

Popular and narrative introductions (Doc. 002, 004, 185, 191 et al.). "The Journey" (narrative introduction without formulas), FFGFT for communications engineers, video scripts. These documents are deliberately not part of the A-series – they are the access point for readers without physics prerequisites and remain so. They are not superseded; they are complementary.

AI and consciousness (Doc. 100, 207 et al.). T0 theory and consciousness; consciousness and bridges. These documents treat a topic that does not appear in the A-series. They are independent philosophical works, not an appendix to the physics.

Methodological and diagnostic documents (Doc. 241, 242, 244, 247, 248, 250, 262, 277, 281, 284, 287, 288, 303–305 et al.). Hawking information paradox, ratio vs. absolute-value pedagogy, document structure maps, embedding protocols. These documents are working materials that arose during the development of the corpus. They are important for following the development history, but are not part of the canonical state.

Physics extensions in the narrow sense (Doc. 021–024, 027–034, 037–040, 054–068, 075–081, 089, 095, 105, 114, 124, 129, 137, 145–170, 172, 178, 179, 182, 184, 187, 192, 206

et al.). Earlier formulations of the mass formula, alternative parametrisations, intermediate stages of the theory, individual calculations that have been absorbed into A-documents or deliberately not taken over. This block is the largest and most heterogeneous: it contains early correct calculations alongside superseded approaches, independent side-lines alongside already-subsumed substance. The correction register (Doc. 190) books the relevant corrections; the A-documents name what they take over. What appears in neither the register nor an A-document is historical record.

1.14 Open Questions

This section appears in every document of the A-series and names what the document does *not* accomplish. The rule is adopted from outside: a completed step that does not name its residual silently claims jurisdiction beyond what it has shown.

For this document, four points remain open.

First, the series does not cover the entire corpus. It orders the geometric core, the unit and constant structure, and the fermion sector; the gauge sector and the dynamic field theory remain in the legacy corpus, as does the early cosmology sector excluded from the series (A240, A250). The commitment to back every computational statement with a verification script has not yet been fulfilled for the series as a whole.

Second, the replacement claim is untested. The blueprint assigns each A-document a set of old numbers whose content it is to subsume. Whether this assignment is complete – whether nothing from the legacy corpus falls through – can only be decided at the finished index A250, not before. Until then, "supersedes" is an intention, not a finding.

Third, the boundary of the cosmological exclusion is not sharp. The exclusion is justified by a difference in standard, not by topic. At three points – the correction factor K_{frak}, the bridge constants, the projection chain – the legacy corpus historically carries cosmological anchors. Whether these points remain valid without such anchors is an open substantive question, not answered by the exclusion decision.

Fourth, the tolerance standard is here only formulated, not applied. The demand to compare every numerical statement against measurement precision and determination tolerance is easy to state. Whether it can be sustained consistently is decided only where numbers appear – that is, from A100 onward.

1.15 Supersedes

The correction register referred to is Doc. 190 (entries K1–K6, P1–P43, R1–R56); the winding-number corrections are in Doc. 210.

This document subsumes the framework and overview sections of: Doc. 086 (document overview), Doc. 201 (FFGFT-all, outline section), Doc. 205 (narrative, framework sections). The narrative part of Doc. 205 remains independently worth reading and is not superseded.

Chapter 2

A015 · The Origin of ξ

2.1 Purpose

[STIPULATION] $\xi = 4/30000$ is a stipulation (A020). This document answers whether this number is *only* empirically anchored, or whether it follows from the structure of the system. The result: ξ is not arbitrary. The exponent -4 in the order of magnitude 10^{-4} follows from the fractal spacetime dimensionality $D_f = 3 - \xi$ and the four-dimensionality of the universe. The mass-scaling exponent $\kappa = 7$ is the *unique self-consistent solution* of the e-p-μ triangle. The prefactor $4/3$ is independently confirmed by the Casimir effect. The order of magnitude of ξ is thus triply anchored – but *not* derivable from the three stipulations of A020 alone; a measured mass anchor remains necessary.

2.2 The Circularity Problem

[S] The original presentation appears circular:

$$\frac{m_p}{m_e} = 245 \times \left(\frac{4}{3}\right)^7 \Rightarrow \xi = \frac{4}{30000}.$$

Objection: Why $\kappa = 7$? Why $K = 245$? Is this not backward fitting?

[B] The answer: the exponent $\kappa = 7$ is not fitted but is the *unique consistent solution* for the complete e-p-μ system. That is the core of the document; the rest demonstrates it.

2.3 The e-p-μ System as Proof

[K] The three fundamental mass ratios:

$$R_{pe} = \frac{m_p}{m_e} = 1836.15267343,$$
$$R_{\mu e} = \frac{m_\mu}{m_e} = 206.7682830,$$
$$R_{p\mu} = \frac{m_p}{m_\mu} = 8.880.$$

Multiplicativity forces $R_{pe} = R_{\mu e} \times R_{p\mu}$. For the ansatz $R_{pe} = K \times (4/3)^\kappa$ one need only test the exponent:

κ	Prediction R_{pe}	Consistent	Error
6	$245 \times (4/3)^6 = 1376.6$	×	25.0 %
7	$\mathbf{245 \times (4/3)^7 = 1835.4}$	✓	**0.04 %**
8	$245 \times (4/3)^8 = 2447.2$	×	33.3 %

[K] $\kappa = 7$ is the unique consistent solution – not fitting but selection. The exponent follows directly:

$$\kappa = \frac{\ln\left(R_{pe}/K\right)}{\ln(4/3)} = \frac{\ln(1836.15/245)}{\ln(1.3333)} \approx 7.000.$$

2.4 The Fundamental Justification of the 10^{-4} Scale

[B] Why 10^{-4}? The argument has three steps:

(1) From the fractal dimension. The fractal dimension $D_f = 3 - \xi \approx 2.9998667$ generates a discrete scale hierarchy. The relative deviation from the integer dimension:

$$\delta = 1 - \frac{D_f}{3} = 1 - \frac{3 - \xi}{3} = \frac{\xi}{3} \approx 1.333 \times 10^{-4},$$

so $\xi = 3\delta$. This makes ξ the relative fractal deviation.

(2) From the spacetime dimensionality. In d-dimensional spaces one expects natural scalings $\xi_d \sim (10^{-1})^d$. For $d = 4$:

$$\xi_4 \sim (10^{-1})^4 = 10^{-4}.$$

This explains the order of magnitude; the exact prefactor $4/3$ is supplied by step (3).

(3) From the Casimir effect. The Casimir effect provides the factor $4/3$ independently of mass fits. The Casimir energy between parallel plates:

$$E_{\text{Casimir}} = -\frac{\pi^2 \hbar c}{720\, a^3} \times \frac{4}{3}.$$

Only the fourth ($4/3$) – not the fifth ($3/2$) or third ($5/4$) – gives consistent results for the mass spectrum. Consistency over five independent ratios:

Ratio	Experiment	T0 with $\kappa = 7$	Error
m_p/m_e	1836.1527	1835.4	0.04 %
m_μ/m_e	206.7683	206.768	0.001 %
m_p/m_μ	8.880	8.880	0.02 %
m_τ/m_μ	16.817	16.817	0.02 %
m_n/m_p	1.001378	1.001333	0.004 %

2.5 What $K = 245$ Is and What It Is Used For

[K] $K = 245$ is a *remainder term*, not a fundamental object. $\ln(R_{pe})/\ln(4/3) = 26.12$ – not an integer. To obtain an integer exponent $\kappa = 7$, one decomposes:

$$K = \frac{R_{pe}}{(4/3)^7} = \frac{1836.152}{7.492} \approx 245.1.$$

K names the remainder after extracting $(4/3)^7$. It appears in no other FFGFT formula. If R_{pe} were exactly $245 \times (4/3)^7 = 1835.43$, K would be exactly an integer; the small remainder $R_{pe} - 245 \times (4/3)^7 = 0.73$ produces the 0.04 % deviation.

[S] The prime factorisation $245 = 5 \times 7^2$ is a post-hoc observation – 7 appears because of $\kappa = 7$, 5 is coincidental. There is no geometric derivation of $K = 245$; a corresponding formula ($245 \approx \phi^{12}/(1-\xi)^2$) from the legacy corpus is numerically wrong ($\phi^{12}/(1-\xi)^2 \approx 322$, deviation 31 %).

[K] Honest assessment. The three anchors – fractal dimension (1), spacetime dimensionality (2), Casimir (3) – explain the order of magnitude 10^{-4} and the prefactor $4/3$. But all three hang on the e-p-μ system as an empirical anchor. There is no derivation of ξ from the three stipulations of A020 alone (without measured masses). What exists is the strongest available argument: the consistency of the unique self-consistent solution over five mass ratios.

2.6 Verification Script

- Numerical verification: $\kappa = 7$ as the unique consistent solution for R_{pe}, $R_{\mu e}$, $R_{p\mu}$; errors for $\kappa = 6, 7, 8$: `python/A_Serie_Skripte/a015_xi_ursprung.py`

2.7 Open Questions

There will be no proof of ξ – and that is not a deficiency. ξ is a fundamental parameter of the theory – analogous to α in the Standard Model or G in gravitation. Fundamental parameters are not proved from deeper principles; they are motivated by plausibility arguments, supported by consistency, and confirmed by measurements. A "purely geometric derivation without empirical input" is in principle impossible for a fundamental parameter – this holds for any theory.

What exists: plausible reasons at several levels.

(1) The fourth as the unique consistent basis. Of all harmonic intervals – fourth $4/3$, fifth $3/2$, third $5/4$ – only the fourth hits the e-p-μ system consistently. This is not a coincidence of the table but the selection mechanism of the experiment.

(2) Fractal deviation. $\xi = 1 - D_f/3$ is the relative deviation of the fractal dimension $D_f = 3 - \xi$ from the integer dimension 3.

The equation $\xi = \xi$ is circular, but the consistency (D_f and ξ determine each other uniquely) is a structural argument.

(3) Natural scaling. In d-dimensional spaces $\xi_d \sim (10^{-1})^d$ is the natural scaling; for $d = 4$ (3 spatial + 1 time) it follows that $\xi_4 \sim 10^{-4}$.

(4) Prime factor structure. $\xi = 4/30000 = 2^2/(3 \cdot 2^4 \cdot 5^4) = 1/(3 \cdot 2^2 \cdot 5^4)$ – factor 3 corresponds to the spatial dimensions, factor 4 to the spacetime dimensions, 5^4 to the fractal structure.

(5) Music theory analogue. Pythagorean 3:4:5 tetrahedra as a structural parallel to the harmonic ratios – not a proof, but a methodological origin (accordion building, acoustics, resonance physics) that suggests the interval 4:3 as a natural unit.

Together these reasons motivate $\xi = 4/30000$ as a plausible, consistent value – not as a proved one. This is the honest state, and it is the same state as for every other fundamental parameter in physics.

$K = 245$ **is a remainder term, not a fundamental parameter.** It follows from R_{pe} and $\kappa = 7$ – an intermediate value, not an independent quantity (A015, section on K=245).

2.8 Supersedes

This document subsumes: Doc. 009 (ξ-origin: circularity problem, e-p-μ system, Casimir confirmation, $K = 245$, complete consistency system), Doc. 180 (complete derivation chain in five steps: T0 Lagrangian → self-consistency condition → uniqueness of $\xi \rightarrow G \rightarrow L_0$, connection to the SI system).

2.9 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 3

A020 · The Three Axioms

3.1 Why the Stipulations Come First

[STIPULATION] This document stands at the beginning of the content section because a theory is measured by the number and nature of its stipulations, not by the number of its results.

FFGFT claims to derive the constants of the Standard Model from a single dimensionless parameter. This claim is worth only as much as the list of what is additionally assumed is short and open. If the list is concealed, the one-parameter claim is worthless, because any amount of information can be hidden in an unstated assumption.

There are three stipulations. They are stated here in full, before any result is mentioned.

3.2 First Stipulation: Time-Mass Duality

[STIPULATION] For the intrinsic time scale $\tilde{T}$ and the mass m of an object:

$$\tilde{T} \cdot m = 1.$$

This is not a definition of mass and not a definition of time. It is the statement that both quantities are two readings of the same thing: a small mass is a long intrinsic time, a large mass a short one. The equation is reciprocal, hence symmetric – neither side is the explanatory one.

In communications engineering terms, this is the Fourier duality between time and frequency, translated into physical language. That is precisely where it came from: the duality was the starting point of the theory, not its result.

[B] From this single equation it follows immediately that time is not an additional independent quantity. An object with mass carries its time scale with it; it needs no externally imposed clock. This is why the geometry manages with four and not five circles (A030), and why time sits inside the state space and not beside it (A060).

What is stipulated and not shown here: that the reciprocity holds exactly and not merely approximately; and that the constant on the right-hand side equals one. The one is a choice of units – any other value could be absorbed into the scales – but the exactness of the reciprocity is a substantive claim that is stipulated, not proved.

This stipulation is also why the theory cannot place itself as a norm above other frameworks: $\tilde{T} \cdot m = 1$ is a commitment, not a discovered law. One who does not share it makes no error but works in a different framework. The distinction between law, norm, and stipulation is developed in A200; it begins here.

3.3 Second Stipulation: The Compact Geometry

[STIPULATION] The underlying space is a four-dimensional compact torus with a Z_3 identification, written T^4/Z_3.

Three sub-decisions are embedded here, to be assessed separately.

Compact instead of infinite. This is the actual stipulation. It has an immediate consequence: a compact space has a discrete spectrum. It follows that masses appear as eigenvalues and need not be inserted as free parameters. This consequence is the entire reason for the stipulation – and it is simultaneously the point at which the stipulation can fail.

Torus instead of sphere. [B] This choice is not free: the torus has a non-trivial fundamental group, $\pi_1(T^4) \cong \mathbb{Z}^4$, the sphere does not. Winding numbers – the discrete, integer, coordinate-free quantities on which the entire information concept of the theory rests (A180) – do not exist on the sphere. Replacing the torus by a simply connected surface loses not a detail but the load-bearing element.

Four dimensions. The accounting here is uncomfortable, and should remain so. That there are *four* and not five follows from the first stipulation: because $\tilde{T} \cdot m = 1$ binds time to mass, no separate fifth circle for time is needed. In this sense the four is derived. But that the experienced structure is precisely $3+1$ – three spatial directions and one time – is **empirical input** and is not derived by the theory. Earlier formulations in the corpus were too strong at this point; the correction is recorded in the register as P34 and incorporated here.

The Z_3 identification. [STIPULATION] The tripling is stipulated. It delivers the three-element fibre on which the three generations appear as the spectrum of a cyclic operator (A110). **[B]** What can be shown: tripling is crystallographically permitted, fivefold symmetry is not – the condition $2\cos(2\pi/n) \in \mathbb{Z}$ permits $n \in \{1, 2, 3, 4, 6\}$. This does not explain why three must be realised; but it excludes an entire class of alternatives.

[STIPULATION] The direction is to be noted: $n = 3$ is *fixed* not by symmetry but by the observed number of generations – the observation selects the symmetry, not the other way around. Z_3 is stipulated, motivated by the economy of the three-point configuration and consistent with the generation count. The reversal – “three generations are not free parameters but the three permitted Z_3 fixed points” – reads the observation as a consequence of the symmetry and is to be avoided. A stipulation does not become a derivation by the presence of reasons.

3.4 Third Stipulation: The Numerical Value

[STIPULATION] The parameter is

$$\xi = \frac{4}{30000} = \frac{1}{3 \cdot 2^2 \cdot 5^4} \approx 1.333 \cdot 10^{-4}.$$

This stipulation is the most vulnerable of the three, and it decomposes into two parts with quite different status. The separation is important enough to make explicit – earlier presentations in the corpus treated both parts jointly as derived, which was not sustainable (register P33).

The form is structural. [B] The factors $4/3$, the 3, and the 2^2 have a geometric origin: they come from the dimension structure and the tripling. In this respect the shape of the expression is not chosen.

The order of magnitude is not. The factor 5^4 – the question of why ξ sits at 10^{-4} and not at 10^{-2} or 10^{-6} – is *not* derived forward. It is justified in two ways, and both are honestly to be named for what they are: intuitively via harmonic ratio structures, and empirically via the Higgs sector. Neither justification is a derivation from the geometry.

[STIPULATION] Therefore: the numerical value of ξ is an input. The theory does not claim to generate it; it claims that everything else follows from it.

A warning belongs at the same place, because it concerns the most common error of reasoning. Every dimensionless quantity can be written as a power of ξ if one is free to choose the exponent. An expression of the form $X = \xi^N$ is in itself *vacuous*. It becomes a statement only when the exponent is fixed independently – by the geometry, by a count, by a recursion depth. Where this is not possible in this series, it will be said.

3.5 What Follows from the Three Stipulations

[B] The three stipulations together fix: a compact space with discrete spectrum, a three-element fibre, a scale. Nothing more. Everything further in this series is either derived from these three, or explicitly marked as a further stipulation.

The test is the count itself. If in the course of the series a fourth or fifth stipulation becomes necessary, it stands at its place with the marker **[STIPULATION]** – and it must then enter the balance of A230. A theory with three openly declared stipulations is better than one with a concealed one; one with seven openly declared would be worse than one with three. The accounting decides, not the rhetoric.

3.6 The Tolerance Standard at This Point

This document deliberately contains not a single comparison number. The reason is methodological: stipulations have no measurement precision. It is meaningless to ask whether $\tilde{T} \cdot m = 1$ holds to six places – the equation is stipulated, not measured.

Stipulations are testable only indirectly, through their consequences, and there the standard from A010 applies: a consequence is considered supported when its residual stays within the determination tolerance and need not be closed by a free parameter. The first such numbers appear in A100.

3.7 Verification Scripts

Two statements of this document are computational and therefore backed by scripts:

- the crystallographic restriction $2\cos(2\pi/n) \in \mathbb{Z}$ with the result $n \in \{1, 2, 3, 4, 6\}$ and the exclusion of fivefold symmetry: `python/A_Serie_Skripte/a020_kristallographie.py`
- the vacuousness of $X = \xi^N$ with a free exponent – shown on arbitrarily chosen target quantities for which a matching N can always be found: `python/A_Serie_Skripte/a020_xi_potenz_trivialitaet.py`

3.8 Open Questions

First, the number three is not proved minimal. That exactly these three stipulations are needed is a claim about the economy of the construction. It has not been shown that none of them can be obtained from the others, nor that two would suffice.

Second, the tripling is permitted but not explained. The crystallographic restriction excludes fivefold symmetry and permits $\{1, 2, 3, 4, 6\}$. Why of the permitted values precisely three is realised, the geometry does not say.

Third, the order of magnitude of ξ is justified but not yet derived within the A-series. The exponent -4 follows from $D_f = 3 - \xi$ and the four-dimensionality of the universe; $\kappa = 7$ emerges as the unique consistent solution of the e-p-μ system; the factor $4/3$ is independently confirmed by the Casimir effect. This derivation is in A015. What *still* lacks: a derivation of the order of magnitude from the three stipulations themselves, without recourse to measured masses as input. Until then, the quantitative side carries a measured anchor.

Fourth, $3+1$ **is an input.** It is marked as such and will not be silently upgraded to a derivation in any later document of the series. Whether this is maintained, the audit script checks.

3.9 Supersedes

This document subsumes: Doc. 003 (foundations), Doc. 008 (ξ and e), Doc. 009 (ξ-origin), Doc. 041 (parameter derivation), Doc. 042 (ξ parameter particles), Doc. 253 (ξ number space). Incorporated register entries: P33 (status of the magnitude), P34 (dimension structure), P35 (ξ^N triviality), R56 (Z_3 as stipulation; direction observation→symmetry, not the reverse).

3.10 Use of AI Tools

This document was created with the assistance of AI tools. The questions, the theoretical stipulations, and all substantive decisions come from the author; the tools formulate, compute, check for contradiction, and create verification scripts. They make no substantive decisions. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 4

A030 · Compact Geometry T^4/Z_3

4.1 Purpose

A020 stipulated the space T^4/Z_3. Here we develop what this stipulation contains – and, equally important, what it does *not* contain.

The distinction is practical. A large part of the objections to the theory target a picture it does not hold: that space is divided into cells, that there is a tiling, that discreteness comes from a partition. None of this is the case, and the reason is explained in this document.

4.2 Compactness and the Discrete Spectrum

[B] The only consequence of compactness that matters is this: the Laplace operator on a compact manifold has a discrete spectrum. The eigenvalues form a countable, upward unbounded sequence; between two adjacent eigenvalues there is nothing.

This is elementary spectral theory and no claim of FFGFT. What FFGFT claims is the assignment: that physical masses are precisely these eigenvalues, and not numbers inserted into the model from outside.

[B] It follows immediately that there is a smallest permitted step. A state cannot lie a little between two modes. This discreteness is spectral, not spatial – it comes from the boundarylessness and finiteness of the space, not from a partition of it.

4.3 Why a Torus and Not a Sphere

[B] The torus has a non-trivial fundamental group,

$$\pi_1(T^4) \cong \mathbb{Z}^4,$$

the sphere does not: $\pi_1(S^n) = 0$ for $n \geq 2$.

The difference is the entire reason for the choice. On the torus there are closed paths that cannot be contracted to a point, and every such path carries four integers – how many times it goes around in each of the four directions. These winding numbers are:

- **integer**: there is no half-winding;
- **coordinate-free**: they change under no reparametrisation;
- **topological**: hence robust against any continuous deformation.

[B] On a simply connected surface none of these quantities exists. Replacing the torus loses not a detail of the construction but the carrier of the entire information concept (A180) and the explanation of Bell correlations as topological connection (A160).

4.4 What the Elementary Objects Are

[STIPULATION] The elementary discrete objects of the theory are not spatial regions but invariants: winding classes and eigenmodes.

This sentence answers an entire family of objections at once and therefore deserves an explicit counterpart.

Against a space-cell ontology the usual objections are correct: where does the tiling originate? Why this orientation? What happens under a coordinate change? How can an entropy depending on an arbitrary partition be physical?

[B] These objections run empty for a spectral-topological individuation. A winding class has no location – it is a property of a closed path, not of a region. An eigenmode has no origin – it is global. An entropy formed as a function of the spectrum is automatically invariant under origin, orientation, and coordinate choice, because the spectrum is.

[B] The torus is moreover boundaryless, $\partial T^4 = \emptyset$. This also excludes a surface-based definition of fundamental units: there is no distinguished surface. A surface in this geometry can only be a projection surface, and projection is lossy (A090).

4.5 The Four Circles

[B] Four circles, not five: because $\tilde{T} \cdot m = 1$ binds time to mass, time needs no circle of its own. The minimum number is therefore four, and in this sense the four is derived, not chosen.

In the compact picture the four circles are *equivalent*. There is no space-time distinction as long as the space is compact – time is written just like the three other directions. The familiar asymmetry between space and time arises only when one circle is unrolled, and that is a measurement process, not a geometric one (A090).

[STIPULATION] That the experienced structure is $3 + 1$ remains empirical input (register P34). The geometry says that exactly one circle is unrolled; it does not say which one, and it does not say why it is one.

4.6 The Z_3 Identification

[STIPULATION] A threefold identification acts on the torus. The quotient T^4/Z_3 is no longer a torus but an orbifold: it has points that remain fixed under the group action.

[B] The practical consequence is that every field quantity decomposes into three sectors that differ by the same phase under the identification. On a three-element fibre such an operator becomes cyclic, and a cyclic operator is diagonalised by the discrete Fourier transform. This is the structural reason why the mass sector later manages without a characteristic polynomial (A110).

[B] On admissibility: the crystallographic restriction $2\cos(2\pi/n) \in \mathbb{Z}$ permits $n \in \{1, 2, 3, 4, 6\}$ and excludes $n = 5$. Threefold symmetry is lattice-compatible, fivefold symmetry is not. This does not justify three – it constrains it.

4.7 Verification Scripts

- Integrality and coordinate-independence of the winding number, shown on a closed path on T^2 under reparametrisation and continuous deformation: `python/A_Serie_Skripte/a030_windungszahl.py`
- Discreteness of the spectrum on the compact torus versus the continuous spectrum in the non-compact case, including smallest permitted step: `python/A_Serie_Skripte/a030_spektrum_diskret.py`

4.8 Open Questions

First, the geometry does not say which circle is unrolled. The selection is deferred to the measurement process (A090). That is a deferral, not a result – as long as what determines the measurement process is not specified, the question is deferred and not answered.

Second, the number of fixed points of the orbifold is not evaluated here. The Z_3 action generates singular points; what physical role they play is not treated in this series.

Third, the assignment eigenvalue-mass remains a claim. That a compact space has a discrete spectrum is mathematics. That *this* spectrum is the particle masses is the physical claim of the theory and is tested against numbers only in A100 and A110.

4.9 Supersedes

This document subsumes: Doc. 181 (torus justification), Doc. 188 (geometry fundamentals), Doc. 189 (torus derivation), Doc. 159

(harmonic structure), Doc. 171 (string/torus), Doc. 204 (fractal boundary condition), Doc. 302 (unit cells and partition invariance). Incorporated: P34, R56 (crystallographic restriction constrains $n = 3$, does not justify it).

4.10 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 5

A040 · Fractal Correction $D_f = 3 - \xi$

5.1 Purpose

The geometry of A030 is described as smooth, but not intended to be smooth. FFGFT assumes that the effective structure carries a dimension deficit: it does not behave like an exactly three-dimensional object, but like one with

$$D_f = 3 - \xi.$$

This document states what follows from this, and – the actual purpose – traces the correction factor K_{frak} back to a purely internal reason.

Why both forms appear here: the closure comma gives the *internal* reading of the factor 100, the scale flow its *derivation* (below). Both describe the same number; the first shows where it sits in the geometry, the second why it is exactly 100.

5.2 The Conformal Form of the Correction

[STIPULATION] The correction acts conformally on the metric:

$$g_{\mu\nu}^{\text{(frak)}} = \eta_{\mu\nu}\left(1 + \xi f\right),$$

with a bounded function f on the torus.

[B] Two properties follow immediately from this form.

First, the intervention is *metric*, not topological. A bounded, ξ-small modulation of the prefactor does not tear a manifold; it changes distances, not connections. Winding numbers and spectral structure remain untouched. This is why the fractal correction leaves everything from A030 in place.

Second, the correction is *small of order* ξ. In the limit $\xi \to 0$ it vanishes and the smooth description is exact. The idealised smooth structure is therefore not wrong, but the limiting case.

[B] Explicitly: fractal here does not mean discrete. The structure remains an uncountable continuum; it is simply not smoothly affine. One who concludes from $D_f < 3$ to a granularity in the sense of smallest building blocks has used a different concept.

5.3 The Correction Factor from the Closure Comma

[B] The recursion (A050) closes in the frozen case after exactly 75 circuits, where

$$75 = \frac{1}{100\,\xi}, \qquad \xi = \frac{4}{30000}.$$

Requiring that 75 rotations each shortened by ε close *exactly*, i.e.

$$75\,(1-\varepsilon) = 74,$$

it follows that

$$\varepsilon = \frac{1}{75} = 100\,\xi,$$

and therefore

$$K_{\text{frak}} = 1 - 100\,\xi = \frac{74}{75} = 0.98\overline{6}.$$

This is the internal form: the correction factor is the ratio between the unrolled and the projected rotation. It is not a fitted value but the closure comma of a winding that falls one rotation short.

The closure comma re-expresses 100ξ; it is the *internal* reading of the factor, not its derivation. The actual derivation of the factor 100 runs through the scale flow.

5.4 Why the Factor 100

[K] The factor 100 is not stipulated but follows from the cumulative action of the sub-dimensionality $D_f = 3 - \xi$ over the entire renormalisation group run. The local deviation from dimension

3 is tiny ($\sim 0.003\,\%$); it becomes physically effective only *cumulatively* over the path from the Planck to the electroweak scale:

$$\ln\left(\frac{\ell_{\mathrm{EW}}}{\ell_P}\right) \approx \ln\left(10^{17}\right) \approx 39.$$

From this run and geometric averaging over scales, an effective amplification factor of around 100 emerges: the cumulative fractal correction is not ξ, but $100\xi \approx 1.3\,\%$.

[K] The equivalent power form confirms this:

$$K_{\mathrm{frak}} = \left(\frac{D_f^{\mathrm{eff}}}{3}\right)^{D_f^{\mathrm{eff}}/2} = 1 - 100\xi, \qquad D_f^{\mathrm{eff}} \approx 2.973.$$

The effective dimension $D_f^{\mathrm{eff}} \approx 2.973$ is *not* the local $D_f^{\mathrm{space}} = 2.9999$; it describes the cumulative action over the RG run. Substituting gives 0.98665 versus $1 - 100\xi = 0.98667$.

[K] And D_f^{eff} itself is uniquely determined, not free: it is the only value for which two independent derivations of the ratio m_e/m_μ – the fractal and the directly-geometric ($5\sqrt{3}/18 \cdot 10^{-2}$) – agree. From this consistency condition $D_f^{\mathrm{eff}} \approx 2.973$ follows, and thereby $K_{\mathrm{frak}} = 1 - 100\xi$ necessarily. The factor 100 is therefore a derived quantity.

5.5 Where the Factor Appears and Where It Does Not

[B] K_{frak} concerns exclusively *absolute* values, never ratios. A ratio of two quantities at the same level is unaffected by the correction, because it cancels.

This is a sharp and testable separation, and simultaneously the most important statement of this document for everything that follows: the dimensionless ratios of the theory are independent of the fractal correction. Deviations attributable to K_{frak} can therefore only appear in the conversion to absolute values (A080), not in the geometric core.

[B] The value $74/75$ is moreover not only derived but independently *confirmed*. In the fine-structure sector (A130) there are two separate paths to the characteristic energy E_0, neither of which uses K_{frak}; their difference is exactly $K_{\text{frak}}^{-1/2}$ and pins the factor to around 10^{-4} at the same value that follows here from the scale run. Two derivations that know nothing of each other hit the same number – this is the strongest support for $74/75$ not being freely chosen.

5.6 The Limit of Resolution

[B] The dimension deficit is a deficit, not a volume-preserving ripple. The difference is computationally accessible: for a volume-preserving modulation, the mean of f over the torus vanishes; for a deficit, it does not.

This decides with what power of ξ the correction enters first-order perturbation calculations – with nonvanishing mean as ξ^1, otherwise only as ξ^2. The verification script computes both cases and shows the exponents numerically.

5.7 Verification Scripts

- Closure comma: $75(1-\varepsilon) = 74 \Rightarrow \varepsilon = 1/75 = 100\xi$, with explicit countercheck that the computation assumes rather than produces the 100: `python/A_Serie_Skripte/a040_schliessungskomma.py`
- Deficit versus volume-preserving ripple: numerical determination of the ξ-power in both cases: `python/A_Serie_Skripte/a040_dimensionsdefizit.py`

5.8 Open Questions

First, the closure comma shows consistency, not necessity. The cumulative RG run derives $K_{\text{frak}} = 1 - 100\xi$ from the accumulated effect of $D_f = 3 - \xi$ over 17 orders of magnitude; the factor 100 emerges as a logarithmic scale ratio. The closure comma ($1 - 100\xi$ hits exactly the empirical value) confirms the construction but does not prove it: a geometric derivation showing that exactly 100 follows and no other factor still stands. This remainder is explicitly smaller than the point as previously formulated.

Second, f is not determined. The form of the correction is stipulated, the function itself is not. Everything depending on details of f – notably the phase of its harmonics – is thereby open.

Third, K_{frak} is a single, generation-independent factor. There are indications that the correction should act generation-dependently on the mass ladder; a constant factor does not close these deviations. The point belongs substantively in A100 and is treated there.

5.9 Supersedes

This document subsumes: Doc. 132 (fractal duality), Doc. 133 (fractal correction derivation), Doc. 300 (two roles of K_{frak}), Doc. 210 (winding-number corrections, insofar as they concern the factor). Incorporated: K1–K6 from the correction register, insofar as they concern the correction form.

5.10 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 6

A050 · Recursion and Non-Closure

6.1 The Recursion

[STIPULATION] The scale structure of the theory is generated by the map

$$\xi_{n+1} = \xi_n \left(1 - 100\,\xi_n\right)$$

starting from $\xi_0 = 4/30000$.

This is a damped feedback: the factor $(1 - 100\xi_n)$ is less than one, the sequence falls monotonically. In communications engineering terms it is a loop with gain below one – it contracts, it does not oscillate. The gain is not adjustable: it depends solely on ξ, which is stipulated. There is no tuning knob in this loop.

6.2 The Non-Closure Theorem

[K] The recursion has exactly one fixed point, namely zero. Every value $c \in (0, 1)$ that the sequence assumes is a transit point, not a target.

The proof is elementary: from $\xi = \xi(1 - 100\xi)$ it follows that $100\xi^2 = 0$, hence $\xi = 0$. And because the sequence is strictly monotonically decreasing but stays positive, it will not revisit any of its values in finitely many steps and will not reach zero in any finite step.

[K] From this follows a statement that is easy to overlook and yet load-bearing: *from the recursion alone, no individual numerical value can be singled out*. For every target c there exists a depth $k^*(c)$ at which the sequence comes close to it – equally so for every c, with identical parameter status. One who tries to justify a number by claiming the recursion passes through it has justified nothing.

[K] This is a result, not a gap. It closes an entire class of derivation attempts: the question of whether a specific constant can be obtained from the recursion depth is *answered*, and the answer is no. What remains open is only whether there is an anchor outside the recursion.

6.3 The Frozen Winding

[B] Holding ξ fixed instead of letting it run, the winding closes after exactly

$$\frac{1}{100\,\xi_0} = 75$$

circuits. The rotation number is then $74/75$, and the missing piece is the closure comma of A040.

This is the *rolled-up*, compact reading: one scale, one closed winding, an integer number of circuits.

6.4 The Unrolled Reading: A Logarithmic Spiral

[K] Letting ξ run, the sequence decays asymptotically as

$$\xi_n \approx \frac{1}{100\,(n+75)},$$

and the cumulative deficit becomes a harmonic sum, hence logarithmic:

$$D(k) \approx \ln\!\left(1 + \frac{k}{75}\right).$$

[K] Setting $\rho = k + 75$ and $\beta = 2\pi D$, we get

$$\rho = 75\, e^{\beta/2\pi},$$

a logarithmic spiral with self-similarity ratio e.

Two points, both important.

The ratio e is *not an additional parameter*. It is the base of the natural logarithm and comes from the harmonic sum – it is not chosen, it falls out. The pole at $n = -75 = -1/(100\xi_0)$ is likewise not fitted but fixed by ξ_0.

The scale invariance is *continuous*, not discrete log-periodic. There are no distinguished steps, no preferred multiples.

[B] Rolled up and unrolled are two readings of the same object, not two objects: the closed 75-winding is the compact single-scale view, the spiral the unrolled across-scales view.

6.5 Why the Recursion Does Not Belong on a Single Axis

[B] The formulation that time rolls up suggests that the time axis is distinguished. It is not. Through $\tilde{T} \cdot m = 1$, every statement about time has a mirror image about mass: if time advances by K_{frak} per step, mass advances by $1/K_{\text{frak}}$. The deficit is the same $100\xi_n$ on both sides.

[B] The factor 100 therefore does not migrate between the axes – it is mirrored. Both projections lead to the same ratio e. The assignment to time is a convention, not a distinction.

6.6 Verification Scripts

- Fixed-point theorem and non-closure: unique fixed point 0, monotonicity, and the construction of $k^*(c)$ for arbitrary targets – as evidence that no value is distinguished: `python/A_Serie_Skripte/a050_nichtschliessung.py`
- Spiral ratio: numerical determination of the ratio from the cumulative deficit, comparison with e, and check for continuous rather than log-periodic scale invariance: `python/A_Serie_Skripte/a050_spirale_e.py`

6.7 Open Questions

First, the recursion depth is not fixed. The theory gives no rule at which k a physical quantity is to be read off. By the non-closure theorem it cannot – this is not a negligence, but proved.

Second, this implies a hard limit for all fine values. Every quantity whose justification would amount to the recursion reaching it is thereby *not* justified. This applies notably to the fine value of the refraction strength δ; it remains open and is booked in A230.

Third, the recursion uses the factor 100, which is derived in A040 (cumulative RG run, consistency condition; A040). It is used here, not re-derived. The non-closure holds independently of this – it follows from the form of the map, not from the numerical value – but the concrete number 75 does not.

6.8 Supersedes

This document subsumes: Doc. 203 (R-operator), Doc. 275 (ξ-recursion), Doc. 295 (time as logarithmic spiral). Incorporated: P37 (three roles of φ separated), P35 (triviality of the power representation).

6.9 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 7

A060 · Time: Duality and Base Tick

7.1 Purpose

[STIPULATION] In FFGFT, time is not an external stage but part of the geometry. From the time-mass duality $\tilde{T} \cdot m = 1$ (A020) it follows in natural units that the native time-energy reciprocity $T \cdot E = 1$ holds – an *equation*, not an inequality. This document develops four things from this: the *location* of time (inside the state space, not beside it), the *reciprocity* and its limits (smallest time increment, black hole as extremum, no empty state), the *fundamental beat* (the permitted modes are commensurate multiples of a single ξ-fixed increment), and the *arrow of time* (geometric, not primarily thermodynamic). The cosmically large sector (scales beyond particle physics) is not treated here (A010); it requires its own independently stipulated scale – why, is explained below.

7.2 The Fork: Time Inside or Beside the State Space

[B] At the start of every quantum-mechanical modelling stands a mostly unstated choice: *Is time a direction inside the state space, or a parameter beside it?* The question is not technical but ontological. Both answers work with the same formalism – self-adjoint operators, spectra, projectors –; the difference lies solely in *where* time sits. And that exactly decides whether physical quantities *fall out* of the structure or must be *attached* to it.

[B] The question reaches back to the roots of quantum theory. Already in 1932 Pauli recognised that a time operator canonically conjugate to a Hamiltonian bounded below cannot exist (Pauli's theorem); this made time as an external parameter seem incontrovertible. Later work (Aharonov–Bohm, Garrison–Wong) showed that under certain conditions – e.g. for non-semi-bounded spectra – time operators can nonetheless be constructed. The dispute remains unsettled and forms the background to the alternative.

7.3 The Unrolled View and Its Price

[B] The familiar formulation places time outside the state space: $\mathscr{H} = L^2(\text{configuration space})$ is the space of states *at one instant*, time the parameter t of the unitary evolution $e^{-i\hat{H}t}$. Observables – position, momentum, spin – are operators *on* $\mathscr{H}$; time is not one (Pauli). The consequence is structural: because time stands outside, the state space alone contains no dynamical scales. Masses, couplings, lifetimes are absent from the pure spatial structure – they are introduced via additional structures, in the Standard Model through Yukawa terms with free parameters. The structure supplies the framework, the numbers come from measurement.

[S] This criticism is old: Weyl and Wigner already saw masses group-theoretically as Casimir invariants – structural labels, not attached parameters –; but the origin of the mass scales remained open. The point is not that the unrolled view is wrong, but that it leaves this question unanswered.

[S] Why attaching is a price. Many programmes with geometric ambition follow a common scheme: (1) a discrete carrier (lattice, cut-and-project structure, periodic graph); (2) a self-adjoint operator family on it; (3) spectral projectors extract sectors; (4) these sectors are *subsequently* assigned meaning – mass often via a parametric map with one or two free parameters. Such a procedure can be mathematically fully controlled; what matters is its *location of time*: the carrier is spatial, the sectors are spatial eigenspaces, not time modes – the mass is not contained but attached. And here an argument applies that is self-supporting: a parametric map with free parameters can hit any finite list of values. If it is not fixed in advance and universally, it merely renames known inputs. To deliver numbers it must be coupled to external values – to measurement, and hence to the Standard Model. An example from established physics: graph-like modelling defines quasiparticles on hexagonal lattices with linear dispersion (massless fermions); a mass always requires additional terms (Haldane, Kekulé) with free parameters. The geometry supplies the kinematics, not the mass scale. The *difficulty* of this path is no coincidence but the direct consequence of the unrolled location of time.

7.4 The Rolled-Up View: Time as a Compact Dimension

[B] The alternative does not leave time outside and does not eliminate it, but draws it *into* the state space – as a closed, periodic dimension. In place of $L^2(\text{space})$ with external t stands a space in which a compact time direction is part of the geometry; in the corpus this is $\mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3$ (A070), one torus direction carries the time. On a closed dimension, the permitted states are *standing modes with discrete winding numbers*: quantisation is not postulated but geometrically forced. Mass becomes the *eigenvalue of an operator on $\mathscr{H}$ itself* – not attached but structurally contained. The mass ratios are fixed because the permitted modes are fixed.

[B] Concretely, one expands a function on a compact time direction of length T in a Fourier series,

$$\Psi(t, \mathbf{x}) = \sum_{n\in\mathbb{Z}} e^{int/T} \psi_n(\mathbf{x}),$$

the discrete frequencies $\omega_n = 2\pi n/T$ are (in natural units) the mass values, generated by the operator $\hat{E} = -i\,\partial_t$ with spectrum $\{2\pi n/T\}$. Mass quantisation is a direct consequence of periodicity – nothing needs to be postulated. The couplings between modes,

$$g_{ijk} \propto \int e^{i(n_i-n_j-n_k)t/T}\, dt = T\,\delta_{n_i,\, n_j+n_k},$$

are exact selection rules for the mode indices – the “Yukawa couplings” appear as conservation rules, not free numbers.

[B] That a compact dimension generates a discrete mode tower with $m_n \propto n/R$ is the Kaluza–Klein mechanism – here for the rolled-up *time* rather than a spatial direction, with mass in the role of the mode index. And Pauli's objection does *not* apply to this view: it holds for an open, downward unbounded time; a compact time behaves like an angular variable, its conjugate spectrum is discrete (as angular momentum to angle), the Pauli argument does not apply. The closed time is the natural seat of a discrete mode structure.

7.5 Time-Mass Duality and Native Reciprocity

[B] The concrete realisation is $\tilde{T} \cdot m = 1$: every massive quantity *is* its own time cycle. In natural units ($E = m$) it follows that

$$\boxed{T \cdot E = 1}$$

This is an *equation*, not an inequality – a sharper statement than Heisenberg's $\Delta E \cdot \Delta t \geq 1/2$, and a native one. It replaces an earlier borrowed justification: the absence of a singular beginning was once argued in the corpus via Heisenberg's energy-time uncertainty. This justification is *replaced*, not refuted – the conclusion stands, the derivation becomes native. Three defects of the borrowed version:

- **Inverted implication.** From $\Delta E \geq 1/(2\Delta t)$, for *finite* Δt a *finite* bound follows, not $\Delta E \to \infty$; the latter follows only from $\Delta t \to 0$, i.e. from the instant $t = 0$, not from a finite beginning.
- **Application error.** By Pauli's theorem there is no self-adjoint time operator conjugate to a bounded Hamiltonian; the strict form (Mandelstam–Tamm) defines Δt as the characteristic *change time of an observable*, not an elapsed duration. An "age" is not a valid Δt of this relation.
- **Style violation.** "Proves irrefutably" is a claim strength that the claim ledger does not know; it knows *derived from* ξ, *restricted*, *blocked*.

[B] The overarching point: Heisenberg's relation is not wrong in the QM framework, but *projected* – it belongs to the description level that FFGFT understands as a projection from the deeper structure (A090). An argument resting on it does not leave the projection but circulates within it.

7.6 The Smallest Increment

[K] FFGFT's time is the cumulative deficit of the unrolled winding,

$$D(k) = \sum_{n\leq k} 100\,\xi_n \;\to\; \ln\!\left(1 + \tfrac{k}{75}\right), \qquad \xi_{n+1} = \xi_n\,(1 - 100\,\xi_n),$$

with the first step

$$d_1 = 100\,\xi_1 = 100 \cdot \tfrac{4}{30000} = \tfrac{1}{75}.$$

$D(k)$ grows without bound for $k \to \infty$ (logarithmically); the decisive statement is not "D never diverges" but: $D(k)$ is finite at every finite point, and no time increment becomes smaller than d_1. From $T \cdot E = 1$ it follows that $E \to \infty \Leftrightarrow T \to 0$; a state with $T = 0$ does not exist, because time is bounded below by the finite, ξ-fixed increment. Hence E is bounded above – an infinite energy density is *structurally* excluded, not by a statement about measurement precision.

7.7 Both Limits: Black Hole as Extremum, Thinnest State

[K] The same chain determines the black hole – and the result is unspectacular: nothing diverges. From $T \cdot E = 1$ with $T \geq \tau_0$ it follows that $E \leq 1/\tau_0$. The extrema are calculated from ξ: with $L_0 = \xi \ell_P$ (A210) the energy-length reciprocity gives

$$E_{\text{max}} = \frac{\hbar c}{L_0} = \frac{E_P}{\xi},$$

$$\tau_0 = \xi t_P \quad \text{(minimal time scale; } c{=}1\text{: same size as } L_0\text{)},$$

and the reciprocity closes *exactly*, because ξ cancels:

$$\tau_0 \cdot E_{\text{max}} = (\xi t_P) \cdot \frac{E_P}{\xi} = t_P \cdot E_P = 1.$$

The same cutoff appears in field theory as the natural UV cutoff at E_P/ξ: the geometry provides the cutoff, a subsequent regulator is not needed (A190).

[K] Scales, not sums. τ_0 and E_{max} are scales, not bounds on totals: τ_0 is the minimal time *increment* of the winding, E_{max} the maximum energy *per excitation* (a cutoff that always bounds a single mode, never a sum). The total energy of a composite system may be arbitrarily large – it consists of many cells, not of a denser one; "the T of the Earth" is not a well-posed quantity. The horizon is accordingly the location at which the *local* density reaches the cell cutoff: slowest time $T = \tau_0$ (not $T \to 0$), maximum concentration $E = 1/\tau_0$ per cell, maximum curvature – all finite and well-defined. It is not a curtain before a singularity but the surface on which time takes its minimum and concentration its maximum. In FFGFT there is therefore *nothing to repair*: models with a bounce or a transition shell fix a defect that FFGFT does not produce.

[S] The other end. $T \cdot m = 1$ has no solution at $m = 0$ – FFGFT has no *empty* state, only a thinnest one. The apparent circularity "vacuum is empty, acts only through curvature, curvature presupposes mass" arises from a causal arrow of general relativity ($T_{\mu\nu}$ *generates* curvature). FFGFT does not have this arrow: $T \cdot m = 1$ is not a cause-effect relation but a reciprocity – mass and temporal curvature are two readings of the same structure, neither generates the other. The circularity dissolves. "Empty" would mean "no winding", and the winding *is* what time is.

[S] The asymmetry, honestly named. The two limits are not equally well determined. The *dense* pole is hard and derived: τ_0 and $L_0 = \xi \ell_P$ follow from ξ. The *thin* pole is structurally excluded, but *uncalculated*: that $m = 0$ is impossible follows from the relation; *how small* m can become minimally does not – and for a structural reason. From $T \cdot m = 1$ every system carries its own upper scale $R_{\text{torus}}(m) = \hbar/mc$; a universal maximum size is not sensibly demanded. For systems beyond the particle scale, a separate, independently stipulated upper bound would therefore have to be introduced – this is not a gap but a consequence of the relation, and it lies outside this document (A010).

7.8 The Boundary Question

[B] The torus is compact and boundaryless, $\partial T^4 = \varnothing$ (A090). The question "what was at $t = 0$?" presupposes a boundary at which a beginning could sit; a compact torus has none. The question is therefore not wrongly *answered*, but wrongly *posed*: it imports the decompactified projection as ontology. What an unrolled description reads as a singular beginning is the point at which the *projection* ends – not the world.

7.9 The Projection Insight

[S] It is natural to arrange systems on a common axis – from maximum to minimum mass concentration – and to find oneself in the middle. But this axis is an *outside view*: it presupposes a standpoint from which all systems appear side by side. No such standpoint exists in FFGFT. $T \cdot m = 1$ holds *per system*; there is no common T against which to plot. More precisely: we *project ourselves into the middle* – the axis is created with the observer as its zero point, and the middle is not a discovery but a stipulation that goes unnoticed because the one stipulating and the one located are the same. The limits appear as "extreme" because they are far from the chosen zero point: *extreme* means *distant*, not *distinguished*.

[S] The intrinsic time is unaffected by this. For an electron it is irrelevant where an outsider places it on the axis: its winding is the same, because $T = 1/m$ and m is the same. What differs is only how another system sees it – a relation between two, not a property of one. The "slowest time" at the horizon too is a statement about comparison, not about experience. The projection is thereby not invalid but a *tool*: without a common axis, nothing could be compared. It is the same move that resolved the question about $t = 0$ – now applied to the scale itself.

7.10 The Fundamental Beat

[K] The permitted states are discrete modes, and they stand in fixed relation to one another – their ratios are exact. From this follows a common *fundamental beat*: through $\tilde{T} \cdot m = 1$ every massive quantity carries its own cycle $T = \hbar/mc^2$, but these cycles are not independent clocks but commensurate multiples of the same ξ-fixed increment $d_1 = 1/75$ – *harmonics of a fundamental beat*, not a continuum of free clocks. This is the precise formulation: not a distinguished base unit against which everything would be measured, but a common beat whose multiples are the modes.

7.11 The Arrow of Time

[S] Why does time pass in one direction? In FFGFT the time direction is *geometric*: it follows from the orientation of the winding on the compact time direction, not from thermodynamics. The thermodynamic arrow of time is *secondary* – an emergent property of the non-compact spatial components, not a fundamental given. This explains the apparent irreversibility of macroscopic physics without resorting to ad hoc assumptions about initial conditions. And the time direction is *scale-dependent*: at the fundamental level it is geometrically fixed, at the macroscopic level it appears as a statistical gradient. Both levels do not contradict each other – one is the projection of the other.

7.12 Granulation and Fundamental Asymmetry

[S] The deeper reason why time "passes" at all is a *fundamental asymmetry* as the principle of motion: the winding is not symmetric, and the deviation – the deficit $100\,\xi_n$ per step – is the motor that generates time. Reality is thereby *granular*: below the fundamental scale there is no continuum but discrete increments; only above certain scales does time *look* continuous, because the granulation $d_1 = 1/75$ vanishes relative to macroscopic times. This is why practical physics correctly computes with continuous time, even though the fundamental structure is discrete: continuity is a good approximation, not a basic property.

7.13 Who Has Already Invoked "Time Not External"

[S] The idea of not treating time as an external parameter is not a novelty but a long-pursued line – and it is honest to locate the rolled-up view within it and to demarcate from it. **Relational time:** Page & Wootters let time arise *within* the state space, as entanglement between a clock subsystem and the rest ("evolution without evolution"); Moreva et al. and Marletto & Vedral refined and illustrated this; Gambini & Pullin developed the Montevideo interpretation. **Timeless quantum gravity:** Wheeler–DeWitt ($\hat{H}\Psi = 0$, the "problem of time"); Kuchař and Isham provided overviews; Rovelli (relational QM, "Forget Time", with Connes the thermal-time hypothesis); Barbour (*The End of Time*). **Compact time:** investigated in string theory; Bars developed a "two-time physics". **Modes as structural eigenvalue:** Wigner (mass as Casimir invariant, a label of irreducible Poincaré representations); Kaluza/Klein (one compactified dimension generates a discrete mode tower with quantised charges).

[S] The demarcation is precise: the overwhelming part of this tradition makes time *relational*, *emergent*, or *timeless* – it *removes* time as a fundamental. The rolled-up view does something subtly different: it *includes* time – as a compact geometric dimension carrying modes. The novelty lies not in "time is not external" (that is old and well established) but in the specific route "compact time generates mass modes".

7.14 The Absolute Scale: Fixed, Not Free

[B] The mass ratios follow from the mode structure and are exact; the absolute scale is thereby not yet expressed in SI numbers – but it is for this reason not a *free* parameter. The ratio scaffold rests on the one dimensionless $\xi = 4/30000$. Absolute values carry their correction not as a fit but via fixed *bridge constants*: v in $m = r\xi^p v$ (A100), E_0 in $\alpha = \xi E_0^2$ (A130), fixed factors in G. In ratios these cancel ($m_\mu/m_e = 206.768$, correction-free). $\hbar$ and c are pure SI conversion; a *measured* dimensional anchor in the standard sense does not exist.

[K] The Planck quantities are not a fundamental input but follow from ξ: the chain is $\xi \to \{m_e, \dots\} \to G \to \ell_P$. The *only* exact ξ-power relation is

$$\frac{L_0}{\ell_P} = \xi, \qquad L_0 = \xi \ell_P \approx 2.15 \times 10^{-39}\ \text{m};$$

the SI factor ℓ_P must *not* be absorbed into a ξ exponent. The causal direction is therefore reversed: the Planck quantities follow from ξ, not ξ from them. Through $\tilde{T} \cdot m = 1$ the cycle is fixed once the mass is fixed by the ξ-geometry – no gauge modulus remains open.

7.15 Causality: Compact Time Is Spectrally, Not Causally, Closed

[B] The obvious concern – does a periodic time generate closed timelike curves? – does not apply. The carrier is a torus ($\pi_1 \neq 0$, stable windings), not a simple circle; the relevant connection is not ontological nonlocality but a *topological* connection on T^4 (A070). The classical CTC problem (Gödel solutions, chronology protection) concerns the *decompactified* Lorentz sector; the compact mass-time direction is *spectrally* closed, not causally.

7.16 Verification Scripts

- Native reciprocity and extrema: numerical verification of $T \cdot E = 1$, the exact closure $\tau_0 E_{\text{max}} = t_P E_P = 1$, and $d_1 = 100\xi_1 = 1/75$: `python/A_Serie_Skripte/a060_reziprozitaet.py`
- Fundamental beat: that the mode cycles are commensurate multiples of d_1 (integer ratios), not independent clocks: `python/A_Serie_Skripte/a060_grundtakt.py`

7.17 Open Questions

First, the length of the compact time direction is not given as a single SI number. That the ratios are exact and the absolute scale is fixed (not free) via bridge constants is the current state; a measured dimensional anchor in the standard sense does not exist.

Second, the *quantum equivalence* of the unrolled and rolled-up views is known, but its methodological consequences are only systematised, not conclusively evaluated. That both use the same formalism is established; which reading is to be preferred remains a stipulation with an ontological price.

Third, the thin pole is structurally excluded but uncalculated. For systems beyond the particle scale the relation demands its own independently stipulated upper bound; its development lies outside this document.

7.18 Supersedes

This document subsumes: Doc. 010 (unit convention, time-energy duality, universal field equation, Lagrange density, characteristic lengths and energy, scale hierarchy), Doc. 069 (time-energy duality, geometric rest-mass definition), Doc. 078 (granulation, fundamental asymmetry, time structure), Doc. 157 (arrow of time), Doc. 296 (time, openness, invariance; observer and projection section), Doc. 306 (native time-energy reciprocity), and Doc. 307 (time in state space, fundamental beat). Incorporated: R50 (the Heisenberg-based singularity argument from Doc. 025/063 is replaced by the native chain $T \cdot E = 1$ – conclusion unchanged, derivation native) and R54 (no N_0 as base unit; modes are commensurate multiples of a common fundamental beat). *Not* taken over: the cosmically large sector (the static ξ-cosmos and background radiation sections from Doc. 069, the speculative sections from Doc. 296 §5, R46) and the scale notation $E_H = E_0\xi^{41/4}$ (calibrated to an externally measured quantity, not a derivation – P35); this area is not treated in the A-series (A010) and requires its own scale. The old $T \cdot m = 1$

singularity proof from Doc. 078 is superseded by the native chain (R50).

7.19 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 8

A070 · Hilbert Space Bridge

8.1 Purpose

[STIPULATION] This document formulates the *complete translatability* between the FFGFT mode formalism and standard Hilbert-space quantum mechanics. The translation is not an approximation and not a specialisation, but a **bijective structural statement**: every QM computation is mappable into FFGFT language, and every FFGFT statement about quantum systems is reproducible in Hilbert-space language.

Layer note. Alongside the algebraic bridges [B] and geometric core statements [K], this document contains a substantial proportion of conceptual and ontological framings [S]. The [S] statements concern interpretational questions (Born rule, determinism, Bloch-sphere position) – they are correctly marked, but the reader should know: the ratio here is K : S $\approx$ 1 : 5. The established core is narrow; the rest is framing. That is not a deficiency – framings belong in a bridge document. But bridge and core are not the same.

This is the methodological point: FFGFT does not *replace* quantum mechanics but *extends* it. Both formalisms deliver the same measurable results – up to a ξ-correction of order 10^{-5}, which currently lies below the NISQ noise floor (A160). They differ only in the ontological reading of some basic concepts. One who computes with Hilbert-space tools can continue to do so; the translation table below is the bridge.

The canonical Hilbert space of the full theory is

$$\boxed{\mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3}$$

– square-integrable functions on the 4-torus, tensored with the three-element Z_3 fibre space (A030). This document builds it in two directions: *downward* (FFGFT → standard Hilbert space, with reductions) and *upward* (which extensions the standard Hilbert space needs to natively carry the full FFGFT).

8.2 Where We Are Geometrically

[B] Before the translation tables, the geometric picture. The full FFGFT geometry is the 4-torus $T^4 = T^3 \times S^1$ (A030): three spatial winding directions and one temporal. For a single qubit – a two-level system at fixed time, without mass dynamics – most of these directions are inactive. Two rotational degrees of freedom remain: one for the position between $|0\rangle$ and $|1\rangle$ (this becomes z), one for the phase (this becomes θ). Their natural space is the 2-torus $T^2 = S^1 \times S^1$.

Pulling one major radius large, T^2 becomes locally the cylinder $\mathbb{R} \times S^1$: one loop is cut, z becomes a *linear* axis. Then contracting the two poles to points, the cylinder becomes the Bloch sphere S^2 of standard QM. This creates a hierarchy of reductions:

Geometry	Dim.	Use
4-torus T^4	4	full FFGFT (masses, α)
3-torus T^3	3	mode space at fixed time
2-torus T^2	2	single qubit, without reduction
Cylinder $\mathbb{R} \times S^1$	2	limit $R \to \infty$, local approximation
Bloch sphere S^2	2	poles contracted, standard QM

[B] What the reductions cost – and where the loss really sits. Each level loses something: $T^4 \to T^3$ the time winding and hence the masses; $T^3 \to T^2$ a spatial winding and hence the third generation; $T^2 \to$ cylinder *one compact loop* – the weakest reduction, and precisely here sits the fractal correction factor $K_{\text{frak}} = 1 - 100\xi \approx 0.9867$ (A040): K_{frak} is the cylinder's memory of its torus origin. The loss is *not* the subsequent SI conversion: the duality-carrying decompactification $S^1_m \to \mathbb{R}_t$ ($\tilde{T} \cdot m = 1$, A060/A090) is a mode-preserving embedding and lossless; compactness survives as a discrete spectrum and as a physical observable (revival time, CHSH$\sim \xi$). The lossy operations are the *reverse direction* $\mathbb{R} \to S^1$, the spatial projections, and – for a readout – the finite, non-periodic measurement window.

8.3 States: the Bijective Bridge

[B] In standard QM a qubit is a vector in $\mathbb{C}^2$:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1, \qquad \alpha, \beta \in \mathbb{C}.$$

In FFGFT the same qubit is a point (z, r, θ) on the cylinder/torus:

$$z \in [-1, 1], \quad r \in [0, 1], \quad \theta \in [0, 2\pi), \qquad z^2 + r^2 = 1,$$

with z as projection onto the computation axis, r as radial superposition amplitude, θ as phase. Both representations are linked by a bridge:

$$\boxed{\alpha = \sqrt{\tfrac{1+z}{2}}\, e^{i\theta/2}, \qquad \beta = \sqrt{\tfrac{1-z}{2}}\, e^{-i\theta/2}}$$

and conversely $z = |\alpha|^2 - |\beta|^2$, $r = 2|\alpha\beta|$, $\theta = \arg(\alpha) - \arg(\beta)$. Consistency is immediate: $|\alpha|^2 + |\beta|^2 = 1$ and $z^2 + r^2 = 1$; $|0\rangle$ corresponds to $(1, 0, *)$, $|1\rangle$ to $(-1, 0, *)$.

[S] The conceptual difference: in QM $|\alpha|^2$ is a *probability* (Born rule), in FFGFT $r^2 = 1 - z^2$ is an *energy density* of the radial configuration. Numerically both coincide at large statistics; FFGFT is a deterministic geometric model whose statistical results agree with the Born rule.

8.4 Operators: Pauli Matrices as Rotations

[B] The three Pauli matrices generate all single-qubit operations, with algebra $\sigma_i\sigma_j = \delta_{ij}\mathbb{1} + i\epsilon_{ijk}\sigma_k$. On the Bloch cylinder they correspond to three geometric rotations: σ_z the phase rotation $\theta \to \theta + \pi$, σ_x a (z, r)-rotation by π (bit flip), σ_y the orthogonal rotation with phase flip.

Operator	Hilbert-space matrix	FFGFT geometry
σ_z	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	Phase rotation $\theta \to \theta + \pi$
σ_x	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	(z, r)-rotation by πK_{frak}
σ_y	$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$	Rotation with phase flip
H	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	swap $z \leftrightarrow r$, $\theta \to \theta + \pi/2$
T	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$	Phase rotation $\theta \to \theta + \pi/4$

[K] The *only* FFGFT-specific deviation is the factor $K_{\text{frak}} \approx 0.9867$ in the σ_x rotation. It follows from the fractal dimension $D_f = 3 - \xi$ (A040) and is a testable prediction: all π-gates deviate systematically by $\Delta \approx 0.013\,\%$ from the ideal value. In the Hilbert-space formalism this would be an ad hoc correction to be built in as hardware noise; in FFGFT it follows from the geometry.

8.5 Gates and Universality

[B] Every unitary single-qubit gate $U \in U(2)$ is a Bloch rotation,

$$U = e^{i\alpha} R_{\hat{n}}(\phi) = e^{i\alpha}\big(\cos(\phi/2)\,\mathbb{1} - i\sin(\phi/2)\,\hat{n}\cdot\vec{\sigma}\big),$$

in FFGFT the rotation of the (z, r, θ) point around axis $\hat{n}$ by ϕ. The CNOT gate translates as geometric conditioning: *if $z_A = -1$ (control qubit on $|1\rangle$), apply bit flip on qubit B* – topologically a linking of two cylinders (entanglement). Since the universal set $\{H, T, \text{CNOT}\}$ (Solovay–Kitaev) is fully geometrically representable, FFGFT is *universal for quantum computation*: there is no QM statement not formulable in FFGFT.

8.6 Tensor Products, Bell States, CHSH

[B] For n qubits the Hilbert space is $\mathscr{H}_n = \bigotimes_{i=1}^{n} \mathbb{C}^2$ with dimension 2^n. In FFGFT n qubits correspond to n mode configurations (z_i, r_i, θ_i) on a shared torus geometry; entanglement is a *topological connection* – knots not reducible to product form. The Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is a topologically connected cylinder configuration. Both formalisms reproduce the exponential growth dim $= 2^n$ with $2 \cdot 2^n - 2$ real parameters.

[K] CHSH prediction. Standard QM gives the Tsirelson bound $|\langle\text{CHSH}\rangle| \leq 2\sqrt{2}$. FFGFT predicts a small, geometrically derived deviation $\Delta\text{CHSH} \approx 10^{-5}$ – the *elementary* deviation per measurement. It lies below the current hardware noise floor and is an open task for high-precision Bell tests (A160/A210). Distinct from this is the cumulative value over many measurements; the two quantities are not directly comparable.

8.7 Measurement as Pole Transition

[S] In standard QM a measurement in the computation basis gives value 0 with probability $|\alpha|^2$, with collapse to $|0\rangle$ or $|1\rangle$. In FFGFT measurement is *not* a probabilistic collapse but a geometric interaction driving the (z, r, θ) point to the nearest stable extreme ($z = +1$ or $z = -1$) – a deterministic motion in mode space that is statistically indistinguishable from Born-rule behaviour because of the unknown phase θ. At large statistics:

$$P_{\text{FFGFT}}(z = +1) = \frac{1+z}{2} = |\alpha|^2 = P_{\text{QM}}(0).$$

The statistical results are identical; the Bell-test debate over local realism dissolves geometrically, because entanglement is represented as topological connection, not as non-local correlation.

8.8 Unitary Evolution: Schrödinger as Low-Energy Limit

[B] The QM time evolution $i\hbar\,\partial_t|\psi\rangle = H|\psi\rangle$ is in FFGFT the deterministic motion of the mode configuration $\varphi_{n,l,j}$ on $T^3 \times \mathbb{R}$. The FFGFT mode equation reduces in the low-energy limit ($E \ll E_\xi$) to the Schrödinger equation:

$$i\hbar\,\frac{\partial\varphi_{n,l,j}}{\partial t} \xrightarrow{E\ll E_\xi} -\frac{\hbar^2}{2m}\nabla^2\varphi_{n,l,j} + V\varphi_{n,l,j}.$$

At higher energies ξ-corrections appear, which in the Hilbert-space formalism appear as Hamiltonian modifications.

8.9 What Is Translatable and What Remains FFGFT-Specific

[B] *All* Hilbert-space statements are mappable into FFGFT: states, operators, gates (with or without K_{frak}), tensor products, entanglement, Bell/GHZ states, measurement, unitary evolution, algorithms (Shor, Grover, VQE). And conversely: the (z, r, θ) configuration corresponds uniquely to the (α, β) vector, geometric rotations to unitary operators, topological entanglements to entangled Hilbert-space vectors. Every computation in one formalism is executable in the other, with identical result.

[K] The translatability holds for *statements about quantum systems*, not for the *input values*. In Hilbert space ξ, the fermion masses, α, and K_{frak} appear as external inputs; in FFGFT they follow from the geometry (ξ as stipulation A020; the masses from $m_i = r_i\,\xi^{p_i}v$, A100/A110; α from A130; K_{frak} from A040). This is the structural asymmetry: Hilbert space has all the tools for QM computations, but no explanation for the input values; FFGFT delivers both from one geometry. In this sense QM is a subset of FFGFT – restricted to the question of *how* systems behave, without the question of *why* the constants have these values.

8.10 Operational Equivalence and Interpretative Divergence

[S] Complete translatability must not be read as ontological identity. *Operationally* FFGFT and standard QM agree in every measurable result, up to ξ-corrections. *Interpretatively* they diverge at points where the standard reading makes statements that in FFGFT appear as artefacts of the chosen carrier assumptions. Three points within QM:

- **Born rule as irreducible randomness.** Standard reading: $|\psi|^2$ as ontological probability, the outcome fundamentally random. FFGFT reading: same distribution, but from deterministic pole-transition dynamics on T^4; stochasticity arises at the measurement apparatus, not in the natural process. Operationally identical.
- **Non-locality as ontological property.** Standard reading of Bell correlations: action at a distance in $\mathbb{R}^3$. FFGFT reading: entanglement is a topological connection on T^4 – geometric proximity in a closed geometry, not action at a distance in open space (A160). Results identical, ontological reading structurally different.
- **$\hbar$ as ontological primitive.** Standard reading: $\hbar$ is a fundamental constant of nature. FFGFT reading: $\hbar$ is an SI conversion factor between energy and time measures, following from the closure condition on T^4 (A080/A085). Numerically identical, ontologically derived.

8.11 The Four Extensions: Hilbert Space Upward

[B] The downward translation loses a structural layer at each reduction. The reverse question is: which extensions must one *add* to the standard Hilbert space so that it natively carries the full FFGFT? Remarkably, each extension corresponds to an established mathematical structure – FFGFT invents no new mathematics, but is a specific combination of existing structures.

Extension 1 – deformed algebra $SU_q(2)$. The isotropic $SU(2)$ algebra $[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k$ becomes anisotropic in FFGFT: $\sigma_x^{\text{FFGFT}} = K_{\text{frak}}\,\sigma_x$, while σ_y, σ_z are undisturbed. This is the q-deformation (Drinfeld–Jimbo 1985) with

$$\boxed{q = e^{i\pi(1-K_{\text{frak}})} = e^{i\pi\cdot 100\xi} = e^{i\pi/75} \approx 0.999 + 0.042\,i}$$

The deformation phase is small ($\pi/75 \approx 0.042$ rad) but structurally significant: it breaks spherical symmetry to an axis-distinguished algebra – the cylinder's memory of the torus. Cost: the Hilbert space remains $\mathbb{C}^2$, the 2×2 matrices remain; only the structure constants change. Consequence: all π-gates deviate by $0.013\,\%$ (testable).

Extension 2 – cyclic winding quantum number. The cylinder has one compact loop; the 2-torus has two. To represent the second, the Hilbert space carries a quantum number $n \in \mathbb{Z}$: from $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ one gets $|\psi; n\rangle = \alpha|0; n\rangle + \beta|1; n\rangle$, and

$$\boxed{\mathscr{H}^{T^2}_{\text{FFGFT}} = \bigoplus_{n\in\mathbb{Z}} \mathbb{C}^2_n}$$

a countably infinite-dimensional space. The winding excitations have energy $E_n \sim n^2\hbar^2/(mR^2)$; at low energies only $n = 0$ is occupied and the space effectively reduces to $\mathbb{C}^2$ – the standard Bloch sphere.

Extension 3 – bundle structure instead of tensor product. In standard QM a spin-$\frac{1}{2}$ particle is $\mathscr{H} = L^2(\mathbb{R}^3)\otimes\mathbb{C}^2$ with $[L_i, S_j] = 0$; spin-orbit coupling $H_{\text{SO}} = \xi(r)\,\vec{L}\cdot\vec{S}$ must be introduced as an additional term with a free parameter. In FFGFT both position *and* spin arise from windings on the same T^3; the correct Hilbert space is *not* a tensor product but the section space of a spinor bundle,

$$\boxed{\mathscr{H}^{T^3}_{\text{FFGFT}} = L^2(T^3, S)}$$

with $[L_i, S_j] \neq 0$ (corrections $\sim \xi$). Spin-orbit coupling is thereby a *geometric* property of the bundle, not a free parameter.

Extension 4 – temporal winding k_t. In standard QM time is a continuous linear axis with $E = \hbar\omega$. In FFGFT it is a compact winding direction with $k_t \in \mathbb{Z}$ and

$$\boxed{E = \hbar f(k_t)}$$

for a periodic function f; in the lowest sector $E \approx \hbar\omega$. The central statement: the *mass* of a mode is its k_t-winding energy, $m_i c^2 = \hbar f(k_t^{(i)})$, which together with the Yukawa form $m_i = r_i \xi^{p_i} v$ (A100) delivers all fermion masses from the winding tuples.

8.12 The Structural Point: Known Mathematics, New Combination

[B] Each of the four extensions corresponds to a well-known structure:

Ext.	Mathematics	Established since
1	Quantum groups $SU_q(2)$	Drinfeld/Jimbo 1985
2	Winding excitations	String theory, Kac–Moody
3	Spinor bundles	Differential geometry 1950s
4	Compact extra dimension	Kaluza 1921, Klein 1926

None of these structures is new. What makes FFGFT specific is the **combination**: all four together, with a single geometric justification (T^4 and ξ).

8.13 The Full Hilbert Space

[B] With all four extensions one obtains a complete FFGFT-equivalent formulation. In the canonical A-series notation the base space is

$$\mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3,$$

where the $\mathbb{C}^3$ factor carries the three-element Z_3 identification (A030); the full formulation supplements it with the spinor bundle S and the deformed $SU_q(2)$ algebra. In this language the FFGFT results are directly derivable from the Hilbert-space structure: the masses from the k_t-windings (A100), α from an eigenvalue relation (A130), spin-orbit coupling from the bundle structure, the K_{frak} corrections from the quantum-group deformation (A040).

[K] On the same space the mass operator acts as a Z_3 circulant; from its diagonalisation follow the Koide condition $Q = 2/3$ and the angle $\theta = 2/9$. This computation is *not* carried out here but belongs substantively in the lepton sector: the operator and its diagonalisation are in A110, the angle $\theta = 2/9$ in A120. This document only provides the space on which they act.

8.14 Quantum Graphity as Sketched Subset

[S] The same subset strategy – Bloch sphere $\subset$ FFGFT through reduction – can be applied to another established, published quantum gravity theory: *Quantum Graphity* (Konopka, Markopoulou, Severini, Smolin; *Phys. Rev. D* 77, 104029, 2008; arXiv:0801.0861) – a background-independent model in which space and geometry emerge from a dynamical weighted graph. It is mathematically clearly defined and a concrete comparison object. It is a comparison with established physics, not a load-bearing argument of FFGFT; and – the crucial difference from the Hilbert-space bijection above – the reduction is *not carried out* but a hypothesis. Hence the consistent $[S]$ marking throughout.

Five reduction steps: (1) continuum to discrete (triangulation of T^4, not proved); (2) topology forgotten (K_N vs. T^4 winding structure); (3) windings to spin labels (physical meaning of (n_x, n_y, n_z, k_t) lost); (4) geometry parameter to phase parameter ($\xi = 4/30000$ fixed vs. free couplings $g_V, g_B, \ldots$); (5) matter sector discarded (all 12 fermion masses absent).

What survives: one genuine correspondence – emergent $U(1)$ from microstructure (Levin–Wen, extended version only). FFGFT contains Hilbert-space QM provably as a subset, and discrete-geometry programmes plausibly as a special case.

8.15 Field Theory: Modes, Lagrangian, Schrödinger Limit

[B] The complete field theory of the Hilbert space – explicit mode functions $\phi_{n,l,j}$, free Lagrangian $\mathscr{L}_0$, bridge formula Lagrangian ↔ mode operator, Dirac line, and Schrödinger limit – is in A075. Here only the key formulas needed for the Hilbert-space connection.

[B] Bridge formula. Lagrangian and mode operator describe the same physics (derivation in six steps: A075):

$$\int \tfrac{1}{2}(\partial\,\delta m)^2\, d^4x = \tfrac{1}{2}\langle\psi|\Phi^2|\psi\rangle, \qquad \Phi = \sum_i m_i P_i.$$

Mass arises not as a Lagrangian parameter but as an eigenvalue of the mode operator.

8.16 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$q = e^{i\pi/75}$	Deformation parameter of the Drinfeld–Jimbo quantum group; follows from $q = e^{i\pi\cdot 100\xi}$ with $\xi = 4/30000$
$U_q(\mathfrak{su}(2))$	q-deformed $\mathfrak{su}(2)$ algebra
$[n]_q$	q-number: $[n]_q = (q^n - q^{-n})/(q - q^{-1})$
$\sigma_x, \sigma_y, \sigma_z$	Pauli matrices (spin operators)
S_{CHSH}	CHSH value, theoretical maximum $2\sqrt{2} \approx 2.828$

8.17 Verification Scripts

- Bijective bridge $(z, r, \theta) \leftrightarrow (\alpha, \beta)$: numerical verification of $|\alpha|^2 + |\beta|^2 = 1$ and $z^2 + r^2 = 1$ over random states, plus operator translation Pauli ↔ rotation: `python/A_Serie_Skripte/a070_tensorfaktor.py`

- q-deformation and factor structure: $q = e^{i\pi/75}$, limit $q \to 1$, and the tensor factor $L^2(T^4) \otimes \mathbb{C}^3$: `python/A_Serie_Skripte/a070_zirkulant_dft.py`

8.18 Open Questions

The ΔCHSH $\sim 10^{-5}$ deviation is calculated, not measured. The CHSH scaling formula is fully derived in A165; the computational framework (ϕ-QFT, Bell damping, T0-Shor) and the hardware status are there. The IBM-Heron-r2 measurement (May 2026) confirms Bell violation ($S = 2.74$, 96.9 % of the Tsirelson bound), but the device-to-device variation ($\sim 1.4\,\%$) exceeds the ξ effect ($\sim 0.001\,\%$) by a factor ~ 1400 – not resolvable with today's NISQ hardware in principle (A165). As a prediction with falsification criteria, ΔCHSH stands in A210.

Top quark, Down-type quarks, and quark r_i deep structure are unclear. The lepton exponents and Up-type quarks (up to Charm) follow the geometric $2/3$ sequence (A075/A100); the top quark ($p_t = -1/3$, no known geometric reason) and the Down-type arithmetic step size remain open (A150).

8.19 Supersedes

This document subsumes: Doc. 230 (translatability FFGFT $\leftrightarrow$ Hilbert-space QM, bijection, operators, gates, measurement, interpretative divergence), Doc. 231 (the four Hilbert-space extensions), Doc. 232 (Quantum Graphity as sketched subset), and the framework section of Doc. 282 (the Hilbert space explicit). Incorporated: R41 (location of losses: the type-I decompactification is lossless; lossy are the reverse direction, spatial projection, and measurement window). *Not* taken over: the mass operator/Koide part of Doc. 282 belongs substantively in A110; the HLV/CP comparison part of Doc. 282 remains independent as an IPI framework comparison (A240).

8.20 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 9

A075 · Field Theory: Mode Functions and Lagrangian

9.1 Purpose

[STIPULATION] This document develops the complete field theory of FFGFT: explicit mode functions on the 4-torus, the free Lagrangian, the bridge formula between Lagrangian and mode operator, the Dirac and Schrödinger equations as low-energy limits, the binding picture of masses, and the non-closure theorem. It is the link between the geometric structure (A030/A040) and the sectors (A100 ff.) – it shows that masses do not appear as parameters in the Lagrangian but emerge as eigenvalues of the mode operator.

9.2 The Basic Geometry of the Torus

[B] The fundamental carrier is the compact 4-torus T^4 with the characteristic length scale

$$L_T = \frac{\ell_P}{\xi} \approx 7500\,\ell_P \approx 6.14 \times 10^{-16}\ \text{GeV}^{-1}.$$

(Not to be confused with the minimal length scale $L_0 = \xi \cdot \ell_P \approx 2.15 \times 10^{-39}$ m from A015.) This is the *fundamental period* of the torus: all particles are modes on L_T – like overtones of a vibrating string in three spatial dimensions. The Compton wavelength $L_i = 2\pi/m_i$ is the wavelength of the respective mode, not the size of a separate torus.

[B] The FFGFT torus carries a fractal correction of the dimension:

$$D_f = 3 - \xi \approx 2.9998667.$$

This minimal deviation is the origin of all particle masses. It generates a non-linear dispersion relation – analogous to an anharmonic crystal lattice. On an exactly three-dimensional torus, masses would be integer-proportional to the wave number; the observed ratios $m_\mu/m_e \approx 206.8$ and $m_\tau/m_e \approx 3477$ force the fractal dispersion.

9.3 Quantum Numbers of the Modes

[B] Each mode on the torus carries three quantum numbers: n (principal quantum number), l (orbital angular momentum, transverse component), j (spin quantum number, $j = l \pm \frac{1}{2}$ for fermions). The wave vector is quantised:

$$\mathbf{k}_{n,l} = \frac{2\pi}{L_0}(k_x, k_y, k_z), \qquad k_x, k_y, k_z \in \mathbb{Z},$$

with $n = k_x^2 + k_y^2 + k_z^2$ and $l = \sqrt{k_x^2 + k_y^2}$. The assignment for the leptons:

Particle	n	l	(k_x, k_y, k_z)	r_i	p_i
Electron	1	0	$(1, 0, 0)$	$4/3$	$3/2$
Muon	2	1	$(1, 1, 0)$	$16/5$	1
Tau	3	2	$(1, 1, 1)$	$25/9$	$2/3$

[K] The lepton exponents $(p_e, p_\mu, p_\tau) = (3/2, 1, 2/3)$ form a geometric sequence with ratio $2/3$ – the inverse of the tetrahedral factor $4/3$ from $\xi = 4/3 \times 10^{-4}$:

$$p_{n+1} = \frac{2}{3}\, p_n.$$

Up-type quarks (up to Charm) follow the same scheme with a step skipped; Down-type quarks an arithmetic sequence ($\Delta p = -1/2$); the top quark ($p_t = -1/3$, $r_t = 1/28$) breaks the scheme – no known geometric reason (open, A150).

9.4 Explicit Mode Functions

[B] The eigenfunctions of the d'Alembertian $\Box = \partial_t^2 - \nabla^2$ on $T^3 \times \mathbb{R}$ are:

$$\boxed{\phi_{n,l,j}(t, \mathbf{x}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}_{n,l}\cdot\mathbf{x}} \cdot Y_{l,j}(\vartheta, \varphi) \cdot e^{-im_{n,l,j}\,t}}$$

with torus volume $V = L_0^3$ and spherical harmonics $Y_{l,j}$. The modes are orthonormal:

$$\int_{T^3} \phi^*_{n,l,j}(\mathbf{x})\, \phi_{n',l',j'}(\mathbf{x})\, d^3x = \delta_{nn'}\delta_{ll'}\delta_{jj'}$$

and satisfy the Klein–Gordon equation:

$$\Box\, \phi_{n,l,j} = -m^2_{n,l,j}\, \phi_{n,l,j}.$$

The spin structure follows from the winding numbers of the torus: spin-$\frac{1}{2}$ corresponds to the winding $(n_\theta, n_\phi) = (1, 1)$; the 720° symmetry follows directly from the topology.

9.5 The Lagrangian

[B] The starting point is the free Klein–Gordon Lagrangian:

$$\mathscr{L}_0 = \tfrac{1}{2}(\partial\,\delta m)^2,$$

the scalar field $\delta m(x)$ describes mass fluctuations on the torus; the equation of motion $\Box\,\delta m = 0$ describes massless oscillations. The fractal geometry adds a correction term generating the non-linear dispersion:

$$\Delta\mathscr{L}_{\text{Torus}} = \varepsilon\,\xi\left[\frac{f(\vartheta)}{2}(\partial\,\delta m)^2 - k^{\mu\nu}(\vartheta)\,\partial_\mu\delta m\,\partial_\nu\delta m\right].$$

The geometric functions $f(\vartheta)$ and $k^{\mu\nu}(\vartheta)$ encode the curvature of the fractal torus. The complete Lagrangian:

$$\boxed{\mathscr{L} = \mathscr{L}_0 + \Delta\mathscr{L}_{\text{Torus}}}$$

[K] Connection to the Standard Model. For $\xi \to 0$, $\Delta\mathscr{L}_{\text{Torus}}$ vanishes; FFGFT contains the SM Lagrangian as a limiting case. The difference from Yukawa: in the SM the coupling y_i is a free parameter; in FFGFT the corresponding term $\xi \cdot m_\ell$ is geometrically fixed.

9.6 The Mode Operator

[B] The Hilbert space of torus modes:

$$\mathscr{H} = \bigoplus_{n,l,j} \mathbb{C} \cdot \mathbf{e}_{n,l,j}, \qquad \langle \mathbf{e}_{n,l,j}, \mathbf{e}_{n',l',j'} \rangle = \delta_{nn'}\delta_{ll'}\delta_{jj'}.$$

The field operator:

$$\Phi(x) = \sum_{n,l,j} m_{n,l,j} \cdot \phi_{n,l,j}(x) \cdot P_{n,l,j},$$

with projectors $P_i = |\mathbf{e}_i\rangle\langle \mathbf{e}_i|$. Since $P_i P_j = \delta_{ij} P_i$, we have $\Phi^2 = \sum_i m_i^2 P_i$.

9.7 The Bridge Formula: Lagrangian and Mode Operator

[B] The core of the field theory: the Lagrangian and the mode operator describe the same physics. The bridge formula in six steps:

$$\boxed{\int \tfrac{1}{2}(\partial\,\delta m)^2 \, d^4x = \tfrac{1}{2}\langle \psi | \Phi^2 | \psi \rangle}$$

Step 1 – Mode expansion: $\delta m(x) = \sum_i a_i \phi_i(x)$.

Step 2 – Substitution: $\int \mathscr{L}_0 \, d^4x = \frac{1}{2} \sum_{i,j} a_i a_j \int (\partial\phi_i)(\partial\phi_j) \, d^4x$.

Step 3 – Orthogonality: $\int (\partial\phi_i)(\partial\phi_j) \, d^4x = 0$ for $i \neq j$. Hence $= \frac{1}{2} \sum_i a_i^2 \int (\partial\phi_i)^2 \, d^4x$.

Step 4 – Klein–Gordon (partial integration): $\int (\partial\phi_i)^2 \, d^4x = m_i^2$ (normalisation).

Step 5 – Summary: $\int \mathscr{L}_0 \, d^4x = \frac{1}{2} \sum_i a_i^2 m_i^2$.

Step 6 – Mode operator: $\langle \psi | \Phi^2 | \psi \rangle = \sum_i a_i^2 m_i^2$. Comparison yields the bridge formula.

[B] Three description levels are thereby equivalent:

Level	Formula
Lagrangian	$\mathscr{L}_0 = \frac{1}{2}(\partial\,\delta m)^2$
Mode operator	$\Phi = \sum_i m_i P_i$
Bridge	$\int \mathscr{L}_0 \, d^4x = \frac{1}{2}\langle \psi \vert \Phi^2 \vert \psi \rangle$

Mass arises not as a Lagrangian parameter but as an eigenvalue of the mode operator – this is the structural difference from the Standard Model.

9.8 Mass Generation as a Binding Phenomenon

[B] The eigenvalue equation on the torus:

$$\left[\Box + \hat{V}(\xi, D_f)\right]\phi_{n,l,j} = -m^2_{n,l,j}\,\phi_{n,l,j}.$$

The mass results as the difference of kinetic energy and potential:

$$m_i^2 = \underbrace{\left(\frac{2\pi n}{L_0}\right)^2}_{\text{kinetic}} + \underbrace{\langle\phi_i|\hat{V}|\phi_i\rangle}_{\approx -(2\pi n/L_0)^2}.$$

[S] The kinetic energy is enormous ($\sim 10^{32}$ GeV^2), the potential almost exactly opposite. The mass is the tiny remainder. This is not fine tuning but the defining property of bound states. *Crystal analogy*: one does not ask why a crystal produces a particular phonon dispersion – the dispersion curve *is* the crystal structure in momentum space. *Hydrogen analogy*: kinetic energy ≈ 13.6 eV, Coulomb potential ≈ -27.2 eV, binding energy -13.6 eV – no one asks why the potential compensates the kinetic energy so precisely, that follows from the Schrödinger equation. In FFGFT: the particles are bound torus modes; the potential $\hat{V}$ is uniquely determined by ξ and D_f – no free parameter.

9.9 The Non-Closure Theorem

[K] For all pairs of distinct torus modes $i \neq j$:

$$\forall i \neq j \;:\; (r_i \cdot r_j,\; p_i + p_j) \notin \mathscr{P},$$

where $\mathscr{P}$ is the set of all pairs (r, p) admitted in FFGFT.

Example: $r_e \cdot r_\mu = \frac{4}{3} \cdot \frac{16}{5} = \frac{64}{15}$, $p_e + p_\mu = \frac{3}{2} + 1 = \frac{5}{2}$. In the complete mode table there is no particle with $r = 64/15$ and $p = 5/2$.

[B] Why does the theorem hold? The cause lies in the orthogonality of the modes: in Hilbert space the product of two distinct basis vectors is structurally meaningless – a category violation. $r_e \cdot r_\mu$ is computable as a number, but belongs to no mode of the torus.

[K] Practical rule: *Permitted* is numerical evaluation ($m_e \cdot m_\mu \approx 53.9$ MeV^2) and the mass formula $m_i = r_i \cdot \xi^{p_i} \cdot v$ for fixed i. *Forbidden* is symbolic cross-multiplication: $(r_i \cdot r_j) \cdot \xi^{p_i+p_j} \cdot v$ for $i \neq j$.

9.10 Dirac Line and Schrödinger Limit

[B] Within the Lagrangian framework the geometric equation replaces the full Dirac equation by:

$$\partial^2 \Delta m = 0.$$

The fundamental structure is the Clifford algebra $\mathbf{e}_\mu \mathbf{e}_\nu + \mathbf{e}_\nu \mathbf{e}_\mu = 2g_{\mu\nu}$; the 4×4 γ-matrices are retained as a representation of the Clifford generators for concrete computations – they are not eliminated but traced back to their geometric meaning. Spin manifests on the fractal torus as a topological property (vacuum-phase winding). A relativistic state:

$$\Phi(x,t) = \rho(x,t)\, e^{i\theta(x,t)} \cdot \chi(\sigma),$$

with spin component $\chi(\sigma)$ from the geometric knot modes.

[B] In the non-relativistic limit, the phase equation of the vacuum field reduces to the standard Schrödinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi,$$

with $\psi \propto \sqrt{\rho}\, e^{i\theta}$, effective mass m from the knot inertia, and potential V from external ρ-perturbations. The Schrödinger equation is *not fundamental* but an effective equation for slow, non-relativistic knot excitations – the low-energy limit of the FFGFT mode structure. This limit follows from the Lagrangian framework via the DVFT polar decomposition $\Phi \to \sqrt{\rho}\, e^{i\theta}$.

[B] The transition Lagrangian → Schrödinger equation in three stages: (1) mode superposition in the low-energy limit:

$\Phi(x) \xrightarrow{\text{low energy}} \sqrt{\rho(x,t)}\, e^{i\theta(x,t)}$ with energy density ρ and coherent phase θ. (2) The phase dynamics of θ satisfies the Schrödinger equation in the non-relativistic limit. (3) Masses follow from k_t-windings via $m_i c^2 = \hbar f(k_t^{(i)})$; the lepton exponents $(3/2, 1, 2/3)$ are the geometric sequence of this tower.

9.11 Algebras, Recursion Operator, and Scale Structure

[B] The recursion operator $\mathscr{R}$ formalises the fractal scaling structure, diagonal on the mode space:

$$\mathscr{R}(\mathbf{e}_{n,l,j}) = \xi^{p(n,l,j)-1} \cdot K_{\text{frak}} \cdot \mathbf{e}_{n,l,j}.$$

The time-mass duality is encoded as a field symmetry: $\mathscr{R} : t \to \xi t,\ m \to \xi^{-1} m,\ \Phi \to \Phi$. The mode algebra for free waves: $\mathbf{e}_{\mathbf{k}_i} \star \mathbf{e}_{\mathbf{k}_j} = \mathbf{e}_{\mathbf{k}_i + \mathbf{k}_j}$, with structure constants $C_{ij}^{k} = \delta_{k,i+j \text{ mod torus period}}$.

[K] The scale running of the coupling – in FFGFT-native language: an $\mathscr{R}$-induced scale correction, not a renormalisation procedure – gives in the QED analogy:

$$\alpha^{-1}(\mu) = \alpha^{-1}(m_e) - \frac{\beta_0}{2\pi} \ln \frac{\mu}{m_e} - \frac{\beta_0 \xi}{4\pi} \Big(\ln \frac{\mu}{m_e} \Big)^2.$$

The ξ-term is the FFGFT-specific addition; ξ is not a fixed point of the β-function but a geometrically fixed constant (A020).

9.12 Methodological Position of the Field Theory

[K] Reduction is not increased precision. Where the Standard Model is experimentally confirmed (QED scattering amplitudes to $\sim 10^{-12}$), FFGFT can only reproduce these plus ξ-corrections of order 10^{-4}, not qualitatively surpass them. The FFGFT mass formulas give values at $\sim 1\,\%$ agreement with PDG values – that is a *structural proof* (masses follow from geometry, not free parameters), not a precision calculation. The reduction from

~ 25 free SM parameters to a single geometric parameter ξ is the quantifiable statement.

[S] Circularity of the comparison basis. PDG values are not model-free raw data; they contain assumptions about units, gauge conventions, and SI-2019-reform definitions. An FFGFT calculation in natural units ($\hbar = c = 1$) avoids the accumulation of model assumptions; as soon as it is compared against PDG data, it inherits their prior corrections. The $\sim 1\,\%$ comparison is therefore to be read in two directions: confirmation of the geometric structure *and* indication of the latitude of the circular measurement basis.

9.13 Open Mathematical Tasks

The Hopf algebra for interactions is not worked out. The free theory has structure constants $C_{ij}^k = \delta_{k,i+j}$ (wave vector addition). Complete Feynman rules from the structure constants and the non-trivial fixed point of the β-function (whether μ^* exists and what it means physically) are open.

Top quark, Down-type step size, quark r_i deep structure. For leptons and Up-type quarks (up to Charm) the geometric $2/3$-sequence of exponents is known; for the Top ($p_t = -1/3$) and Down-type (arithmetic sequence $\Delta p = -1/2$) the geometric justification is missing. The α-π-$\sqrt{3}$ deep structure of the lepton r_i is partially present (A130); the analogue for quark r_i is missing (A150).

The test condition is structural, not quantitative. The test $\langle\phi_i|\hat{V}|\phi_i\rangle = r_i^2\xi^{2p_i}v^2 - (2\pi n/L_0)^2$ is the bottleneck: explicitly deriving $\Delta\mathscr{L}_{\text{Torus}}$ from the torus geometry and showing that the eigenvalues agree with the mass table – as phonon dispersion from lattice forces of a crystal.

9.14 Verification Scripts

- Mode functions and orthonormality: verification $\int \phi_i^* \phi_j\, d^3x = \delta_{ij}$ for $i, j \in \{e, \mu, \tau\}$ numerically: `python/A_Serie_Skripte/a075_modefunktionen.py`

- Bridge formula: $\int \frac{1}{2}(\partial\delta m)^2\, d^4x = \frac{1}{2}\langle\psi|\Phi^2|\psi\rangle$ for random coefficients (a_i): `python/A_Serie_Skripte/a075_bruecke.py`
- Non-closure theorem: verification that $(r_e r_\mu, p_e + p_\mu)$ does not lie in $\mathscr{P}$: `python/A_Serie_Skripte/a075_nichtabschluss.py`

9.15 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$\square = \partial_t^2 - \nabla^2$	d'Alembertian (wave operator)
$\mathscr{L}$	Lagrangian density
$\phi_{n,l,j}(x)$	Mode function on T^4; quantum numbers as in A100
$m_{n,l,j}$	Mass of mode (n, l, j)
Ψ_ℓ	Lepton spinor field
$g_T^\ell = m_\ell \xi$	Mass-proportional coupling of the time field

9.16 Open Questions

The complete form of $\Delta\mathscr{L}_{\textbf{Torus}}$ – the geometric functions $f(\vartheta)$ and $k^{\mu\nu}(\vartheta)$ – is not derived in closed form from the torus geometry. The correction factor $K_{\text{frak}} = 1 - 100\xi \approx 0.9867$ is known (A040), the complete tensor function is not.

The quark-sector extension of the mode table and the non-closure theorem to all twelve fermions is worked out for leptons and Up-type quarks (up to Charm), but not for Down-type and Top (A150).

The Hopf algebra structure for interactions and complete Feynman rules are outstanding.

9.17 Supersedes

This document subsumes: Doc. 202 (complete FFGFT field theory: fundamental scale L_0, quantum numbers of modes, explicit mode functions $\phi_{n,l,j}$, free Lagrangian $\mathscr{L}_0$, torus correction term, mode operator $\Phi = \sum m_i P_i$, bridge formula in six steps, non-closure theorem, binding picture of masses, Dirac and Schrödinger limits, recursion operator $\mathscr{R}$, scale structure). *Not* taken over: the large-scale sector from Doc. 202 (DVFT line on vacuum field effects at galactic scales) – that does not belong in the A-series core (A010); the section on circularity of the maximal-formula notation is left to the correction programme.

9.18 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 10

A080 · Units and Bookkeeping

10.1 The Separation That Everything Depends On

[K] The theory predicts ratios, not absolute values.

This sentence is the most important methodological statement of the entire block, and it cuts the error discussion into two halves that have nothing to do with each other.

A dimensionless ratio of two quantities at the same level is intrinsic: it depends on the geometry and nothing else. It needs no unit, no bridge constant, no correction factor – these cancel (A040).

An absolute value, by contrast – a mass in megaelectronvolts, a length in metres – additionally requires a conversion, and this conversion is not part of the geometry.

[B] From this follows a sharp diagnostic rule: *every deviation between theory and measurement must first be examined as to whether it sits in the geometric core or in the translation layer.* Treating both the same obscures the statement.

10.2 What the Physical Constants Are Here

[STIPULATION] In the natural system of the theory, $\hbar$, c, and k_B equal one. They are not physical quantities but unit bridges.

The word bookkeeping, which appears in the corpus for this, is regularly misunderstood as dismissive. The opposite is meant. Bookkeeping here means *necessary*, not *dispensable*:

- c is the bridge between units of space and time,
- $\hbar$ the bridge between units of mass and time,
- k_B the bridge between temperature and energy.

[B] The SI separates metres, seconds, kilograms, and kelvins into distinct units. As soon as *computation* is performed in this system, the conversion factors are indispensable – without k_B an energy in kelvins would result. Ontologically they are secondary; operationally indispensable.

10.3 The Four Dresses

[K] A single quantity appears in four guises:

$$m_0c^2 \;=\; E_0 \;=\; k_BT_0 \;=\; \hbar/\tilde{T}.$$

Dividing by c^2 gives a mass; by $\hbar$, a time; by k_B, a temperature. These are four expressions of the same fact, not four quantities. This identity is *exact* – it is the definition of the conversion, and it contains no approximation.

[B] E_0 itself is an empirical bridge: $E_0 = \sqrt{m_e m_\mu} \approx 7.398$ MeV *[Note 3 August 2026: E_0 is the calibration of the unit system – by A085 there is exactly one – and the value carried is the one that provides the anchoring in SI: 7.398 MeV matches the measured α to 5×10^{-6} via $\alpha = \xi E_0^2$. The geometric mean $\sqrt{m_e m_\mu} = 7.348$ MeV (A130, route 1) names the origin of the order of magnitude but is unsuitable as an anchor: inserted, it would give $1/138.9$, i.e. $-1.35\,\%$ in α, and that deviation would propagate into every derived absolute value and accumulate in comparisons. The formula notation here is abbreviated; the anchor value governs (A130, route 2: $E_0^2 = 4\sqrt{2}\,m_\mu/\xi^p$, without m_e).]* follows from measured lepton masses, not from ξ alone. The geometric derivation is in A130; the unit check in A085. A reader of only this document must know: E_0 is treated here as a given bridge constant, not as a derived quantity.

[B] Where then do deviations arise? Not in this identity. They arise where a theoretically determined dimensionless number is held against a measured quantity – and there they sit in the dress, not in the conversion.

10.4 Bridge Constants

[STIPULATION] To state absolute values, the theory needs at least one quantity with dimension. This bridge constant is a declared input.

This is not a weakness of the approach, but a necessity of any theory that seeks to generate absolute numbers from dimensionless structures: from pure numbers no mass in kilograms follows. What matters solely is that the number of bridge constants is stated openly and remains small.

[B] At the same time: choosing a bridge constant is a calibration, not a test. An agreement produced by the choice of bridge must not be counted as confirmation. This rule is applied in A130 at a concrete case, where it makes the difference between a success and a self-confirmation.

10.5 The Tolerance Standard in Units

[B] From the separation follows how to read a deviation. Three quantities belong together: the value, the measurement precision of the comparison quantity, and the determination tolerance of the theoretical quantity.

A deviation of one percent can be meaningless if the theoretical quantity is determined only to two places – and it can be devastating if both sides are fixed to eight places. The percentage alone says nothing. This rule is applied consistently from A100 onward.

10.6 Verification Scripts

- Exactness of the four-dress identity $m_0c^2 = E_0 = k_BT_0 = \hbar/\tilde{T}$ in SI numbers, with display of the relative deviation (expected: zero up to rounding): `python/A_Serie_Skripte/a080_vier_kleidungen.py`
- Invariance of ratios under change of bridge constant, against the non-invariance of absolute values: `python/A_Serie_Skripte/a080_verhaeltnis_invarianz.py`

10.7 Open Questions

First, the number of bridge constants is not tallied here. It belongs in the overall balance A230, not here – but it must appear there, otherwise the one-parameter claim of the theory is not testable.

Second, it is not shown that the bridge constant could follow from the geometry. As long as it is declared, the absolute scale remains an input. This is consistent with A020, where the same holds for the order of magnitude of ξ, and it is the same gap in a different guise.

10.8 Supersedes

This document subsumes: Doc. 013 (SI), Doc. 014 (natural/SI), Doc. 015 (systematics of natural units), Doc. 122 (ratio and absolute), Doc. 134 (unit conventions), Doc. 261 (static natural units). Incorporated: P40 (ratios exact, deviations at the conversion).

10.9 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 11

A085 · Natural Units

11.1 Why This Document Is Inserted Here

A080 treated units as bookkeeping: bridge constants, the separation of ratio and absolute value. That is the *operative* side. This document provides the *fundamental* side that A080 presupposes but does not develop: why a natural unit system exists in this theory at all, in which all dimensionful constants take the value one.

It precedes the gravitation document (A145) and quantum mechanics (A160) because both use it: G is obtained from ξ and a mass, and that is a statement about structure rather than numerical conventions only in natural units.

11.2 The Origin Lies in the Duality

[K] The natural unit system of the theory is not a choice but a consequence of the first stipulation. From $\tilde{T} \cdot m = 1$ (A020) it follows that time and mass are reciprocal – the same quantity in two readings. The conversion between them is therefore not a physical constant but the identity.

[B] In standard physics three constants perform the conversion between four fundamental dimensions:

$$\begin{aligned} c &: \text{ length} \leftrightarrow \text{time}, \\ \hbar &: \text{ mass} \leftrightarrow \text{time}, \\ k_B &: \text{ temperature} \leftrightarrow \text{energy}. \end{aligned}$$

But once time and mass are the same quantity, the system collapses: there is only one independent dimension left, and these three conversion factors become one.

[STIPULATION] A word of demarcation that belongs here: this mechanism concerns the *dimensionful* bridge constants

$c, \hbar, k_B$ (and, via the mass-length connection, G; A145). The fine-structure constant α is *not* among them. It is dimensionless, and its reduction to one has a different reason – electromagnetic duality, not time-mass duality. That is a separate chapter and is treated in A130, not here. This document claims nothing about α; it says only why $c, \hbar, k_B$ become one.

[K] In natural units therefore

$$[M] = [E] = [T^{-1}] = [L^{-1}],$$

mass, energy, inverse time, and inverse length are a single dimension. This is not set simplifyingly but is the direct consequence of the duality.

11.3 Why This Is More Than a Convention

[B] One can set $\hbar = c = 1$ in any theory; that alone is inconsequential. The difference here is that the *reduction to a single dimension* is a substantive claim, not a notation.

In standard physics, after $\hbar = c = 1$ one mass dimension remains – energy is a genuine unit that must be chosen. Here *nothing* remains to choose: the one remaining dimension is scaled by ξ (A020), and the absolute scale is the only remaining input (A080, A130).

[STIPULATION] The bookkeeping of A080 is thus bound to its ground: the bridge constants are not equal to one because one sets them so, but because the duality identifies their dimensions. Setting $= 1$ is reading off an identity, not a normalisation.

11.4 The Four Dresses, Read Fundamentally

[K] A080 introduced the identity

$$m_0 c^2 = E_0 = k_B T_0 = \hbar / \tilde{T}$$

as an exact conversion. Here is its reason: the four expressions are not four quantities that happen to be equal, but one quantity in four dresses, because the duality places mass, energy, temperature, and inverse time on the same dimension.

The conversion factors c^2, k_B, $\hbar$ are the seams between the dresses. In natural units the seams disappear, because there is only one garment.

11.5 What Follows for the Following Documents

[B] Three consequences that A145 and A160 presuppose:

First, every mass is an inverse length and vice versa. This allows a gravitational coupling linking masses with lengths to be expressed through ξ alone – the content of A145.

Second, every dimensionful constant is either one or a calibration. There is no third case. Where in the corpus such a constant carries another value, that is an SI artefact, not a fact of nature.

Third, ratios are invariant under the choice of scale (A080), and the choice of scale is the *only* remaining degree of freedom in the unit system. Everything dimensionless is thereby scale-free-determined or not at all.

11.6 Verification Script

- That $\tilde{T} \cdot m = 1$ reduces the four-dimension system to one dimension, shown in the dimensional bookkeeping; and that the Planck quantities are exactly the conversion factors of the reduced system: `python/A_Serie_Skripte/a085_natuerliche_einheiten.py`

11.7 Open Questions

First, the absolute scale remains an input. The reduction to one dimension is derived; what size this one dimension has is not. This is the same gap as in A020 (order of magnitude of ξ) and A130 (calibration of E_0), here in its most general form.

Second, the identification of temperature with energy is an additional thermodynamic assumption. $k_B = 1$ presupposes that temperature is an energy per degree of freedom; that is standard, but it does not follow from the duality, it stands beside it.

11.8 Supersedes

This document supplements A080 with the fundamental justification and subsumes the corresponding parts of: Doc. 013 (SI), Doc. 014 (natural/SI), Doc. 015 (systematics of natural units), Doc. 012 (units section). Incorporated: P40.

11.9 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 12

A090 · Projection Chain $T^4 \to T^0$

12.1 Purpose

[STIPULATION] FFGFT describes fundamental physics on a compact 4-torus T^4 with the single parameter $\xi = 4/30000$. The observable world is $\mathbb{R}^3 \times \mathbb{R}_t$. This document addresses the question of how both are connected – and shows that the chain $T^4 \to T^0$ consists of *two qualitatively different* operations that must not be confused: a *duality decompactification* (nearly reversible, no mode losses) and a *geometric projection* (lossy, not reversible). A third operation, going in the opposite direction, is added: the *representational translation* to the Hilbert space (bijective, lossless).

The Standard Model is situated therein as a *decompactified projection*: its constants (α, v, Yukawa) are recursion outputs from ξ; whoever takes them as inputs works in the unrolled picture, without naming the compact whole. This is not a refutation of the Standard Model – it computes correctly in its domain; it is a location.

12.2 The Tension in the Question

[B] Three observations about the transition $T^4 \to \mathbb{R}^3 \times \mathbb{R}_t$ appear in mutual contradiction:

- **Loss picture.** Every projection $D \to D-1$ is not injective, loses information. The chain is not reversible.
- **Compactness picture.** T^4 is fully compact; the observable spacetime is decompactified.
- **Upward picture.** There is already a *bijective* translation of T^4 into a higher-dimensional space – the Hilbert space (A070) – which is *lossless*.

[B] These three pictures cannot be naively overlaid: if the first step were pure loss, no decompactified time direction could arise from it (forgetting does not generate a new dimension); and it would remain puzzling how an upward movement can be lossless while the downward movement destroys information. The resolution: there are not one but *three* dimension-changing operations of *two kinds*.

12.3 The Structure of T^4

[B] The fundamental level of FFGFT is the compact 4-torus

$$T^4 = S^1_x \times S^1_y \times S^1_z \times S^1_m,$$

with three *spatial* circles and one *mass circle* S^1_m. The fourth direction S^1_m is not a mass dimension *beside* a separate time, but the *substitute time direction* itself. Mass and time are two *readings* of one and the same dimension; the fundamental relation $\tilde{T} \cdot m = 1$ is the translation dictionary:

$$\underbrace{S^1_m \text{ (compact)}}_{\text{mass reading, discrete}} \quad \xleftrightarrow{\tilde{T}\cdot m=1} \quad \underbrace{\mathbb{R}_t \text{ (open)}}_{\text{time reading, continuous}} .$$

[K] Kaluza–Klein reading. A compact circle of radius R carries a discrete mass tower $m_n \sim n/R$. The four directions carry two radius classes: the three spatial circles ($R_{x,y,z} \sim R_{\text{large}}$) generate a quasi-continuous mode tower with ~ 0 gap – this becomes space; the mass circle ($R_m \sim \xi \ell_P$) has a huge gap and a discrete, heavy spectrum – this becomes the particle mass spectrum. The fundamental relation $\tilde{T} \cdot m = 1$ is thereby structurally the Kaluza–Klein relation for the mass dimension.

12.4 Type I – Duality Decompactification: Mass Becomes Time

[B] The first, qualitatively distinguished operation is the *unrolling* of the mass circle via its universal cover:

$$S^1_m \xrightarrow{\text{cover}} \mathbb{R}_t, \qquad p : \mathbb{R}_t \to S^1_m, \quad t \mapsto e^{2\pi i t/R_m},$$

governed by $\tilde{T} \cdot m = 1$. This operation is *embedding*, not projection: it generates a new direction ($\mathbb{R}_t$) instead of discarding one. Periodic structures on S^1_m lift uniquely to $\mathbb{R}_t$; no mode content is lost. The "price" of decompactification is not information loss but the appearance of the time direction itself.

[B] What does not remain directly visible: the *compactness*. In the mass reading the topology is manifest – the spectrum is discrete, the identification $t \sim t + R_m$ directly expressed. In the time reading the same compactness is only *latent*: from continuous time alone the topological identification is not uniquely reconstructible (the winding remains ambiguous). Precisely there the recursion ξ^n sets in – it *is* the spectral signature by which the hidden compactness remains recognisable.

[S] What makes unrolled time continuous and uniform. Three questions, three answers: (1) *Why is* $\mathbb{R}_t$ *continuous?* Because the unrolling is the universal cover $p : \mathbb{R} \to S^1$ – a local homeomorphism; the cover of a circle *is* the continuous line. (2) *Why does time flow uniformly?* Because of the homogeneity of S^1_m (translation invariant, no distinguished point) and the single fixed eigenvalue $m = 1/\tilde{T}$. (3) *Why is it unrolled at all?* Not the geometry decides this – in the compact T^4 all four circles are equivalent, there is no time. The *measurement process* (the embedded observer) chooses one circle as time and unrolls it. Continuity is the forgotten winding: keeping it would give back the discrete circle.

12.5 Type II – Geometric Projection: Forget Spatial Directions

[B] All further steps ($D3 \to D2 \to D1 \to D0$) are genuine geometric projections of spatial directions – a projection $\pi : \mathbb{R}^k \to \mathbb{R}^{k-1}$ with $\ker(\pi) = \mathbb{R}$. It collapses a complete fibre; the information about the forgotten direction is not contained in the image. Without additional data (holographic completeness, boundary conditions) not reconstructible. The recursion ξ^n does *not* act on these steps – spatial projection losses are genuine.

[B] The step $D3 \to D2$ is mathematically the holographic principle; in the horizon case the Bekenstein–Hawking bound $S \leq A/(4\ell_P^2)$ appears here.

12.6 Type III – Representational Translation: Bijective Upward

[B] Downward (geometric) and upward (representational) are *opposite* information properties:

$$T^4 \xleftrightarrow{\text{bijective}} \mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3$$

The translation (A070) is not an approximation but a bijective structural statement: every QM computation is mappable into FFGFT language and vice versa. *No* physical extra dimensions are added – the same 4D content is expressed in a higher-dimensional *language* in which it fits without reduction. Losslessness follows precisely from the representational character.

[B] A geometric extension $T^4 \to T^5 \to \ldots$ is excluded in FFGFT: complexity arises from *fractal depth* ($D_f = 3 - \xi$, self-similarity), not from *more* dimensions. This is the deliberate contrast to string theory (10/11D, Calabi–Yau, landscape problem). "Higher" in FFGFT always means type III (representational), never geometric.

12.7 The Formal Chain, Level by Level

[B] The chain $T^4 \to T^0$ separated by type:

Level	Operation	Type	Invariant	ξ^n
$D4 \to D3{+}t$	$S^1_m \to \mathbb{R}_t$ (unroll)	I	$\tilde{T}m = 1$	spectral signature
$D3 \to D2$	forget space	II	area A	—
$D2 \to D1$	forget space	II	length L	—
$D1 \to D0$	forget space	II	$n \in \mathbb{Z}$	—

$$\underbrace{T^4 \longrightarrow \mathbb{R}^3 \times \mathbb{R}_t}_{\text{Type I: embedding, nearly reversible}} \qquad \longrightarrow \qquad \underbrace{\mathbb{R}^2 \longrightarrow \mathbb{R} \longrightarrow \{\text{pt}\}}_{\text{Type II: projecting, not reversible}}$$

12.8 Three Operations at a Glance

[B]

Type	Operation	Character	Information
I	mass ↔ time ($\tilde{T}m = 1$)	geometric	nearly reversible
II	spatial projection (down)	geometric	lossy
III	$T^4 \leftrightarrow \mathscr{H}$/Hilbert space (up)	representational	**bijective**

The bijectivity of type III is the mirror image of the non-reversibility of type II: geometrically one loses when reducing, representationally one loses nothing when translating.

12.9 The Role of the Recursion ξ^n

[K] The recursion ξ^n is *not* a universal compensation for projection losses – its function is narrowly limited. For type I, ξ^n generates the discrete mass spectrum (particle ladder) – that is the compact mode content of S^1_m, dual as a time-frequency structure. For type II, ξ^n does not act – spatial projection losses are genuine and not touched by the recursion.

12.10 Reversibility, Precisely Per Type

[B] The type-I step is invertible in mode content as a duality (Fourier/KK); the winding remains ambiguous – "nearly reversible". The spatial sub-chain $\mathbb{R}^3 \to \{\text{pt}\}$ is not invertible: there is no continuous map $\rho : \{\text{pt}\} \to \mathbb{R}^3$ with $\rho \circ \Pi = \text{id}$ – every π_k has $\ker = \mathbb{R} \neq \{0\}$, the collapsed fibre structure is no longer present in the image.

12.11 Why Exactly Four Circles

[B] No theory derives the dimension number of spacetime; FFGFT does *not derive the 4 forward* – it takes $3+1$ as empirical input. The question is therefore not "why is the world four-dimensional?" but: *given* $3+1$ *– what is the minimal structure that carries everything?*

[K] Naively one would need *five* building blocks: three for space, one for a mass spectrum, one for time evolution. But the fundamental relation $\tilde{T} \cdot m = 1$ identifies mass reading and time reading of *one* dimension (type I): the fourth circle S^1_m carries as a KK tower the mass spectrum *and*, decompactified, time. The duality lets one circle do the work of two – the naive 5 collapses to **4**. Fewer than four (T^3 alone): no mass spectrum, no time. More than four (a second dual circle): a second independent mass tower – superfluous, without correlate. "Optimal" means here *minimally sufficient*: the smallest dimension count in which space, full mass spectrum, and time together arise from one geometric object.

12.12 In the Compact Picture: the Time Circle Is Chosen by the Measurement

[S] In the compact T^4 all four circles are identical: dimensionless cyclic coordinates, no signature, no space/time distinction. The asymmetry (one time, signature $(3,1)$) appears only under decompactification (type I): exactly one circle unrolls via its universal cover into non-compact time, the other three remain compact (discrete spectrum) or project (space). *Which* circle becomes time is no intrinsic geometric fact – the geometry is symmetric. Time is *relational*: the selection is the measurement process itself. The embedded observer reading from inside experiences one direction as duration; this experiencing constitutes the time circle. The symmetry breaking is the measurement process, not the geometry.

12.13 The Topological Identification $t \sim t + R_m$ as Physical Observable

[K] The topological identification is not a mere accompanying structure but a physical observable. In the time reading its period is the fundamental revolution time

$$T_{\text{rev}} = \frac{R_m}{c} = \frac{\xi \ell_P}{c} \approx 7.2 \times 10^{-48}\ \text{s},$$
$$\Delta\phi = 2\pi\xi \approx 8.4 \times 10^{-4}\ \text{rad/revolution}.$$

[K] Two testable readings: (1) **Mass reading (mature, quantitative).** The discrete mode spectrum generates particle masses as torus overtones; from this follow Koide, $g-2$, and three generations (A100/A110/A120). Compactness is numerically redeemed. (2) **Time reading (falsifiable).** The periodicity generates a CHSH deviation from the Tsirelson bound of order ξ; for 73 qubits $\Delta \approx 5.4 \times 10^{-4}$ (resolvable at $\sigma \sim 10^{-4}$). The IBM Heron r2 hardware test (May 2026) confirms Bell violation ($S = 2.74$); the ξ effect is below the NISQ noise floor and not yet resolvable (A160). No unfalsifiable accompanying structure.

[S] Scaling: linear vs. logarithmic. A naive *coherent linear accumulation* $r \cdot \Delta\phi$ over r circuits would give physically absurd phase drifts ($\sim 10^{44}$ rad/s) and would contradict functioning atomic clocks and deep circuits. The laboratory prediction scales *logarithmically*,

$$\delta\phi \sim \frac{\xi \ln(r)}{D_f} \approx 2 \times 10^{-4} \quad (D_f = 3 - \xi),$$

not linearly. The linear form is the worst-case bound of the RSA argument (coherence over $r \approx N$ cycles), not the laboratory prediction.

12.14 The Standard Model as Decompactified Projection

[K] Every computation using α, v, or Yukawa couplings carries the K_{frak} recursion *indirectly* from the FFGFT perspective:

$\alpha = \xi E_0^2$ (A130), where E_0 already absorbs the K_{frak} factor. The masses are the ξ^p-ladder ($m = r\xi^p v$, A100); a Yukawa coupling $g = m/v$ is already a ξ-ladder ratio. The Standard Model takes these quantities as measured inputs; FFGFT *derives* them. Working with α, v, or Yukawa in continuous fields on $\mathbb{R}$ is structurally working in the *unrolled* picture – the type-I region of decompactification. The compact origin of the constants is carried along, but not named.

[S] What is established, what remains programme. Established is the containment relation for the sector FFGFT actually derives: α, the lepton masses, the Koide scalar $Q = 2/3$, the phase $\theta = 2/9$ (A120/A130) – for these it is shown that they are recursion outputs. Programme remains the *full* Yukawa matrix: quark masses, mixing angles (CKM/PMNS), CP phase. This separation is the same as throughout the corpus: core derivations (proved from ξ), bridges (algebraically proved), reductions (plausibility sketches).

12.15 Open Questions

First, the large-scale outer scale is treated here as a declared external input, not derived from ξ. The two-type structure depends only on the radius-class separation large/small, not on the numerical value. The connection of the large-scale outer scale to ℓ_1 remains open.

Second, holographic completeness for type II ($D3 \to D2$) is not worked out. To what extent can the type-II loss be reconstructed exactly (not just partially) via boundary data?

Third, the full Yukawa matrix remains programme. The SM-subset relation is a core derivation for the lepton sector; for quark masses/CKM/PMNS/CP-phase it is an open obligation.

12.16 Verification Scripts

- Type-I reversibility: Fourier bijection $S^1_m \leftrightarrow \mathbb{R}_t$ numerically, and winding ambiguity: `python/A_Serie_Skripte/a090_projektion.py`

- CHSH deviation: ΔCHSH $\approx$ $\xi/(2\pi)$ numerically and logarithmic vs. linear regime scaling: `python/A_Serie_Skripte/a090_chsh.py`

12.17 Supersedes

This document subsumes: Doc. 270 (projection chain $T^4 \to T^0$, type I/II/III, reversibility, role of ξ^n, why four circles, time circle as measurement process, $t \sim t + R_m$ as observable, linear/logarithmic scaling), Doc. 298 (SM as decompactified projection, Copernicus analogy, what is established/programme). *Not* taken over from Doc. 263 (fractal holography): the bulk with 31 cosmology hits, notably H_0 as projection artefact, vacuum energy, UIFT bridge – all do not belong in the A-series core (A010). Likewise not taken over from Doc. 298: the HLV-YUKAWA section (IPI framework). The holographic projection $D3 \to D2$ (Doc. 263/268/254) is named but not carried out (open point).

12.18 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 13

A095 · Torus Chirality and $g_R = 0$

13.1 Purpose

[STIPULATION] This document closes the gap between half-integrality on the torus (A263) and chiral projection into flat space (A090, A264). The core: a torus has an inside and an outside with topologically different coupling directions. After decompactification these appear as two chiral states – and the projection is not symmetric.

13.2 Half-Integrality and Torus Geometry

[B] Spin-$\frac{1}{2}$ follows from the winding $(n_\theta, n_\phi) = (1, 1)$ on T^4 (A263): two rotations (720°) bring the spinor back to its initial state. This is pure topology.

[B] A torus with winding $(1, 1)$ has an intrinsic distinction: inside and outside. For an embedded torus in $\mathbb{R}^3$: the Gaussian curvature is positive on the outside ($K > 0$) and negative on the inside ($K < 0$). For the flat identification torus T^4, $K = 0$ everywhere – but the poloidal/toroidal circulation structure nevertheless defines a topological inside/outside distinction through the orientation of the winding.

[B] The field direction is topologically fixed by the winding geometry: the electric field vector points inward on the outside (for negative charge) and outward (for positive charge). This direction is not freely choosable – it follows from the quantised winding number $\Phi = n \cdot h/e$.

13.3 Chiral Projection

[B] The decompactification $T^4 \to \mathbb{R}^3 \times \mathbb{R}_t$ (type-I projection, A090) is not symmetric with respect to inside and outside. The compact torus carries both sides simultaneously; the decompactified flat space sees only one projection.

[B] What remains after projection:

- The *outer state* projects to left-handed coupling (positive rotation phase $+\xi/2$).
- The *inner state* projects to right-handed coupling (negative rotation phase $-\xi/2$).

The projection selects one side – this is the geometric origin of chiral asymmetry.

[B] Why this explains the chiral structure. In the compact T^4, both sides couple with equal strength – there is no preferred direction. After decompactification, only the outer perspective exists as an observable state. The inner state is not directly accessible through the type-I projection – it is the "other branch" that is lost in the dimensional reduction (A090, type-II loss).

13.4 Connection to the Weak Interaction

[B] Fourier projector and $g_R = 0$. The Hilbert space on T^4 decomposes into sectors by the winding index m. The physical projector P_+ selects the $m \geq 0$ sector (outer state):

$$P_+ = \sum_{n,\, m \geq 0} |n, m\rangle\langle n, m|.$$

Action on the two chiral states:

$$\begin{aligned} P_+|1, +1\rangle &= |1, +1\rangle && \text{(outer state survives)}, \\ P_+|1, -1\rangle &= 0 && \text{(inner state in kernel)}. \end{aligned}$$

After projection the inner state is $|0\rangle$ – a null vector. The weak coupling:

$$g_L = \langle 1, +1|H_{\text{weak}}|1, +1\rangle \neq 0, \qquad g_R = \langle 0|H_{\text{weak}}|0\rangle = 0.$$

This is not a sign argument ($g_R = -g_L$) but a kernel argument: the inner state lies in the kernel of P_+ and has no physical existence after projection. $g_R = 0$ follows algebraically.

[S] What remains open: why P_+ and not P_-? The projector P_+ corresponds to the outer side ($K > 0$, positive orientation, electric flux inward). P_- would be the inner-state projector. That P_+ is the physical projector follows from orientation arguments (Gaussian curvature, field rotation $\nabla \times \mathbf{E}_{\text{field}}$) – but not from a derivable symmetry breaking. This is the one remaining open edge.

13.5 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$(n_\theta, n_\phi) = (1,1)$	Winding numbers for spin-$\frac{1}{2}$ on T^4
K	Gaussian curvature; outside $K > 0$, inside $K < 0$
$\Phi = n \cdot h/e$	Quantised flux lines, topologically fixed field direction
g_L, g_R	Left- and right-handed weak coupling
Type-I projection	Information-preserving decompactification (A090)
Type-II loss	Non-reversible spatial projection, information loss

13.6 Verification Script

- Topological consistency: winding $(1,1)$, 720° symmetry, Gaussian curvature inside/outside, phase difference $\pm\xi/2$: `python/A_Serie_Skripte/a095_torus_chiralitaet.py`

13.7 Open Questions

$g_R = 0$ **is now at [B] level:** given P_+ as the physical projector, $g_R = 0$ follows algebraically from $P_+|1,-1\rangle = 0$.

What remains [S]: why P_+ is the physical projector. The orientation argument (Gaussian curvature, field rotation) is geometrically plausible, but not derived from a deeper symmetry breaking. This is the remaining open edge – and it is precisely named.

Connection to the gauge sector (A190) missing. The chiral projection explains the $SU(2)_L$ coupling qualitatively; whether it reproduces the full electroweak gauge structure is not shown.

13.8 Supersedes

This document systematises the inside/outside chirality mechanism from the legacy corpus (particle mass torus model, Dirac winding structure, projection chain framework). It supplements A263 (Dirac, half-integrality), A090 (projection chain, type-I/II), and A264 (asymmetry, CP) with the missing link.

13.9 Use of AI Tools

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Part II

Sectors

Chapter 14

A100 · Lepton Sector I: the ξ-Ladder

14.1 The Ladder

[B] The lepton masses can be written as rungs of a ξ-ladder:

$$m_i = r_i\,\xi^{p_i}\,v,$$

with rational structure numbers r_i, rational exponents p_i, and a bridge constant v. For the leptons these are

$$(r_e, p_e) = \left(\tfrac{4}{3}, \tfrac{3}{2}\right), \quad (r_\mu, p_\mu) = \left(\tfrac{16}{5}, 1\right), \quad (r_\tau, p_\tau) = \left(\tfrac{25}{9}, \tfrac{2}{3}\right).$$

[K] In the *ratios* v cancels, leaving parameter-free expressions:

$$\frac{m_\mu}{m_e} = \frac{12}{5}\,\xi^{-1/2}, \qquad \frac{m_\tau}{m_\mu} = \frac{125}{144}\,\xi^{-1/3}, \qquad \frac{m_\tau}{m_e} = \frac{25}{12}\,\xi^{-5/6}.$$

According to A080 these are the intrinsic quantities: they depend on the geometry and nothing else. They are precisely what is to be measured against.

14.2 Where the Exponents Come From

[K] The exponents p_i are not free. They are winding numbers on a *logarithmic* torus with step size $\Delta p = 1/3$. This step size is the fundamental third of the harmonic series – the same geometry of periodic boundary conditions that generates overtones on a vibrating string. Half-integer values are geometric means between two winding levels, negative values counter-windings.

[K] The three lepton exponents lie on this ladder:

$$p_e = \tfrac{3}{2}, \qquad p_\mu = 1, \qquad p_\tau = \tfrac{2}{3},$$

with steps

$$p_e - p_\mu = \tfrac{1}{2} \text{ (geometric mean)}, \qquad p_\mu - p_\tau = \tfrac{1}{3} \text{ (fundamental third)}.$$

These are the same winding numbers (n_θ, n_ϕ) that also fix the prefactors $r_i = n_\phi^2/n_\theta$ (A110): mass prefactor and exponent come from one source, not two. The ladder is therefore not a fit but the reading of the resonance structure of the torus.

14.3 What the Ladder Delivers

[K] The calculation against PDG values gives:

Ratio	Ladder	PDG	Deviation
m_μ/m_e	207.85	206.77	+0.52 %
m_τ/m_μ	16.992	16.817	+1.04 %
m_τ/m_e	3532	3477	+1.57 %

[K] The tolerance standard from A010 and A080 requires three figures side by side. The measurement precision of the comparison quantities is at m_μ/m_e in the range 10^{-9}, at m_τ about $7{\cdot}10^{-5}$. The determination tolerance of the ladder values is orders of magnitude coarser: the structure numbers are declared rational, not determined to significant figures.

Assessment: the ladder carries the *hierarchy* and the per cent level. It does not carry precision. A one per cent deviation is not a finding against the structure here, because the structure claims nothing more precise at this level – but it is also not a success that could be reported to five figures.

14.4 The Reverse Direction: What the Data Return for ξ

[K] Solving the three ratios for ξ, the data give

$$\xi_{\mu/e} = 1.347 \cdot 10^{-4}, \quad \xi_{\tau/\mu} = 1.375 \cdot 10^{-4}, \quad \xi_{\tau/e} = 1.358 \cdot 10^{-4},$$

against $4/30000 = 1.3333 \cdot 10^{-4}$, i.e. $+1.1$ to $+3.2\,\%$. The three values scatter among themselves by about $2\,\%$.

This is the honest empirical status: the ladder is consistent in the ratios at the per cent level, not exactly. It does not return a uniform ξ.

[K] The same from the other side: solving for the bridge constant per lepton gives 248.9, 247.6, and 245.1 GeV, scattered around 246.22 GeV. The residual deviation is therefore absorbed into the bridge constant and does *not* cancel completely in the ratios.

14.5 The Countercheck That Fails

[K] The obvious conjecture, that the residual deviations follow the fractal correction factor, is *denied* by the calculation. $K_{\text{frak}} = 1 - 100\xi$ corresponds to $-1.333\,\%$, the half and three-half powers to -0.665 and $-1.995\,\%$. The measured deviations are $+0.52$ and $+1.04\,\%$ – they fit neither in magnitude nor in *sign* a single K_{frak} power.

This section is here because a failed countercheck belongs to the finding just as much as a successful one. The obvious explanatory attempt has been tested and has failed; it is not rescued by a free exponent.

The striking doubling is arithmetically forced – and does not evidence a correction. The τ/μ deviation is almost exactly double the μ/e deviation ($1.04\,\%$ vs. $0.52\,\%$), and the τ/e deviation is their sum ($0.52 + 1.04 = 1.56\,\%$). This is not a new law and not a coincidence but a pure bookkeeping necessity: along a multiplicative ladder in which $m_\tau/m_e = (m_\tau/m_\mu)(m_\mu/m_e)$ holds, the logarithmic deviations add. Of the three deviations therefore only *one* is independently measured (the ratio 1.98); the third is fixed as a sum.

Importantly, this step does *not* establish anything about the fractal factor that the countercheck just rejected – on the contrary, it shows that the doubling singles out no correction at all: it already follows from the ladder form alone. One who reads a law from the doubling reads a single measured number three times.

[K] The factor that carries the ladder per generation (its generation-counting step, order $N_0 \approx 39$) is a *different* quantity from the fixed scale 100 in K_{frak}: the one is the ladder step, the other the RG scale of the absolute correction (A040). The value of this ladder step is not derived – that is the one open point named below.

14.6 Two Mutually Exclusive Computation Modes

[STIPULATION] Before any of these deviations is booked, the computation mode must be fixed. There are two, and they exclude each other.

Mode 1, reference-free. All ratios are computed purely from ξ, without anchor. Then all three stand independently against the data, each 0.5 to $1.6\,\%$ off, and all three are legitimate *check points*.

A note on terminology, because otherwise a contradiction with A010 arises: these three ratios are *not* prediction quantities of the series. The prediction list comprises four quantities, and the lepton masses appear there solely via the τ-mass from the circulant (A110), not via the ladder. The ladder provides check points at the per cent level – less than a prediction and more than nothing.

Mode 2, anchored. m_μ/m_e is declared a reference point. Then this ratio is *spent*: its deviation may no longer be booked as a test result, otherwise the same quantity is used twice – once as anchor, once as test.

[B] Mixing the two modes is the most common error in this sector. The A-series consistently uses mode 1, unless something else is explicitly stated.

14.7 Verification Script

- All numbers in this document – ladder ratios, reverse-ξ, bridge constants per lepton, the failed K_{frak} countercheck: `python/A_Serie_Skripte/a100_leiter.py`

14.8 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$r_i \in \mathbb{Q}$	Rational prefactor in $m_i = r_i \cdot \xi^{p_i} \cdot v$
$p_i \in \mathbb{Q}$	Exponent; ladder step size $\Delta p = 1/3$
(n, l, j)	Principal, orbital, total angular momentum of the mode
$v = 246$ GeV	Higgs vacuum expectation value as energy unit

14.9 Open Questions

First, the per cent scatter is intended, not a defect. The ladder is a statement at the per cent level; a ξ back-computed from it scatters accordingly by $2\,\%$. For this reason the sector anchors the absolute scale not in the ladder but in a single reference point (A110): the ladder carries the hierarchy, the reference point the precision.

Second, the exponents p_i are derived, not merely motivated. They are winding numbers on the logarithmic torus (above), with step size $1/3$ and geometric means at half-integer values – the same source from which the prefactors r_i follow. The condition from A020 is thereby met: the exponent is fixed independently, the ξ power is a statement.

Third, the doubling is clarified. It is arithmetically forced (above): along a multiplicative ladder the logarithmic deviations add, only one is independent. No unbooked residual remains.

Fourth, exactly one point remains open – and it is named. The one independently measured deviation corresponds to a generation-counting ladder step of order $N_0 \approx 39$, whose value is not derived. It is to be separated from the fixed RG scale 100 in K_{frak} (A040) and is not explained by the bridge constant in which the residual deviation is absorbed (A080). The value of this step and the candidates for it are collected in A105; whether structure sits behind it is the only genuinely open question of the sector.

14.10 Supersedes

This document subsumes: Doc. 006 (mass formula), Doc. 046 (mass calculation), Doc. 016 (particle masses), Doc. 189 (winding numbers of exponents), Doc. 190 (R55 additivity, P42 reference point), and the ladder sections from Doc. 292. Incorporated: K2 (exponent p_τ), P40 (level separation), P42 (reference point), P35, R55 (evidential breadth: only one point independently measured).

14.11 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 15

A105 · Ladder Base Unit N_0

15.1 What N_0 Is

[B] A100 showed that the rough ξ-ladder of the lepton masses carries a small, generation-dependent correction. Writing the correction required per rung as an effective factor N_g via $K = 1 - N_g\,\xi$, the three factors stand in the ratio

$$N_1 \;:\; N_2 \;:\; N_3 \;=\; 1 \;:\; 1.981 \;:\; 2.971 \;\approx\; 1 \;:\; 2 \;:\; 3.$$

The correction thus grows *linearly with generation number*,

$$\boxed{N_g \approx g \cdot N_0}, \qquad N_0 \approx 38.5,$$

and the three individual estimates N_1, $N_2/2$, $N_3/3$ lie at 38.89, 38.53, 38.52 – consistent with each other to under one per cent. That is the core: N_0 is *not a fixed constant of the theory* but the *ladder step per generation*. At $g = 2$ it is already 77, at $g = 3$ about 116.

[K] This triple consistency is, however, weaker than it looks: only *two* of the three estimates are independent. Because a consistently multiplicative correction $K^{p_3} = K^{p_1}K^{p_2}$ and hence $p_3 = p_1 + p_2$ forces $N_3 \approx N_1 + N_2$ arithmetically (checked: 115.95 vs. 115.55, difference $0.35\,\%$, second order in ξ) – in the ratio $1 : 1.981 : 2.971$ only 1.981 is measured, the third number is fixed as $1 + 1.981$. The generation-linear law therefore rests on *one* observation (the μ/e rung), not three; it is a candidate on one measurement, not a triply supported regularity. The standard from A220 – "do not count dependent as independent" – applies immediately here.

[K] Two things follow immediately. First, N_0 is to be sharply separated from the fixed number 100 in $K_{\text{frak}} = 1 - 100\xi$ (A040): the 100 is the *one* RG scale of a corpus-wide constant, N_0 is a generation counter in a different sector. That both lie near powers of ten does not make them the same quantity. Second, with the generation-linear law the real question is not the *value* of N_0 but why the ladder is corrected in a *generation-counting* way at all. A closed form for the base alone would leave this law untouched.

15.2 The Right Standard: Determination Tolerance Instead of Last Digit

[K] N_0 is not known to arbitrary precision. In the referenced mode its value follows solely from the two τ rungs, and their precision is limited by the τ mass (± 0.12 MeV). Propagating that gives

$$N_0 = 38.52 \pm 0.21,$$

a relative determination uncertainty of about half a per cent. From this follows the standard against which every candidate expression is to be measured: not whether it hits 38.5 to the last decimal – that is in principle unreachable – but whether it *falls within this interval*.

[K] This standard has a converse that carries the whole section: *the band is wide enough that many expressions fall into it*. Hitting a ± 0.21 window is easy. Hence bare compatibility with the band *cannot* be the test that turns a candidate into a derivation. Compatibility says the number fits; it does not say the number *follows*.

15.3 The Candidates

[K] Three expressions lie in or near the band and are to be recorded as *permissible, unrefuted* candidates – none is a derivation:

Expression	Value	Distance
$100/\varphi^2$	38.20	$-1.5\,\sigma$
$3 \cdot (\pi^2/2) \cdot \varphi^2$	38.76	$+1.1\,\sigma$
$\ln(M_{\mathrm{Pl}}/v)$	38.44	$-0.4\,\sigma$
$100/e$	36.79	$-8.3\,\sigma$

with $\varphi = \frac{1+\sqrt{5}}{2}$ and $V_{S^4} = \pi^2/2$ the volume of the unit 4-ball. The first three are compatible; $100/e$ lies far outside and is excluded.

The middle expression $3\cdot(\pi^2/2)\cdot\varphi^2$ lies even closer to the centre than $100/\varphi^2$.

[K] And here the standard from the previous section applies: that three seemingly well-motivated expressions fall into the band is precisely the diagnosis. A fourth would do the same. Band hits do not separate candidates because the band is wide.

15.4 Why None of Them Is a Derivation

[K] The objection falls equally on all three expressions, and it concerns not the *value* but the *form*. For none is there a justification of why precisely this form should be the form of N_0: why the 4-ball volume $\pi^2/2$, why φ^2, why the factor 3 and not 9. What is missing is a chain that begins at the T^4/Z_3 geometry and has this structure *as output at the end* – rather than an expression assembled backwards to hit an already known value.

[K] The role of φ is to be precisely stated, because φ is not foreign to the mass sector. In A120 φ appears *forward and parameter-free* as an eigenvalue of a named construction – the golden inflation $\varphi^2=\varphi+1$ in the icosahedral redistribution. In *this* role φ is permissible and derived. Strictly to be separated from this is φ as a *power inserted into an expression to hit a scale* – φ^N as a fitting base. That is exactly the warning example: it carries no forward content. Both $100/\varphi^2$ and $3\cdot(\pi^2/2)\cdot\varphi^2$ are of the second kind – φ^2 assembled, not produced by a construction. That is why both remain candidates and do not become derivations, even though φ itself is at home in the theory when it appears as output of a named symmetry.

15.5 A Warning Example: Target First, Justification After

[K] How easily a fit dresses itself as a derivation is shown by a concrete case. To the expression $3\cdot(\pi^2/2)\cdot\varphi^2=38.76$, which lies $+1.1\sigma$ above the centre, an "anomaly" term Δ was subsequently added to close the residual. Its value:

$$\Delta=\frac{3}{4\pi}=0.2387,$$

and the gap to be closed is $38.758 - 38.52 = 0.238$ – the same number to three figures. The correction was therefore not derived and then found to fit; the gap was known first, and a term of the right size was *subsequently* given a name. Two physical features confirm this independently of the coincidence: a one-loop contribution would carry the factor $1/(4\pi)^2$, not linear $1/(4\pi)$; and a genuine localised anomaly would be *quantised* – an integer index, not a fractional 0.2387. The physical dress does not fit the number underneath it. That does not refute 38.52 as a candidate; it only shows that the anomaly narrative is not the reason why the value sits there.

[K] From this follows the test rule that A220 records generally and that applies particularly here: a derivation is not to be assumed in prose but to be written as a script that begins at the geometry, contains *no target value*, and one checks what number it prints. Prose can hide a fitted step behind a secure sentence; code cannot. A value hitting the centre of a ± 0.21 band to the last digit is not a triumph but a warning: nothing fundamental lands exactly on the mean of a measurement unless it was built to do so.

15.6 Verification Script

- The tolerance $N_0 = 38.52 \pm 0.21$, the σ distances of the four candidates, the target back-computation $\Delta = 3/(4\pi)$ vs. the gap, and the generation run $N_g = g N_0$: `python/A_Serie_Skripte/a105_n0.py`

15.7 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
N_0	Recursion limit; value at which the iteration converges
δ	Refraction strength; deviation from ideal torus lattice
Δ_{rec}	Recursion deficit; difference to next closure

15.8 Open Questions

First, N_0 is open. It is well bounded as a value within its tolerance, but not derived from the geometry. No expression presented so far is more than a permissible candidate.

Second, the standard is named. The question is closed by a construction that begins at the T^4/Z_3 geometry, fixes invariant, exponent, and scales, and delivers N_0 at the end – a construction that could have missed and did not –, not by another well-looking expression that falls into the band.

Third, the generation law remains the deeper task. Even a perfect closed form for the base does not explain why the correction $N_g = g\,N_0$ runs generation-counting. That is the real open question, and it lies beneath the question of the numerical value.

15.9 Supersedes

This document subsumes: Doc. 292 (generation-linear coupling, $N_g = g\,N_0$, tolerance $N_0 = 38.52 \pm 0.21$, candidates $100/\varphi^2$, $100/e$), Doc. 190 (R54/R55: N_0 open, separation K_{frak}-100 / N_0; P35). Incorporated: the external candidate $3 \cdot (\pi^2/2) \cdot \varphi^2$ and the $\Delta = 3/(4\pi)$ target back-computation from the IPI correspondence (July 2026), booked as permissible-but-not-derived candidate and as warning example respectively.

15.10 Use of AI Tools

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Chapter 16

A110 · Circulant and Koide: $Q = 2/3$

16.1 Purpose: Lepton Masses from One Parameter

[K] The lepton sector is where the theory performs most strongly and is most sharply testable. From the single parameter $\xi = 4/30000$ and a rational assignment per lepton, the three charged lepton masses follow, their ratios, the Koide relation $Q = 2/3$, and – the sharpest point – a prediction of the τ-mass. This document develops both: the *ladder* that generates the masses, and the *circulant* that sharpens the Koide structure.

16.2 The Yukawa Ladder

[K] The basic mass formula is

$$m_i = r_i \, \xi^{p_i} \, v, \qquad v = 246.22 \text{ GeV (Higgs-VEV)},$$

with rational prefactors r_i and rational exponents p_i per lepton:

Lepton	r_i	p_i	m_i (calculated)
Electron	$4/3$	$3/2$	0.505 MeV
Muon	$16/5$	1	105.1 MeV
Tau	$25/9$	$2/3$	1785 MeV

[K] The exponents form a ladder with fixed steps:

$$p_e - p_\mu = \tfrac{3}{2} - 1 = \tfrac{1}{2}, \qquad p_\mu - p_\tau = 1 - \tfrac{2}{3} = \tfrac{1}{3}.$$

These steps $\frac{1}{2}, \frac{1}{3}$ are the core: they generate the $\sqrt{m}$-structure on which the Koide formula rests, and they are what recurs throughout the following sections.

16.3 Origin of the Prefactors: Resonances on the Torus

[K] The prefactors r_i are not freely chosen. Each lepton corresponds to a standing field mode on the torus, characterised by two integer winding numbers (n_θ, n_ϕ) – the number of circuits along each cycle. This is the resonance condition: only modes with integer winding are permitted, as with any standing wave on a ring.

[K] The prefactor follows directly from the winding numbers:

$$r_i = \frac{n_{\phi,i}^2}{n_{\theta,i}}.$$

For the three leptons:

Lepton	n_θ	n_ϕ	$r_i = n_\phi^2/n_\theta$
Electron	3	2	$4/3$
Muon	5	4	$16/5$
Tau	9	5	$25/9$

[K] This explains the "structure in the numerators" visible in the r_i: the numerators $4, 16, 25$ are the squares $n_\phi^2 = 2^2, 4^2, 5^2$, the denominators $3, 5, 9$ are the n_θ. The r_i are simple ratios of two resonance quantum numbers, not fitted values. The masses of the sector thus follow from ξ, the ladder exponents, and a pair of integer winding numbers per lepton.

[K] The same r_i carry a second, equivalent interpretation: they are the consonant intervals of just intonation. In just (5-limit) tuning, consonant intervals are ratios of small integers with prime factors $2, 3, 5$ – and that is exactly what the r_i are:

Lepton	r_i	Prime factors	Just interval
Electron	$4/3$	$2^2/3$	Perfect fourth (498 ct)
Muon	$16/5$	$2^4/5$	Min. sixth + octave (814 ct)
Tau	$25/9$	$5^2/3^2$	Square of the major sixth

[K] This is not an analogy but the same fact seen from two sides: a standing mode on a torus cycle *is* a harmonic resonance, like the overtones of a vibrating string. The winding numbers are the order numbers of these resonances; the $r_i = n_\phi^2/n_\theta$ are the corresponding interval ratios.

[K] The three charged leptons are the *simplest* of these resonances – the low 5-limit consonances. The series does not stop there: higher modes are the remaining fermions. The quarks are further overtones of the same torus: the top is the only overtone above the Higgs ground tone, the bottom with $r = 3/2$ – the perfect fifth – the harmonically simplest quark. There is therefore no cutoff at three modes; there is an ascending resonance ladder, and the leptons are its lowest part.

[K] The *ordering* is therefore necessary, not chosen: resonances are ordered by their order number, ascending winding number means ascending mass. In a theory where particles are resonances, "lightest resonance = electron, next = muon, then tau" is not an additional rule but the resonance structure itself.

[K] The tau prefactor $r_\tau = 25/9 = (5/3)^2$ is the square of the major sixth. An earlier iteration carried $r_\tau = 8/3$ (giving $m_\tau \approx 1712$ MeV, $-3.6\,\%$); the correct value $25/9$ gives $m_\tau \approx 1783$ MeV and is recorded corpus-wide.

16.4 The Three Mass Ratios from ξ

[K] Because the masses are products $r_i\,\xi^{p_i}\,v$, the absolute v cancels in every ratio, leaving pure ξ-geometry:

$$\frac{m_e}{m_\mu} = \frac{r_e}{r_\mu}\,\xi^{p_e-p_\mu} = \frac{4/3}{16/5}\,\xi^{1/2} = \frac{5}{12}\,\xi^{1/2} = 0.004811,$$
$$\frac{m_\mu}{m_\tau} = \frac{r_\mu}{r_\tau}\,\xi^{p_\mu-p_\tau} = \frac{16/5}{25/9}\,\xi^{1/3} = \frac{144}{125}\,\xi^{1/3} = 0.05885,$$
$$\frac{m_e}{m_\tau} = \frac{r_e}{r_\tau}\,\xi^{p_e-p_\tau} = \frac{12}{25}\,\xi^{5/6} = 0.000283.$$

[K] Against experiment (PDG):

Ratio	from ξ	PDG	dev.
m_e/m_μ	0.004811	0.004836	0.51 %
m_μ/m_τ	0.05885	0.05946	1.02 %
m_e/m_τ	0.000283	0.0002876	1.6 %

All three to roughly one percent, from a single parameter and rational numbers.

16.5 Koide as Consequence: The Independent Route

[K] The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2}$$

is not an independent empirical symmetry but follows from the ladder. Inserting the three calculated masses – *without* using Koide anywhere – gives

$$Q_{\text{ladder}} = 0.66774,$$

which is $0.16\,\%$ from $2/3$. The geometric ladder thus predicts the Koide quantity on its own. That the experimental value then hits exactly $2/3$ is an *independent* confirmation of the exponent structure, not its prerequisite.

[B] The reason Q comes out so accurately is ratio-invariance: since only exponent *differences* enter and v cancels, Q depends solely on $\xi^{p_i - p_j}$ and is free of any absolute scale. This is the purest form in which ξ appears.

16.6 Three Routes to 2/3

[B] At exactly $D = 3$ (the topological dimension of the T^4 sector) three different structures meet at the same value $2/3$:

$$1 - \frac{1}{D} = \frac{2}{3}, \qquad \frac{2}{D} = \frac{2}{3}, \qquad \frac{1}{2}\left(1 + \frac{1}{D}\right) = \frac{2}{3}.$$

The first is the complement step, the second the Koide equilibrium condition between democratic and orthogonal projection components ($A = B$), the third the ladder ratio. A formal deformation $D = 3 + \varepsilon$ separates the three again – they coincide only at integer $D = 3$. This binds $2/3$ to the *integer* topological dimension, not to the fractal $D_f = 3 - \xi$.

16.7 The Circulant: Sharpening to the Tau Mass

[K] Alongside the ladder there is a second, complementary description: the three $\sqrt{m_i}$ as discrete Fourier transforms of a Z_3-circulant. In this representation two structural quantities appear, an amplitude and a phase angle θ, and the corpus finds

$$\theta = \frac{2}{9}.$$

[K] Two conditions then fix m_τ: the Koide condition $Q = 2/3$ and the phase condition $\theta = 2/9$. Both are formally distinct, but at the current measurement precision they coincide (distance $< 0.05\,\sigma$), delivering the same m_τ; the sharp prediction is

$$m_\tau^{\text{pred}} = 1776.968 \text{ MeV}.$$

[K] The bookkeeping of the sector is decisive here. The lepton masses are *not* inputs to the theory: the structural numbers ($\sqrt{2}$ from $Q = 2/3$, $2/9$ as address) are theory-side. To anchor to the measured pole masses, exactly *one* declared reference point is used: the ratio m_μ/m_e. It fixes the operative angle $\theta_{\text{ref}} = 0.2222220471$ and is thereby consumed as a calibration quantity.

[K] The gain of this one anchor choice is the sharp prediction:

$$m_\tau = 1776.9690 \pm 0.0001 \text{ MeV},$$

input-error-free, $+0.91\,\sigma$ against PDG 1776.86 ± 0.12 (Belle-II world average decides). This is the series' actual prediction (A210, A220): a numerical value fixed *before* measurement – not a bookkeeping value and not a scale choice like α or G, but a genuinely measurable quantity, because the ratio structure forces the third mass from the two light ones.

[B] The Q-route and the θ-route are *two* formally distinct conditions, not one. At current measurement precision they coincide (distance $< 0.05\,\sigma$), delivering the same m_τ; they are separated only by a measurement that determines m_τ more sharply than this bound.

16.8 The xi-Cascade: Why the Masses Are Most Precise

[B] The lepton sector stands at the pure end of a hierarchy that orders the whole theory:

$$\xi \xrightarrow{\text{ratios}} Q = \tfrac{2}{3} \xrightarrow{\text{scale}} \alpha \xrightarrow{\text{SI metric}} G.$$

Each step adds a layer of complexity, and precision falls accordingly:

Quantity	requires	precision
Q (Koide)	ratios only	$0.00003\,\%$
α	+ energy scale E_0	$0.006\,\%$
G	+ Planck length	$0.5\,\%$

[K] This explains the accuracy classes from A220 from *one* principle: the closer a quantity is to pure ratios, the more accurately it follows from ξ; the more absolute scale it requires, the more bookkeeping enters. The lepton masses are the purest form and therefore the most precise and the only genuine prediction. α and G are the same ξ-geometry, only in increasingly dressed form (A130, A145).

16.9 Verification Script

- The Yukawa ladder $m_i = r_i \xi^{p_i} v$ with the three mass ratios from ξ (range check against PDG); the ladder prediction $Q_{\text{ladder}} = 0.6677$ without Koide input; the two-route sharpening to m_τ (prediction, $0.90\,\sigma$); and the ξ-cascade with falling precision: `python/A_Serie_Skripte/a110_zirkulant.py`

16.10 Symbol Glossary

Symbols used throughout the A-Series are explained in A010. Specific to this document:

Symbol	Meaning
C	3×3-circulant matrix; diagonalisable via DFT
$r=\sqrt{2}$	eigenvalue ratio; condition for $Q=2/3$
$Q=(\sum m_i)/(\sum\sqrt{m_i})^2$	Koide operator; measured $Q\approx 2/3$
$\omega=e^{2\pi i/3}$	primitive third root of unity

16.11 Open Questions

First: the mode ordering is not an open point. The three charged leptons are the simplest torus resonances, and their ordering follows necessarily from the ascending resonance order (above). Higher modes are the remaining fermions; nothing cuts off at three. What is set – and exclusively that – is the *cardinality* three via the order Z_3 (A020). Z_3 is an axiom, not a derivation, and an axiom needs no written derivation to be valid: every theory enters its own framework at some point. Z_3 is motivated by the parsimony of the three-point configuration in the two-dimensional cross-section and by agreement with the generation number – but these reasons do not make the axiom a derivation, and their absence would not be a deficiency. What is to be avoided is only the reversal: reading the three generations as a *consequence* of Z_3. In fact the observed number three motivates the choice Z_3, not vice versa.

Second, the percent scatter of the ladder is not an open point but the reason for the reference-point booking. The ladder is a percent-level statement; a ξ back-calculated from it scatters accordingly. That is exactly why the sector anchors the scale not in the ladder but in *one* reference point m_μ/m_e. The circulant with the pair $(\sqrt{2},2/9)$ misses m_μ/m_e at pole-mass level by $1.0\cdot 10^{-5}$ – which at this level excludes reading the pair as *exact*; as a rational address to seven digits it remains valid. The booking is thus resolved, not open.

Third, $\theta=2/9$ is precisely located, not unmotivated. $2/9$ is the rational address of the reference angle θ_{ref} to seven digits ($|\theta_{\mathrm{ref}}-2/9|=1.75\cdot 10^{-7}$). It lives in the angular phase sector and corresponds to the phase position $\chi=\pi/2$; the radial mass sector remains rigid. What stands as a candidate – not an achieved result – is only the *dynamic selection*: whether a skew-adjoint generator outside the static geometry locks the phase onto $\chi = \pi/2$. FFGFT's own real recursion carries no such generator; whether an external dynamics supplies it is the only remaining open point – the location and address of $2/9$ are fixed.

Fourth, the dependence on m_τ is the falsifiability, not a deficiency. After the reference-point booking, m_μ/m_e is the consumed anchor and $m_\tau = 1776.9690 \pm 0.0001$ the one sharp test quantity. That the sector stands or falls by it is intentional: a measurement placing m_τ clearly away from 1776.97 refutes the exponent structure (A210). That is the purpose of a prediction, not its weakness.

16.12 Supersedes

This document subsumes: Doc. 116 (Koide from ξ, Yukawa ladder, ξ-cascade), Doc. 258 (three routes to $2/3$, $D = 3+\varepsilon$), Doc. 259 (cross-terms), Doc. 060 (musical resonance interpretation of prefactors), Doc. 189 (torus derivation of winding numbers), Doc. 190 (register K2 on r_τ, R56 on Z_3), Doc. 282 (circulant, HLV bridge), Doc. 291 (dynamic location of $\theta = 2/9$, $\chi = \pi/2$), Doc. 292 (empirical lepton check). Incorporated: P40 (ratios/bridge constants), P42 (one reference point m_μ/m_e).

16.13 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 17

A120 · $\theta = 2/9$: Location, Status and Boundary

17.1 The Question

$\theta = 2/9$ is empirically confirmed to seven figures (A110). This document shows three things: *where* the value resides (solely in the dynamic phase channel, not in the magnitude), *that* it follows exactly and parameter-free from the icosahedral doubling geometry, and *what* remains open as a derivation step (the bridge to the mass ratios).

The central finding is positive: two mutually independent paths – the phenomenological Koide angle from the measured masses and the purely geometric redistribution of modes – hit exactly the same rational number $2/9$, without knowing about each other.

17.2 The Magnitude Channel Is Blind

[K] With amplitudes $a_k(\theta) = 1 + \sqrt{2}\cos(\theta + 2\pi k/3)$ the Koide invariant is

$$Q(\theta) = \frac{\sum_k a_k^2}{\left(\sum_k a_k\right)^2} = \frac{6}{9} = \frac{2}{3} \qquad \text{for } \textit{every } \theta.$$

The proof is two lines: three cosines offset by 120° sum to zero, hence $\sum_k a_k = 3$; and with $\sum_k \cos^2 = 3/2$ it follows that $\sum_k a_k^2 = 6$.

[K] From this follows a statement to be recorded, because it limits the reach of A110: *the Koide relation says nothing about* θ. $Q = 2/3$ holds for every angle. The value $2/9 = 0.2222$ also lies *below* the range $[1/3, 1]$ that Q traverses for positive masses – $2/9$ is therefore neither a limit, nor an extremum, nor a fixed point of the symmetric functional.

In the magnitude channel $2/9$ is completely invisible.

17.3 The Elimination Chain

[K] Systematically checked where θ does *not* live. The orbit members $\{1/9, 2/9, 4/9\}$ are to be distinguished; the question is which quantity achieves this.

Channel	Behaviour	Distinguishes?
Magnitude $\lvert B_\theta \rvert$	$= 1$ to 10^{-15} for all θ	no
Winding number	$= 3$ exactly for all θ	no
Symmetric functional	identical for all θ	no
Continuous phase profile	0.093 / 0.171 / 0.249	**yes**

[K] Noteworthy is the second row. The integer winding – precisely the topological quantity that, according to A030, carries the information concept of the theory – does *not separate* $2/9$ from $1/9$ and $4/9$. The reason is that $2/(9\pi)$ is irrational; θ is not a flat topological invariant.

[K] Exactly one channel remains: the continuous, dynamic phase profile. There and only there is $2/9$ distinguishable from its orbit companions.

17.4 Forced or Contingent

[K] This raises the decisive question: is $2/9$ forced like the speed of light, or contingent?

The test is the variational behaviour. Coupling a second triplet with strength δ, the extremum moves away from the Z_3 point: θ^* rises smoothly and monotonically from 0.099 at $\delta = 0.10$ through 0.223 at $\delta = 0.24$ to 0.501 at $\delta = 1.0$. The achieved range $[0.05; 0.52]$ contains $2/9$ *in the interior and undistinguished*.

[K] This test sets δ as a *free* coupling parameter. Under this assumption $2/9$ would lie undistinguished in the achieved range and the value would be contingent. But the assumption is not that of the theory: the icosahedral geometry fixes δ (next section). Once δ is no longer free but the total leak $(7 - 3\varphi)/9$ of the same redistribution that also carries θ, the contingency falls away. The variation test therefore shows that θ *lives in the dynamic channel* – it does not decide forced/contingent, because it assumes δ to be free, which the geometry does not.

17.5 The Geometric Source: C_3 in A_5

[B] The dynamic channel has a geometric source. The three lepton modes are the Fourier eigenmodes of the Z_3 circulant: the trivial mode $v_0 = \frac{1}{\sqrt{3}}(1,1,1)$ and two non-trivial, Galois-conjugate modes. The doubling places the C_3 structure into the icosahedral group A_5; its fivefold generator is a rotation R_5 by 72° about an icosahedral fivefold axis. Because the threefold and fivefold axes are differently oriented, R_5 mixes the C_3 modes.

[B] The complete redistribution of the electron mode has weights $p_j = |\langle v_j \,|\, R_5\, v_e\rangle|^2$:

$$\underbrace{p_0 = \tfrac{2}{9}}_{\text{trivial}}, \qquad p_2 = \tfrac{5-3\varphi}{9}, \qquad p_1 = \tfrac{2+3\varphi}{9}, \qquad p_0 + p_1 + p_2 = 1,$$

with $\varphi = \frac{1+\sqrt{5}}{2}$. *All three weights carry the denominator* $9 = 3^2$ – the same Z_3 structure in which $2/9$ sits as orbit address. The fraction that transitions into the trivial mode under the fivefold rotation is

$$p_0 = \left|\langle v_0 \,|\, R_5\, v_e\rangle\right|^2 = \frac{2}{9}$$

exactly and parameter-free (numerically confirmed to 40 places). No mass ratio, no Koide parametrisation, and no $2/9$ input enters this value.

[K] The value is icosahedron-*specific*, not generic:

- 200 random rotation axes at 72° give the leak in the range $[0.036; 0.452]$; *not one* hits $2/9$. Only the icosahedral fivefold axes deliver it.
- The angle scan shows $2/9$ exactly at 72°; the n-fold series gives $2/9$ exclusively at $n = 5$ (at $n = 3$ one gets $2/5$, at 144° one gets $4/9$).
- Of the six fivefold axes, three give $2/9$, three give $4/9$; the embedding selects the nearest axis and hence $2/9$. That is a discrete binary choice, not a free parameter.

[B] The same redistribution also delivers the second quantity: the total leak $p_0 + p_2 = (7 - 3\varphi)/9 = 0.2384$ is the parameter-free value of the coupling δ and agrees with the $\delta^* = 0.2389$ back-computed from θ to $0.18\,\%$. Angle *and* amplitude therefore arise from a single fivefold redistribution; the earlier separate treatment as two harmonics was too narrow a view.

[K] Thus two independent paths coincide exactly: the phenomenological Koide angle (from the masses, without geometry) and the geometric redistribution (from A_5, without masses and without $2/9$ input) both give $2/9$. That is a convergence finding. $2/9$ is an invariant of the translation between torus geometry and spectral representation – it belongs to the common structure, not to one representation, analogously to $\hbar$ and c being conversion factors and not ontological primitives.

[B] The connection to the dynamic channel closes here: the fivefold rotation R_5 sets a phase $\pi/2$ – precisely the value $\chi = \pi/2$ that marks the one distinguishing channel above. The geometric source and the dynamic location are the same.

17.6 Verification Script

- θ-independence of Q (exact, for arbitrary angles), the elimination chain through the three blind channels, and the contingency test $\theta^*(\delta)$ with the proof that $2/9$ lies undistinguished in the achieved range: `python/A_Serie_Skripte/a120_theta.py`

17.7 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$\theta = 2/9$	Geometric fixed-point angle; dynamic locus on T^4
$p_0 = 2/9$	Exact θ value as C_3-in-A_5 identity
Ω_{res}	Resonance frequency of the θ-mode
$\Delta\omega$	Frequency spacing to the next mode

17.8 Open Questions

First, the value $2/9$ is derived, not open. It follows exactly and parameter-free from the icosahedral redistribution and independently coincides with the phenomenological Koide angle. The earlier classification as structurally unjustified is thereby superseded.

Second, δ is not free. It is the total leak $(7 - 3\varphi)/9$ of the same redistribution that carries θ. The contingency impression arose solely from treating δ as a free parameter; the geometry fixes it.

Third, one derivation edge remains open – and is precisely named. The explicit mechanism connecting the geometric redistribution weights $(2/9, (5 - 3\varphi)/9, (2 + 3\varphi)/9)$ with the physical mass ratios has not been written out. This bridge need not run via Koide; it can be led directly through the masses or the mode structure. That is the subject of further work – and explicitly not an objection to the convergence finding, but the next stage.

Fourth, the underlying doubling structure A_5 is presupposed. The value $2/9$ is exact within the established icosahedral doubling; whether this doubling itself necessarily follows from the three stipulations of A020 or enters as an additional structure is the remaining structural question – one that goes beyond the five-fold counting but no longer touches the value $2/9$.

17.9 Supersedes

This document subsumes: Doc. 291 (dynamic locus), Doc. 293 (icosahedral approach), Doc. 286 (dynamic sector), Doc. 290 (magnitude/phase, insofar as concerning θ). Incorporated: P42, P35.

17.10 Use of AI Tools

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Chapter 18

A130 · Fine Structure Constant α

18.1 The Thesis: In Natural Units $\alpha = 1$

[K] The strongest statement about the fine structure constant is the simplest: in natural units

$$\alpha = 1.$$

The famous value $1/137.036$ is *not* a mystery of nature, but the bookkeeping that arises when converting to SI units – just as $c = 3 \cdot 10^8$ is the bookkeeping of the metre-second choice.

This is not an invention of this theory. Setting α to one is a *known* unit convention: one normalises the charge so that $e^2 = 4\pi$ (rationalised Heaviside-Lorentz units with $\varepsilon_0 = \mu_0 = 1$). In these units $\alpha = 1$ by construction. The theory adopts this convention; it does not re-justify it.

But the reason is different from $c, \hbar, k_B$. For $c, \hbar, k_B$, the conversion factors become one because time-mass duality identifies their dimensions (A085). α is dimensionless; this justification does *not* apply here. The convention $\alpha = 1$ rests on the electromagnetic charge normalisation – a separate reason having nothing to do with time-mass duality and therefore handled here separately, not under the natural-units topic.

18.2 How $\alpha = 1$ Is Derived

[B] The derivation runs through the charge normalisation and Maxwell duality, not through an axiom. In SI:

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c}.$$

With $\hbar = c = 1$ and the natural choice $\varepsilon_0 = 1$, the charge normalisation $e^2 = 4\pi$ follows from $\alpha = 1$. The Maxwell relation $c^2 = 1/(\varepsilon_0\mu_0)$ with $c = 1$ and $\varepsilon_0 = 1$ then gives $\mu_0 = 1$. Substituting confirms:

$$\alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} = \frac{4\pi}{4\pi \cdot 1 \cdot 1 \cdot 1} = 1.$$

[K] The core is electromagnetic duality $\frac{1}{\varepsilon_0 c} = \mu_0 c$, which is perfectly satisfied in natural units ($1 = 1$). $\alpha = 1$ is the expression of this balance between electric (ε_0) and magnetic (μ_0) coupling.

Honest assessment. The step itself is standard: the charge normalisation $e^2 = 4\pi$ is the rationalised convention in which $\alpha = 1$ comes out. What is non-standard is the *interpretation*: that the SI value $1/137$ is thereby not a fundamental mystery of the coupling but measures the historical charge unit. Not everyone shares this interpretation – the classical Duff position holds α to be absolute on account of its dimensionlessness. The theory's actual contribution lies not here, however, but in the next section: that the SI value not only measures charge units but, via $\alpha = \xi E_0^2$, carries the geometric parameter ξ.

This removes the mystification. The question "why exactly $1/137$?" is a question for the SI system, not for nature – of the same kind as "why exactly $299{,}792{,}458$ metres per second?". The numerical value measures the historical charge unit, not a property of the coupling itself.

18.3 What Remains: ξ and One Scale

[K] When all constants are one in natural units, the only genuine content remaining is the single dimensionless parameter ξ and one scale. The SI value of α carries precisely these two quantities:

$$\alpha_{\mathrm{SI}} = \xi \cdot \left(\frac{E_0}{1\,\mathrm{MeV}}\right)^2.$$

Here α is no longer the native one but its SI dress. In this dress one sees that α and ξ encode the same quantity, connected by the scale E_0: ξ is the geometric reading (packing ratio, A020), α_{SI} the electromagnetic one. Two sides of one coin, one degree of freedom – just as mass, energy, temperature, and inverse time are one quantity in four dresses (A085).

18.4 Why the Measured Value Is Irrelevant

[K] The theory works with ratios (A080). Ratios are unit-free and depend only on the geometry. The absolute value of α – $1/137.036$ – is an absolute value and therefore belongs to the bookkeeping layer, not the core.

[K] The *structural relations* at issue are already known and confirmed in SI units and in the Standard Model. The theory does not invent them, it orders them: $\alpha = \xi E_0^2$ connects known quantities. What it asserts is the *structure* of this connection – that α and ξ are the same degree of freedom – not a new numerical value for α.

[B] It follows how the numbers in the following sections are to be read: *not* as a test the theory passes or fails. They show the precision with which the bookkeeping reproduces the SI value – a statement about the bridge constant, not about the structure. As long as one works with ratios, they are irrelevant.

[STIPULATION] The test of the theory lies elsewhere. α is not one of the four prediction quantities of the series (A010); those are ratios and structural statements (m_τ, the ξ-order of the CHSH deviation, D_f, L_0). Whether α approaches $1/137$ to three or five digits does not test the theory – it tests the SI conversion.

18.5 Why the Circularity Is Not a Weakness

[STIPULATION] As soon as one works in SI dress, the system is circular, and rightly so. Every system that fixes its own constants is circular – the only question is where one begins. (In natural units the question does not arise at all: there $\alpha = 1$ and nothing to determine.)

One can start from ξ (geometric perspective): then ξ is primary and $\alpha = \xi E_0^2$ derived. One can start from α (electromagnetic perspective): then α is primary and $\xi = \alpha/E_0^2$ derived. Both routes give the same values for all other constants. They are mathematically equivalent.

[B] The question "can α be derived from ξ without circularity?" is therefore wrongly posed. In a closed system there is no exterior from which one could derive. There is an entry point one declares, and the rest that follows from it. Classical physics conceals this by placing several constants side by side as "fundamental"; this theory makes explicit that only one entry point is needed.

A mistake would still be possible here, namely presenting the entry point as a derivation from nothing. Whoever substitutes $E_0 = \sqrt{\alpha/\xi}$ and then announces "exactly the measured value of α" confuses reading consistency with proof. The E_0 determination avoids exactly this mistake, because it does *not* run via $\sqrt{\alpha/\xi}$: the exact E_0 is reached by Route 2 independently of α (previous section), and the agreement of the two α-free routes makes the α point over-determined rather than merely consistent. The circularity that always threatens in a closed system is therefore broken at this point by two separate approaches.

18.6 The Scale E_0 as Entry Point

[K] What separates ξ from α is only the scale E_0. For it there are *two mutually independent* routes, both not using α and confirming each other.

[K] *Route 1 (empirical).* The geometric mean of the two lightest lepton masses,

$$E_0 = \sqrt{m_e m_\mu} = 7.348 \text{ MeV},$$

gives when substituted $\alpha = \xi \cdot (7.348)^2 = 1/138.9$, thus $-1.35\,\%$ from the measured value.

[B] *Route 2 (geometric).* A construction from ξ and m_μ alone,

$$E_0^2 = 4\sqrt{2}\,\frac{m_\mu}{\xi p},$$

gives $E_0 = 7.398$ MeV – *without* using the geometric mean, *without* m_e, and *without* back-calculation from α. Substituted, this hits α to $5 \cdot 10^{-6}$.

[K] Decisive is that the two routes *meet* without knowing each other. Their ratio is

$$\frac{7.398}{7.348} = 1.00682 = K_{\text{frak}}^{-1/2} \quad \text{to } 9 \cdot 10^{-5},$$

with the same fractal factor $K_{\text{frak}} = 1 - 100\xi$ derived in A040 and acting in the corpus at an entirely different point. Since E_0 enters α quadratically, the $K_{\text{frak}}^{1/2}$ bridge at E_0-level corresponds to a full K_{frak} correction at α-level. Two separate derivations thus meet to $8 \cdot 10^{-5}$, and the depth of the correction is not free but structurally fixed.

[K] The α point is thereby *not a calibration but over-determined*. The exact $E_0 = 7.398$ is not "formed backwards from α" but reached forward by Route 2; and that precisely the derived K_{frak} closes the gap between the two routes is a cross-sector consistency, not a fit. Because Route 2 does not use m_e, the agreement with Route 1 is even a genuine statement about m_e: the remaining residual of $8 \cdot 10^{-5}$ is too large for mass measurement noise and is thus a test of the pole mass against the geometry.

18.7 The Two Paths Also Confirm K_{frak}

[K] The over-determination says more than has been exploited so far. Neither Route 1 nor Route 2 uses K_{frak} anywhere: Route 1 is the geometric mean of two measured masses, Route 2 a construction from ξ and m_μ. Their ratio is therefore a *measurement* of K_{frak} that does not presuppose the factor:

$$K_{\text{frak}} = \left(\frac{E_0^{(2)}}{E_0^{(1)}}\right)^{-2} = \left(\frac{7.398}{7.348}\right)^{-2} = 0.9862.$$

[K] This value read from the two routes hits the factor independently derived in A040:

$$K_{\text{frak}} = 1 - 100\xi = \frac{74}{75} = 0.98667$$

to $9 \cdot 10^{-5}$. This is remarkable because the two origins have nothing to do with each other: A040 obtains K_{frak} from the cumulative RG run and the consistency condition on the effective fractal dimension; the α sector obtains *the same* value from the difference between two energy-scale routes. A factor originating from an RG argument that simultaneously closes the gap of two independent E_0 determinations is not freely chosen; it is fixed from two sides.

[B] K_{frak} thus has a *second witness* here. The α sector confirms not only the value of E_0 and α, but via the route difference also that $K_{\text{frak}} = 74/75$ has the correct value – without inserting this factor anywhere. What in A040 is a derived constant is here additionally *measured*.

18.8 A Coincidence That Must Be Resisted

[K] It is tempting to search for a *more elegant* expression for E_0 than the mass route, and there is a striking one: Euler's number squared,

$$\mathrm{e}^2 = 7.389\ \text{MeV}\ (?), \qquad \alpha = \xi \cdot (7.389)^2 = \frac{1}{137.37} \quad (-0.24\,\%),$$

which hits the SI value even better than the mass route. This expression should be *resisted*, for a principled reason: it equates a pure number (Euler's number) with an energy in the arbitrary unit MeV. One MeV is a human convention; no pure number can fix a physical scale *in* MeV without the agreement depending on the unit choice. In a different energy unit a different number would appear, and the coincidence would vanish.

[K] This is exactly the pattern the corpus warns against elsewhere (A105): a number fitted backwards to an elegant expression without forward content. That e^2 hits closer than $\sqrt{m_e m_\mu}$ does *not* make it the better candidate – the mass route carries a physical reason (the lepton masses), the Euler coincidence carries none. It is recorded here only to explicitly *not* book it as an anchor. The elegant expression is not needed anyway: the exact E_0 is already hit by Route 2 with forward content, while e^2 merely holds a number to a unit.

18.9 Verification Script

- The equivalence of both perspectives ($\alpha = \xi E_0^2$ and $\xi = \alpha/E_0^2$ deliver identical values); the α-free harmonic route with $\kappa = 4$;

the two numerical routes; and the explicit check that substituting $\sqrt{\alpha/\xi}$ is a consistency and not an independent proof: `python/A_Serie_Skripte/a130_alpha_kette.py`

18.10 Open Questions

First, the exact E_0 is α-free and over-determined – not open. The exact $E_0 = 7.398$ is reached by Route 2 from ξ and m_μ, without α and without m_e; it agrees with the mass route (Route 1) up to the derived factor $K_{\text{frak}}^{-1/2}$. The earlier classification that the exact value was formed only backwards from α is thereby superseded. The Euler coincidence $E_0 = \mathrm{e}^2$ remains explicitly *not* a derivation and is not needed.

Second, the absolute anchor remains tied to a measured mass. Route 2 fixes E_0 from ξ and m_μ; the absolute scale therefore enters via *one* measured mass anchor. A determination of the exact E_0 from purely dimensionless geometry without any mass anchor does not exist – but this is the general structure of the SI bridge (A080), not a deficiency specific to the α sector. Since the theory tests via ratios, the remaining residual carries no weight for its evaluation.

Third, a lower bound applies. The relative description $\alpha \leftrightarrow \xi$ is secured down to the sub-Planck bound $L_0 = \xi \ell_P$ (A210); below this the theory makes no statement.

18.11 Supersedes

This document subsumes: Doc. 011 (fine structure constant), Doc. 043 (resolution of constants), Doc. 044, Doc. 101 (circularity of constants, core thesis: one degree of freedom, ξ and α equivalent). Incorporated: P40.

18.12 Use of AI Tools

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Chapter 19

A135 · What Can Be Set to Unity

19.1 The Diagnosis

[K] From A085 (the constants of nature $c, \hbar, k_B$) and A130 (α), a single sharp statement can be drawn that until now appeared only scattered:

> *Every constant that can be set to one through a consistent unit or normalisation choice is thereby proved to be* not fundamental. *It is a conversion factor, not a degree of freedom.*

The settability to one is therefore not merely a convenient notation but a *test*. What can be brought to one without losing any physical statement was never content of nature but a property of the measurement system used. The famous numerical value – $c = 299\,792\,458$, $1/\alpha = 137.036$ – then measures the *historical unit*, not nature.

19.2 Two Paths to One, One Result

[K] There are two different reasons why a constant can be brought to one, and it is important to distinguish them – because the result is the same in both cases, but the reason is not.

[B] *First, via a physical identity.* c, $\hbar$, k_B become one because the time-mass duality *identifies* their dimensions (A085): mass, energy, temperature, and inverse time are one quantity in four guises. The $= 1$ is here the reading of an identity. The reason is substantive.

[K] *Second, via a normalisation.* α becomes one because one normalises the charge so that $e^2 = 4\pi$ (A130). The reason is a unit choice, not a physical statement.

[K] The crucial point: *for the diagnosis the distinction is irrelevant*. Whether a constant falls to one via an identity or via a normalisation – in both cases the settability shows that it is not an independent degree of freedom. The reason decides *how deeply* the bookkeeping is anchored (for the duality, in a physical identity; for α, in a convention), not *whether* the constant is fundamental. Neither is fundamental.

19.3 What Cannot Be Set to One

[K] The statement is sharp only if its converse holds: there must be quantities that *no* unit choice can bring to one – and precisely those are the content.

[B] Two kinds survive the test:

- **The geometric parameter ξ.** It is a pure numerical ratio of the packing geometry (A020). No change of the metre, the second, or the charge unit changes it; it is $4/30000$ in every system of units. ξ cannot be normalised to one without changing the geometry itself.
- **Dimensionless ratios.** $m_\mu/m_e = 206.77$ is the quotient of two quantities of *the same* dimension; no unit choice cancels it. Such ratios, unlike α, are not reachable by a normalisation – α conceals a charge unit, a mass ratio conceals nothing.

[STIPULATION] Thus the dividing line is: settable to one $\Rightarrow$ convention; not settable $\Rightarrow$ content. α stands, despite its dimensionlessness, on the convention side, because it carries a charge unit; m_μ/m_e stands on the content side. The only remaining content of the theory is ξ and a single absolute scale.

19.4 Two Sides of a Coin – Yet Only One Fundamental

[K] Here a contradiction threatens that must be cleared up. A130 calls α and ξ "two sides of a coin", a single degree of freedom, connected by

$$\alpha_{\text{SI}} = \xi \cdot E_0^2.$$

This document simultaneously says α is not fundamental, but ξ is. How do both fit together?

[K] They fit as soon as one reads "one degree of freedom" correctly: the statement concerns the *information content*, not the *rank*. α_{SI} and ξ carry *the same* geometric information – one who knows ξ and the scale E_0 knows α_{SI}, and vice versa. In this sense it is one quantity, not two. That is the coin.

[B] But the two sides are not equally ranked. ξ is this information *bare*: a pure ratio, $4/30000$ in every system of units. α_{SI} is *the same* information *dressed*: $\alpha_{\text{SI}} = \xi \cdot E_0^2$, where the factor E_0^2 is pure scale – the unit dress. The famous value $1/137$ is therefore not ξ beside α but ξ *in the costume* of E_0^2.

[K] The diagnosis of this document confirms exactly that. Strip the units – set $\alpha = 1$ in natural units – and α carries nothing more: the information has not disappeared, it sits entirely in ξ and the one scale. ξ survives every unit change; $\alpha = 1$ is empty afterwards. The coin therefore remains intact, but it *is* ξ: α_{SI} is how this coin looks in SI light, not a second coin.

[STIPULATION] In short: "two sides" means same content, not same rank. One degree of freedom – and its fundamental form is ξ, its dressed form α_{SI}. The apparent contradiction arises only when one confuses "equivalent" with "equally fundamental".

19.5 Why SI Computation at All?

[K] If in natural units everything except ξ and one scale falls to one, the question arises why one needs the more elaborate SI calculations, the bridge constants, and the multiple derivation paths at all. The answer separates three things that are easily confused.

[K] *First: translation.* Measurement happens in the laboratory, and laboratories compute in SI. The SI dress is the *translation* of the geometric core into readable numbers – indispensable, but not itself physics. One needs it because the measuring device speaks in metres and seconds, not because the content demands it. A result in natural units is complete; the SI number is its pronunciation in a particular language.

[B] *Second: overdetermination as test.* The multiple derivation paths add nothing to the content – they *test* it. If two paths lead to the same energy scale and their difference is exactly the derived fractal factor (A130), or if three derivations of the lepton masses hit the same value (A110), that is not an additional degree of freedom but a cross-check. A single path could be a disguised fit; several independent paths that coincide cannot be. The redundancy is the hardening, not the content.

[K] *Third: connectivity.* Different paths bind the one ξ-content to different known physics – Yukawa coupling, Koide relation, RG running. This makes the same core testable from several familiar directions and prevents it from appearing consistent only in its own language. The content remains one; the approaches are many so that it can be controlled from outside.

[STIPULATION] In short: the SI calculation is translation, the multiple paths are control. Both are needed to connect the geometric core to measurement and to harden it – neither adds a degree of freedom to the core. The core remains ξ and one scale.

19.6 Verification Script

- The diagnosis automated: for $c, \hbar, k_B, \alpha$ it is shown that a unit/normalisation choice brings them to one (hence convention); for ξ and m_μ/m_e, that no unit choice achieves this (hence content); and the cross-check that two paths hit the same value without adding a degree of freedom: `python/A_Serie_Skripte/a135_eins_diagnose.py`

19.7 Open Questions

First, the one scale remains an input. The statement reduces everything to ξ and one absolute scale; what value this scale has is itself not derived (A130) but the one remaining calibration point. The scale is *not* settable to one in the sense that its value carries content – but its numerical value is the one input the system needs.

Second, the diagnosis is not a proof of completeness. That everything *so far* either falls to one or is ξ shows the parsimony of the theory; it does not prove that no further independent quantity will ever be needed. The statement orders what is present; it does not exclude what is future.

19.8 Supersedes

This document subsumes: Doc. 043 and Doc. 044 (dissolution of constants), Doc. 101 (circularity of constants, one degree of freedom). It draws together the diagnosis carried out case by case in A085 (duality) and A130 (α), and answers the question of the purpose of SI computation. Incorporated: P40.

19.9 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 20

A138 · Anomalous Magnetic Moments

20.1 Purpose

[STIPULATION] This document treats the anomalous magnetic moments of the leptons (g-2) in the FFGFT framework. The central distinction: *ratios* of g-2 differences are exactly derivable from the fractal structure, because all conversion factors cancel. *Absolute values* in SI units depend on the projection factor k_{geom} and on SI bridge constants – they have, after correction by $K_{\text{frak}}^{3/2}$, an accuracy of 0.01–$0.15\,\%$, but the absolute value itself is not a parameter-free geometric prediction. That is the structurally honest status.

20.2 Physical Foundations

[B] The anomalous magnetic moment is the deviation from the Dirac value:

$$a_\ell = \frac{g-2}{2}.$$

For a point particle Dirac predicts $g = 2$, hence $a = 0$. Quantum effects (vacuum polarisation, vertex corrections) generate $a \neq 0$.

[B] In FFGFT leptons are winding structures in the 4D torsion lattice; the three generations carry different fractal Hausdorff exponents: electron (simple winding, 1st generation), muon (fractally branched with exponent $p_\mu = 5/3$), tau (more branched, $p_\tau = 4/3$). The anomalous moment follows from rotation of the winding, charge distribution on it, and projection $4D \to 3D$.

20.3 The Geometric Formulas for Absolute Values

[K] The fundamental parameters of the FFGFT g-2 calculation:

$$\varphi = \frac{1+\sqrt{5}}{2} = 1.618\ldots \qquad \text{(golden ratio)},$$
$$f = 7500 \qquad \text{(sub-Planck factor)},$$
$$\xi = \frac{4}{3} \times 10^{-4}.$$

[K] Electron. From the 3D surface of the 4D winding $S_3 = 2\pi^2 = 19.739$ and the geometric projection factor

$$k_{\text{geom}} = \frac{2}{\sqrt{\varphi}} \times \sqrt{2} = 2.224$$

follows the ideal formula:

$$a_e^{\text{(ideal)}} = \frac{S_3/f}{k_{\text{geom}}} = \frac{19.739/7500}{2.224} = 1.184 \times 10^{-3}.$$

Experiment: $a_e = 1.1597 \times 10^{-3}$. Deviation ideal: $2.03\,\%$.

[K] Muon. The fractal additional winding:

$$\Delta a_{\text{fractal}} = \frac{4\pi}{f^{p_\mu}} = \frac{4\pi}{7500^{5/3}} = 4.373 \times 10^{-6},$$
$$a_\mu^{\text{(ideal)}} = a_e + \Delta a_{\text{fractal}} = 1.188 \times 10^{-3}.$$

Experiment: $a_\mu = 1.1659 \times 10^{-3}$. Deviation ideal: $1.89\,\%$.

20.4 Resolution of the Systematic Deviation: $K_{\textbf{frak}}^{3/2}$

[K] The $\sim 2\,\%$ deviation of the ideal formulas is not an open question but is explained by the fractal correction. The effective projection factor:

$$k_{\text{eff}} = \frac{k_{\text{geom}}}{K_{\text{frak}}^{3/2}} = \frac{2.224}{(1-100\xi)^{3/2}} = \frac{2.224}{0.98007} = 2.2692.$$

The exponent $3/2 = D/2$ describes how the fractal structure acts on the projection across three spatial dimensions. $K_{\text{frak}} = 1-100\xi$ is the same correction as in A040 – no new parameter.

[K] With the corrected factor:

Lepton	T0 ideal	T0 corrected	Experiment	Dev. ideal	Dev. corr.
Electron	1.184×10^{-3}	1.1598×10^{-3}	1.1597×10^{-3}	2.03 %	**0.01 %**
Muon	1.188×10^{-3}	1.1642×10^{-3}	1.1659×10^{-3}	1.89 %	**0.15 %**
Tau	1.269×10^{-3}	1.2454×10^{-3}	(not measured)	–	Prediction

[B] Methodological clarification. The corrected absolute value is a consequence of the FFGFT-internally consistent K_{frak} – not a fit. The correction agrees with the experimentally reconstructed value $k_{\text{rec}} = S_3/(f \cdot a_e^{\text{exp}}) = 2.2696$ to $0.014\,\%$. But: k_{geom} and f depend on SI bridge constants and measured quantities. An absolute SI value without any measured anchor – as the SM can achieve with α – is not available at the same level in FFGFT (the same holds for masses, A130, A080).

20.5 Ratios: Parameter-Free and Exact

[B] In contrast to absolute values, in ratios of g-2 differences all conversion factors cancel:

$$\frac{\Delta a(\tau-\mu)}{\Delta a(\mu-e)} = \frac{4\pi/f^{4/3} - 4\pi/f^{5/3}}{4\pi/f^{5/3}} = \frac{f^{5/3}}{f^{4/3}} - 1 = f^{1/3} - 1.$$

Numerically: $f^{1/3} = 7500^{1/3} = 19.57$, hence

$$\boxed{\frac{\Delta a(\tau-\mu)}{\Delta a(\mu-e)} = f^{1/3} - 1 = 18.57}$$

No k_{geom}, no K_{frak}, no SI factor, no α. The ratio depends exclusively on $f = 7500$ – and f is the sub-Planck factor $30000/4$, following directly from $\xi = 4/30000$.

[S] Structural parallel to the Koide formula. The Koide formula $(m_e+m_\mu+m_\tau)/(\sqrt{m_e}+\sqrt{m_\mu}+\sqrt{m_\tau})^2 \approx 2/3$ is confirmed to $0.001\,\%$ because there ratios rather than absolute values are compared.

The same structure operates here: the exponent step $\Delta p = 1/3$ of the masses (A100) appears directly as $q_\mu = 5/3$, $q_\tau = 4/3$, $\Delta q = 1/3$ in the g-2 exponents. No coincidence – the same geometric origin.

20.6 The Bridge Formula: Masses and g-2 Connected

[B] From the mass formulas: $m_\tau/m_\mu = (125/144)\cdot f^{1/3}$. From the g-2 formulas: $\Delta a(\tau - e)/\Delta a(\mu - e) = f^{1/3}$. Combining gives the bridge formula:

$$\boxed{\frac{\Delta a(\tau - e)}{\Delta a(\mu - e)} = \frac{144}{125}\cdot\frac{m_\tau}{m_\mu}}$$

The parameter f cancels. No α, no k_{eff}, no K_{frak}. The prediction follows solely from the shared $1/3$ step size of masses and g-2.

[K] Tau prediction from the bridge formula. With measured values $a_e^{\text{exp}} = 1.160\times10^{-3}$, $\Delta a(\mu - e)^{\text{exp}} = 6.269\times10^{-6}$, $m_\tau/m_\mu = 16.817$:

$$\begin{aligned}\Delta a(\tau - e) &= \frac{144}{125}\cdot 16.817\cdot 6.269\times10^{-6} = 1.214\times10^{-4},\\ a_\tau^{\text{(bridge)}} &= a_e + \Delta a(\tau - e) = \boxed{1.282\times10^{-3}}.\end{aligned}$$

The SM calculates $a_\tau^{\text{SM}} = 1.177\times10^{-3}$. Difference: $9\,\%$. Testable at Belle II (~ 2027).

20.7 Fair Comparison: SM and FFGFT

[S] The SM circularity. The SM extracts α from the a_e measurement and then computes a_e back. For electron and muon this is a circular fit – a precision calculation, not an independent prediction. FFGFT derives α from ξ alone: $\alpha = \xi\cdot E_0^2$ with $E_0 = 7.398$ MeV gives $1/\alpha_{\text{T0}} = 137.035 - 0.0005\,\%$ from experiment (A130). That is a genuine, independent prediction, though at a different precision level than the SM perturbation calculation.

[S] Where the theories diverge: Not at e or μ (both theories lie close to experiment there), but at the *tau*. T0: $a_\tau \approx 1.282\times$

10^{-3}, SM: 1.177×10^{-3} – factor 7 larger deviation than the muon discrepancy. That is the decisive test.

20.8 Verification Script

- Verification of all formulas: ideal and corrected absolute values, ratio $f^{1/3} - 1$, bridge formula with experimental values, tau prediction: `python/A_Serie_Skripte/a138_g2.py`

20.9 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$a_\ell = (g-2)/2$	Anomalous magnetic moment of lepton ℓ
$f = 30000/4 = 7500$	Sub-Planck factor; follows directly from $\xi = 4/30000$
$k_{\text{geom}} = (2/\sqrt{\varphi})\sqrt{2}$	Geometric projection factor
$k_{\text{eff}} = k_{\text{geom}}/K_{\text{frak}}^{3/2}$	Effective projection factor after fractal correction
$S_3 = 2\pi^2$	Surface of the 3-sphere (volume of the 4D unit ball)
$p_\mu = 5/3$, $p_\tau = 4/3$	Fractal Hausdorff exponents of the generation structure

20.10 Open Questions

The absolute SI values depend on SI bridge constants and measured quantities. The correction $K_{\text{frak}}^{3/2}$ explains the systematic deviation; but the geometric projection factor k_{geom} is not derived at the same level as the pure ratios.

The tau prediction ($a_\tau \approx 1.282 \times 10^{-3}$) has not yet been measured. Belle II will test it. Until then it is a prognosis – precisely formulated, but not a result.

The exponent $3/2$ for the fractal correction is geometrically motivated ($D/2 = 3/2$) but not uniquely proved. That exactly $D/2$ and not some other exponent appears is a consistency observation, not a derivation.

20.11 Supersedes

This document subsumes: Doc. 018 (all three versions 018-10/11/12, anomalous magnetic moments of the leptons: geometric formulas, $K_{\text{frak}}^{3/2}$ correction, ratio predictions, bridge formula masses $\leftrightarrow$ g-2, tau prediction, fair SM comparison). Not taken over: narrative passages and cosmological generalisations from Doc. 018.

20.12 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 21

A140 · Gravity: G as Bridge

21.1 The Claim

[STIPULATION] The gravitational constant G is in this theory not an independent constant of nature but a unit bridge – like c, $\hbar$, and k_B (A080).

From this follows the stronger and better-known claim of the corpus: the problem of quantum gravity does not arise in this form. This document examines what supports this claim and what does not.

21.2 Why G Is a Bridge

[B] G connects a mass with a length. In the natural system of the theory, in which $\tilde{T} \cdot m = 1$ holds and time and mass are the same quantity in two readings, this connection does not need to be separately established – it is already in the duality.

This is the same statement as for $\hbar$: there the bridge between mass and time is meant, here the one between mass and length. Both are, in a system that does not carry time, mass, and length as independent units, conversions and not natural laws.

[STIPULATION] Operationally G remains indispensable. One who computes in SI needs it. Ontologically it is secondary. This is exactly the dual role that A080 describes for all bridge constants, and it is not a demotion.

21.3 Why the QG Problem Does Not Arise in This Form

[B] The usual problem runs: quantise gravitation as a field and thereby obtain a renormalisable theory. It arises because the geometry is treated as a dynamic field that must in turn be quantised.

In this theory the geometry is *pre-ordered* and not dynamic in the same sense: the T^4/Z_3 is stipulated (A020), the quantum structure arises as its spectrum (A030, A070). There is therefore nothing left that would need to be subsequently quantised.

Honest assessment of this statement. It is not a solved problem but a *dissolved* one – and dissolved by a stipulation. One who does not stipulate the pre-ordered geometry still has the problem. The theory does not answer the question; it does not pose it.

This is the *negative* argument. A positive argument stands alongside it, which derives the numerical value of G from ξ: $G = \xi^2/(4\,m_{\text{char}})$ (A145). Both belong together – the negative explains why quantisation is unnecessary, the positive explains where the value comes from.

This is a legitimate move, but it must not be presented as a solution. The difference is the same as between a derivation and a calibration (A130): both are permissible, but they are not the same.

21.4 What Follows from This and What Does Not

[B] It follows: no gravitons as necessary constituents, no renormalisation problems of gravitation, no singularities from a field quantisation.

[STIPULATION] It does *not* follow: that the classical gravitational dynamics is reproduced. The A-series shows nowhere that the observed gravitational interaction with its strength and inverse-square law follows from the compact geometry. This is a major open point and is not minimised here.

21.5 Verification Script

- The bridge role of G: that G can be absorbed in natural units just like c, $\hbar$, and k_B, shown by the consistency of the unit equations: `python/A_Serie_Skripte/a140_g_bruecke.py`

21.6 Open Questions

First, gravitational dynamics has not yet been treated as an A-series document – it is taken up in A142. The complete field-equation framework (time-field Lagrange, scalar/fermion coupling, modified Dirac equation, Higgs coupling, ξ from Higgs matching) is available in the legacy corpus and is incorporated in A142. What remains after that: (a) the complete quantisation of these field equations, (b) the transition to measurable predictions beyond the Schwarzschild case, (c) the connection to the projection chain (A090).

Second, the strength of the interaction is not derived. Treating G as a bridge does not explain why gravitation is forty orders of magnitude weaker than electromagnetism.

Third, the dissolution of the QG problem is a consequence of the stipulation. See above; booked as such, not as a result.

21.7 Supersedes

This document subsumes: Doc. 012 (gravitational constant), Doc. 127 (gravitation), Doc. 163 (quantum gravity). Incorporated: P40.

21.8 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 22

A142 · Gravitational Dynamics: Time-Field Lagrangian

22.1 Purpose

[STIPULATION] This document develops the complete Lagrangian formulation of gravitational dynamics in FFGFT: the intrinsic time field $\mathcal{T} = 1/\max(m, \omega)$ as the fundamental object, the field equation for spherical and extended systems, the coupling of all matter fields to the time field through conformal transformation, the modified Dirac equation, and the complete total Lagrangian density. The document thus provides the framework designated as "gravitational dynamics" in A140 and recorded there only as a bridge result. What remains open after this document is explicitly named.

22.2 The Intrinsic Time Field

[B] The fundamental object of gravitational dynamics in FFGFT is not the metric but the intrinsic time field:

$$\boxed{\mathcal{T}(x,t) = \frac{1}{\max(m, \omega)}}$$

in natural units ($\hbar = c = 1$). $[\mathcal{T}] = [E^{-1}]$. The field is reciprocal to the local energy – heavy objects have small $\mathcal{T}$ (rapidly oscillating), light ones have large $\mathcal{T}$. The time-mass duality $\tilde{T} \cdot m = 1$ (A020/A060) is the zero-field equation of this structure.

[B] Three fundamental field geometries:

Geometry	Parameters
Localised spherical	$\beta = 2Gm/r$ (dimensionless), $\xi = 2\sqrt{G} \cdot m$ (dimensionless), $\mathcal{T}(r) = (1/m_0)(1-\beta)$
Localised non-spherical	$\beta_{ij} = r_{0ij}/r$ (tensor), $\xi_{ij} = 2\sqrt{G} \cdot I_{ij}$
Infinite homogeneous	$\nabla^2 m = 4\pi G \rho_0 m + \Lambda_T m$, $\xi_{\text{eff}} = \xi/2$

[K] For quantum computing applications and atomic scales ($R/r \gg 10^{12}$) only the localised spherical geometry is relevant; the others are for theoretical completeness.

22.3 The Field Equation

[B] For a point-mass source $\rho = m\delta^3(\mathbf{r})$ the complete geometric solution is:

$$m(r) = m_0\left(1 + \frac{2Gm}{r}\right) = m_0(1+\beta), \qquad \mathcal{T}(r) = \frac{1}{m_0}(1+\beta)^{-1} \approx \frac{1}{m_0}(1-\beta).$$

The factor 2 in $r_0 = 2Gm$ follows from the relativistic field structure and coincides exactly with the Schwarzschild radius. The field equation:

$$\nabla^2 m = 4\pi G\,\rho(x,t)\cdot m. \qquad [E^3 = E^{-2}\cdot E^4\cdot E = E^3\ ]\checkmark$$

For infinite systems with background consistency:

$$\nabla^2 m = 4\pi G\rho_0 m + \Lambda_T m, \qquad \Lambda_T = -4\pi G\rho_0.$$

The Λ_T term generates the cosmic shielding effect $\xi_{\text{eff}} = \xi/2$.

[B] No new objects. Neither $r_0 = 2Gm$ nor $\Lambda_T = -4\pi G\rho_0$ are independent quantities beside the scale structure from ξ. r_0 is the characteristic length *of the respective mass* – the same reciprocity as L_0, only with m instead of the Planck mass as reference. Λ_T is the same field equation, computed from the homogeneous background instead of the point source. Both are reference-point choices, not new physics – the same logic as the reference-point table in A262.

22.4 The Universal Scale Parameter from Higgs Matching

[K] The fundamental scale parameter ξ is uniquely determined by the Higgs physics:

$$\boxed{\xi = \frac{\lambda_h^2\,v^2}{16\pi^3\,m_h^2} \approx 1.33\times 10^{-4}}$$

with Higgs self-coupling $\lambda_h \approx 0.13$, Higgs VEV $v \approx 246$ GeV, Higgs mass $m_h \approx 125$ GeV. Dimensional verification: $[\xi] = [\lambda_h^2 v^2]/[16\pi^3 m_h^2] = [E^2]/[E^2] = [1]$. The condition $\beta_T = 1$ in natural units ($\hbar = c = \alpha_{\text{EM}} = \beta_T = 1$) fixes ξ uniquely and eliminates all free parameters.

22.5 The Lagrangian Formulation

[B] The time-field Lagrangian density:

$$\mathscr{L}_{\text{time}} = \sqrt{-g}\left[\tfrac{1}{2}\, g^{\mu\nu}\partial_\mu\mathscr{T}\partial_\nu\mathscr{T} - V(\mathscr{T})\right].$$

Dimensional consistency: $[\mathscr{L}_{\text{time}}] = [E^{-4}]([E^2] + [E^4]) = [E^0]$.

[B] Matter field coupling through conformal transformation. All matter fields couple to the time field via:

$$g_{\mu\nu} \;\to\; \Omega^2(\mathscr{T})\, g_{\mu\nu}, \qquad \Omega(\mathscr{T}) = \frac{\mathscr{T}_0}{\mathscr{T}} \quad \text{[dimensionless]}.$$

Scalar fields:

$$\mathscr{L}_\phi = \sqrt{-g}\,\Omega^4(\mathscr{T})\left(\tfrac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi - \tfrac{1}{2}m^2\phi^2\right). \qquad [\mathscr{L}_\phi] = [E^{-4}][1][E^4] = [E^0]\checkmark$$

Fermion fields:

$$\mathscr{L}_\psi = \sqrt{-g}\,\Omega^4(\mathscr{T})\left(i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi\right). \qquad [\mathscr{L}_\psi] = [E^{-4}][1][E^4] = [E^0]\checkmark$$

Higgs time-field coupling:

$$\mathscr{L}_{\text{Higgs-T}} = |\mathscr{D}_{\text{Higgs-T}}|^2 - V(\mathscr{T}, \Phi),$$
$$\mathscr{D}_{\text{Higgs-T}} = \mathscr{T}(\partial_\mu + igA_\mu)\Phi + \Phi\,\partial_\mu\mathscr{T}.$$

This delivers $\mathscr{T} = 1/(y\langle\Phi\rangle)$ – the connection between time field and Higgs vacuum expectation value.

[B] Complete total Lagrangian density:

$$\mathscr{L}_{\text{total}} = \mathscr{L}_{\text{time}} + \mathscr{L}_{\text{gauge}} + \mathscr{L}_\phi + \mathscr{L}_\psi + \mathscr{L}_{\text{Higgs-T}},$$

with gauge term $\mathscr{L}_{\text{gauge}} = \sqrt{-g}(-\frac{1}{4}F_{\mu\nu}F^{\mu\nu})$. Each summand is dimensionally $[E^0]$.

22.6 The Modified Dirac Equation

[B] In the FFGFT time-field framework the Dirac equation reads:

$$\left[i\gamma^{\mu}(\partial_{\mu} + \Gamma_{\mu}^{(T)}) - m(x,t)\right]\psi = 0,$$

with the time-field connection:

$$\Gamma_{\mu}^{(T)} = \frac{1}{\mathcal{T}}\partial_{\mu}\mathcal{T} = -\frac{\partial_{\mu} m}{m^2}.$$

This equation reduces in the limit $\mathcal{T} \to$ const ($\Gamma_{\mu}^{(T)} \to 0$) to the standard Dirac equation. The connection to the Schrödinger equation (A075) remains via the polar decomposition of the vacuum field.

[B] Modified Schrödinger equation:

$$i\,\mathcal{T}\frac{\partial\Psi}{\partial t} + i\,\Psi\left[\frac{\partial\mathcal{T}}{\partial t} + \mathbf{v}\cdot\nabla\mathcal{T}\right] = \hat{H}\,\Psi.$$

In the limit $\partial_{\mu}\mathcal{T} \to 0$ this reduces to the standard form $i\hbar\,\partial_t\Psi = \hat{H}\Psi$ with $\mathcal{T} = 1/m \leftrightarrow \hbar$.

22.7 Dimensional Consistency Overview

[K]

Equation	LHS	RHS	Status
Time-field definition	$[E^{-1}]$	$[1/\max(m,\omega)] = [E^{-1}]$	✓
Field equation	$[E^3]$	$[4\pi G\rho m] = [E^3]$	✓
β parameter	$[1]$	$[2Gm/r] = [1]$	✓
ξ from Higgs	$[1]$	$[\lambda_h^2 v^2/(16\pi^3 m_h^2)] = [1]$	✓
Lagrangian density	$[E^0]$	all terms	✓

22.8 Testable Predictions

[K] The Lagrangian framework gives specific predictions:

$$a_e^{(T0)} = \frac{\alpha}{2\pi}\cdot\xi^2\cdot I_{\text{loop}} \approx 2.34\times 10^{-10}$$

(QED correction to the anomalous magnetic moment). Rotation curves with modified potential: $v^2(r) = GM/r + \kappa r^2$. Energy-dependent time delay: $\Delta t = (\xi/c)(1/E_1 - 1/E_2)(2Gm/r)$.

22.9 Verification Script

- Dimensional consistency of all terms numerically; limiting case Dirac → standard; ξ from Higgs matching: `python/A_Serie_Skripte/a142_gravitation_lagrange.py`

22.10 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$\mathcal{T}(x,t)$	Intrinsic time field (dynamic scalar field)
Ω	Conformal transformation factor
$\Gamma^{(T)}_\mu = \partial_\mu m/m^2$	Time-field connection (connection coefficient)
$\mathcal{L}_{\text{time}}$	Lagrangian density of the time field
$G_{\mu\nu}$	Einstein tensor
$T_{\mu\nu}$	Energy-momentum tensor

22.11 Open Questions

A separate quantisation of gravitation is not required in FFGFT. G is not an independent field quantity but follows intrinsically from ξ: $G = \xi^2/(4m_{\text{char}})$. Gravitation is the projection of the time-field geometry, not an independent quantum field. The question "when is gravitation quantised?" does not arise in this framework – it is already answered by the basic structure. What remains: the translation of the classical field equations into explicit loop corrections for scenarios beyond the Schwarzschild limit.

The transition to measurable predictions beyond the Schwarzschild case is not complete. The QED correction (above) is computable; predictions for strong fields, gravitational waves, or the post-Newtonian sector are not systematically worked out.

The connection to the projection chain (A090) is missing. How the gravitational sector fits into the type-I/II/III structure – whether it is a type-II loss or a type-I equivalent – is not spelled out.

22.12 Supersedes

This document subsumes: Doc. 067 (time-field Lagrangian density with dimensional consistency, matter-field coupling through conformal transformations, spherically symmetric solutions, Higgs coupling, ξ from Higgs matching, complete total Lagrangian density, modified Dirac equation, experimental predictions). *Not* taken over: the section on a static universe from Doc. 067 – it does not belong in the A-series core (A010).

22.13 Use of AI Tools

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Chapter 23

A145 · Gravitational Constant G

23.1 The Central Formula

[K] In this theory the gravitational constant is not an independent quantity but follows from ξ and a mass:

$$G = \frac{\xi^2}{4\,m_{\text{char}}} \qquad \text{(natural units)}.$$

This is a *derivation*, not a bridge. G is not absorbed as a unit factor (that would be A080), but expressed as a function of the single parameter and a characteristic mass. The difference from A130 is decisive: there α was in the end calibrated; here the *structure* of G is determined from ξ.

[B] The equivalent form

$$\xi = 2\sqrt{G m_{\text{char}}}$$

shows the statement from the other side: ξ, the tetrahedral packing quantity from A020, *is* up to the factor the square root of the product of gravitational strength and characteristic mass. Gravity and geometry are thereby not two things but two readings of ξ.

This is a statement about structure only in natural units (A085). In SI it would be an equation between incommensurable units.

[K] For context: in natural units $G = 1$, exactly as $c = \hbar = k_B = \alpha = 1$ (A085, A130). The SI value $6.674 \cdot 10^{-11}$ is bookkeeping. The formula $G = \xi^2/(4m)$ therefore does not say what native value G has – that is one – but how the *SI dress* of G composes from ξ and a mass. This is the same bookkeeping as for α, only for a dimensioned rather than a dimensionless quantity.

23.2 Why This Is a Particularly Direct Derivation

[K] As with α (A130), there is no circularity at the core of the formula here either – for α the initial circularity is broken by the two-route over-determination of E_0; for G it is absent from the outset: $G = \xi^2/(4m)$ does not use G anywhere to determine G. The only difference is directness – α needs the meeting of two routes, G does not because the formula simply does not contain G. A hidden calibration is not in the core here either, only in the SI dress (E_{char}, see below). The formula says: choose the characteristic mass – say m_e – and the gravitational strength follows from ξ.

[B] This is the actual resolution of the gravity problem in this theory, and it is stronger than what A140 states. A140 says: the QG problem does not arise because the geometry is pre-ordered. That is a *negative* argument. A145 says: G has a *positive* expression from ξ. The negative argument explains why quantisation is unnecessary; the positive explains where the numerical value comes from. This makes G the most direct ξ-manifestation in the corpus – not the most precisely tested (that is the lepton ladder A110 and the α over-determination A130), but the most immediate in form.

23.3 The Dimension Bridge E_{char}

[K] The basic formula $G = \xi^2/(4m)$ is not yet dimensionally complete: in natural units G has dimension $[E^{-2}]$, but the right-hand side has $[E^{-1}]$. A factor $1/E_{\text{char}}$ is missing:

$$G_{\text{nat}} = \frac{\xi_0^2}{4\,m_e} \cdot \frac{1}{E_{\text{char}}}, \qquad E_{\text{char}} \approx 28.4.$$

[K] The corpus provides a structural decomposition of E_{char}:

$$E_{\text{char}} = E_0 \cdot \left(\frac{4}{3}\right)^2 \cdot \frac{\pi}{\sqrt{2}} \cdot K_{\text{frak}} = 7.40 \cdot 1.778 \cdot 2.221 \cdot 0.986 = 28.8,$$

against the target value 28.4, i.e. $+1.5\,\%$.

The assessment – and it is reassuring. E_{char} carries two quantities that are not purely geometric elsewhere: E_0 (A130) and K_{frak} (A040). The factor $(4/3)^2$ is geometric, $\pi/\sqrt{2}$ likewise; E_0 and K_{frak} are not. But this concerns only the *SI numerical value* of G – and that is bookkeeping. How precisely E_{char} hits 28.4 is a statement about the scale conversion, not about the structure. As long as one works with ratios, the measured value of G is irrelevant; the structural relations are known and confirmed in SI and in the Standard Model. The *basic structure* $G = \xi^2/(4m)$ is unaffected by this and is the actual statement.

Cleanly separated: the statement "G is proportional to ξ^2 and inversely proportional to a mass" is **[K]**. The precise numerical value via E_{char} is **[B]** with a calibrated component.

23.4 The Same Diagnosis as for α, More Complex Dress

[K] G passes the same test by which A135 decides whether a quantity is fundamental: *can it be set to one by a consistent unit choice?* For G the answer is yes – in Planck units $G = c = \hbar = 1$. By the A135 diagnosis this means: G is not content of the theory but a conversion factor. It thereby stands on the same side as c, $\hbar$, k_B (physically one via duality) and α (one via normalisation) – all convention. The SI value $6.674 \cdot 10^{-11}$ is what this convention conceals, not a fundamental numerical value.

[K] And as with α, the content lies not in this value but in ξ. The bare coin is ξ; G is – seen in SI light – the same coin, only dressed. That is the meaning of $G = \xi^2/(4m_{\text{char}})$: not a second degree of freedom but ξ in dimensioned form. Equivalent information content, not equal rank – exactly the resolution of the apparent tension from A135.

[K] The difference from α is only the *richness of the dress*, and it has a dimensional reason. In natural units α is dimensionless, $[E^0]$; its SI form therefore needs exactly one scale:

$$\alpha_{\text{SI}} = \xi \cdot E_0^2 \qquad \text{(one scale, squared).}$$

G by contrast has dimension $[E^{-2}]$ in natural units (it is M_{Planck}^{-2}) and thereby ties mass to length to time. Its SI form can therefore not make do with one scale; it needs a mass *and* the intermediate scale E_{char}:

$$G_{\text{nat}} = \frac{\xi^2}{4\,m_{\text{char}}} \cdot \frac{1}{E_{\text{char}}}.$$

E_{char} is itself multi-layered – $E_0 \cdot (4/3)^2 \cdot (\pi/\sqrt{2}) \cdot K_{\text{frak}}$ (see above) – whereas α makes do with the one E_0. This is the "more complex SI dress": not more content but more layers of the same content, forced by the richer dimension of G.

23.5 What Counts: Exact Reconstructibility

[K] The central statement of this document is not how precisely the $0.5\,\%$ is hit, but that G in natural units can be set to one – and from this something exact follows. If G is a convention (one in Planck units), then its SI value is not an independent numerical value but *completely* fixed by the conversion:

$$G_{\text{SI}} = \frac{\hbar c}{M_{\text{Planck}}^2}.$$

Given ξ and one absolute scale, G_{SI} is thereby *exactly* determined – without a free parameter, without approximation. This is the same structure as $\alpha_{\text{SI}} = \xi E_0^2$: the dimensioned constant is the SI expression of ξ plus scale, and the reconstruction is exact, not approximate.

[K] The E_{char} decomposition is strictly separate from this. The exact E_{char} – the one that hits G_{SI} – is already fixed by ξ and the scale. What is $0.5\,\%$ off is not this exact quantity but the *attempt to express it by a closed formula* from known quantities ($E_0 \cdot (4/3)^2 \cdot (\pi/\sqrt{2}) \cdot K_{\text{frak}}$). The deviation measures the quality of this *analytical approximation of the dress*, not the reconstructibility itself. The reconstruction is exact; only its shortcut to a simple expression is not yet.

[STIPULATION] The priority is thereby correct: what distinguishes G from a genuinely fundamental constant is the settability to one and the resulting exact reconstructibility – and that is shown. The remaining E_{char} approximation is a question of closed form, not of structure, and carries no weight for the central statement.

23.6 The Scale Hierarchy

[B] The formula orders three energy scales:

$$E_0 = 7.40 \text{ MeV} \ll E_{\text{char}} = 28.4 \ll E_{T0} = \frac{1}{\xi_0} = 7500.$$

E_0 is the electromagnetic scale (A130), $E_{T0} = 1/\xi$ the fundamental scale of the theory, E_{char} the intermediate scale at which gravity couples. That gravity sits at an *intermediate scale* is the structural statement behind its weakness: it couples far below the fundamental scale.

[S] Whether this hierarchy *quantitatively* explains the observed weakness of gravity – the forty orders of magnitude relative to electromagnetism – is not shown by this. It is a direction, not a calculation.

23.7 Verification Script

- $G = \xi^2/(4m_e)$ and the reverse direction $\xi = 2\sqrt{Gm}$; the dimensional gap and its closure by E_{char}; the structural decomposition of E_{char} with identification of the calibrated component; and the scale hierarchy: `python/A_Serie_Skripte/a145_gravitationskonstante.py`

23.8 Symbol Glossary

Symbols used throughout the A-Series are explained in A010. Specific to this document:

Symbol	Meaning
m_{char}	Characteristic mass; limiting case of the self-consistency condition, $G = \xi^2/(4m_{\text{char}})$
G	Newtonian gravitational constant

23.9 Open Questions

First, the characteristic mass is not uniquely fixed. The corpus uses m_e; why the electron and not another mass is the characteristic one is not justified.

Second, E_{char} carries a calibrated component (see above). The basic structure $\xi^2/(4m)$ is unaffected by this; the precise numerical value is not. *This residual is explicitly not a measurement effect.* It is tempting to attribute the $\sim 0.5\,\%$ to the notorious measurement weakness of G – but the opposite is the case: G is measured to $2 \cdot 10^{-5}$ (about $0.002\,\%$), and even the large scatter between individual experiments only reaches $\sim 0.05\,\%$. The theory's residual is thus roughly 200 times larger than the measurement uncertainty and about 10 times larger than the experimental scatter; the inputs to the dress (E_0 via masses to $\sim 10^{-8}$, K_{frak} derived, the geometry factors exact) do not carry it either. The residual is thus structural – a residual of the E_{char} approximation – and must remain named as such; attributing it to measurement imprecision would be the same target-first assignment that A105 and A130 warn against.

Third, the residual is entangled with a Standard Model assumption – the mass definition. *Measured* lepton masses enter the dress ($E_0 = \sqrt{m_e m_\mu}$, $m_{\text{char}} = m_e$), and "mass" in the Standard Model is scheme-dependent: the measured values are *pole masses*, whereas a purely geometric construction could equally correspond to a running (MS-bar) mass. The difference between both schemes is of order $\alpha/\pi \approx 0.23\,\%$ per mass; since G depends on the masses via $G \sim 1/(m_e^{3/2} m_\mu^{1/2})$ with total mass power 2, this propagates to $\sim 0.46\,\%$ on G – *the same order as the residual*. This cuts both ways: it shows that the $0.5\,\%$ *cannot* be cleanly read as a geometric defect because part of it may be a convention on the SM input side; but it does *not* show that the scheme effect closes the gap. That would only be shown by computing the pole-to-running shift for precisely these masses and this formula with sign and coefficient *forward* – an order-of-magnitude argument is not a derivation. Only this remains: the residual lies between geometric approximation and mass scheme, and both are of the same order.

Fourth, the gravitational dynamics are still missing. A145 gives the *value* of G from ξ; it does not give the field equations from which the inverse-square law and motion would follow. This remains the large open point already named in A140.

23.10 Supersedes

This document subsumes: Doc. 012 (gravitational constant, derivation part), Doc. 127 (gravity, Planck perspective). It supplements A140 with the positive derivation; A140 retains the bridge and QG argumentation. Incorporated: P40. New: the measurement of G by the one-test (Planck units $G = c = \hbar = 1$) and the dress comparison with α (dimensional reason for the richer SI dress).

23.11 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 24

A150 · Quarks and Neutrinos

24.1 Purpose: Marking the Boundary

The leptons are treated in A100 to A120, with clear layers and clear tolerances. For quarks and neutrinos the situation is different, and this document says above all *how* different.

[STIPULATION] This sector has two layers: the up-type quark ladder is at [K] level (geometric exponent sequence, numerically verified). Down-type quarks and neutrinos are [S]: sketch, candidate, plausibility. There is no statement there of the rank of the Koide relation or the up-type ladder.

24.2 Quarks

[STIPULATION] The quark sector has two different layers that must be kept separate: the up-type ladder (u, c, t) is verified at [K] level; down-type (d, s, b) and neutrinos remain [S].

24.2.1 Up-Type Quarks: Geometric Ladder at [K] Level

[K] The up-type quarks follow the same geometric sequence as the leptons – exponent $p_{n+1} = (2/3) \cdot p_n$, starting value $p_u = 3/2$:

$$p_u = \frac{3}{2}, \quad p_c = \frac{2}{3}, \quad p_t = -\frac{1}{3}.$$

The prefactors r_i follow from the experimental masses; the exponents are geometrically determined. Numerical verification:

Quark	p_i	r_i	m_{FFGFT}	Exp. (PDG)
Up	$+3/2$	5.703	2.16 MeV	2.16 MeV
Charm	$+2/3$	1.978	1270 MeV	1270 MeV
Top	$-1/3$	$1/28$	172.0 GeV	172.76 GeV

Top quark deviation: $0.45\,\%$. The top is the only fermion with a negative exponent – inverse mode rather than suppression, $\xi^{-1/3} \approx 19.6$ instead of ξ^{+p}.

[K] Where the ladder has [K] level and where not. The p_i values follow geometrically; that is shown. The r_i prefactors are back-calculated from the masses – they do not have the same [K] level as the r_i of the leptons, for which the circulant and the Koide relation give an independent structural justification. Open: why precisely these rungs are occupied (u, c, t) and not others.

24.2.2 Down-Type Quarks: Isospin Step Size

[K] The down-type quarks follow an arithmetic sequence:

$$p(n) = \tfrac{3}{2} - n \cdot \tfrac{1}{2}, \quad n = 0, 1, 2 \;\Rightarrow\; (p_d, p_s, p_b) = \left(\tfrac{3}{2},\ 1,\ \tfrac{1}{2}\right).$$

Numerically (all deviations $< 1\,\%$):

Quark	p_i	r_i	m_{FFGFT}	Exp. (PDG)
Down	$3/2$	12.33	4.67 MeV	4.67 MeV
Strange	1	2.85	93.4 MeV	93.4 MeV
Bottom	$1/2$	1.47	4.18 GeV	4.18 GeV

[B] Derivation of the step size from torus geometry and isospin. The down quarks are the only fermion group with a uniform d_{akt} step per generation:

$$\Delta d_{\text{akt}}(\text{Down}) = -\tfrac{3}{2} \quad (-\tfrac{3}{2}, -1 \text{ leptons};\ -\tfrac{5}{2}, -3 \text{ up-quarks}).$$

The uniformity explains the arithmetic p-sequence. The step $-3/2 = -D_{\text{space}} \cdot |T_3|$ follows from two components:

1. $D_{\text{space}} = 3$: spatial dimensions after decompactification $T^4 \to \mathbb{R}^3 \times \mathbb{R}_t$ (A090).
2. $T_3(\text{Down}) = -1/2$: weak isospin in the $SU(2)_L$ doublet (A192).

Thus:

$$\Delta p = \frac{\Delta d_{\text{akt}}}{D_{\text{space}}} = \frac{-D_{\text{space}} \cdot |T_3|}{D_{\text{space}}} = -|T_3| = -\tfrac{1}{2}.$$

Steps 1, 2 and the division are **[B]**. What remains **[S]**: the connection rule "$\Delta d_{\text{akt}} = -D_{\text{space}} \cdot |T_3|$ per generation" is geometrically plausible (each generation loses a T_3-weighted spatial dimension of active mode), but has not yet been formally derived from the projection geometry.

Why this is worth less than the up-type ladder – three reasons:

First, quark masses are not directly measured quantities. They are scheme-dependent and run with the energy scale. A percent-level comparison presupposes fixing scheme and scale – and this choice is itself a degree of freedom.

Second, there are six quarks instead of three leptons, but no ordering corresponding to the Z_3 structure that would lead them to three eigenvalues of an operator. The circulant layer from A110 has no counterpart here.

Third, the number of structural numbers to be adjusted is correspondingly larger. By A020, a ξ-power is a statement only when its exponent is independently fixed. For the quarks this is not shown.

[STIPULATION] The A-series therefore treats the quark sector as a descriptive classification, not as a result, and does not count it among the prediction quantities.

[S] The sharpest individual case: the sign of the top quark. A concrete example shows where the boundary lies exactly. The top quark carries in the ladder representation a *negative* exponent $p_t = -\frac{1}{3}$, which gives an amplification $\xi^{-1/3} \approx 19.6$ instead of a suppression – the top is the only fermion that sits not below but at the electroweak scale (its Yukawa coupling is ≈ 1, all others are $\ll 1$). So far the reliable observation.

[S] The sign is structured, not arbitrary. The exponents are rungs of a logarithmic resonance ladder with fixed step size $\Delta p = 1/3$; each step corresponds to the geometric factor $\xi^{1/3} \approx 1/19.6$. This ladder follows from $p = d_{\text{akt}}/3$, the number of active torus dimensions – a connection that is *linear* in d_{akt} and therefore admits negative rungs without difficulty. In this sense $-\frac{1}{3}$ is not an invented number but the rung one unit *below* the Higgs scale ($p = 0$), the opposite direction of the $+\frac{1}{3}$ step. A common short-circuit interprets the same instead as a "counter-winding" via the Laplace operator ($\lambda = n_1^2 + n_2^2 + n_3^2$); this does not hold because this operator is *even* in each winding number ($n = -1$ and $n = +1$ give the same eigenvalue) and thus cannot generate any sign. The supportable justification is the linear dimension ladder, not the counter-winding.

[STIPULATION] If $\frac{1}{3}$ is only the *step size* of the ladder, then $-\frac{1}{3}$ is simply its continuation one step *below* the Higgs reference ($p = 0$) – not a mirror that would need a partner at $+\frac{1}{3}$, but the next regular position downward. The occupation of individual rungs is a different question from their existence. Noteworthy is that even the bottom at $p = \frac{1}{2}$, i.e. $d_{\text{akt}} = 1.5$, occupies a *non-integer* rung: the interpretation of d_{akt} as a number of active dimensions therefore does not even hold throughout the positive range. Primary is the resonance axis with step $\frac{1}{3}$; the "dimensions" are only an illustration for the integer rungs. On this axis $-\frac{1}{3}$ is a regular position. What remains open is therefore not the *sign* but why the top is on precisely this rung and which rungs are occupied at all – the assignment, not the scale. The value therefore remains on the $[S]$ layer: a regular position on a justified axis, whose occupation has not yet been derived.

24.3 Neutrinos

[S] The neutrino sector is the weakest of the entire corpus, and that should be stated clearly here.

The absolute masses are experimentally unknown, only differences of mass squares. The mass ordering is open. Whether these are Dirac or Majorana particles is open. A theory that delivers numbers into this situation can hardly be refuted by the data – and what cannot be refuted confirms nothing either.

[S] The view: neutrinos as nearly massless photons. FFGFT treats the neutrino not like the charged leptons but by analogy with the photon. The photon is exactly massless,

$$\text{Photon:} \quad E^2 = (pc)^2 + 0,$$

the neutrino carries a tiny residual mass from a *double* ξ suppression,

$$\text{Neutrino:} \quad E^2 = (pc)^2 + \left(\sqrt{\frac{\xi^2}{2}}\, mc^2\right)^2.$$

The first ξ factor makes the particle "nearly massless" like a photon, the second leaves a small mass. As an estimate this gives

$$m_\nu = \frac{\xi^2}{2}\, m_e = \frac{(4/3 \cdot 10^{-4})^2}{2} \cdot 0.511 \text{ MeV} \approx 4.5 \text{ meV}.$$

[S] What the estimate achieves and what it does not. It captures the *qualitative* character: the neutrino is quasi-massless and lies many orders of magnitude below the charged leptons – the photon analogy explains *why* this is so without a free mass parameter. But it does not hit the *scale*. The oscillation data require model-independently that the heaviest neutrino weighs at least $\sqrt{\Delta m^2_{\text{atm}}} \approx 50$ meV; the estimate gives only ~ 5 meV, roughly one order of magnitude too small (sum ~ 14 meV against the required $\gtrsim 60$ meV: factor ~ 4). The approach thus explains the *smallness*, not the *magnitude* of the neutrino mass – it is a picture with the right direction and the wrong scale, and remains explicitly on the $[S]$ layer.

[STIPULATION] The sector is therefore tracked but not weighted. It belongs to the points where the theory has not yet reached its range.

24.4 What the Sector Still Achieves

[B] One statement remains supportable and is independent of the numerical values: the generation number three follows from the Z_3 structure (A030, A110) and applies to all fermions, not only the leptons. That there are three quark families and three neutrino species need not be assumed additionally in this structure.

This is a structural statement without free parameters. It is weaker than a statement about numerical values – and it is one of the two reliable things this sector offers. It does not belong to the four prediction quantities of the series.

[B] The second reliable statement concerns not the masses but the *mixing*, and it is also parameter-free. FFGFT separates a magnitude from a phase sector (A120): the magnitude is the mass spectrum, the phase the mixing. In this reading the most striking difference between quarks and neutrinos is precisely a *magnitude-versus-phase* contrast:

- **Quarks:** strong mass hierarchy (large magnitude), small mixing – the CKM matrix is close to the identity.
- **Neutrinos:** flat, nearly degenerate magnitude, large mixing – the PMNS matrix is far from the identity.

This can be quantified. Measuring the off-diagonal weight of the $|U|^2$ matrix (per row) gives

$$\text{CKM: } 0.035 \qquad \text{versus} \qquad \text{PMNS: } 0.554,$$

a factor ~ 16. The famous contrast "small versus large mixing" is thus structurally exactly the contrast "magnitude versus phase" – one family sits in the magnitude sector, the other in the phase sector. This statement uses no derived masses and no adjusted exponent; it only reads the measured mixing contrast into the magnitude/phase separation that FFGFT already carries.

[STIPULATION] This contrast is also not one of the four prediction quantities. It is a classification of the known, not a new numerical value – but one that makes do without free parameters and therefore stands above the $[S]$ layer of the rest of the sector.

24.5 Verification Script

- The CKM/PMNS contrast from measured mixing matrices: off-diagonal weight 0.035 versus 0.554, factor ~ 16 – without derived masses, without adjusted exponents: `python/A_Serie_Skripte/a150_quarks_neutrinos.py`

24.6 Open Questions

First, the quark exponents are not independently justified.

Second, the neutrino sector is largely not testable – the absolute masses, the ordering, and the Dirac/Majorana character remain open. The one point that is testable goes against the theory: the ξ^2 estimate lies an order of magnitude below the scale required by oscillations. This does not disqualify the sector –

an estimate is not a model – but it shows that no supportable mass scale yet stands here.

Third, a counterpart to the circulant layer is missing. Without one, the sector remains at the accuracy level of the ladder, i.e. at percent level, and nothing lies below that.

24.7 Supersedes

This document subsumes: Doc. 005 (quarks), Doc. 007 (neutrinos, photon analogy), Doc. 047 (masses), Doc. 289 (magnitude/phase, CKM/PMNS contrast), Doc. 189 (torus windings, sign picture), Doc. 202 and Doc. 006 (top special case $p_t = -1/3$ as inverted hierarchy without derivation). Incorporated: P35, P40.

24.8 Use of AI Tools

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Chapter 25

A155 · Meson Sector

25.1 Purpose and Context

[STIPULATION] This document treats meson masses within the FFGFT framework using a self-contained additive ansatz. It establishes the ansatz's domain of validity, evaluates it on the pion, documents the connection to the Z_3 sector structure (A270), and states the structural boundary at which the additive ansatz ends.

A150 treats the quark sector as descriptive classification, not as a predictive quantity. The same caution applies here: quark masses are scheme-dependent (A150, §Quarks, first reason); all numerical values use PDG figures ($\overline{\text{MS}}$, 2 GeV) and carry their uncertainty ($\sim$1–2 %).

25.2 External Inputs

[STIPULATION] Besides the A-series quantities ξ and $K_{\text{frak}} = 1 - 100\xi$ (derivation: A040), this document uses three external standard quantities of QCD; they are declared here, not derived:

Quantity	Value	Role
Λ_{QCD}	0.217 GeV	QCD scale (PDG range 0.21–0.22 GeV); sole dimensionful input of the binding term
m_u, m_d, m_s	2.16; 4.67; 93.4 MeV	current quark masses ($\overline{\text{MS}}$, 2 GeV, PDG)
Meson masses	PDG	comparison values

25.3 The Additive Ansatz

[STIPULATION] For a meson of two constituents we set:

$$m_M = m_{q_1} + m_{q_2} + \Lambda_{\text{QCD}} \cdot K_{\text{frak}}^{n_{\text{eff}}}.$$

Rationale for the form: constituent masses enter additively; the binding part is taken as the QCD scale times a fractal attenuation $K_{\text{frak}}^{n_{\text{eff}}}$ — the same K_{frak} power structure that carries scale transitions throughout the corpus (A040; level overview P40, Doc. 190). The exponent n_{eff} is initially free and is evaluated against experiment; its topological interpretation is supplied by A270.

25.4 Domain of Validity

[K] Since $K_{\text{frak}} < 1$, for every $n_{\text{eff}} > 0$:

$$\Lambda_{\text{QCD}} \cdot K_{\text{frak}}^{n_{\text{eff}}} < \Lambda_{\text{QCD}}.$$

The additive ansatz can structurally carry only binding contributions below Λ_{QCD}. This is a property of the functional form, not an approximation issue.

[K] Numerical check:

Meson	m_M (GeV)	$\sum m_q$	Binding	n_{eff}	Status
π^0 $(u\bar{u}/d\bar{d})$	0.1350	0.0043	0.1307	37.79	within domain
$\pi^\pm$ $(u\bar{d})$	0.1396	0.0068	0.1327	36.62	within domain
$K^\pm$ $(u\bar{s})$	0.4937	0.0956	0.3981	—	binding $> \Lambda$
η	0.5479	0.1868	0.3611	—	binding $> \Lambda$
ρ $(u\bar{u})$	0.7753	0.0043	0.7709	—	binding $> \Lambda$
ω $(u\bar{u})$	0.7827	0.0043	0.7783	—	binding $> \Lambda$

Only the pions lie within the domain. The failure for K, η, ρ, ω is not a numerical defect but the declared boundary of the functional form.

25.5 Sector Connection (A270)

[K] From the Z_3 sector structure, A270 predicts for mesons (1 twisted sector) the exponent $36 + 1 = 37$; the exact reference value of the bulk relation is $n = 35.99$ ($K_{\text{frak}}^{-36} \approx 16/\pi^2$, A270 §Key Relation).

Evaluating the pion data yields:

$$n_{\text{eff}}(\pi^0) = 37.79, \qquad n_{\text{eff}}(\pi^\pm) = 36.62.$$

Both values lie within the window of the sector prediction $36 + 1$. **The meson step of the A270 sector table is thereby empirically anchored at the pion** — the exponent was determined here from the mass data independently of the sector argument.

25.6 GMOR Check and the Limit of the Additive Ansatz

[K] For pseudo-Goldstone bosons the Gell-Mann–Oakes–Renner relation of chiral perturbation theory holds (external standard physics):

$$m_P^2 = (m_{q_1} + m_{q_2}) \cdot B, \qquad B = \frac{-2\langle \bar{q} q \rangle}{f_\pi^2}.$$

Evaluation with the external inputs:

$$B_\pi = \frac{m_{\pi^\pm}^2}{m_u + m_d} = 2.852 \text{ GeV},$$
$$B_K = \frac{m_K^2}{m_u + m_s} = 2.550 \text{ GeV},$$
$$\frac{B_\pi}{B_K} = 1.12.$$

Pion and kaon share the same chiral structure (B constant up to the known SU(3) breaking). **The additive ansatz thus fails for the kaon not because of different physics but because of the wrong functional form**: linear instead of quadratic (GMOR).

[S] Candidate: B carries the sector factor. Numerically one notes:

$$\frac{B_\pi}{\Lambda_{\text{QCD}} \cdot K_{\text{frak}}^{-37}} = 7.999 \approx 8, \qquad \text{i.e.} \qquad B \approx 8\,\Lambda_{\text{QCD}} \cdot K_{\text{frak}}^{-37}.$$

The 8 admits two corpus readings: $\dim \mathrm{SU}(3) = 8$ (number of gluons — the chiral condensate is gluonically carried) or the D4 ratio Weyl order/kissing number $= 192/24 = 8$ (A270). **P35 applies**: the central deviation ($0.02\,\%$) lies far below the scheme uncertainty of the quark masses ($\sim$2 %); the hit is within uncertainty, but the precision is not robust. Booked as candidate, not as result.

Were the candidate viable, a GMOR form with sector factor would follow,

$$m_P^2 = (m_{q_1} + m_{q_2}) \cdot 8\,\Lambda_{\text{QCD}} \cdot K_{\text{frak}}^{-37},$$

covering pion *and* kaon (kaon then to $\sim$6 % via SU(3) breaking in B). A [K]-level test of this form requires scheme-fixed quark masses — open.

[STIPULATION] Vector mesons (ρ, ω) are not Goldstone bosons; neither the additive ansatz nor GMOR applies to them. Their masses sit at the chiral symmetry-breaking scale itself ($\sim 4\pi f_\pi$). An FFGFT treatment of the vector sector does not exist and is not claimed here.

25.7 Baryon Candidate: Fully Geometric Proton Formula

[S] The additive meson ansatz and the sector structure (A270) raise the question whether the proton mass, too, admits a geometric decomposition. Candidate:

$$m_p = \underbrace{\frac{\pi^2}{2}}_{\text{Vol}(B^4)} \cdot \underbrace{\frac{\pi}{6}}_{\eta_{\text{sc}}} \cdot \underbrace{N_c\,(1+\alpha_s)}_{\text{color/coupling}} \cdot \underbrace{e^{-3\xi/4}}_{\xi\text{-damping}} \cdot \underbrace{\frac{\Lambda_{\text{QCD}}}{2}}_{\text{scale}} = 0.9402\ \text{GeV}$$

against PDG 0.93827 ($\Delta = +0.21\,\%$). Every factor is interpreted:

Factor	Value	Interpretation
$\pi^2/2$	4.9348	volume of the unit 4-ball (T^4-native)
$\pi/6$	0.5236	packing density of the simple cubic lattice, Vol$(B^3)/2^3$ — 3D analogue of the D4 density $\pi^2/16$ (4D) that carries the bulk exponent (A270)
$N_c(1+\alpha_s)$	3.354	color number times coupling correction (external QCD quantities, §External Inputs)
$e^{-3\xi/4}$	0.9999	ξ-damping (N_c-fold)
$\Lambda_{\text{QCD}}/2$	0.1085 GeV	halved QCD scale; sole dimensionful carrier

Absorbability of the residual: $+0.21\,\%$ closes exactly with $\alpha_s = 0.1156$ (instead of 0.118; $-2\,\%$) or $\Lambda_{\text{QCD}} = 0.2165$ (instead of 0.217; $-0.2\,\%$) — both deep within the uncertainty of the external inputs.

[S] Structural link to the Higgs formula. The two geometry factors combine to

$$\frac{\pi^2}{2}\cdot\frac{\pi}{6} = \frac{\pi^3}{12}, \qquad \text{and} \qquad \frac{16\pi^3}{\pi^3/12} = 192$$

— the D4 Weyl order. Proton geometry and the Higgs formula $16\pi^3$ (Doc. 190, K1) thus stand in the exact ratio of the Weyl number 192. Noted under P35, not booked.

P35 caveat (competing candidate): The product of free factors fixed by the calculation is $X = 0.52248$; $\pi/6 = 0.52360$ hits at $+0.21\,\%$. Numerically closer would be $K_{\text{frak}}^{48.4}$ ($-0.05\,\%$), but without integer exponent, without interpretation, and without an absorbability argument (K_{frak}^{48} misses at $+0.49\,\%$). Two candidates, no forward derivation: [S], not [K]. The historical occasion of this analysis (correction of a legacy-corpus calculation) is registered in Doc. 190, R62.

25.8 Layer-Status Summary

Statement	Status
Additive ansatz $m_M = \sum m_q + \Lambda\, K_{\text{frak}}^n$	[STIPULATION] (ansatz)
External inputs Λ_{QCD}, quark masses	[STIPULATION] (declared)
Domain: only binding $< \Lambda_{\text{QCD}}$	[K] (form property)
Pion within domain, $n_{\text{eff}}(\pi^0) = 37.79$	[K]
Sector connection $n_{\text{eff}} \approx 37 = 36 + 1$ (A270)	[K]
GMOR consistency $B_\pi \approx B_K$	[K] (standard ChPT)
$B \approx 8\,\Lambda \cdot K_{\text{frak}}^{-37}$	[S] (candidate, P35)
Vector mesons outside	[STIPULATION]
Proton formula $\frac{\pi^3}{12} N_c(1+\alpha_s)e^{-3\xi/4}\frac{\Lambda}{2}$	[S] (candidate, P35)
$16\pi^3/(\pi^3/12) = 192$ (Higgs link)	[S] (noted, P35)

25.9 Open Points

1. Scheme-fixed quark masses for a [K]-level test of the $8\Lambda K_{\text{frak}}^{-37}$ candidate.
2. Origin of the 8: gluon number or D4 ratio — both readings unconnected; unbooked under P35 until a forward derivation exists.
3. $\pi^\pm/\pi^0$ difference (36.62 vs. 37.79): does the electromagnetic self-energy of the charged pion carry a sector contribution of its own?
4. Vector meson sector: fully open.
5. Forward derivation of the baryon candidate: why $\pi/6$ (3D packing) alongside Vol(B^4) (4D volume)? Decision between the candidates $\pi/6$ and a K_{frak} power pending.
6. For the correction of the historical legacy-corpus baryon calculation see Doc. 190, register R62.

25.10 Notation

Symbols used throughout the A-series are explained in A010. Specific to this document:

Symbol	Meaning
m_M, m_P	meson mass in general resp. pseudoscalar
m_{q_i}	current quark masses ($\overline{\text{MS}}$, 2 GeV, PDG; §External Inputs)
$\Lambda_{\text{QCD}} = 0.217$ GeV	QCD scale (§External Inputs)
n_{eff}	exponent of the fractal binding correction $K_{\text{frak}}^{n_{\text{eff}}}$
B	GMOR parameter, $B = -2\langle\bar{q}q\rangle/f_\pi^2$; from $m_P^2 = (m_{q_1} + m_{q_2})\,B$
$\langle\bar{q}q\rangle$, f_π	chiral condensate; pion decay constant
$\eta_{\text{sc}} = \pi/6$	packing density of the simple cubic lattice
$\text{Vol}(B^4) = \pi^2/2$	volume of the unit 4-ball
$N_c = 3$, $\alpha_s = 0.118$	color number; strong coupling (external QCD quantities)

Verification script: `python/A_Serie_Skripte/a155_meson.py` — all numerical statements of this document are checked there with assertions.

25.11 References

Document	Reference
A010	A-series notation
A040	derivation of $K_{\text{frak}} = 1 - 100\xi$
A110	circulant (Z_3 algebra of the untwisted sector)
A150	quark-sector caution: scheme dependence, no predictive quantity
A270	Z_3 sector structure, meson exponent $36 + 1 = 37$
Doc. 190	P35, P40; register R62 (baryon correction, legacy corpus)

Chapter 26

A160 · Quantum Mechanics on T^4

26.1 The Thesis

[STIPULATION] Bell correlations are on T^4 a *topological* phenomenon: two measurement sites that lie far apart in the unrolled picture are connected in the compact picture by a closed path.

What appears as action at a distance in the projected picture is proximity in the compact picture. Non-locality is thereby an artefact of the projection (A090), not an ontological property of the world.

[B] This is not a circumvention of Bell's theorem. Bell rules out local hidden variables *in the projected space*. A topological connection in the pre-ordered space is not a hidden variable in the sense of the theorem – it is not an additional quantity that fixes the outcome, but a statement about the geometry in which both measurement sites lie.

26.2 What Does *Not* Change

[K] The predictions of quantum mechanics remain. The CHSH value for the singlet state is

$$S = 2\sqrt{2} = 2.8284271\ldots$$

and the Tsirelson bound holds unchanged.

[B] In particular, the mass extension changes nothing: the $\mathbb{C}^3$ tensor factor is invisible for correlation quantities, because the fibre state is normalised and the observable acts there as the identity (A070). This is computed, not claimed.

A theory that changed something here would already be refuted – CHSH is one of the best-tested quantities in physics.

26.3 The Residual of Order ξ

[K] Compactness has not completely disappeared after the projection. It leaves a residual, and this residual is the only place where the theory can show itself in the quantum domain at all.

The order of magnitude is fixed by ξ: the relative deviation is about 10^{-4} to 10^{-5}, depending on how many ξ-factors enter.

What this means for measurability – without gloss. The effect is *not resolvable* with current hardware. The systematic uncertainties of real Bell experiments – detector efficiency, angle calibration, state preparation – are orders of magnitude larger. A fit of the ξ parameter to measured CHSH values is therefore *pointless*: it would adjust hardware systematics, not measure geometry.

This finding is explicitly part of the result. A prediction whose signal lies below the noise is a prediction for later, not a confirmation for today.

26.4 Why the Residual Is Nevertheless a Prediction Quantity

[K] The ξ-order of the CHSH deviation is one of the four prediction quantities of the A-series (A010). The reason is not that it is testable today, but that it is *in principle* testable and the theory can fail at it.

Three properties make it a genuine prediction:

- The *direction* is fixed: the residual reduces the correlation, it does not increase it.
- The *order of magnitude* is fixed and not adjustable – it depends solely on ξ, which is stipulated in A020 and not readjustable.
- The *scale dependence* is fixed: the residual grows logarithmically with the number of degrees of freedom involved, not linearly.

[STIPULATION] If a CHSH value is measured that *exceeds* the Tsirelson bound, or one that falls below it by significantly more than ξ-order without hardware systematics explaining it, the theory is refuted at this point.

26.5 The Geometric Reading of Entanglement

[B] In the compact picture, an entangled pair is not a pair of separated objects with a connection but *one* object with two projection images. The correlation is then not a transmission but identity.

From this it follows immediately why no information is transmitted and Relativity remains unviolated: there is nothing that travels from A to B. Signal-freedom is in this reading not an additional condition to be imposed, but a consequence of the construction.

[S] The assignment of which closed paths correspond to which concrete entanglement states has not been worked out. It is a sketch and is marked as such.

26.6 Verification Script

- CHSH in the singlet vs. the Tsirelson bound, the invisibility of the $\mathbb{C}^3$ factor, and the order of magnitude of the ξ-residual compared to typical experimental uncertainties: `python/A_Serie_Skripte/a160_chsh.py`

26.7 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
S_{CHSH}	CHSH correlation value; classical ≤ 2, Tsirelson $\leq 2\sqrt{2}$
Δ_{CHSH}	Deviation from Tsirelson value; T0 prediction $\sim \xi$
$\varepsilon_{\text{CHSH}}$	Relative CHSH correction from T^4 geometry

26.8 Open Questions

First, the prefactor $1/(2\pi)$ is now geometrically justified. ξ is the fractal correction to the arc length on the torus. A CHSH measurement is an angle measurement over a full circle (circumference 2π). The correction per angle unit is $\xi/(2\pi)$ – the circumference is the conversion factor, not a free parameter. **[B]**

[B] Connection to the cumulative formula (A165). The elementary formula $\xi/(2\pi)$ and the cumulative formula $\text{CHSH}^{\text{T0}}(n) = 2\sqrt{2} \cdot \exp\left(-\xi \ln n/D_f\right)$ describe different observables:

- **Elementary:** phase correction per single measurement (one circumference 2π) – hence $\xi/(2\pi)$.
- **Cumulative:** damping over $\ln n$ entanglement layers (logarithmic entropy scale) – hence $\xi \ln n/D_f$. The 2π does not appear because the relevant integration length is $\ln n$, not the circumference.

Why D_f in the denominator? The fractal correction ξ acts in $D_f = 3 - \xi$ effective dimensions simultaneously. Per dimension: ξ/D_f. The cumulative formula integrates over all dimensions – that is fractal self-consistency: ξ/D_f instead of $\xi/3$.

Both formulas are now geometrically justified. What remains: the measurement. The deviation lies below the NISQ noise (A165).

Second, the path assignment is open (see above, **[S]**).

Third, the claim that Bell is thereby ontologically defused is an interpretation. It is consistent, but it does not follow compellingly – the same mathematics can be read without this interpretation. The A-series does not claim to have decided the interpretational question.

26.9 Supersedes

This document subsumes: Doc. 020–023 (quantum mechanics, Bell), Doc. 035 (non-locality), Doc. 073 (entanglement), Doc. 142 (CHSH), Doc. 055 (dynamic mass photons, energy-dependent non-locality), Doc. 074 (no-go theorems and T0 answer: Bell's theorem, Kochen–Specker, locality argument), Doc. 131 (apparent instantaneity, local vs. global relations, Green's function and causality in the T0 formalism). Incorporated: R41.

26.10 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 27

A165 · Bell Tests and Hardware

27.1 Purpose

[STIPULATION] This document develops the quantum-computing framework of FFGFT: energy-field qubits, the ϕ-hierarchical quantum Fourier transform (ϕ-QFT), Bell-test modifications with the CHSH scaling formula $\text{CHSH}^{\text{T0}}(n) = 2\sqrt{2} \cdot \exp\left(-\xi \ln(n)/D_f\right)$, and the hardware validation on IBM quantum computers (May 2026). The document provides evidence that T0 circuits are executable on production hardware and show Bell violation ($S = 2.74$, $96.9\,\%$ of the Tsirelson bound), while the ξ effect itself is not resolvable with today's NISQ hardware.

27.2 Energy-Field Qubits and the T0 Representation

[B] Standard qubits are represented as complex vectors $\alpha|0\rangle + \beta|1\rangle \in \mathbb{C}^2$ in Hilbert space. T0 instead uses cylindrical coordinates (z, r, θ):

- $z \in [-1, 1]$: projection onto the computation axis ($z = +1 \Leftrightarrow |0\rangle$, $z = -1 \Leftrightarrow |1\rangle$)
- $r \in [0, 1]$: superposition amplitude (radial distance), $z^2 + r^2 = 1$
- $\theta \in [0, 2\pi)$: phase (azimuthal angle)

[S] The decisive conceptual shift: r^2 is not a probability but represents the *energy density* of the superposition state. This permits deterministic evolution while retaining quantum-mechanical interference. The connection to standard form is the same bridge as in A070: $\alpha = \sqrt{(1+z)/2}\, e^{i\theta/2}$.

[B] The cylindrical representation is, for atomic systems ($R/r \gg 1$), the approximation of the full torus geometry. For proton-scale systems the aspect ratio is $R/r \sim 2.5 \times 10^{18}$ – the approximation is exact in the limit. Numerical comparisons of cylindrical and toroidal calculations deviate by less than $0.02\,\%$ for $R/r > 10^{12}$.

[K] The modified Hadamard gate with Bell damping for an n-qubit system:

$$H_{\mathrm{TO}}^{(n)} \;:\; (z, r, \theta) \;\mapsto\; \big(r \cdot e^{-\xi \ln(n)/D_f},\; z \cdot e^{-\xi \ln(n)/D_f},\; \theta + \tfrac{\pi}{2}\big).$$

The transform $(z, r) \to (r, z)$ implements the basis change; the damping factor $\exp\!\big(-\xi \ln(n)/D_f\big)$ stabilises the multi-qubit entanglement.

27.3 The ϕ-Hierarchical QFT

[B] The ϕ-hierarchical QFT applies phases $2\pi/\phi^k$ instead of $2\pi/2^k$, with $\phi = (1 + \sqrt{5})/2$:

$$\phi\text{-QFT} \;:\; |x\rangle \;\mapsto\; \frac{1}{\sqrt{Q_\phi}} \sum_{y=0}^{Q_\phi - 1} e^{2\pi i x y / Q_\phi} |y\rangle, \qquad Q_\phi = \phi^n.$$

[B] Equivalence theorem. For Shor's algorithm for factoring $N < 2^{20}$ with error probability $\delta < 10^{-6}$:

$$\begin{aligned} &P_{\text{success}}(\text{Standard-QFT}) \\ &\quad \le\; P_{\text{success}}(\phi\text{-QFT}) \\ &\quad \le\; P_{\text{success}}(\text{Standard-QFT}) + \xi. \end{aligned}$$

The proof proceeds in three steps: (1) For every period $r \in [2, N]$ there exists $k \in \mathbb{Z}$ with $|r - \phi^k \cdot c| < 1/(2r^2)$ for a rational number c with small denominator (Weyl equidistribution theorem, Dirichlet pigeonhole). (2) This perturbation satisfies the continued-fraction stability $|y/Q - s/r| < 1/(2r^2)$. (3) Bell damping increases the signal-to-noise ratio: $\text{SNR}_{\phi\text{-QFT}} = \text{SNR}_{\text{std}} \cdot (1 + \xi \ln(r)/D_f)$, ensuring $P_{\phi\text{-QFT}} \ge P_{\text{std}}$.

[K] Decoherence suppression. Under phase noise $\epsilon \cdot \sigma_z$ with $\epsilon \sim \mathcal{N}(0, \sigma^2)$, for $\epsilon < 0.1$:

$$\text{Fidelity}_{\phi\text{-QFT}} = \text{Fidelity}_{\text{std}} \cdot \exp\!\big(\xi \epsilon^2 / D_f\big) > \text{Fidelity}_{\text{std}}.$$

The error term is quadratic in ϵ instead of linear – this is the structural stability advantage.

27.4 Bell-Test Modifications and CHSH Scaling

[K] The T0-modified Bell correlation for two qubits with measurement angles a and b:

$$E^{\mathrm{T0}}(a,b) = -\cos(a-b)\cdot(1-\xi\cdot f(n,l,j)),$$

with $f(n,l,j) = (n/\phi)^l \cdot (1+\xi j/\pi)$. For photon-like qubits ($n=1, l=0, j=1$): $f \approx 1.000042$.

[K] CHSH scaling formula. For n entangled qubits:

$$\boxed{\mathrm{CHSH}^{\mathrm{T0}}(n) = 2\sqrt{2}\cdot\exp\left(-\frac{\xi\ln n}{D_f}\right)}$$

Derivation: With T0 modification and n-qubit Bell damping, $E_i^{\mathrm{T0}} \approx E_i^{\mathrm{QM}}\cdot(1-\xi\ln n/D_f)$; summing over the four CHSH terms gives the formula. For 73 qubits: $\mathrm{CHSH}^{\mathrm{T0}}(73) = 2.828427\cdot e^{-\xi\cdot 4.290/D_f} = 2.827888$, deviation $\Delta = 5.39\times10^{-4}$.

n qubits	QM CHSH	T0 CHSH	Δ	Testable
10	2.828427	2.828138	1.0×10^{-3}	Yes
20	2.828427	2.828051	1.3×10^{-3}	Yes
50	2.828427	2.827935	1.7×10^{-3}	Yes
73	2.828427	2.827888	1.9×10^{-3}	Yes
100	2.828427	2.827848	2.1×10^{-3}	Yes

[K] Spatial correlation delay. For Bell tests over distance d, T0 predicts a measurable delay:

$$\Delta t = \xi\cdot\frac{d}{c}.$$

The correlation field propagates causally; ξ introduces a phase delay. For $d = 1000$ km: $\Delta t \approx 445$ ns (measurable with atomic clocks, precision ~ 10 ns).

27.5 Application: T0-Shor Algorithm

[B] T0-Shor supplements the standard algorithm with two search stages: (1) **ξ-resonance scan**: For $r \in [2, \min(N, 100)]$, compute $a^r \bmod N$ and evaluate the resonance strength $\xi_{\text{mod}}/(|a^r - 1| + 1)$ with $\xi_{\text{mod}} = e^{-\xi r^2/D_f}$; if $a^r \equiv 1 \pmod N$, return r directly. (2) **ϕ-hierarchy search**: Test $r = \text{round}(\phi^k)$ for $k \in [0, 20]$.

[K] Complexity. The ξ scan has cost $O(\min(\sqrt{N}, 100) \cdot \log^2 N)$; the ϕ search $O(\log N \cdot \log^2 N)$. Total complexity remains $O(\log^3 N)$ – the additional term $O(\xi \log N)$ is negligible for practical N.

[K] Benchmark results (Python implementation, 100 % success rate):

N	Factors	Period r	Method	Time (s)
15	3×5	4	ξ-resonance	0.033
21	3×7	2	ξ-resonance	0.0003
33	3×11	10	ξ-resonance	0.0003
77	7×11	30	ξ-resonance	0.0003
143	11×13	60	ξ-resonance	0.0003

27.6 Hardware Validation: IBM Quantum May 2026

[K] Hardware validation was carried out on 28 May 2026 on `ibm_kingston` (Heron r2, 156 physical qubits) with 50 independent runs and on `ibm_marrakesh` with 10 runs; 2048 shots each.

[K] Bell-state fidelity. Circuit: standard Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ from Hadamard + CNOT. Result of the 50-run measurement (ibm_kingston): mean fidelity 0.9876 ± 0.0028, variance $P(|00\rangle) = 1.249 \times 10^{-4}$. The chi-squared test against the shot-noise expectation (1.221×10^{-4}) gives $\chi^2 = 50.14$, $p = 0.856$ – the variance is fully consistent with pure quantum shot noise.

[K] Important correction. An earlier evaluation (3 runs, 2025) reported a variance of 0.000248 and interpreted this as "40 times more deterministic than QM". The extended 50-run measurement refutes this: the variance is indistinguishable from shot noise (ratio 1.02). Three runs were statistically insufficient; the claimed deterministic variance reduction does not hold.

[K] CHSH measurement. Angles: Alice $0^\circ/45^\circ$, Bob $22.5^\circ/-22.5^\circ$; correlators $E = (N_{00} - N_{01} - N_{10} + N_{11})/N$.

Device	Runs	Mean S	SE	% Tsirelson	Bell violation
ibm_kingston	15	2.7396	0.0071	96.9%	$105\,\sigma$
ibm_marrakesh	10	2.7016	0.0138	95.5%	$51\,\sigma$
Pooled	**25**	**2.7244**	**0.0078**	**96.3%**	—

The measured value $S = 2.74$ lies $105\,\sigma$ above the classical bound (2.0): local realism is excluded; the generated entanglement is genuine.

[K] Deviation from the Tsirelson bound – decoherence, not ξ. The value lies $3.14\,\%$ below $2\sqrt{2}$. The mean correlator strength is 0.685 instead of ideal 0.707 – corresponding to $\sim 1.5\,\%$ gate and readout error (typical for NISQ). This deviation is *decoherence*. The predicted ξ effect lies at $\sim 10^{-5}$ ($\approx 0.001\,\%$ of S), roughly $3000\times$ smaller than the measured gap.

[K] Cross-platform finding. Two identically built Heron-r2 chips differ in the measured CHSH value by 0.038 ($\approx 1.4\,\%$; Welch-t: $t = 2.46$, $p = 0.028$), due solely to calibration and noise differences. Since the sought ξ effect is $\sim 1400\times$ smaller than this device-to-device variation, the ξ prediction is *in principle not resolvable* with today's NISQ hardware – a directly measured, not merely estimated, finding. The ξ test requires error-corrected qubits with sub-ppm stable, device-stable precision.

27.7 Falsification Criteria

[K] The theory makes precisely testable statements:

CHSH scaling law: Measurement of $S(n)$ for $n = 10, 20, 50, 100, 200$ qubits. T0 predicts $S(n) = 2\sqrt{2} \cdot \exp\left(-\xi \ln n/D_f\right)$. Falsified if systematic deviation $> 3\sigma$ from the scaling law.

73-qubit Bell test: Predicted value $S(73) = 2.827888 \pm 0.0001$ (measurable with $\sigma \sim 10^{-4}$). Falsified if $S = 2.8284 \pm 0.0001$.

Spatial satellite Bell test: $\Delta t = 445 \pm 50$ ns for $d = 1000$ km. Falsified if $|\Delta t| < 50$ ns.

27.8 Proof of ϕ-QFT Equivalence: Complete Derivation

[B] Lemma (Period coverage by ϕ-powers). For every period $r \in [2, N]$ with $N < 2^{20}$ there exist $k \in \mathbb{Z}$ and a rational number c with small denominator such that $|r - \phi^k \cdot c| < 1/(2r^2)$.

Proof: The golden ratio $\phi = (1 + \sqrt{5})/2$ is a Pisot number. By Weyl's equidistribution theorem, the sequences $\{\phi^k \bmod r\}$ are equidistributed modulo r for finite k ranges. By Dirichlet's pigeonhole principle, for every $r \in [2, N]$ there exists k with $|\phi^k - r| < r/M$ for $M = \lceil \log_\phi(2r^2) \rceil + 1$. Rearranging gives a rational approximation c with the required error $< 1/(2r^2)$. For $N < 2^{20}$, k values up to $\lceil 20/\log_2 \phi \rceil \approx 36$ suffice.

[B] Main theorem. The proof proceeds in five parts:

(A) Signal analysis. After measuring the function register in Shor's algorithm the state collapses to $|\psi_3\rangle = M^{-1/2} \sum_{j=0}^{M-1} |jr + \ell\rangle$ with $M = \lfloor Q/r \rfloor$. The QFT gives amplitude $\alpha(y) = (QM)^{-1/2} e^{2\pi i \ell y/Q} \sum_{j=0}^{M-1} e^{2\pi i j r y/Q}$. Peaks arise when ry/Q is close to an integer s: $y_{\text{peak}} \approx s \cdot Q/r$.

(B) ϕ-QFT modification. Replace $Q = 2^n$ by $Q_\phi = \phi^n$; the Bell damping $\mathcal{D}(\theta) = \exp\left(-\xi\theta^2/(\pi^2 D_f)\right)$ modifies the amplitude. This suppresses off-peak contributions (where $\Delta\phi$ significantly departs from an integer multiple of 2π) and acts as a filter.

(C) Peak positions. Main peaks at $y_{\phi,\text{peak}} \approx s \cdot Q_\phi/r$. By the Lemma there exists k with $|Q_\phi/r - 2^n/r \cdot \phi^k/2^k| < 0.2 \cdot 2^n/r$.

(D) Continued-fraction stability. The continued-fraction algorithm extracts r from y/Q under the condition $|y/Q - s/r| < 1/(2r^2)$. Our error from (C) is $0.2/r < 1/(2r^2)$ for $r \geq 2$ – the condition is satisfied.

(E) Success probability. The standard formula gives $P_{\text{std}} = 4/\pi^2 - 1/(3r) + O(r^{-2})$. Bell amplification gives $\text{SNR}_{\phi\text{-QFT}} = \text{SNR}_{\text{std}} \cdot (1 + \xi \ln(r)/D_f)$, so $P_{\phi\text{-QFT}} \geq P_{\text{std}}$. The upper bound $P_{\text{std}} + \xi$ follows since ϕ-QFT cannot exceed perfect success.

27.9 Decoherence Suppression: Complete Proof

[K] Under phase noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$, for Standard-QFT: $|\alpha_{\mathrm{std}}(y)| \to |\alpha_{\mathrm{std}}(y)| \cdot (1 - |\epsilon|) + O(\epsilon^2)$ (linear degradation). For ϕ-QFT with Bell damping $\mathscr{D}(\theta) = \exp\left(-\xi\theta^2/(\pi^2 D_f)\right)$:

$$|\alpha_\phi(y)| \;\to\; |\alpha_\phi(y)| \cdot \exp\left(-\frac{\xi}{D_f}\epsilon^2\right) \;\approx\; |\alpha_\phi(y)| \cdot \left(1 - \frac{\xi}{D_f}\epsilon^2\right).$$

Since $\xi\epsilon/D_f \ll 1$ for $\epsilon < 0.1$, we have $1 - \xi\epsilon^2/D_f > 1 - \epsilon$. The error term is quadratic in ϵ instead of linear. For $\sigma = 0.1$: error standard-QFT $\approx 10\,\%$, error ϕ-QFT $\approx \xi\sigma^2/D_f \approx 4.4 \times 10^{-7}$.

27.10 Implementation: Key Code

[K] ξ-resonance period finding. The core function:

```
def find_period_xi_resonance(a, N, max_r=100):
    XI, D_F = 4/30000, 3 - 4/30000
    for r in range(2, min(N, max_r) + 1):
        power = pow(a, r, N)
        xi_mod = exp(-XI * r * r / D_F)
        if abs(power - 1) < 1e-10:
            return r            # exact resonance
    return None
```

[K] Bell damping in T0 qubit.

```
class T0Qubit:
    def apply_bell_damping(self, n_qubits):
        d = exp(-self.XI * log(n_qubits) / self.D_F)
        self.z *= d; self.r *= d
        norm = sqrt(self.z**2 + self.r**2)
        self.z /= norm; self.r /= norm

    def apply_hadamard_t0(self, n_qubits):
        # basis change: swap z,r
        self.z, self.r = self.r, self.z
```

```
        self.apply_bell_damping(n_qubits)
        self.theta = (self.theta + pi/2) % (2*pi)
```

[K] CHSH calculation (Monte-Carlo loop). For each of the four angle pairs $(a, b) \in \{(0^\circ, 22.5^\circ), (0^\circ, 67.5^\circ), (45^\circ, 22.5^\circ), (45^\circ, 67.5^\circ)\}$:

```
damping = exp(-XI * log(n) / D_F)
E = -cos(a - b) * damping     # T0 correlator
```

CHSH $= |E_1 - E_2 + E_3 + E_4|$ plus shot noise $\mathcal{N}(0, 1/\sqrt{\text{shots}})$.

27.11 Position in the A-Series

[B] The bijection T0-qubit↔Hilbert-space-qubit is carried out in A070 (bridge $(z, r, \theta) \leftrightarrow (\alpha, \beta)$). The CHSH deviation ΔCHSH $\sim \xi$ follows from the torus geometry (A090). As a prediction quantity, ΔCHSH is listed in A210. This document provides the concrete computational framework (ϕ-QFT, Bell damping, Shor) and the hardware status: measured $S = 2.74$ confirms executability; the ξ effect itself is not detectable with today's hardware.

27.12 Verification Scripts

- CHSH scaling law numerically for $n = 2 \ldots 200$, comparison with T0 formula: `python/A_Serie_Skripte/a165_chsh_skalierung.py`
- ϕ-QFT equivalence: period finding for $N = 15, 21, 33, 77, 143$, success rate and complexity: `python/A_Serie_Skripte/a165_phi_qft.py`

Measurement protocol for hardware runs. The IBM Heron-r2 measurements ($S = 2.74$ on `ibm_kingston`, May 2026, 50 independent runs) are the author's own measurement campaigns, not literature values. Raw data, job IDs and evaluation scripts are in the corpus directory for quantum-computing runs; the claim is verifiable from there. A070 and A090 cite the same campaign via this document.

27.13 Symbol Glossary

Symbols used throughout the A-Series are explained in A010. Specific to this document:

Symbol	Meaning
S_{CHSH}	CHSH correlation value; classical ≤ 2, Tsirelson bound $2\sqrt{2}$
ϕ-QFT	Golden-ratio-structured QFT on T^4
$S_{\mathrm{T0}}(n)$	T0 scaling formula for n qubit pairs
IBM Heron r2	Superconducting quantum processor (reference architecture)

27.14 Open Questions

The ξ effect is not measured. The predicted deviation $\Delta\mathrm{CHSH} \approx 5.4 \times 10^{-4}$ (at 73 qubits) requires error-corrected qubits with sub-ppm stability – beyond today's NISQ hardware.

The CHSH scaling is not fully verified. The table above contains predictions; measurements beyond $n > 2$ are missing.

Scalability to RSA-relevant N. Benchmark tests go up to $N = 143$; RSA-2048 requires further analysis.

Integration with quantum error correction. Surface codes and ϕ-QFT are not connected.

27.15 Supersedes

This document subsumes: Doc. 147 (quantum computing on T0/FFGFT basis: energy-field qubits, ϕ-QFT equivalence theorem, Bell-test modifications, ξ-resonance-Shor, IBM hardware validation May 2026). *Not* adopted: the section "Cosmic corrections for quantum computing" from Doc. 147 (daily/lunar/annual cycle effects on qubits – speculative appendix, not core content; outside the A-series framework A010). The English appendix duplicates from Doc. 147 are also not adopted.

27.16 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 28

A180 · Information: Winding Quantum $\Delta w = 1$

28.1 The Unit

[K] The elementary information unit of this theory is the *winding quantum*

$$\Delta w = 1.$$

It is the smallest difference between two winding classes on T^4. Half a winding quantum does not exist – the fundamental group $\pi_1(T^4) \cong \mathbb{Z}^4$ is integer-valued (A030).

[B] Three properties follow immediately, and they are the reason why this choice is to be preferred over any cell-based definition:

- **integer-valued** – no arbitrary fineness;
- **coordinate-free** – independent of origin, orientation, and parametrisation;
- **energy-independent** – the unit is the same at every scale, because it is topological, not metric.

The last property is the strongest. An information unit that depended on the energy would not be a unit but a scale choice.

28.2 Why No Unit Cells

[B] The obvious alternative – dividing space into cells and counting information – fails at the same objections listed in A030: origin, orientation, coordinate dependence, and an entropy that depends on an arbitrary partition.

For the winding quantum all these objections are empty, because a winding class has no location. It is the property of a closed path, not of a region. And the torus is boundaryless, $\partial T^4 = \emptyset$ – a surface-based definition is likewise excluded because no surface is distinguished.

28.3 Magnitude and Phase

[K] The mass operator splits into two channels (A070, A120): magnitudes and phases of the spectral coefficients. For the concept of information the separation is essential.

[B] A magnitude spectrum fixes the phase only up to an all-pass factor. There is therefore information that is *in principle* not visible in the magnitude channel – the proof stands in A120, where it is shown that θ is captured by no symmetric, counting functional.

[STIPULATION] From this follows a booking rule for the whole series: a statement that uses only magnitudes can make no claim about phase information – neither its existence nor its absence.

28.4 The Connection to Landauer

[B] The Landauer principle links the erasure of one bit to a minimum energy $k_B T \ln 2$. In the language of this theory this is a statement of the *translation layer*: it contains k_B and T, two quantities that are bridge and dress by A080.

The winding quantum lies beneath it. It is the topological unit; the Landauer energy is what it costs in a particular thermal dress.

[B] The derivation. On the 4-torus $\pi_1(T^4) \cong \mathbb{Z}^4$: the first homotopy group is $\mathbb{Z}^4$, each direction carries discrete winding numbers. Erasing a bit corresponds to the transition $\Delta w = 1$ – a change of winding number by one in one direction. This transition is irreversible (type-II loss, A090): the winding number is a topological invariant; its loss cannot be undone without expenditure of energy.

The energy expenditure follows from the thermodynamic dress of this topological transition. A system in thermal equilibrium at temperature T must for an irreversible bit transition dissipate at least entropy $\Delta S = k_B \ln 2$ (two states, one erased). The minimum energy is:

$$E_L = T \cdot \Delta S = k_B T \ln 2.$$

The winding quantum $\Delta w = 1$ is the geometric carrier of this step; the Landauer energy is its thermodynamic dress. Beneath: a topological transition. Above: a thermal energy amount.

[K] Limits. The derivation presupposes that the bit transition may be identified with the winding quantum – that is the physical claim that is not reduced further. The Landauer energy $k_B T \ln 2$ is a lower bound; actual erasure operations cost more. Whether exactly $\Delta w = 1$ is the only possibility to erase a bit is the second open question (see below).

28.5 The Reversal Test

[B] The projection chain (A090) is lossless in one direction and not in the other. For the concept of information this means: the question "how much information is lost?" has an answer only for type I and type II; for type III the answer is zero.

In unrolling (type I) the loss can be exactly named – it is the winding number, specifically one integer per direction. This is the only place in the series where an information loss can be *exactly* quantified, and it is, because the lost quantity is integer-valued.

28.6 Verification Script

- Integer-valuedness, coordinate-freedom, and scale-independence of Δw; countercheck that a cell-based entropy depends on the partition and a spectral one does not: `python/A_Serie_Skripte/a180_windungsquant.py`

28.7 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$S_L = k_B \ln 2$	Landauer energy; minimum cost per erased bit
Δw	Winding change; topological discreteness of information change
$I_{\max}$	Maximum information at given energy
$E_L = k_B T \ln 2$	Landauer bound at temperature T

28.8 Open Questions

First, the derivation presupposes that bit transition and winding quantum may be identified. That is the core physical claim; it is not reduced further. The derivation is carried out in A180 (above), but the claim that exactly $\Delta w = 1$ carries a bit is a stipulation – not forced by the topology alone.

Second, it has not been shown that $\Delta w = 1$ is the *only* invariant unit. Shown is that it is invariant and that cell-based alternatives are not. That is less.

Third, an entropy formula is missing. This series contains no written-out entropy as a function of the spectrum – only the finding that such a function would automatically be invariant. That is a programme, not a result.

28.9 Supersedes

This document subsumes: Doc. 243, 255, 257 (information), Doc. 290 (magnitude/phase), Doc. 301 (winding quantum), Doc. 302 (unit cells and partition invariance).

28.10 Use of AI Tools

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Chapter 29

A190 · Standard Model as Projection

29.1 The Thesis

[STIPULATION] The Standard Model is in this theory not a competing framework but the *decompactified projection* of the compact structure.

What in the Standard Model appears as a list of independent inputs – masses, couplings, mixing angles – are in this reading projection images of a structure with fewer degrees of freedom.

29.2 What the Projection Explains

[B] Three features of the Standard Model become, in this reading, consequences and not assumptions:

The number three. Three generations follow from the Z_3 structure (A030, A110). In the Standard Model it is an observation without reason.

The hierarchy. The orders of magnitude of the lepton masses follow the ξ-ladder (A100), at the per cent level. In the Standard Model the Yukawa couplings are free parameters without ordering among themselves.

The form of the mass relation. The Koide relation $Q = 2/3$ is an operator condition, not coincidence (A110). In the Standard Model it has no standing.

29.3 What the Projection Does Not Explain

[STIPULATION] This section is the longer one, and that is the honest finding.

The gauge group – three convergent arguments.

[B] Argument 1: Charge quantisation from torus flux. The electric charge arises from topologically quantised flux lines through the torus: $\Phi = n \cdot h/e$ with $n \in \mathbb{Z}$. The quantisation is topologically protected – the torus has two non-contractible loops (toroidal and poloidal), the flux through them is invariant under continuous deformations. $U(1)_{\text{EM}}$ follows from the torus topology, not from a postulated gauge symmetry.

[B] Argument 2: Colour charge from linking number. The colour charge of QCD arises from the topological linking (linking number) of exactly three flux threads around the torus. Three colours (red, green, blue) = three threads: $SU(3)_C$ is topologically forced by the triple-linking structure of $T^4/\mathbb{Z}_3$.

[S] Argument 3: $SU(27)$ group chain decomposition. From $27 = 3^3$ ($\mathbb{Z}_3$-torus) follows the chain $SU(27) \supset SU(9)^3 \supset SU(3)^9 \supset SU(3)\times SU(2)\times U(1)$. This makes the SM gauge group the canonical subgroup of the symmetry breaking.

What is still missing: Arguments 1 and 2 deliver $U(1)$ and $SU(3)$ from the torus topology. $SU(2)_L$ follows from the chiral projection (A095). What is not written out: an explicit mapping of the torus topology to the covariant derivatives $D_\mu = \partial_\mu + igA^a_\mu T^a$ – i.e. the transition from the topological charge structure to the gauge Lagrangian $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$.

The coupling constants. Their values and their running are not the subject of the A-series. For α the status is clarified in A130: calibrated, not derived.

The mixing matrices. CKM and PMNS do not appear.

The Higgs sector – consistency check near proof. A complete 1-loop EFT-matching calculation (carried out in the legacy corpus) derives:

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2}$$

with $\lambda_h = m_h^2/(2v^2)$, $v = 246$ GeV, $m_h = 125.09$ GeV. Numerically: $\xi_{\text{EFT}} = 1.303 \times 10^{-4}$, deviation from $\xi_0 = 4/30000$ only $2.3\,\%$. The $16\pi^3$ structure is the natural result of the 1-loop suppression and NDA normalisation – no fitting.

[S] Two restrictions that together are decisive.

First, the EFT calculation delivers a lepton-specific ξ ($\xi_e \approx 6.7 \times 10^{-14}$, $\xi_\mu \approx 2.9 \times 10^{-9}$), not the universal ξ_0. The numerically close value $\xi_{\text{EFT}} \approx 1.303 \times 10^{-4}$ is the value for a reference fermion at $\mu = m_h$ in the $\overline{\text{MS}}$ scheme.

Second, the $2.3\,\%$ deviation is structurally expected – from two independent reasons that act together:

(a) Unrolled SI values. The formula $\xi_{\text{EFT}} = m_h^2/(64\pi^3 v^2)$ contains $v = 246$ GeV and $m_h = 125.09$ GeV – SI measured values in the $\overline{\text{MS}}$ scheme, scheme-dependent and only at 1-loop level. A $2.3\,\%$ deviation corresponds to a $1.1\,\%$ deviation in v, which lies within the theoretical uncertainty of 1-loop matchings (typically 1–$5\,\%$ without NNLO corrections).

(b) Fractal corrections are not cleanly separable in v. The measured VEV $v = 246$ GeV absorbs all SM quantum corrections – the fractal torus structure $D_f = 3 - \xi$ is not separately visible therein but is implicitly contained in the measurement and mixed with the SM loops. The EFT matching in flat space with integer dimension $D = 3$ cannot cleanly extract $K_{\text{frak}} = (1 - 100\xi)$, leading to a systematic inaccuracy. Numerically: $\xi_{\text{EFT}}/K_{\text{frak}}^{3/2} \approx 1.330 \times 10^{-4}$, deviation from ξ_0 only $0.3\,\%$ – but that is a post-hoc correction, not a proof.

In short: the deviation is structurally expected, because two different description levels – flat perturbative QFT and fractal torus geometry – do not see the same effective v value.

ξ_{EFT} must therefore not be used to calculate α or E_0. That would be a circularity trap: one would run SI measured values (v, m_h) through the FFGFT formulas and count the result as confirmation, even though the SI values themselves carry the deviation. The formulas of the A-series use exclusively $\xi_0 = 4/30000$.

ξ_{EFT} is a Wilson coefficient (scheme- and scale-dependent); ξ_0 is the geometric ratio L_0/ℓ_P (scheme-independent). That both agree to $2.3\,\%$ – in a calculation using SI measured values with built-in theoretical uncertainty – is the consistency finding. No more, but also no less.

[B] The balance reads: the theory explains the *ordering* of the fermion sector and not the gauge sector. That is a partial statement, and presenting it as an explanation of the Standard Model would be an overreach.

29.4 Why the Projection Loses Information

[B] According to A090, the spatial projection (type II) is not reversible. From this it follows that the Standard Model *cannot* fully reproduce the compact structure – regardless of how precisely one measures it.

This is testable, though difficult: what does not completely disappear in the projection leaves a residual of order ξ, and the four prediction quantities of the series are precisely these residuals (A090, A210).

29.5 Verification Script

- Numerical verification of projection-chain steps and mass ratios from A090/A100: `python/A_Serie_Skripte/a090_proj.py` (shared with A090).

29.6 Open Questions

First, the gauge sector is completely missing. Without it the claim that the Standard Model is a projection is untested for the largest part of the Standard Model.

Second, the projection mapping is not written out. There is in this series no explicit mapping from T^4/Z_3 to the fields of the Standard Model – only the claim that it exists, and examples in the fermion sector.

Third, the statement in its present form is not falsifiable. As long as the mapping is missing, the thesis can be neither confirmed nor refuted. It is a programme.

29.7 Supersedes

This document subsumes: Doc. 019 (Standard Model), Doc. 097 (QFT), Doc. 141 (SM parameters), Doc. 298 (SM as projection).

29.8 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 30

A192 · Gauge Sector: Torus Topology

30.1 Purpose

[STIPULATION] A190 names the gauge sector as the largest remaining gap. This document carries out the derivation: from torus topology to the covariant derivative and the gauge Lagrangian, for all three factors of the SM gauge group $SU(3)_C \times SU(2)_L \times U(1)_Y$.

30.2 Starting Point: Torus and Flux Quantisation

[B] On T^4 there exist non-contractible loops. The flux of a $U(1)$ connection through such a loop C is quantised: every continuous deformation of C leaves the flux invariant. This is the basic structure of all three gauge groups.

30.3 $U(1)_Y$: Electromagnetic Charge

[B] The wave function ψ accumulates a phase upon traversing the torus loop C:

$$\psi \longrightarrow \psi \cdot \exp\left(i\,\frac{e}{\hbar}\,\Phi\right), \qquad \Phi = \oint_C A_\mu\, dx^\mu = n\,\frac{h}{e}, \quad n \in \mathbb{Z}.$$

The monodromy $\exp(i \cdot 2\pi n) = 1$ is satisfied for all $n \in \mathbb{Z}$. This is the $U(1)$ gauge structure: the phase $e^{i\alpha(x)}$ with $\alpha(x) = (e/\hbar)\int A_\mu dx^\mu$.

[B] The covariant derivative and the field-strength tensor follow:

$$D_\mu = \partial_\mu + ieA_\mu, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \qquad \mathscr{L}_{\mathrm{EM}} = -\tfrac{1}{4}\,F_{\mu\nu}F^{\mu\nu}.$$

The gauge Lagrangian follows from minimal coupling: $F_{\mu\nu}$ is the curvature of the connection A_μ, $\mathscr{L}_{\mathrm{EM}}$ is its norm. No free step.

30.4 $SU(3)_C$: Colour Charge from Linking Number

[B] Three flux tubes on $T^4/\mathbb{Z}_3$ carry linking numbers $\ell k(i,j) \in \mathbb{Z}$ for $i, j \in \{1,2,3\}$ (tubes i and j). The independent non-trivial states are:

Linking state	Generator	Physics
$\ell k(1,2) = \pm 1$, rest 0	λ_1, λ_2	Gluon $r \leftrightarrow g$
$\ell k(1,3) = \pm 1$, rest 0	λ_4, λ_5	Gluon $r \leftrightarrow b$
$\ell k(2,3) = \pm 1$, rest 0	λ_6, λ_7	Gluon $g \leftrightarrow b$
$\ell k(1,1) - \ell k(2,2)$	λ_3	Gluon neutral (1)
$\ell k(1,1) + \ell k(2,2) - 2\,\ell k(3,3)$	λ_8	Gluon neutral (2)

[B] Count: $\binom{3}{2} = 3$ pairs, each with real and imaginary part, gives 6 off-diagonal; plus 2 traceless diagonal generators (rank-1 reduction of the 3×3 tracelessness condition) gives $6 + 2 = 8$ independent generators – exactly dim$(\mathfrak{su}(3))$. The gauge group $SU(3)_C$ is topologically forced, not postulated.

[B] Covariant derivative and Lagrangian analogous to $U(1)$:

$$D_\mu = \partial_\mu + i g_s G_\mu^a \lambda_a/2, \qquad G_{\mu\nu}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a + g_s f^{abc} G_\mu^b G_\nu^c,$$

$$\mathscr{L}_{\text{strong}} = -\tfrac{1}{4} G_{\mu\nu}^a G^{a,\mu\nu}.$$

The non-commutativity $f^{abc} \neq 0$ follows from the non-commutativity of the linking-number operations for three tubes.

30.5 $SU(2)_L$: Weak Isospin Group from Chiral Projection

[B] The chiral projection P_+ (A095) selects the $m \geq 0$ sector of the torus. The spinor-doublet character of the weak interaction follows from the spin-$\frac{1}{2}$ structure (winding $(1,1)$, A263): the two components of the isospin doublet (u_L, d_L) and (ν_L, e_L) respectively are the two independent $m = 0$ and $m = +1$ sectors after projection.

[B] The three generators τ^+, τ^-, τ^3 (Pauli matrices) of $SU(2)_L$ correspond to the three independent transitions between the two projected states. Covariant derivative:

$$D_\mu = \partial_\mu + ig_w W^a_\mu \sigma_a/2, \qquad W^a_{\mu\nu} = \partial_\mu W^a_\nu - \partial_\nu W^a_\mu + g_w \varepsilon^{abc} W^b_\mu W^c_\nu,$$

$$\mathscr{L}_{\text{weak}} = -\tfrac{1}{4} W^a_{\mu\nu} W^{a,\mu\nu}.$$

30.6 Overall Picture: Gauge Lagrangian from Torus Topology

[B] The complete Standard Model gauge Lagrangian:

$$\mathscr{L}_{\text{gauge}} = -\tfrac{1}{4} G^a_{\mu\nu} G^{a,\mu\nu} - \tfrac{1}{4} W^a_{\mu\nu} W^{a,\mu\nu} - \tfrac{1}{4} F_{\mu\nu} F^{\mu\nu}$$

follows from three topological structures of $T^4/\mathbb{Z}_3$:

- $U(1)_Y$: flux quantisation through non-contractible loop
- $SU(3)_C$: linking number of three $\mathbb{Z}_3$ flux tubes ($6 + 2 = 8$ generators)
- $SU(2)_L$: chiral projection of the spin-$\frac{1}{2}$ sector (A095)

Group	Topological origin	Generators	Gauge bosons
$U(1)_Y$	Flux quantisation	1	Photon (γ)
$SU(2)_L$	Chiral projection	3	W^+, W^-, Z^0
$SU(3)_C$	Linking number	8	8 gluons
Total		12	12 SM gauge bosons

30.7 Symbol Glossary

Symbols used throughout the A-Series are explained in A010. Specific to this document:

Symbol	Meaning
$\ell k(i,j)$	Linking number of tubes i and j on $T^4/\mathbb{Z}_3$
λ_a	Gell-Mann matrices, generators of $SU(3)$
σ_a	Pauli matrices, generators of $SU(2)$
G^a_μ, W^a_μ, A_μ	Gauge fields of $SU(3)$, $SU(2)$, $U(1)$
f^{abc}	Structure constants of $SU(3)$
ε^{abc}	Levi-Civita symbol, structure constants of $SU(2)$
g_s, g_w, e	Coupling constants (not derived from ξ)

30.8 Verification Script

- Count of linking states; Gell-Mann matrix identification; tracelessness condition; $6+2=8$: `python/A_Serie_Skripte/a192_eichsektor.py`

30.9 Open Questions

The coupling constants g_s, g_w, e are not derived from ξ. The group structure follows from the topology; the strength of the couplings does not. This is the remaining gap.

The running of the couplings ($g_s(\mu)$, $g_w(\mu)$ as functions of the energy scale μ) is not derived from FFGFT.

The Higgs mechanism – the spontaneous symmetry breaking $SU(2)_L \times U(1)_Y \to U(1)_{\text{EM}}$ – is not derived from the torus geometry. The Higgs-EFT consistency check (A190, $2.3\,\%$) is a hint, not a proof.

30.10 Supersedes

This document systematises the gauge-sector content of the old corpus (charge quantisation and colour charge from torus topology, Lagrangian structure) self-contained without old-corpus citations. It supplements A190 (SM as projection), A095 (chiral projection) and A075 (field theory).

30.11 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.

Part III

Methodology and Assessment

Chapter 31

A200 · Order Without Ranking

31.1 The Four Layers Are Not a Ranking

[STIPULATION] The entire series marks every statement with one of four layers: **[K]** core (derived from ξ), **[B]** bridge (algebraically proved), **[S]** sketch (plausibility), **[STIPULATION]** declared. This document says what this marking means and what it does not mean.

It is an *ordering*, not a *ranking*. The layers say with what right a statement stands – not how valuable it is. A **[STIPULATION]** is not inferior to a **[K]**; it is of a different kind. Without stipulations there would be no core from which anything could follow.

[B] The confusion of ordering with ranking is the most common misreading of layered presentations. One who reads **[S]** as "not yet quite [K]" assumes that everything strives toward the core. It does not. A sketch marked as a sketch is complete in its place – it claims no more than it shows, and that is its strength.

31.2 What Each Layer Accomplishes

[STIPULATION] The four layers in their own language:

[K] Core. From the three stipulations (A020) the statement follows by computation or geometric necessity. Examples: the non-closure theorem (A050), $Q = 2/3$ from $r = \sqrt{2}$ (A110), $\Delta w = 1$ (A180). The core is what the theory stands or falls by.

[B] Bridge. The statement is algebraically proved, but uses a translation or a known structure that does not itself follow from ξ. Example: the Hilbert space representation (A070). A bridge carries everything on both its sides and proves nothing new – it connects.

[S] Sketch. The statement is plausible but not computed through. Examples: the Landauer connection (A180), the path assignment of entanglement (A160). A sketch is an honest placeholder, not a half-truth.

[STIPULATION] Declaration. The statement is stipulated, not derived. Examples: the three fundamental stipulations (A020), the choice of the characteristic mass (A145). A stipulation is the entry point without which a closed system cannot begin (A130).

31.3 The Self-Application

[STIPULATION] The ordering applies to itself. The fundamental relation $\tilde{T} \cdot m = 1$ is a **[STIPULATION]**, not a **[K]**. It is not proved; it is the beginning.

This is not a weakness but the condition of possibility of the whole. Every theory that derives something derives it from something it does not derive. The difference between a clean and an unclean theory is not whether it has stipulations – every theory does – but whether it *declares them as such*.

[B] That is exactly what the layer marking does: it makes the stipulations visible instead of disguising them as results. The error that A130 identifies in the older α presentation – presenting a calibration as a derivation – is a violation of precisely this rule. The marking is the means to avoid it.

31.4 Why No Ranking Is Possible

[B] One might try to rank the layers: [K] better than [B] better than [S] better than [STIPULATION]. The attempt fails at the self-application. The most valuable statement of the theory – $\tilde{T} \cdot m = 1$ – would then be the least. That is absurd, and the absurdity shows that the ordering is horizontal, not vertical.

The layers are four ways of being true, not four degrees of truth.

31.5 Open Questions

First, the assignment in individual cases is not always sharp. Some statements lie between [B] and [S]. The series decides such cases conservatively – when in doubt, the weaker layer – but the boundary is not compelling in every case.

Second, the marking does not replace judgment. It says with what right a statement stands; whether this right is correctly assigned in the individual case must be checked. The verification scripts do this for the computable cases.

31.6 Supersedes

This document takes up the methodological note “Ordnung ohne Rangordnung / Order without Ranking” and makes its distinction (law, norm, stipulation) the throughgoing principle of the series. It stands alongside A010, which names the rules; A200 justifies them.

31.7 Use of AI Tools

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Chapter 32

A210 · Falsifiability

32.1 How the Theory Can Fail

[STIPULATION] A theory that cannot fail at anything says nothing. The A-series therefore carries exactly four quantities at which it can be refuted. They are ratios or structural statements – never pure absolute values, since absolute values are bookkeeping (A080, A130). The first, the τ-mass, is the sharpest: a genuine prediction of a measurable value, because the ratio structure of the leptons binds so tightly.

That there are only four is not a lack of courage, but a consequence of the projection chain (A090): only few quantities survive the projection from the compact structure into the observed world. What disappears in the projection is not measurable – not through ineptitude, but structurally.

32.2 The Four Prediction Quantities

[K] First: the τ-mass.

$$m_\tau^{\text{pred}} = 1776.968 \text{ MeV},$$

from (m_e, m_μ) and the structural numbers $(\sqrt{2}, 2/9)$ (A110). Against PDG 1776.86 ± 0.12 that is $+0.90\,\sigma$. A measurement to 0.05 MeV decides: if the value lands near ≈ 1776.97, the pair $(\sqrt{2}, 2/9)$ is confirmed to 10^{-5}; significantly below, it is refuted. This is the sharpest point of the series and its actual prediction core: the theory *does* predict the **lepton masses**. Unlike α or G, here a numerical value stands *before* the measurement – no bookkeeping, no anchor choice, but a genuine prediction, because the ratio structure forces the third mass from two.

[K] Second: the ξ-order of the CHSH deviation. The Bell value falls short of the Tsirelson bound by a residual of order

ξ, i.e. 10^{-4} to 10^{-5} (A160). Direction fixed (reduction), order of magnitude fixed (not adjustable), scaling fixed (logarithmic). Not resolvable today; a CHSH value above the bound or significantly below without hardware explanation refutes the theory.

[K] Third: the fractal dimension.

$$D_f = 3 - \xi = 2.99986667.$$

A geometric structure with a measurably different dimension from 3 in exactly this size – or its absence where the theory requires it – decides. D_f is ξ in geometric dress; the deviation from 3 *is* ξ.

[K] Fourth: the fundamental length as anchor point. The definition

$$L_0 = \xi \ell_P$$

places the fundamental length of the theory in relation to the Planck length. Below this sub-Planck scale the theory makes no statement (A130). L_0 is the lower bound of validity of the relative description; a structure that showed ξ-ratios even below it would speak against it.

The numerical value is an anchor matter, not a test. $L_0 = \xi \ell_P$ is the *only* anchor point at which the theory is bound to an absolute scale – and like any anchor, it can be set differently depending on one's viewpoint (against the Planck length, against the Planck time, against $1/\xi$). *Which anchor* is chosen is convention; falsifiable is solely the ξ-scaling: that the boundary lies at ξ times the Planck scale and not at another power.

The numerical value itself does, however, require care – it is not arbitrary. At the standard anchor (Planck length):

$$L_0 = \xi \ell_P = \frac{4}{30000} \cdot 1.616 \cdot 10^{-35}\,\text{m} = 2.155 \cdot 10^{-39}\,\text{m}.$$

An occasionally appearing value $5.39 \cdot 10^{-39}$ m is *not* a valid anchor variant but a transcription error: the mantissa 5.391 is that of the Planck *time* ($t_P = 5.391 \cdot 10^{-44}$ s), and the value implies $\xi = 3.34 \cdot 10^{-4}$ instead of $4/30000 = 1.333 \cdot 10^{-4}$ – a factor 2.5 too large. Convention is the *choice* of anchor; the numerical value at a given anchor is not.

32.3 Why They Must Be Ratios

[B] All four are dimensionless or traceable to ξ. This is no coincidence but a principle: only ratios are intrinsic (A080). An absolute value – e.g. "$G = 6.674 \cdot 10^{-11}$" – is bound to a unit choice and can neither confirm nor refute the theory, because it depends on the scale that is itself an input.

That is why α is not on the list, even though A130 treats it in detail: its numerical value is bookkeeping. And that is why the τ-mass is on it, even though it is a mass: what is tested is not its absolute value but the *ratio* that $(\sqrt{2}, 2/9)$ forces between the three lepton masses.

32.4 What Is Not a Test Quantity

[STIPULATION] Explicitly *not* counted as falsification:

The agreement of α with $1/137$ (bookkeeping, A130). The numerical value of G (bookkeeping, A145). The ladder ratios at the per cent level (too coarse, check points not predictions, A100). The neutrino sector (not testable, A150). The thesis "SM is a projection" as long as the mapping is missing (A190).

This honesty belongs to falsifiability: listing as a test quantity something one cannot sharply test simulates testability.

32.5 Verification Script

- The four prediction quantities with value, comparison quantity, and decision criterion; and the demonstration that each is dimensionless or traceable to ξ: `python/A_Serie_Skripte/a210_prognosegroessen.py`

32.6 Open Questions

First, three of the four are not sharply measurable today. Only the τ-mass is within reach of running experiments. The other three are *in principle* testable, but not yet in practice. That is honest to name.

Second, the completeness of the list is not proved. That exactly four quantities survive the projection is an inventory (A090), not a derivation. A fifth is not excluded.

32.7 Supersedes

This document collects the test quantities from A090, A110, A120, A130, A160 and presents them as a closed list. It subsumes: Doc. 292 (τ-prediction), Doc. 142 (CHSH), Doc. 101 (L_0, D_f), and the falsification and no-go statements from Doc. 249, Doc. 074 (no-go), and Doc. 193. Incorporated: R52 (numerical value of L_0 at the standard anchor corrected; $5.39 \cdot 10^{-39}$ m rejected as transcription error).

32.8 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 33

A220 · Tolerance Scale

33.1 Three Numbers, Not One

[STIPULATION] For every comparison number the series gives three quantities, never one: the value, the measurement precision of the comparison quantity, and the determination tolerance of the theory quantity. A percentage deviation alone says nothing.

The reason is simple and used throughout: a one per cent deviation is meaningless when the theory quantity is only determined to two significant figures – and damning if both sides are fixed to eight figures. Without both tolerances the percentage is unreadable.

33.2 We Compute Nothing

[STIPULATION] The most important statement about all numbers in this series: the theory *computes nothing*. It is not a calculation engine producing predictions to many decimal places. It orders geometric structures and shows that they fall in the right range.

From this follows how the verification scripts are to be read. When a script reports agreement "to 10^{-15}", that is *not* a result. It is either an **identity** – a quantity equal by definition or construction ($\alpha = 1$ after charge normalisation, $L_0/\ell_P = \xi$ by definition) – or it is a **range check**: the test of whether we are in the right order of magnitude. The many decimal places are floating-point arithmetic, not the precision of a statement.

[B] *Three* types of test, strictly to be distinguished:

An *identity* verifies a definition. It cannot fail as long as the algebra is correct. Passing it is no achievement; failing it would be a programming error. Examples: $\alpha = 4\pi/4\pi = 1$ (A130), $L_0 = \xi\ell_P$ (A210), the reverse direction $\xi = 2\sqrt{Gm}$ (A145).

A *range check* compares a structure with measurement and asks only: correct order of magnitude, correct sign, correct scaling? Whether E_{char} is near 28 rather than 3 or 300, whether κ is near 4 rather than 9, whether the CHSH residual is at 10^{-4} rather than 10^{-1}.

A *prediction* is the case where the theory *genuinely* predicts something measurable: the **lepton masses**. The structure $(\sqrt{2}, 2/9)$ fixes m_τ from m_e and m_μ – a sharp, falsifiable prediction, $m_\tau = 1776.968$ MeV against PDG 1776.86 ± 0.12, i.e. $+0.90\,\sigma$. This is not a range check: here a numerical value stands *before* the measurement, and the measurement decides it.

[STIPULATION] "We compute nothing" therefore applies to the *bookkeeping values* – $\alpha = 1/137$, the SI value of G, the anchor choice L_0 – but not to the lepton masses. For the bookkeeping values the question "is the result exactly correct?" is wrongly posed; for m_τ it is exactly right. The difference: the lepton masses are bound so tightly by a *ratio* structure that the third mass follows from two – that is a genuine prediction, not an absolute value and not a calibration.

33.3 The Four Precision Classes of the Series

[B] Across the whole series four classes appear. Distinguishing them is the actual work of the tolerance standard.

Class 1 – five figures and more. The circulant (A110): $Q = 2/3$ to six figures, $\theta(\mu/e)$ to seven. Here a deviation of 10^{-6} is a serious signal, because both sides are fixed that precisely.

Class 2 – per cent level. The ladder (A100): μ/e to $0.52\,\%$, τ/μ to $1.04\,\%$. Here one per cent is *not* a signal against the structure, because the structure claims nothing more precise at this level – and no success that one could report to five figures.

Class 3 – bookkeeping precision. The SI value of α (A130, $-0.24\,\%$) and of G (A145, E_{char} to $1.5\,\%$). These deviations concern the scale conversion, not the structure. They are not a test (A210); the percentage measures the bridge, not the theory.

Class 4 – undetermined. The neutrino sector (A150), the quark masses (A150, scheme-dependent). Here the theory quantity is so little fixed that any agreement is inconsequential. What cannot be refuted confirms nothing.

33.4 The Most Common Errors

[B] The tolerance standard exists to avoid five errors that all occurred at least once in the corpus:

Confusing classes. Taking a class-2 deviation (ladder, per cent) against a class-1 precision (circulant, five figures) – that is the level error from A080/A110. Both describe the same masses, but at different levels.

Reading bookkeeping as a test. Counting a class-3 deviation as a confirmation or refutation. The error A130 names in the α presentation.

Counting undetermined as confirmed. Booking a class-4 quantity as a success because the theory "lies in the range" – even though the range is so wide that anything fits.

Inventing figures. Giving a class-2 quantity five significant figures because the calculation provides them. The floating-point precision of a calculation is not the determination tolerance of the quantity.

Counting dependent as independent. Presenting a quantity as independent evidence or as a second figure, even though it *follows arithmetically* from one already counted. Example from the ladder (A100): in the ratio $N_1 : N_2 : N_3 \approx 1 : 1.981 : 2.971$, only $N_2 = 1.981$ is measured – the third number is forced as $1 + 1.981$ and carries no independent evidence. The standard requires booking such forced figures for what they are – consequence, not finding.

33.5 The Standard in One Sentence

[STIPULATION] An agreement counts only as far as the *worse* determined of the two sides reaches. A deviation weighs only as heavily as the *better* determined of the two sides supports it.

That is the whole standard. Everything else is its application.

33.6 Verification Script

- Classification of the comparison numbers of the series into the four classes, with presentation of measurement precision and determination tolerance per case; and the countercheck that a class-2 number reported to five figures generates spurious precision: `python/A_Serie_Skripte/a220_toleranzmassstab.py`

33.7 Open Questions

First, the determination tolerance of some theory quantities is itself not sharp. For the ladder exponents (A100) it is unclear to how many figures they are determined at all. Where that is the case, the standard is to be applied conservatively.

Second, the standard does not replace error propagation. It orders orders of magnitude; it does not propagate uncertainties through formulas. A full error propagation the series does not provide.

33.8 Supersedes

This document makes the tolerance rule from A010 and A080 into its own principle and applies it to all sectors. It subsumes the corresponding parts of Doc. 122 (ratio and absolute) and Doc. 292 (tolerance analysis leptons). Incorporated: R55 (evidential breadth of the ladder coupling: one observation instead of three; “factor 300” not as a figure).

33.9 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 34

A230 · Open Points

34.1 Purpose: Residuals in One Place

[STIPULATION] Every document of the series closes with "Open Questions". This document collects these residuals in one place, so that the open frontier of the theory is visible as a whole – not scattered, not glossed over.

A residual is not a weakness to be hidden. It is the edge at which work continues. A theory without named open points is either finished or dishonest; this one is neither.

Note: This edition reflects the state after the A-series extension 2026-07 (Blocks 0–3 complete, A095 and A192 new, several points raised or closed). Points previously listed as open but since closed are no longer carried here.

34.2 Closed Points (Since Last Edition)

[K] The following points were open in earlier editions of A230 and have been closed in this session:

- **$g_R = 0$ from torus projection.** Was [S]. Now [B]: the Fourier projector P_+ annihilates the $|1, -1\rangle$ state algebraically; $g_R = \langle 0|H|0\rangle = 0$ follows from the kernel of P_+ (A095).
- **CHSH prefactor $\xi/(2\pi)$.** Was unexplained. Now [B]: 2π is the circumference of the unit circle; the phase correction per measurement is $\xi/(2\pi)$ (A160).
- **Cumulative CHSH formula CHSH$^{\mathbf{TO}}(n)$.** Now [B]: $\xi \ln n/D_f$ integrates over D_f fractal dimensions; 2π is absent because the relevant integration length is $\ln n$, not the circumference (A160).
- **Gauge sector $U(1)$ and $SU(3)$.** Was fully open. Now [B]: $U(1)_{\text{EM}}$ from flux quantisation $\Phi = n \cdot h/e$ on the torus; $SU(3)_C$

from linking number of three threads ($\dim(su(3)) = 8$ geometrically proved). $SU(2)_L$ from chiral projection (A095). Open edge: covariant derivatives D_μ not written out (A192).

- **Higgs-EFT 2.3% deviation.** Now explained [S]: SI values are scheme-dependent (1-loop) and fractal corrections in v not separable. $\xi_{\text{EFT}}/K_{\text{frak}}^{3/2} \approx 1.330 \times 10^{-4}$, residual only 0.3 % – but post-hoc correction, not proof (A190).
- **Gravitational quantisation.** Was a demand. Now declared: not required; $G = \xi^2/(4m_{\text{char}})$ is intrinsic (A140, A142).
- **ξ-proof demand.** Was open. Now declared: no proof possible or necessary; ξ is a fundamental parameter with five plausibility reasons (A015).

34.3 The One Large Open Point

[STIPULATION] Above everything stands a residual that generates several others: the *measurement process* is not modelled (A090).

Which compact direction is unrolled and which becomes space is deferred to the measurement process and not resolved there. From this follow the openness of the unrolled direction (A060), the interpretational openness of Bell (A160), and the question of what distinguishes the observer as an embedded object (A090). Whoever closes this one point closes several.

34.4 Residuals of the Foundation (A020–A090)

[STIPULATION] Absolute scale (A020, A085, A100, A130). The reduction to one dimension is derived; what size this dimension has is not. The same gap appears as the order of magnitude of ξ (A020), as the calibration of E_0 (A130), as the bridge constant v (A100) – one gap in several guises, not several gaps.

[STIPULATION] Factor 100 in K_{frak} (A040). Any integer factor closes the Pythagorean comma equally well; 100 is consistency, not necessity.

[STIPULATION] Non-closure quantity δ (A050). The recursion cannot fix it; free.

[S] $\mathbb{C}^3$ fibre and generations (A070). The assignment is interpretation, not derivation.

[STIPULATION] Holographic completeness type II (A090). Whether type-II loss is exactly reconstructible from boundary data is open.

[STIPULATION] Why P_+ and not P_- (A095). $g_R = 0$ is now [B]; why nature selects P_+ (orientation argument) remains [S].

[STIPULATION] Domain of $\tilde{T} \cdot m = 1$ for composite systems (A030, A060). The relation is declared for a single object. How it propagates across subsystem boundaries — which m and which $\tilde{T}$ pertain to an interacting subsystem, and whether the relation is preserved under composition — is not formalized. Named by J. T. Guevara Calderón (block review of the A-series v1.0, July 2026).

34.5 Residuals of the Sectors (A100–A192)

[STIPULATION] $\theta = 2/9$ (A120). Geometric source $C_3 \subset A_5$ known as a second witness; derivation from the stipulations is missing. Empirically sharp to seven places, structurally contingent.

[STIPULATION] Ladder exponents p_i (A100). Geometric sequence $p_{n+1} = \frac{2}{3} p_n$ established [K] for leptons and Up-type quarks (up to Charm); for Top ($p_t = -1/3$) and Down-type ($\Delta p = -1/2$) the geometric justification is missing.

[STIPULATION] $r = \sqrt{2}$ as stipulation (A110). Algebraic consequence: $Q = 2/3$. No derivation from ξ alone.

[STIPULATION] $\kappa = 4$ in $E_0 = e^2$ (A130). Whether the four follows from dim(T^4) or is an entry point remains open.

[STIPULATION] Characteristic mass in G (A145). $G = \xi^2/(4m_{\text{char}})$: m_{char} not uniquely fixed. Gravitational dynamics and inverse-square law completely missing (A140).

[STIPULATION] Quark sector (A150, A075). No counterpart to the circulant level; Hopf algebra and Feynman rules for interactions missing.

[STIPULATION] Neutrino sector (A150). Not testable.

[S] CHSH ξ-residual measured (A160, A165). IBM Heron r2 (May 2026) confirms Bell violation ($S = 2.74$); device-to-device variation $\sim 1.4\,\%$ exceeds ξ effect by factor 1400. Not resolvable in principle with NISQ hardware.

[S] Path assignment of entanglement (A160). Which closed paths correspond to which entanglement states: sketch.

[S] Landauer connection (A180). $\Delta w = 1 \leftrightarrow k_B T \ln 2$: connection justified [B], but that $\Delta w = 1$ is the *only* invariant unit has not been shown.

[STIPULATION] Gauge sector (A190, A192). $U(1)$ and $SU(3)$ from torus topology [B]; $SU(2)_L$ from chiral projection [B]. Open edge: covariant derivatives $D_\mu = \partial_\mu + igA_\mu^a T^a$ not written out; coupling constants g_s, g_w, e not from ξ; Higgs mechanism $SU(2) \times U(1) \to U(1)$ not from torus topology; CKM/PMNS missing.

[STIPULATION] Block-3 additions. Exponent $41/4$ in A265 calibrated to H_0, not derived. $\alpha = 1$ stipulation (A267) worked out for EM only. Mass-free Dirac formulation (A263) for free fields only.

34.6 Findings for the Correction Register

[STIPULATION] For booking in Doc. 190 (author's responsibility):

Symbol collision κ: Mass exponent $\kappa = 7$ (Doc. 009, correct) vs. energy-scale exponent (Doc. 101, erroneously 7 instead of 4). In Doc. 101, $\kappa = 7$ with $E_0 \approx 33.12$ gives $1/\alpha = 6.84$; correct is $\kappa = 4$, $E_0 = e^2$.

α **labelling:** "Derivation" (Doc. 011, 043) to be refined to "calibration / SI bookkeeping"; substance correct (A130).

34.7 What the Residual Picture Shows

[B] Since the last edition the picture has shifted: the gauge sector is no longer fully open but at [B] level for $U(1)$ and $SU(3)$; chirality has been raised from [S] to [B]; the CHSH prefactor is geometrically justified. The two large construction sites remain – measurement process and complete gauge sector – but the gauge-sector site has shrunk.

The pattern is stable: a load-bearing core (geometry, units, lepton ratios, three gauge groups topologically), two large construction sites (measurement process, covariant derivatives and coupling constants), and a series of smaller declared stipulations.

34.8 Open Questions

First, the collection is as complete as the sources. A230 bundles the residuals of the other documents; an open point not named there is also missing here.

Second, this document does not weight. Exception: the measurement process is explicitly the largest; all others stand without ranking (A200).

Third, “open” is not “unsolvable”. The list says where the theory is incomplete, not whether the gaps are closable.

34.9 Supersedes

This document collects the “Open Questions” sections of all A-documents (state: A-series extension 2026-07). It generates no new statements, but bundles existing ones and records points closed since the last edition.

34.10 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 35

A240 · Delimitation

35.1 Purpose: The External Boundary

[STIPULATION] The A-series describes *one* theory. It stands alongside other frameworks – the Standard Model, General Relativity, and various independent works. This document draws the boundary: what the series claims, what it does not, and how it relates to established frameworks.

The purpose is demarcation, not comparison. The series does not appear as a competitor that refutes something; it appears as a reordering that reads the same things differently.

35.2 On the Standard Model

[B] The series *contradicts* the Standard Model at no tested point. The CHSH value remains $2\sqrt{2}$ (A160), the masses agree within their tolerances (A100, A110), the constants are the same in a different notation (A130).

[STIPULATION] What it adds is an ordering: three generations from Z_3, the mass hierarchy from ξ, the Koide relation as an operator condition (A190). What it adds beyond the fermion sector: $U(1)_{\mathrm{EM}}$ from flux quantisation and $SU(3)_C$ from linking number are derived from the torus topology [B] (A192). What is missing: covariant derivatives, coupling constants, CKM/PMNS – the operational gauge sector is not written out (A190, A192). The series is therefore not a replacement of the Standard Model but an attempt to justify its fermion and gauge sectors – the latter topologically established but not fully executed.

The formula connections on which everything rests are those of the Standard Model and SI metrology; the series computes within them, it does not replace them (A130).

35.3 On General Relativity

[STIPULATION] The series gives G an expression from ξ (A145) and argues that the quantum gravity problem does not arise in its form (A140). But it delivers *no* field equations and does not reproduce the classical gravitational dynamics.

It therefore stands not in competition with General Relativity but beneath it: it says something about the *constant* G, nothing about the *dynamics* that Relativity describes. Where Relativity is a theory of gravitation, the series is at this point a programme.

35.4 On Independent Frameworks

[STIPULATION] In the vicinity of the theory there exist further independent works – among others on icosahedral selection structures, on projector-based memory models, and on golden-ratio-based approaches. The A-series does not treat these as parts of itself. Detailed comparisons stand where they belong: in independent comparison documents outside this series, not in the canonical FFGFT text.

[B] Where such a work shares a structure – for example the φ-icosahedral object in which $2/9$ appears forward (A120) – that is a *point of contact*, not evidence. A shared object does not confirm a number; it only shows that two frameworks stand at the same point. The series records such points of contact for what they are, not as derivations.

[STIPULATION] This restraint is methodological. Supporting a thesis through agreement with a second, likewise unproved framework shifts the burden of proof rather than bearing it. The A-series rests on its own stipulations and their consequences, not on external frameworks.

35.5 What the Demarcation Is Not

[B] The external boundary is not a valuation. That the series does not treat the gauge sector does not make the Standard Model superior and the series inferior – they address different questions. That the series gives no field equations does not make it wrong, but incomplete at a named point (A140).

Demarcation means: saying what one speaks about and what one does not. This is the prerequisite for the series' claims to be testable at all – one must know against what to measure them.

35.6 Verification Script

This document contains no computational statements – it draws conceptual boundaries to external frameworks. A numerical verification script is not applicable.

35.7 Open Questions

First, the relation to the gauge sector is unresolved. As long as the projection mapping is missing (A190), it is not decided whether the gauge sector *could* follow from the geometry or whether it stands beside it.

Second, the points of contact with independent frameworks have not been evaluated. That several frameworks stand at $2/9$ or the φ-icosahedron is noted, but not examined as to whether a common deeper structure lies behind it.

35.8 Supersedes

This document demarcates the series from external frameworks. It subsumes the corresponding parts of Doc. 019 (Standard Model), Doc. 127 (gravitation), Doc. 293 (icosahedral point of contact). The detailed framework comparisons (Doc. 246, 251, 252, 269, 271, 272) remain independent and are *not* incorporated into the series.

35.9 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Chapter 36

A250 · Reference Table

36.1 Purpose and Limits of This Document

[STIPULATION] This document is a *reference table*, not a register. It assigns to each A-document the legacy documents it subsumes. It generates no correction entries – the A-series maintains no register of its own (A010).

The distinction matters. A register books corrections with date and status; that is solely the role of Doc. 190, and remains the author's responsibility. This table only refers – it says where in the legacy corpus the material resides that an A-document reorganises. References to legacy documents and existing register entries are permitted; new entries are not.

36.2 The Mapping of A-Series to Legacy Corpus

[B] The following table is compiled from the "Supersedes" sections of the A-documents. "Subsumes" means: the material is reorganised, checked, and where necessary corrected – the legacy documents remain unchanged.

A-document	subsumes (Dok. numbers)
A010 Purpose/Structure	086, 190, 201, 205, 210
A020 Stipulations	003, 008, 009, 041, 042, 253
A030 Compact Geometry	159, 171, 181, 188, 189, 204, 302
A040 Fractal Correction	132, 133, 210, 300
A050 Recursion	203, 275, 295
A060 Time	025, 069, 078, 157, 296, 306, 307
A070 Hilbert Space	230, 231, 232, 282
A080 Units	013, 014, 015, 122, 134, 261
A085 Natural Units	012, 013, 014, 015
A090 Projection Chain	263, 270, 298

A-document	subsumes (Dok. numbers)
A095 Torus Chirality	(new 2026-07; renamed from A268)
A100 Lepton Ladder	006, 016, 046, 292
A110 Circulant/Koide	116, 258, 259, 282, 292
A120 $\theta = 2/9$	286, 290, 291, 293
A130 Fine Structure	011, 043, 044, 101
A140 Gravitation	012, 127, 163
A145 Grav. Constant	012, 127
A150 Quarks/Neutrinos	005, 007, 047, 289
A160 Quantum Mechanics	020, 021, 022, 023, 035, 073, 142
A180 Information	243, 255, 257, 290, 301, 302
A190 Standard Model	019, 097, 141, 298
A192 Gauge Sector	(new 2026-07)
A200 Order w/o Ranking	(methodological note)
A210 Falsifiability	101, 142, 292
A220 Tolerance Standard	122, 292
A230 Open Points	(compilation; state 2026-07)
A240 Demarcation	019, 127, 293
A250 Reference Table	(this document)

36.3 What the Series Reorganises – and What Remains Unchanged

[STIPULATION] The A-series covers the geometric core, the unit and constant structure, and the fermion sector. Large parts of the legacy corpus are *not* transferred to A-documents and remain solely in the legacy corpus: the cosmological sector (excluded by A010), the operational gauge sector (coupling constants, CKM/PMNS), the dynamic field-theory documents, and the numerous comparison and diagnostic works.

That a legacy document does not appear here therefore does not mean it is outdated – it means the A-series has not (yet) reorganised it.

36.4 Findings That Belong in Doc. 190

[STIPULATION] The series has produced two findings to be booked in the correction register (Doc. 190). They are repeated here as *findings*, not as entries – the booking is the author's responsibility:

First, the symbol collision κ: mass exponent $\kappa = 7$ (Doc. 009, correct) vs. energy-scale exponent (Doc. 101, there erroneously 7; correct is 4).

Second, the α labelling: "derivation" (Doc. 011, 043) to be refined to "calibration / bookkeeping"; the substance is correct.

36.5 Open Questions

First, the coverage is incomplete. The A-series orders part of the corpus; the rest remains in the legacy. A complete transfer is not the goal of this edition.

Second, the mapping is not bijective. Some legacy documents are subsumed by several A-documents (e.g. 292, 101), some A-documents subsume several. The table is a reference structure, not a renaming.

36.6 Supersedes

This document is the reference table of the series and supersedes nothing. It refers to the entire legacy corpus without modifying it.

36.7 Use of AI Tools

This document was created with the assistance of AI tools. The questions, stipulations, and all substantive decisions come from the author; the tools formulate, compute, and create verification scripts. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.

Part IV

Extensions

Chapter 37

A260 · Casimir Effect in FFGFT

37.1 Purpose

[STIPULATION] This document treats the Casimir effect within the FFGFT framework. The Casimir effect is in the legacy corpus the only experimentally accessible bridge between the fundamental length scale $L_0 = \xi \cdot \ell_P$ and measurable vacuum forces. The document shows: (1) The modified Casimir formula reproduces the standard formula exactly – no new physics at currently accessible separations. (2) The prefactor $4/3$ in $\xi = 4/3 \times 10^{-4}$ is confirmed by the Casimir effect independently of mass fits (A015). (3) There is a characteristic vacuum length scale $L_\xi \approx 100\ \mu$m at which Casimir energy density and a specific vacuum quantity coincide – this is here explicitly excluded.

37.2 Fundamental Length Scales

[B] FFGFT defines a hierarchy of characteristic length scales:

$$L_0 = \xi \cdot \ell_P \approx 2.155 \times 10^{-39}\ \text{m},$$
$$\ell_P = \sqrt{\hbar G/c^3} \approx 1.616 \times 10^{-35}\ \text{m},$$
$$L_\xi \approx 100\ \mu\text{m}.$$

Sub-Planck (spacetime granulation), Planck length, and characteristic vacuum length scale (measurable). The coupling constant $\xi = 4/3 \times 10^{-4}$ is the same parameter as in A020. It can be derived from a Lagrangian describing the dynamics of the time field (A142).

[B] Spacetime granulation. At L_0 all vacuum fluctuations are fully active. For $d > L_0$ only parts of these forces are visible – this is the physical origin of the $1/d^4$ dependence of the Casimir force. Due to the extremely small size of $L_0 \approx 2 \times 10^{-39}$ m, direct measurement is not possible.

37.3 The Modified Casimir Formula and Its Consistency

[K] The modified Casimir energy density in the FFGFT framework:

$$|\rho_{\text{Casimir}}(d)| = \frac{\pi^2}{240\,\xi}\,\rho_{\text{vac}}\cdot\left(\frac{L_\xi}{d}\right)^4,$$

where $\rho_{\text{vac}} = \xi\hbar c/L_\xi^4$ is a characteristic vacuum density. Substituting:

$$|\rho_{\text{Casimir}}(d)| = \frac{\pi^2}{240\,\xi}\cdot\frac{\xi\hbar c}{L_\xi^4}\cdot\frac{L_\xi^4}{d^4} = \frac{\pi^2\hbar c}{240\,d^4}.$$

This is exactly the standard Casimir formula. No contradiction, no new parameter – the FFGFT formulation is numerically equivalent to the established QED results.

[K] Numerical verification (Casimir energy density vs. plate separation):

Separation d	ρ_{Casimir} (J/m³)	Ratio to ρ_{vac}
$100\ \mu$m	4.17×10^{-14}	1.00
$10\ \mu$m	4.17×10^{-10}	10^4
$1\ \mu$m	4.17×10^{-2}	10^{12}

[K] Characteristic scale hierarchy:

$$\frac{L_0}{\ell_P} = \xi = 1.333\times10^{-4}, \qquad \frac{\ell_P}{L_\xi} \approx 1.6\times10^{-31}.$$

37.4 Confirmation of the Factor $4/3$

[B] The prefactor $4/3$ in $\xi = 4/3\times10^{-4}$ appears independently in the Casimir context: the factor $4/3$ appears in the Casimir energy term as $E_{\text{Casimir}} \propto \frac{4}{3}$ (the fourth – musical interval $4:3$) and is therefore not only a mass-fit quantity. This independence is recorded in A015 as Casimir confirmation: no other prefactor (fifth $3/2$, third $5/4$) gives consistent results for the mass spectrum *and* the Casimir structure simultaneously.

37.5 Experimental Tests

[K] Three classes of tests:

Precision measurements at $d \approx L_\xi$: At plate separation $\approx 100\,\mu$m the Casimir energy density reaches the order of ρ_{vac}. This regime is experimentally accessible.

Scaling behaviour: The $1/d^4$ dependence should be exactly satisfied down into the nanometre range. Deviations at $d \lesssim 10$ nm could indicate spacetime granulation.

Experimental L_ξ values from the literature: Parallel plates: ~ 228 nm; sphere-plate geometry: $\sim 1.75\,\mu$m to $\sim 18\,\mu$m. The scatter reflects geometric differences ($F \propto 1/d^4$ vs. $F \propto 1/d^3$).

37.6 Verification Script

- Numerical verification of the modified Casimir formula (reproduction of the standard formula), length scale hierarchy, comparison with L_ξ values: `python/A_Serie_Skripte/a260_casimir.py`

37.7 Open Questions

Direct measurement of L_0 is not possible. The minimal length scale $L_0 \approx 2 \times 10^{-39}$ m lies far beyond any experimental reach. The characteristic scale L_ξ is experimentally accessible, but its precise connection to L_0 presupposes the FFGFT framework.

Deviations from the $1/d^4$ law at $d \sim 10$ nm are not measured. The prediction of a scale change near the sub-Planck boundary is qualitative, not quantitatively worked out.

37.8 Supersedes

This document subsumes: Doc. 091 (modified Casimir theory, length scale hierarchy, numerical verification, experimental predictions and tests). *Not* taken over: the Casimir vacuum scaling from Doc. 091 (excluded sector, outside A-series core) and the macroscopic vacuum implications.

37.9 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.

Chapter 38

A261 · Scale Hierarchy

38.1 Purpose

[STIPULATION] This document develops the complete scale hierarchy of FFGFT: from the single geometric constant ξ to the characteristic energy scales, to the lepton masses, to the fine-structure constant α, and to E_0 as the logarithmic mean. The core: all these quantities follow from ξ – but the connection is not always achievable by direct cancellation. Ratios (mass ratios, E_0) are correction-free; absolute values need K_{frak}.

38.2 The Starting Point: A Single Constant

[B] FFGFT begins with a single dimensionless constant:

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333 \ldots \times 10^{-4}.$$

From it all fundamental scales follow in natural units ($\hbar = c = \ell_P = 1$).

[B] Characteristic T0 scales:

$$L_0 = \xi \cdot \ell_P \approx 1.333 \times 10^{-4} \, \ell_P, \qquad E_0^{\text{scale}} = L_0^{-1} = \xi^{-1} \cdot E_P.$$

38.3 Lepton Masses from Geometry

[K] The mass formulas (A100) in compact form:

$$m_e = c_e \cdot \xi^{5/2}, \quad m_\mu = c_\mu \cdot \xi^2, \quad m_\tau = c_\tau \cdot \xi^{3/2},$$

where the coefficients have geometric origin:

$$c_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}, \qquad c_\mu = \frac{9}{4\pi\alpha}, \qquad c_\tau = \frac{27\sqrt{3}}{8\pi\alpha^{3/2}}.$$

[K] Important: apparently simple fractions. In the short form the coefficients appear as rational numbers ($r_e = 4/3$, $r_\mu = 16/5$, $r_\tau = 25/9$ from A100). These fractions are *not* arbitrary and must not be directly cancelled: they encode spatial geometry (π, $\sqrt{3}$) and electromagnetic coupling (α). Both representations (Yukawa short form and extended form) are the same theory in two notations (A130).

38.4 E_0 as Logarithmic Mean

[B] The characteristic energy E_0 is the geometric mean of electron and muon mass:

$$\boxed{E_0 = \sqrt{m_e \cdot m_\mu}}.$$

Logarithmically: $\log E_0 = \frac{1}{2}[\log m_e + \log m_\mu]$ – E_0 lies exactly in the middle between m_e and m_μ on a logarithmic scale. Numerically: $E_0 = \sqrt{0.511 \times 105.658}$ MeV ≈ 7.35 MeV (geometric mean). The value from the ξ-path (A130) gives $E_0 \approx 7.398$ MeV; this difference is where K_{frak} enters.

[B] From the mass formulas:

$$E_0 = \sqrt{c_e c_\mu} \cdot \xi^{9/4}.$$

E_0 is *correction-free* in the sense: because K_{frak} appears as a common factor in both masses, it does not cancel in the geometric mean – $E_0^{\text{exp}} = K_{\text{frak}} \cdot E_0^{\text{bare}}$, so E_0 itself is an observable that already contains the fractal correction.

38.5 The Fine-Structure Constant from ξ

[K] The fundamental relation:

$$\alpha = \xi \cdot E_0^2 = c_e c_\mu \cdot \xi^{11/2}.$$

Substituting the geometric coefficients:

$$\alpha^{5/2} = \frac{27\sqrt{3}}{8\pi^2} \cdot \xi^{11/2}, \qquad \text{hence} \qquad \alpha = \left(\frac{27\sqrt{3}}{8\pi^2}\right)^{2/5} \cdot \xi^{11/5}.$$

With fractal correction (A040):

$$\alpha = \left(\frac{27\sqrt{3}}{8\pi^2}\right)^{2/5} \cdot \xi^{11/5} \cdot K_{\text{frak}}, \qquad K_{\text{frak}} = 1 - 100\xi \approx 0.9867.$$

Numerically: $\alpha^{-1} \approx 137.036$ – agreement to $0.0005\,\%$ (A130).

38.6 Why α Does Not Follow Directly from $\xi^{11/2}$ Without Unit Convention

[S] $\alpha \propto \xi^{11/2}$ means: if ξ changed, α would change with exponent $11/2 = 5.5$. That is a strong coupling. The coefficients c_e, c_μ are of order 10^{19} in natural units – that is not a free parameter but the conversion factor between the Planck scale and MeV. In natural units (Planck energies) $c_e \sim 9.67$ and $c_\mu \sim 98.1$ – order 1 to 100, geometrically understandable.

38.7 Ratios Are Correction-Free

[B] The mass ratio:

$$\frac{m_\mu}{m_e} = \frac{K_{\text{frak}} \cdot m_\mu^{\text{bare}}}{K_{\text{frak}} \cdot m_e^{\text{bare}}} = \frac{m_\mu^{\text{bare}}}{m_e^{\text{bare}}}.$$

K_{frak} cancels. The same holds for the Koide ratios (A110) and the g-2 ratios (A138). This is no coincidence: multiplicative corrections always cancel in ratios.

[K] Absolute values, by contrast, need K_{frak}: without correction $\alpha^{-1} \approx 138.9$ instead of $137.04 - 1.4\,\%$ deviation. With K_{frak}: exact. Hence K_{frak} is not a post-correction but part of the structure (A040).

38.8 The Complete Derivation Chain

[B] All fundamental quantities from ξ in one chain:

$$\xi \xrightarrow{\text{geometry}} r_0, E_0^{\text{scale}} \xrightarrow{\text{quantum numbers}} m_e, m_\mu, m_\tau$$
$$\xrightarrow{\text{geom. mean}} E_0 \xrightarrow{\xi \cdot E_0^2} \alpha \xrightarrow{\text{K-corr.}} \alpha_{\text{exp}}.$$

Additionally: $\xi \to L_0 \to G$ (gravitational constant, A142) and $\xi \to L_0$ (A015).

38.9 Verification Script

- Numerically: $E_0 = \sqrt{m_e m_\mu}$, $\alpha = \xi E_0^2$, ratios correction-free vs. absolute values with K_{frak}: `python/A_Serie_Skripte/a261_skalenhierarchie.py`

38.10 Open Questions

The coefficients c_e, c_μ, c_τ are not fully independently derived. They contain α explicitly – and α in turn follows from the masses. The circularity is named as a methodological problem in A130; the T0 resolution (common geometric origin) is a claim, not a proof.

The quark coefficients are missing. The c_i structure for quarks is not worked out at the same level as for leptons (A150).

38.11 Supersedes

This document subsumes: Doc. 070 (mathematical structure of T0 theory: circular ratios, scale hierarchy from ξ, lepton masses, α derivation, E_0 as logarithmic mean, geometric constant C, why ratios are correction-free). Doc. 093 (β-parameter derivation, self-consistency condition).

38.12 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.

Chapter 39

A262 · c-Convention: $E = m$

39.1 Purpose

[STIPULATION] This document addresses two related points: (1) $E = mc^2$ is in natural units identical to $E = m$ – the only difference lies in the convention of setting c as a constant. (2) The three-clock experiment (following Sci. Rep. 2024) shows that all physical quantities – lengths, masses – can be traced back to a single time standard, which metrologically supports the T0 duality $\tilde{T} \cdot m = 1$.

39.2 E = mc² = E = m

[B] In natural units ($c = 1$):

$$E = mc^2 = m \times 1^2 = m.$$

This is not an approximation – it is the same equation. The difference between Einstein and T0 lies not in the physics but in the convention.

[B] What is c really? $c = L/T$ is a ratio of length to time. Einstein's setting $c = 299\,792\,458$ m/s freezes this ratio and makes the equation $E = mc^2$ necessary. T0 theory allows c to be a dynamic ratio – then $E = m$ is the more fundamental form. Both formulations are physically equivalent; what differs is the ontological status of c.

[B] The constant illusion. Setting $c =$ const. is a legitimate normalisation choice, not a constant of nature. With it:

$$E = mc^2 \text{ (in SI units)} \quad \equiv \quad E = m \text{ (in natural units)}.$$

The apparent complexity of c^2 is a consequence of the unit choice, not of the physics. The same holds for $\hbar$ and k_B (A085).

[S] The time dilation paradox. In the standard view, time is relative, mass is constant. In T0, time is absolute, mass is variable. Both views describe the same measurements; which to prefer remains an ontological choice (A060).

39.3 The Three-Clock Experiment (Sci. Rep. 2024)

[B] The experiment (Sci. Rep. 2024, DOI: 10.1038/s41598-024-71907-0) shows: a single time standard suffices as a starting point to define and measure all physical quantities – time intervals, lengths, masses.

[B] Length measurement from time. Lengths are determined exclusively from time differences: $L = c \cdot \Delta t$. This sets $c = 1$ (convention) and gives lengths in units of time – exactly what the T0 duality requires.

[B] Mass measurement from frequencies. Via the Compton wavelength $\lambda_C = \hbar/(mc)$ and metrological procedures (Kibble balance), masses can be traced to the time standard: $m \propto 1/T$ with T as the Compton period. This is the metrological realisation of $\tilde{T} \cdot m = 1$.

[K] Connection to T0 duality. The reduction to a single time standard is not new – T0 already achieves this in the units documents (A080/A085). The three-clock experiment is an external, independent confirmation of the metrological consistency of this reduction. It shows that the programme is technically feasible, but presupposes the same convention choice ($c = 1$).

39.4 Reference-Point Revolution

[S] T0 theory makes the implicit reference-point shift explicit:

Epoch	Reference	Consequence
Antiquity	Earth as centre	Epicycles
Newton	Sun as reference	Kepler's laws
Einstein	$c =$ const, space as reference	Time dilation
T0	Time as reference	Mass variation

Every shift is a convention choice – none is "more correct" than another. What counts: which describes the physics more simply.

39.5 Verification Script

- Numerical equivalence $E = mc^2 = E = m$ for various c values; length and mass measurement from a single time standard: `python/A_Serie_Skripte/a262_c_konvention.py`

39.6 Open Questions

The ontological question is not decided. Which view is preferable – time relative and mass constant, or time absolute and mass variable – is a question of interpretation, not physics. The A-series makes no judgement (A060).

The three-clock experiment does not cover the full T0 claim. It confirms the metrological reduction; it does not confirm the geometric derivation of ξ or the mass formulas.

39.7 Supersedes

This document subsumes: Doc. 077 ($E = mc^2 = E = m$, c-convention, time dilation paradox, reference-point revolution), Doc. 029 (three-clock experiment Sci. Rep. 2024, length and mass measurement from time standard, connection to T0 duality).

39.8 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.

Chapter 40

A263 · Dirac Equation in FFGFT

40.1 Purpose

[STIPULATION] This document develops the Dirac equation in FFGFT completely: the Clifford algebra as fundamental structure, the transition from the standard Dirac to the T0 formulation, mass elimination (physics formulable without mass parameters), spin as topological property, and the modified Dirac equation from the time-field Lagrangian (A142). The document supplements A075 (field theory, Schrödinger limit) with the complete relativistic structure.

40.2 The Clifford Algebra as Fundamental Basis

[B] The standard Dirac equation $(i\gamma^\mu\partial_\mu - m)\psi = 0$ uses 4×4 γ-matrices as a representation. In FFGFT the fundamental structure is the Clifford algebra:

$$\mathbf{e}_\mu\mathbf{e}_\nu + \mathbf{e}_\nu\mathbf{e}_\mu = 2g_{\mu\nu}.$$

The γ-matrices are one representation – they are not eliminated but traced back to their geometric origin. That is the difference: in the SM γ-matrices are the foundation; in FFGFT it is the Clifford algebra, with γ-matrices being only one representation.

[B] What changes in T0? The standard Dirac equation has m as a fixed parameter. In T0 the mass $m(x,t)$ is a dynamic field:

$$\left[i\gamma^\mu(\partial_\mu + \Gamma_\mu^{(T)}) - m(x,t)\right]\psi = 0,$$

with time-field connection $\Gamma_\mu^{(T)} = \partial_\mu m/m^2$ (A142). In the limit $\partial_\mu m \to 0$ (constant mass) this reduces to the standard Dirac equation.

40.3 Spin as Topological Property

[B] In FFGFT spin is not an intrinsic property but a topological one: spin-$\frac{1}{2}$ corresponds to the torus winding $(n_\theta, n_\phi) = (1, 1)$. The 720° symmetry (two rotations return a spinor to its initial state) follows directly from the topology of the torus – it need not be postulated.

[B] A relativistic spinor state in the torus representation:

$$\Phi(x,t) = \rho(x,t)\, e^{i\theta(x,t)} \cdot \chi(\sigma),$$

with spin component $\chi(\sigma)$ from the geometric knot modes (A075). The quantum number $j = l \pm \frac{1}{2}$ for fermions follows from the winding structure.

40.4 Mass Elimination: Physics Without Mass Parameters

[K] In the T0 framework all observables can be formulated as ratios that require no absolute mass parameter. The basic idea: instead of carrying m in SI units as a parameter, everything is expressed through ξ and time intervals.

[K] Systematic mass elimination. From the T0 duality $\tilde{T}{\cdot}m = 1$ it follows that $m = 1/\tilde{T}$. Substituting into the Dirac equation:

$$\left[i\gamma^\mu \partial_\mu - 1/\tilde{T}(x,t)\right]\psi = 0.$$

The mass parameter disappears from the equation; what remains is the time field $\tilde{T}(x,t)$ as a dynamic quantity. This is the fully mass-free T0 formulation.

[S] Philosophical consequence. If all physics is formulable in ratios (without absolute masses), then masses are not fundamental objects but auxiliary constructs of measurement. What is fundamental: time intervals and the geometry of the torus. This is the same thesis as A085 (natural units) in relativistic language.

40.5 Mass-Proportional Coupling

[K] In the T0 Lagrangian the time field couples mass-proportionally: $g_T^\ell = m_\ell \cdot \xi$. This is the mechanism behind the g-2 calculation (A138): the coupling is not a free Yukawa constant but fully determined by the mass and ξ. The one-loop formula for the anomalous magnetic moment follows directly from this mass-proportional coupling.

40.6 Ratios vs. Absolute Values in the Dirac Structure

[B] The same distinction as in A261 and A138: ratios of Dirac corrections (e.g. a_μ/a_e) are correction-free and follow from the structure of the Clifford algebra and the quantum numbers. Absolute values of the g-2 terms require the SI bridge constants and K_{frak} (A138).

40.7 Verification Script

- Numerical verification: standard Dirac equation as limiting case of the T0 form ($\partial_\mu m \to 0$); spin-$\frac{1}{2}$ from winding $(1, 1)$; mass-free formulation for simple cases: `python/A_Serie_Skripte/a263_dirac.py`

40.8 Open Questions

The complete mass-free formulation for all interactions is missing. Mass elimination is shown for free fields; for interaction terms it has not been systematically carried out.

The quark sector is missing. The Dirac equation for quarks with dynamic mass and colour charge is not developed at the same level as for leptons (A150).

The relation to QED renormalisation is not worked out. How the T0 Dirac equation relates to perturbative QED (Feynman rules, loop integrals) is not systematically developed (A075, open Hopf algebra).

40.9 Supersedes

This document subsumes: Doc. 051 (T0 Dirac equation, Clifford algebra, spin as topology, mass-proportional coupling, ratios vs. absolute values), Doc. 052 (systematic mass elimination, complete mass-free T0 formulation), Doc. 053 (mass elimination in the Dirac Lagrangian, ratio-based physics), Doc. 050 (simplified Dirac equation).

40.10 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.

Chapter 41

A264 · Structural Asymmetries

41.1 Purpose

[STIPULATION] This document treats three structural "ugliness" of the Standard Model and their T0 explanation: (1) chirality of the weak interaction, (2) gravitation as a separate tensor term, (3) absence of magnetic monopoles. In all three cases FFGFT provides a geometric explanation from the energy field $E_{\text{field}}(x,t)$ and the parameter ξ. Methodological restriction: parts of the derivation are still in development; what counts as proof and what as sketch is explicitly marked.

41.2 Chirality: Left-Handedness of the Weak Interaction

[B] In the Standard Model the exclusive left-handedness of the weak interaction is an input. In FFGFT it has a geometric basis that follows from half-integrality on the torus.

Half-integrality and inside/outside asymmetry. Spin-$\frac{1}{2}$ follows from the winding number $w = n_\phi/n_\theta = \frac{1}{2}$ (winding (1,1) on T^4, 720° symmetry). The decisive consequence: the torus has an *inside and outside* with different effective couplings. The electric field on the outside points topologically fixed inward (electron) or outward (positron) – the direction is prescribed by the winding geometry, not freely choosable.

[B] After the projection $T^4 \to \mathbb{R}^3 \times \mathbb{R}_t$ (A090), inner and outer coupling appear as different chiral states. The projection is not symmetric: the compact topology selects one coupling direction. Left-handed coupling is the outer perspective, right-handed the inner perspective – and the weak interaction couples exclusively to the projected outer state.

[S] What has not yet been carried out. The qualitative mechanism is clear: half-integrality ⇒ inside/outside asymmetry ⇒ chiral projection. What is missing: an explicit computation showing that $g_R = 0$ (not $g_R = -g_L$) follows from this projection. That is the open quantitative edge – not the basic idea.

41.3 Gravitation as Field Gradient

[S] In the SM gravitation is a separate tensor field ($G_{\mu\nu}$). In FFGFT gravitation is the gradient of the same energy field that also describes particle masses. The nonlinear field equation:

$$\Box E_{\text{field}} + \xi E_{\text{field}}^3 = 0.$$

[K] Equivalence in the weak field. For $E_{\text{field}} = E_0(1 + h)$ with small perturbation h, the field equation linearises to a form that is equivalent to the linearised Einstein equation. The full covariant tensor formulation $g_{\mu\nu}(E_{\text{field}})$ with Riemann and Ricci tensors is worked out in the legacy corpus. This is a **[K]** result: equivalence in the weak field is shown; completeness of the equivalence in the strong field is not at proof level.

41.4 Magnetic Monopoles

[S] Magnetic monopoles are absent in the SM; their absence is unexplained. In FFGFT they are topological excitations of the energy field and satisfy the Dirac quantisation condition $q_e q_m = 2\pi n\hbar$. Their rarity is a consequence of the high energy threshold $\sim E_P/\xi \approx 7.5 \times 10^3\, E_P$ – far beyond any currently achievable energy.

[S] Status. The topological argument is plausible; a direct proof that the T0 energy-field monopole exactly satisfies (and not merely reproduces) the Dirac quantisation condition is not fully worked out.

41.5 Objection-Refutation Table

[K] Standard objections and their status in FFGFT:

Objection	Status	Comment
Dimensional inconsistency	Refuted	Normalisation with E_P, L_0 shown
No GR equivalence	Partial	Weak field rigorous; strong field sketch
Missing nonlinearity	Refuted	E^3 term present, Lagrangian shown
No covariance	Refuted	Tensor $g_{\mu\nu}$ constructed
Monopole problem	Sketch	Topological interpretation, threshold plausible
No renormalisability	Refuted	$[E^4] = 4$, renormalisable

41.6 Position in the A-Series

[B] The chirality question is worked out in A095: the inside/outside topology of the torus and the chiral projection connect half-integrality (A263) with the left-handedness of the weak coupling geometrically.

[B] The gravitation-as-field-gradient thesis is related to A142 (time-field Lagrange): there gravitation is derived from the time field; here from the energy field. Both approaches need to be checked for consistency – this has not yet been done.

41.7 Verification Script

- Chirality phase difference for left/right-handers; weak-field linearisation; monopole threshold from E_P/ξ: `python/A_Serie_Skripte/a264_asymmetrie.py`

41.8 Open Questions

Open quantitative edge: $g_R = 0$ from the projection. The mechanism half-integrality ⇒ inside/outside asymmetry ⇒ chiral projection is geometrically clear. What is still missing: the explicit computation showing that the weak coupling of the projected inner state is exactly zero – and not merely smaller than the outer coupling. That is a quantitative task, not a conceptual one.

Gravitation in the strong field is not at proof level. The weak-field equivalence is shown; the full nonlinear equivalence is missing.

The connection to the projection chain (A090) is not worked out. Whether the weak interaction follows from the decompactification is open.

41.9 Supersedes

This document subsumes: Doc. 143 (chirality from energy-field rotation, gravitation as field gradient, magnetic monopoles as topological excitations, field equation $\Box E + \xi E^3 = 0$, covariant tensor), Doc. 144 (objection-refutation table, renormalisability, SM as effective low-energy theory). *Not* taken over: excluded applications from Doc. 143/144 (Λ_{cosmo} value, tension parameter).

41.10 Use of AI Tools

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Chapter 42

A265 · Redshift: Static and Achromatic

42.1 Purpose

[STIPULATION] This document develops three related theses:

(1) Redshift as a static geometry effect. In FFGFT z arises not from space expansion but from energy loss of the photon in the ξ-field along the light path: $1 + z = e^{\xi x}$.

(2) Achromaticity. The mechanism is purely geometric – it stretches all wavelengths equally. Earlier presentations of chromatic T0 redshift in the corpus are recorded as a correction in the correction programme.

(3) The SI anchor of the sector. Both FFGFT and the Standard Model need an empirical anchor for the transition from natural units to measurable SI quantities in cosmology. In FFGFT this anchor is the ratio $m_e c^2/k_B T_{\text{CMB}}$ – a measured number, not pure geometry. The SM has the analogue: H_0 in km/s/Mpc presupposes c, k_B, and the megaparsec definition, which are themselves historical SI conventions.

42.2 The Static Redshift

[B] A photon loses energy linearly with distance travelled in the static ξ-field:

$$\frac{dE}{dx} = -\xi E \;\Rightarrow\; E(x) = E_0\, e^{-\xi x}.$$

The redshift:

$$\boxed{1 + z = e^{\xi x}, \qquad z \approx \xi x \;\text{ (small } z\text{)}}$$

No scale factor $a(t)$, no Big Bang, no expanding space. Light travel time to an object with redshift z:

$$\tau(z) = \frac{\ln(1+z)}{H_0^{\text{T0}}}.$$

The function grows without bound – there is no Big Bang ceiling.

[K] Comparison with the Hubble law. For small z: $z \approx \xi x$. Hubble law: $z = H_0 d/c$. Identification: $H_0 = \xi c$. The T0 formulation uses the Hubble energy $E_H = E_0 \cdot \xi^{41/4}$, from which $H_0^{\mathrm{T0}} = E_H/\hbar \approx 66.2$ km/s/Mpc ($-1.9\,\%$ vs. Planck). The exponent $41/4$ is not derived from ξ alone – that is the named open point.

42.3 Achromaticity: Correction of the Earlier Presentation

[B] The mature FFGFT mechanism is *fractal path elongation*: the path through the recursive time geometry is stretched by a factor that depends solely on the distance – not on the wavelength of the photon. Hence $1 + z = e^{\xi x}$ holds for *all* wavelengths with the same factor.

[K] Physical reason. A *geometric* stretch factor acts on wavelength and pulse duration in the same ratio – just as gravitational redshift and standard cosmological redshift are achromatic. A wavelength-dependent term would arise only from a dispersive medium effect, not from pure geometry.

[K] Consequence. The observed cosmological redshift is achromatic to high precision – FFGFT gets this right. The earlier presentation of chromatic T0 redshift in earlier versions of the legacy corpus is superseded; the correction is recorded as a correction entry.

42.4 The SI Anchor: Where the Circularity Sits

[B] This is the methodologically most important section. FFGFT derives H_0 from ξ and E_0. The derivation ends with:

$$H_0^{\mathrm{T0}} = \frac{E_H}{\hbar}, \qquad E_H = E_0 \cdot \xi^{41/4}.$$

Apparently a pure ξ-derivation. But $E_0 = \sqrt{m_e m_\mu} = 7.398$ MeV *[Note 3 August 2026: E_0 is the calibration of the unit system – by A085 there is exactly one – and the value carried is the one that* *provides the anchoring in SI:* 7.398 *MeV matches the measured* α *to* 5×10^{-6} *via* $\alpha = \xi E_0^2$*. The geometric mean* $\sqrt{m_e m_\mu} = 7.348$ *MeV (A130, route 1) names the origin of the order of magnitude but is unsuitable as an anchor: inserted, it would give* $1/138.9$*, i.e.* $-1.35\,\%$ *in* α*, and that deviation would propagate into every derived absolute value and accumulate in comparisons. The formula notation here is abbreviated; the anchor value governs (A130, route 2:* $E_0^2 = 4\sqrt{2}\,m_\mu/\xi^p$*, without* m_e*).]* is a *measured* quantity (A130), and the transition to km/s/Mpc needs $\hbar$ and the megaparsec definition – both SI conventions.

[K] The remaining anchor. After cancelling the $\hbar c$ factors (which appear in both length scales L_ξ and $\bar{\lambda}_e$) what remains is:

$$\boxed{\frac{m_e c^2}{k_B T_{\mathrm{CMB}}} = 2.176 \times 10^9}$$

Here c^2 (mass → energy) and k_B (kelvin → energy) sit – that is a ratio of two *measured* quantities: electron rest energy to CMB temperature. The bridge formula is thus geometry form ($K_{\mathrm{geo}} \cdot \xi^{41/4}$) **times** SI anchor ($m_e c^2/k_B T_{\mathrm{CMB}}$) – not a pure geometry identity.

[B] Where the exponent changes. The same ratio sits in the exponent itself: referenced to the Planck scale instead of E_0, the exponent becomes $63/4 = 41/4 + 11/2$, where $11/2$ encodes the SI scale conversion $\ell_P \leftrightarrow E_0$. The factor never disappears – it only changes location.

Reference scale	Exponent	SI factor explicit as
E_0 (electron/CMB)	$41/4$	Ratio $m_e c^2/k_B T_{\mathrm{CMB}}$
ℓ_P (Planck)	$63/4 = 41/4 + 11/2$	Additional exponent $11/2$

42.5 The SM Has the Same Problem

[B] The Standard Model gives H_0 in km/s/Mpc. This unit presupposes: c (speed of light in km/s), the megaparsec definition (1 Mpc $= 3.0857 \times 10^{22}$ m, historically from parallaxes), k_B for temperature measurements, and the SI 2019 reform that fixes c, $\hbar$,

k_B, and e by definition. These quantities are not independent of one another – the measurement chain is self-referential.

[B] Concretely: $T_{\text{CMB}} = 2.7255$ K (Fixsen 2009) is linked to energy via k_B; $H_0 \approx 67.4$ km/s/Mpc is linked to a length via c and Mpc; the ratio $m_e c^2/k_B T_{\text{CMB}}$ connects particle physics with cosmology – and this ratio has no place in the SM basic theory. The SM calls it a measurement; FFGFT attempts to understand it from ξ, and comes closer, but not all the way (the exponent $41/4$ is missing).

[S] The honest comparison. Both frameworks – ΛCDM and FFGFT – need an empirical anchor for cosmology that lies outside the theoretical core. In the SM that is H_0 itself (a measurement, not a derivation). In FFGFT it is $m_e c^2/k_B T_{\text{CMB}}$ plus the not-yet-derived exponent $41/4$. FFGFT is one step further here – but not yet completely parameter-free.

42.6 Cosmological Degeneracy

[K] FFGFT and ΛCDM are *observationally degenerate* on the standard tests: both reproduce the same fluxes, angular sizes, and time dilations. The key points:

Supernova time dilation: DES measures $b = 1.003 \pm 0.011$ (White et al. 2024, Dark Energy Survey, MNRAS). The fractal path elongation stretches wavelength *and* pulse duration in the same ratio $\Rightarrow b = 1$. The test only excludes non-dilating tired light ($b = 0$) – FFGFT predicts $b = 1$ and is not affected.

Hubble tension: In FFGFT H_0 is not an expansion rate but the geometric energy-loss coefficient. Different measurement methods probe different scales; systematic variations are geometrically expected. The tension 67.4 (Planck 2018) vs. 73.0 km/s/Mpc (SH0ES) is not a fundamental problem in FFGFT.

[S] Honest situation. The degeneracy cuts both ways: the data *fit* FFGFT (no wrong prediction), but they do not *force* it. The case for FFGFT rests on parsimony (no Λ, no dark energy) and on the ξ-derivation of the constants – not on a single discriminator.

42.7 Verification Script

- Numerical verification: $1 + z = e^{\xi x}$, achromaticity (same stretch factor for $\lambda_1 \neq \lambda_2$), SI anchor $m_e c^2/k_B T_{\text{CMB}}$, exponent table $41/4$ vs. $63/4$: `python/A_Serie_Skripte/a265_rotverschiebung.py`

42.8 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
z	Cosmological redshift; $1 + z = \lambda_{\text{obs}}/\lambda_{\text{emit}}$
$E_H = E_0 \cdot \xi^{41/4}$	Hubble energy in FFGFT
$H_0^{\text{T0}} \approx 66.2$ km/s/Mpc	FFGFT value of the Hubble constant
$\tau(z) = \ln(1 + z)/H_0^{\text{T0}}$	Light travel time in FFGFT
$m_e c^2/k_B T_{\text{CMB}}$	Empirical SI anchor of the cosmological sector

42.9 Open Questions

The exponent $41/4$ is not derived from ξ alone. It is calibrated to the measured H_0. As long as it is missing, FFGFT in the cosmological sector is not parameter-free.

The anchor $m_e c^2/k_B T_{\text{CMB}}$ needs a geometric explanation. Why precisely the ratio of electron rest energy to CMB temperature appears as the natural transition is an open structural question.

BAO and CMB peaks have not yet been derived from FFGFT. The degeneracy claim presupposes that the same angular scales are reproducible; the proof in FFGFT's own formalism is outstanding.

42.10 Supersedes

This document subsumes: Doc. 026 (static redshift as geometric path elongation, FEM simulation, Hubble energy $E_H = E_0\xi^{41/4}$, derivation $H_0 \approx 66.2$ km/s/Mpc), Doc. 280 (static redshift, achromaticity as correction K6, z →time as ΛCDM transfer, cosmological degeneracy, no Big Bang), Doc. 279 (Casimir–H_0 bridge formula, SI anchor $m_e c^2/k_B T_{\text{CMB}}$, exponent $41/4$ vs. $63/4$, location of SI factor).

42.11 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.

Chapter 43

A266 · Unit Check and SI Back-Calculation

43.1 Purpose

[STIPULATION] In FFGFT all formulas are set up in natural units ($c = \hbar = k_B = 1$). When back-computing to SI quantities, the physical constants reappear with specific powers. These powers are *not* freely choosable – they follow necessarily from dimensional analysis. An error in the power of c is an error of factors $\sim 3 \times 10^8$ to $(3 \times 10^8)^4$ – not noticeable when looking only at orders of magnitude, fatal when comparing with a measurement.

This document gives the systematic procedure: first dimensional analysis in natural units, then automated back-computation, then verification. It is not a teaching document on unit systems (that is A080/A085) – it is a practical toolbox for concrete calculations.

43.2 Basic Principle: All Dimensions as Energy Powers

[B] In natural units $c = \hbar = k_B = 1$. Every physical quantity thereby has a unique dimension $[E]^n$ with rational n:

Quantity	Nat. dim.	SI origin	n
Energy	$[E]^1$	—	$+1$
Mass	$[E]^1$	$E = mc^2 \Rightarrow m = E/c^2$	$+1$
Momentum	$[E]^1$	$p = E/c$	$+1$
Length	$[E]^{-1}$	$\ell = \hbar c/E$	-1
Time	$[E]^{-1}$	$t = \hbar/E$	-1
Velocity	$[E]^0$	$v = \ell/t = c = 1$	0
Force	$[E]^2$	$F = E/\ell = E^2/(\hbar c)$	$+2$
Pressure/energy density	$[E]^4$	$\rho = E/\ell^3 = E^4/(\hbar c)^3$	$+4$
Charge	$[E]^0$	from $\alpha = e^2/(4\pi\varepsilon_0\hbar c)$ dimensionless	0
Temperature	$[E]^1$	$T = E/k_B$	$+1$
Entropy	$[E]^0$	$S = k_B \cdot$ (dimensionless)	0
Gravitational constant G	$[E]^{-2}$	$G = \ell_P^2 c^3/\hbar$	-2

[B] Rule. From n the back-computation factor follows directly: a quantity with $[E]^n$ in natural units has the SI value

$$X_{\text{SI}} = X_{\text{nat}} \cdot \left(\frac{\hbar c}{1\,\text{SI-length}}\right)^n.$$

In practice: assemble the SI factor from $\hbar$, c, k_B so that the SI dimension comes out.

43.3 The c-Power in Back-Computation: Systematic Derivation

[B] The question "which power of c?" is answered from the SI dimension of the target quantity. The method in three steps:

Step 1: Write down the SI target dimension. E.g. energy density: $[\rho] = \text{J/m}^3 = \text{kg}\,\text{m}^{-1}\text{s}^{-2}$.

Step 2: Express SI dimension through $\hbar^a c^b k_B^d$. $\hbar$ has SI dim [J s], c has [m/s], k_B has [J/K]. Find $a, b, d \in \mathbb{Z}$ such that $[\hbar^a c^b k_B^d] =$ target dim$/[E_{\text{nat}}^n]$.

Step 3: Verification by substitution. Insert the found (a, b, d) and check against the measured SI quantity.

[K] Table of most common back-computation factors:

Quantity	Nat. unit	SI factor	Derivation
Mass	$[E]$ in MeV	$\div c^2$	$m_e = 0.511\,\text{MeV}/c^2$
Length	$[E]^{-1}$	$\times \hbar c$	$\ell_P = 1/E_P \cdot \hbar c$
Time	$[E]^{-1}$	$\times \hbar$	$t_P = 1/E_P \cdot \hbar$
Velocity	dimensionless	$\times c$	$v = 0.5 \cdot c$
Force	$[E]^2$	$\div(\hbar c)$	$F[\text{N}] = F_{\text{nat}} \cdot E^2/(\hbar c)$
Energy density	$[E]^4$	$\div(\hbar c)^3$	$\rho[\text{J/m}^3] = \rho_{\text{nat}} \cdot E^4/(\hbar c)^3$
Temperature	$[E]$	$\div k_B$	$T[\text{K}] = E[\text{J}]/k_B$
Frequency f	$[E]^1$	$\div \hbar$	$f[\text{s}^{-1}] = E[\text{J}]/\hbar$
G	$[E]^{-2}$	$\times \hbar c^3$	$G[\text{m}^3/\text{kg s}^2] = G_{\text{nat}} \cdot \hbar c^3$

43.4 Concrete Examples from FFGFT

43.4.1 Mass: One Power of c^2

[K] From $E = mc^2$ it follows that $m = E/c^2$. A value in natural units (MeV) becomes a SI mass by dividing by c^2:

$$m_e = 0.511\,\frac{\text{MeV}}{c^2} = \frac{0.511 \times 10^6 \times 1.602 \times 10^{-19}\,\text{J}}{(2.998 \times 10^8\,\text{m/s})^2} = 9.109 \times 10^{-31}\,\text{kg}.$$

Error c^1 instead of c^2: factor 3×10^8 off. Error c^0 (no c): factor $(3 \times 10^8)^2 \approx 10^{17}$ off.

43.4.2 Planck Length: $\hbar c$ Together, Not c Alone

[K] ℓ_P in natural units is an inverse energy. Back-computation to metres needs $\hbar c$:

$$\ell_P = \frac{\hbar c}{E_P} = \frac{(1.055 \times 10^{-34}\,\text{J s})(2.998 \times 10^8\,\text{m/s})}{1.956 \times 10^9\,\text{J}} = 1.616 \times 10^{-35}\,\text{m}.$$

Only $\hbar/E_P$ (without c) gives the Planck *time*, not the Planck *length* – a typical error.

43.4.3 Energy to Inverse Time: Only $\hbar$, No c

[K] An energy E (dimension $[E]^1$) is back-computed to inverse time $[\mathrm{s}^{-1}]$ by dividing by $\hbar$:

$$f = \frac{E}{\hbar}, \qquad [f] = \frac{[\mathrm{J}]}{[\mathrm{J\,s}]} = [\mathrm{s}^{-1}].\checkmark$$

Example: $E = 1.41 \times 10^{-33}$ eV $= 2.26 \times 10^{-52}$ J gives $f = E/\hbar = 2.14 \times 10^{-18}\ \mathrm{s}^{-1}$ (A265). Dividing by $\hbar c$ instead gives an inverse length (wrong dimension). The c-power is zero – it does not appear.

43.4.4 Energy Density: Four Powers of c in the Denominator

[K] Casimir energy density ρ has dimension $[E]^4$ in natural units. Back-computation:

$$\rho[\mathrm{J/m^3}] = \rho_{\mathrm{nat}}\,[E^4] \cdot \frac{1}{(\hbar c)^3}.$$

Three powers of $\hbar c$ in the denominator – because length contributes $[E]^{-1}$ three times. c appears as c^3 (from the three $\hbar c$).

43.4.5 Gravitational Constant: c^3 in the Numerator

[K] In natural units G has dimension $[E]^{-2}$. The SI constant $G = 6.674 \times 10^{-11}\ \mathrm{m^3kg^{-1}s^{-2}}$ requires:

$$G_{\mathrm{SI}} = G_{\mathrm{nat}} \cdot \hbar c^3, \qquad [G_{\mathrm{SI}}] = [E]^{-2} \cdot [\hbar][c]^3 = \frac{\mathrm{J\,s} \cdot \mathrm{m^3/s^3}}{\mathrm{J^2}} = \frac{\mathrm{m^3}}{\mathrm{kg\,s^2}}.\checkmark$$

Error: c^2 instead of c^3 gives wrong dimension immediately recognisable.

43.5 The General Procedure: Dimension Matrix

[B] For every formula $X = f(\xi, E_0, \ldots)$ in natural units:

1. Determine n from the table above (or by counting lengths/masses/times in the SI dimension of X).

2. Write the SI factor:

$$X_{\text{SI}} = X_{\text{nat}} \cdot \hbar^a \cdot c^b \cdot k_B^d$$

with a, b, d from the linear system:

$$\begin{aligned} [\text{kg}]: &\quad -a + b = n_{\text{kg}} \\ [\text{m}]: &\quad a + b = n_{\text{m}} \\ [\text{s}]: &\quad -a - b = n_{\text{s}} \end{aligned}$$

(where n_{kg}, n_{m}, n_{s} are the exponents of the SI dimension of X; k_B is added when kelvin appears.)

3. Verification: insert $\hbar = 1.055 \times 10^{-34}$ J s, $c = 2.998 \times 10^{8}$ m/s, $k_B = 1.381 \times 10^{-23}$ J/K and check the SI unit.

[K] When k_B appears and when not. k_B appears exactly when the target quantity contains kelvin (temperature, entropy/kelvin) or when converting from an energy to a kelvin quantity. In all other cases (masses, lengths, times, forces) k_B does not appear.

43.6 Unit Check of a Formula

[B] Every formula set up in FFGFT should pass the unit check: both sides of the equation must have the same energy power $[E]^n$ in natural units, and on back-computation the same SI unit must emerge.

[K] Example: $T \cdot E = 1$ (A060). In natural units: $[T] = [E]^{-1}$ (time) and $[E] = [E]^{+1}$. Product: $[E]^0$ – dimensionless. The product $T \cdot E = 1$ is a correct equation between a time and an inverse time – the unit check is passed.

[K] Example: $\alpha = \xi \cdot E_0^2$ – unit check of the α formula. The formula is *correct*. The unit check shows why – and simultaneously makes the implicit reference scale visible.

α is dimensionless, ξ is dimensionless. The equation therefore holds only if E_0^2 is also dimensionless. That is the case when E_0 appears as a pure numerical ratio – i.e. measured in MeV and divided by 1 MeV:

$$\alpha = \xi \cdot \left(\frac{E_0}{1\,\text{MeV}}\right)^2.$$

The $1\,\mathrm{MeV}$ in the denominator is the *implicit reference scale* – it makes $E_0/1\,\mathrm{MeV}$ dimensionless and the equation correct. It does not stand explicitly in the short notation, but is indispensable.

Numerical verification:

$$\alpha = \frac{4}{30000} \cdot \left(\frac{7.398\,\mathrm{MeV}}{1\,\mathrm{MeV}}\right)^2 = 1.333 \times 10^{-4} \times 54.73 = 7.297 \times 10^{-3} = \frac{1}{137.0}.$$

Agreement with experiment to $0.0005\,\%$ – the formula is correct.

What the unit check reveals: the choice of MeV as reference scale is not free – it is the empirical SI anchor of the FFGFT sector. A different reference scale (e.g. GeV) would give ξ a different numerical value, but α remains the same physical number. This is the statement from A261: $E_0 = 7.398\,\mathrm{MeV}$ carries the anchor, and therefore one may not simply insert different energy units in this formula without simultaneously changing ξ.

[S] Converting to natural units. Translating a SI formula to natural units: replace all $\hbar$, c, k_B by 1 and drop the units. The resulting formula is then valid only in natural units; on back-computation all factors return.

43.7 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$[X]$	Dimension of quantity X
$[E]^n$	Energy dimension to power n
$n_{\mathrm{kg}}, n_{\mathrm{m}}, n_{\mathrm{s}}$	Exponents of the SI unit of X in $\mathrm{kg}^{n_{\mathrm{kg}}}\mathrm{m}^{n_{\mathrm{m}}}\mathrm{s}^{n_{\mathrm{s}}}$
a, b, d	Exponents of $\hbar^a c^b k_B^d$ in the back-computation factor
$E_P = \sqrt{\hbar c^5/G} \approx 1.956 \times 10^9\,\mathrm{J}$	Planck energy

43.8 Verification Script

- Numerical verification of all back-computation examples; dimension matrix solver for arbitrary SI target dimensions; verification for m_e, ℓ_P, inverse time, ρ_{Casimir}, G: `python/A_Serie_Skripte/a266_einheitenpruefung.py`

43.9 Open Questions

The dimension matrix presupposes that the formula is set up in pure Planck units. Mixed systems (e.g. MeV for energy, fm for length) require additional conversion steps.

Non-integer energy powers. Some FFGFT expressions have rational n (e.g. charge $[E]^{1/2}$). The procedure applies in principle, but the linear system then gives rational solutions.

43.10 Supersedes

This document subsumes: Doc. 013 (dimensional analysis, gravitational constant from ξ, Planck length, conversion factors), Doc. 014 (natural units, dimension table, scaling factor S_{T0}, derivation vs. back-computation), Doc. 015 (unit conversions, energy dimension table, force/pressure/charge). *Not* taken over: pedagogical derivation of the individual constants from ξ (A015, A145) and the unit convention as such (A080, A085).

43.11 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.

Chapter 44

A267 · Stipulation $\alpha = 1$

44.1 Purpose

[STIPULATION] In FFGFT α follows from the geometry (A130). This document treats the complementary question: how does one set $\alpha = 1$ *as a computational tool* in a way that is fully reversible and makes the coupling structure visible? The answer is the Heaviside–Lorentz stipulation $e^2 = 4\pi\alpha$ combined with $\alpha = 1$ – not a unit choice but a *stipulation* with preserved trace.

44.2 Stipulation: Heaviside–Lorentz with $\alpha = 1$

[STIPULATION] In natural units ($\hbar = c = 1$) and the Heaviside–Lorentz system, the *defining equation* is

$$e^2 = 4\pi\alpha.$$

The fine-structure constant is set to $\alpha = 1$. This is not a *norm* (unit choice) – a dimensionless number is invariant under rescaling of base units – but a *stipulation*: the coupling is adopted as a computational unit.

[B] The charge carries the 4π; what is set to 1 is α, not e. From $e^2 = 4\pi\alpha$ with $\alpha = 1$ follows the value $e = \sqrt{4\pi} \approx 3.545$ – not $e = 1$. The relation $e^2 = 4\pi\alpha$ remains as a defining equation and cannot be dissolved by the stipulation.

44.3 Back-Computation as Substitution

[B] Because $e^2 = 4\pi\alpha$ remains as a relation, the coupling power is automatically carried along. Back-computation is pure substitution

$$\alpha \longrightarrow \frac{1}{137.036},$$

applied to the α-order of the process:

- per vertex a factor $e = \sqrt{4\pi\alpha}$; compared to the α=1 value $\sqrt{4\pi}$, that is a $\sqrt{\alpha}$ back,
- amplitude with V vertices $\propto (4\pi\alpha)^{V/2} \Rightarrow \times\alpha^{V/2}$,
- rate/cross section $\propto (4\pi\alpha)^V \Rightarrow \times\alpha^V$,
- compact: process of rate order $\alpha^n \Rightarrow \times\alpha^n$.

[B] The 4π is not double-counted: it is already in e^2 and in the phase-space normalisation. Only α^n is reset, not $(4\pi\alpha)^n$.

44.4 Conversion Examples

[K] All quantities in $\hbar = c = 1$, Heaviside–Lorentz. Back-factor = the α-order of the expression.

Quantity	Physical form	With α=1	Back-factor
Coulomb potential	$V = \alpha/r$	$1/r$	$\times\alpha$
Class. electron radius	$r_e = \alpha/m_e$	$1/m_e$	$\times\alpha$
Compton wavelength	$\lambda_C = 1/m_e$	$1/m_e$	$\times1$
Bohr radius	$a_0 = 1/(\alpha m_e)$	$1/m_e$	$\times\alpha^{-1}$
Rydberg / binding	$E = \frac{1}{2}\alpha^2 m_e$	$\frac{1}{2}m_e$	$\times\alpha^2$
Thomson cross section	$\sigma_T = \frac{8\pi}{3}\alpha^2/m_e^2$	$\frac{8\pi}{3}/m_e^2$	$\times\alpha^2$

44.5 The 137-Ladder: What $\alpha = 1$ Makes Visible

[B] The three atomic length scales stand in the ratio

$$r_e : \lambda_C : a_0 = \alpha : 1 : \alpha^{-1}.$$

At $\alpha = 1$ they all collapse to $1/m_e$: they are not three independent scales but *one* scale with three α-powers. a_0 is exactly $137\times$ larger than λ_C, and λ_C exactly $137\times$ larger than r_e.

[B] With $\alpha = 1/137$ written out everywhere, this common origin remains hidden – three different awkward numbers appear and the α-power as ordering principle is not visible. The $\alpha = 1$ stipulation reveals it.

44.6 Why This Stipulation – Comparison with Alternatives

[S] Versus Gaussian ($e^2 = \alpha$). Also works, but places the 4π in the field equations rather than in Coulomb's law. Heaviside–Lorentz places the 4π where they geometrically belong (sphere surface, Coulomb) and leaves the Maxwell equations 4π-free. α is then the pure coupling, not a mixture of coupling and geometry.

[S] Versus $c = \hbar = 1$ unit choice. The setting $c = \hbar = 1$ destroys the powers – the trace is gone, back-computation runs only via dimensional analysis (A266). The defining equation $e^2 = 4\pi\alpha$, by contrast, *retains the trace*: the coupling power is carried along because the relation stands. Stipulation with preserved trace, not norm with erased trace – hence constructively reversible, not merely justifiably reconstructable.

44.7 FFGFT Connection: Back-Computing α Means Inserting ξ

[B] α is in the FFGFT corpus not a free input but follows from the geometry (A130):

$$\alpha^{-1} = 137.036 \quad \text{from} \quad \xi = \frac{4}{30000}, \quad D_f = 3 - \xi, \quad E_0 = \sqrt{m_e m_\mu}.$$

The stipulation $\alpha = 1$ isolates precisely the point at which the ξ-geometry enters. Every reset α^n is a location at which $D_f = 3 - \xi$ becomes visible.

[B] Within the corpus, "back-computing α" therefore does not mean "inserting $1/137$" but "inserting the ξ-structure" – and the α=1 system makes countable at how many places and in which power that occurs.

44.8 Symbol Glossary

Symbols used throughout the entire A-series are explained in A010. Specific to this document:

Symbol	Meaning
$e^2 = 4\pi\alpha$	Heaviside–Lorentz defining equation
$e = \sqrt{4\pi} \approx 3.545$	Elementary charge at $\alpha = 1$
$r_e = \alpha/m_e$	Classical electron radius
$\lambda_C = 1/m_e$	Compton wavelength
$a_0 = 1/(\alpha m_e)$	Bohr radius
V	Number of Feynman vertices in the process
α^n	α-order / back-factor of the process

44.9 Verification Script

- Numerical verification of all table rows with $\alpha = 1/137.036$ and $m_e = 0.511\,\text{MeV}$; ratio $r_e : \lambda_C : a_0 = \alpha : 1 : \alpha^{-1}$: `python/A_Serie_Skripte/a267_alpha_eins.py`

44.10 Open Questions

The stipulation $\alpha = 1$ is fully worked out only for electromagnetic processes. For weak and strong coupling (α_W, α_s) an analogous table would be possible but has not been worked out.

The connection α^n as locus of $D_f = 3 - \xi$ is plausibly shown for simple QED processes but not for multi-loop or non-abelian processes.

44.11 Supersedes

This document subsumes: Doc. 011 (individual formulas r_e, λ_C, a_0, dimensional analysis $\alpha = r_e/\lambda_C$, SI numerical values of atomic scales), Doc. 015 (Heaviside–Lorentz line $e^2/(4\pi) = 1$ in the unit systematics). The coherent conversion table with explicit α^n back-factors and the 137-ladder $r_e : \lambda_C : a_0 = \alpha : 1 : \alpha^{-1}$ as a structural statement are not present as a unit in the legacy corpus and are newly systematised here. Related: A130 (geometric derivation α from ξ), A266 (SI back-computation).

44.12 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.

Chapter 45

A270 · Z_3 Sector Structure

45.1 Purpose and Status

[STIPULATION] This document investigates the sector structure of the T^4/Z_3 orbifold with regard to a structural distinction between elementary particles (leptons) and composite particles (hadrons). Central result: the three Z_3 sectors assign to each particle class a topologically fixed K_{frak} exponent — leptons 0, mesons $+1$, baryons $+2$ — in addition to a common bulk exponent 36.

Status: hypothesis at consistency level [K]. The meson anchor is passed via the pion binding (A155). The baryon step is a prediction; its direct test requires a geometric baryon base mass (open).

45.2 Starting Point: Bulk Exponent 36

45.2.1 The Key Relation

[K] Two independently determined FFGFT quantities meet in one number. First, the D4 packing density $\pi^2/16$, whose reciprocal is

$$\frac{16}{\pi^2} \approx 1.62114.$$

Second, the scale-recursion depth from Doc. 275,

$$k^* = \frac{\ln\varphi}{\xi} \approx 3609.1 \quad \Rightarrow \quad \frac{k^*}{100} = 36.09.$$

Numerically:

$$K_{\text{frak}}^{-36} = \left(\frac{75}{74}\right)^{36} \approx 1.62130 \approx \frac{16}{\pi^2} \quad (\Delta = 0.010\,\%).$$

45.2.2 Statistical Safeguard

[K] Monte-Carlo check (100 000 trials): the hit probability $\leq 0.010\,\%$ for a random target in the value range is $1.56\,\%$ per candidate. Of 21 tested physical constants (π, e, φ, $\sqrt{2}$, $\sqrt{3}$, ln 2, ...) *only* $16/\pi^2$ hits at $\leq 0.010\,\%$; the next-best candidate lies at $0.066\,\%$. The independent support by $k^*/100$ (Doc. 275 was determined without reference to masses) considerably weakens the chance hypothesis. P35 remains noted: a forward derivation of the identity $K_{\text{frak}}^{-36} = 16/\pi^2$ does not exist.

[STIPULATION] The exponent 36 is henceforth called the *bulk exponent*: the common fractal correction of the untwisted sector in which all particles live.

45.3 T^4/Z$_3$ Orbifold Structure

45.3.1 Fixed-Point Geometry

[B] Z$_3$ acts on T$^3 = S^1_x \times S^1_y \times S^1_z$ by cyclic permutation:

$$(x, y, z) \xrightarrow{g} (y, z, x) \xrightarrow{g} (z, x, y) \xrightarrow{g} (x, y, z)$$

Fixed points: $x = y = z = k \cdot L/3$, $k = 0, 1, 2$ — exactly **3 fixed points**, isomorphic to Z$_3$.

Local geometry at each fixed point:

Direction	Vector	Z$_3$ sector
$\mathbf{e}_0 = (1, 1, 1)/\sqrt{3}$	invariant	sector 0 (untwisted)
$\mathbf{e}_1 = (1, -1, 0)/\sqrt{2}$	phase ω	sector g (twisted)
$\mathbf{e}_2 = (1, 1, -2)/\sqrt{6}$	phase ω^2	sector g^2 (twisted)

At each fixed point: 1 untwisted + 2 twisted degrees of freedom — exactly the Z$_3$ sector structure $\{e, g, g^2\}$.

45.3.2 Quark Localization and Confinement

[H] In FFGFT, $\tilde{T} \cdot m = 1$ holds; the mass m sits on S^1_m, one of the three spatial circles. Under Z$_3$ permutation, S^1_m has two images on the other circles.

Elementary particle (lepton): sits on S^1_m alone, transforms as a 1-dimensional Z_3 representation, remains in the untwisted sector 0.

Composite particle of 3 constituents (baryon): constituent 1 on S^1_x, constituent 2 on S^1_y, constituent 3 on S^1_z. The localization condition "all 3 at the same spatial point" reads:

$$x = y = z \quad \Leftrightarrow \quad \textbf{fixed-point condition}$$

Quark confinement in T^4/Z_3 is thereby *topologically enforced*: free quarks would violate the fixed-point condition. Leptons (1 circle) have no fixed-point condition and are free.

45.3.3 Sector Exponents

[H] K_{frak} is a property of the sector partition function. Each of the 3 Z_3 sectors contributes K^1_{frak}. The bulk exponent 36 covers the untwisted sector (already contained in the ladder formulas). Twisted sectors contribute an additional K^1_{frak} each. Hence the sector table:

Particle class	Const.	Active sectors	Exponent
Lepton (e, μ, τ)	1	$\{e\}$	$36 + 0$
Meson ($q\bar{q}$)	2	$\{e, g\}$	$36 + 1 = 37$
Baryon (qqq)	3	$\{e, g, g^2\}$	$36 + 2 = 38$

The exponents are topologically fixed: $Z_3 = \{e, g, g^2\}$ has exactly two non-trivial elements, independently of any mass.

45.4 Empirical Anchor: the Pion

[K] A155 introduces the additive formula $m_M = m_{q_1} + m_{q_2} + \Lambda_{\text{QCD}} \cdot K^{n_{\text{eff}}}_{\text{frak}}$ for mesons. Evaluating the pion binding yields (A155):

$$\boxed{n_{\text{eff}}(\pi^0) = 37.79} \qquad n_{\text{eff}}(\pi^\pm) = 36.62.$$

Both values lie within the window of the sector prediction $36 + 1 = 37$ for mesons. **The meson step of the sector table**

is thereby empirically anchored at the pion — the exponent was determined in A155 (from the legacy-corpus formula) independently of this sector argument.

The domain of validity of the additive formula (only binding $< \Lambda_{\text{QCD}}$; kaon, ρ, ω structurally outside) and the GMOR extension are treated in A155.

45.5 Consistency Evidence

45.5.1 Koide Relation

[K] The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}$$

holds exactly for leptons (A110: the geometric ladder predicts Q within $0.16\,\%$ of $2/3$ without using Koide; the experimental value hits $2/3$ exactly with $\Delta \approx 3 \times 10^{-5}$).

For quarks, according to A150, *no* corresponding structure exists:

> “Second, there are six quarks instead of three leptons, but no ordering corresponding to the Z_3 structure that would map them onto three eigenvalues of one operator. The circulant plane of A110 has no counterpart here.” (A150, §Quarks)

This is consistent with the sector hypothesis — and sharpens it: the Z_3 circulant (A110), which carries the Koide structure of the leptons, *is* the algebra of the untwisted sector. Leptons in the untwisted sector → circulant acts → Koide holds. Quarks couple to twisted sectors → the circulant ordering has “no counterpart” (A150) → no Koide analogue exists. The sector hypothesis thus supplies a candidate for the *reason* of the structural deficit recorded in A150.

45.5.2 K_{frak} Distribution in the Corpus

[K] K_{frak} appears in the corpus base formulas exactly where the sector table demands it:

Class	K_{frak} in base formula?	Sector table
Leptons	no ($m_L = r\xi^p v$, A100)	untwisted: no extra
Mesons	yes ($\Lambda \cdot K_{\text{frak}}^n$, A155)	$+1$ sector
Baryons	yes; corrected legacy calculation with anchor $\pi^2/2$ (Doc. 190, R62; $\Delta = 0.05\,\%$)	$+2$ sectors

45.6 Prediction: Baryon Step

[H] From the sector table, the exponent for baryons is $36 + 2 = 38$:

$$K_{\text{frak}}^{-38} = 1.6654.$$

The corresponding statement: a geometric baryon base mass (before sector correction) relates to the observed one by K_{frak}^{-38}; equivalently, the baryon exponent lies exactly one unit above the meson value measured at the pion.

Direct test open: it requires an independently derived geometric baryon base mass, which does not exist in the corpus (A150: the quark sector is descriptive classification, no predictive quantity). Until then, the baryon step rests on the topology (two twisted sectors) and the pion anchoring of the meson step.

45.7 Open Points and Falsification

1. **Forward derivation of $K_{\text{frak}}^{-36} = 16/\pi^2$** (P35): the identity is supported numerically and via $k^*/100$, but not derived.
2. **Baryon sector**: the legacy calculation is corrected by register entry (Doc. 190, R62). A fully geometric candidate for the proton mass — $m_p = (\pi^3/12)\, N_c(1 + \alpha_s)\, e^{-3\xi/4}\, \Lambda_{\text{QCD}}/2$, every factor interpreted, $\Delta = +0.21\,\%$ absorbable — is kept in A155 §Baryon Candidate ([S]; forward derivation open). The direct test of the $+2$ step awaits this derivation.
3. **$\pi^{\pm}/\pi^0$ difference** (36.62 vs. 37.79): does the EM self-energy carry a sector contribution of its own? (A155)

Falsification hierarchy:

1. Meson exponent $36 + 1$: confirmed at the pion (37.79, A155); extension to the kaon via the GMOR form in A155 as [S] candidate.

2. Koide analogue for quarks: must *not* hold — consistent with A150 ("circulant plane of A110 has no counterpart") ✓
3. Baryon exponent $36 + 2$: falsifiable as soon as a geometric baryon base mass exists.

45.8 Notation

Symbols used throughout the A-series are explained in A010. Specific to this document:

Symbol	Meaning
$Z_3 = \{e, g, g^2\}$	cyclic group; g: permutation $(x, y, z) \to (y, z, x)$
$\omega = e^{2\pi i/3}$	primitive third root of unity (twist phase)
sectors $0, g, g^2$	untwisted sector; the two twisted sectors (eigenvalues $1, \omega, \omega^2$ of the permutation)
$\mathbf{e}_0, \mathbf{e}_1, \mathbf{e}_2$	eigenbasis at the fixed point: invariant direction $(1, 1, 1)/\sqrt{3}$; twisted directions
fixed point	point with $x = y = z$ on T^3; three of them ($k \cdot L/3$)
bulk exponent 36	common K_{frak} exponent of the untwisted sector; $K_{\text{frak}}^{-36} \approx 16/\pi^2$
sector exponent	addition per particle class: lepton $+0$, meson $+1$, baryon $+2$
$k^* = \ln \varphi/\xi$	scale-recursion depth to $1/\varphi$ (Doc. 275); $k^*/100 = 36.09$
$\pi^2/16$	D4 packing density; reciprocal $16/\pi^2$
n_{eff}	empirical binding exponent from A155 (pion anchor)
Q	Koide quantity $(m_e+m_\mu+m_\tau)/(\sqrt{m_e}+\sqrt{m_\mu}+\sqrt{m_\tau})^2$

Verification script: `python / A_Serie_Skripte / a270_z3_sektoren.py` — all numerical statements of this document are checked there with assertions.

45.9 References

Document	Reference
A015	origin of ξ, Casimir confirmation
A100	lepton ladder (r_i, p_i)
A110	circulant and Koide: $Q = 2/3$, Z_3 algebra of the leptons
A150	quark ladder, lepton/quark distinction
A155	meson formula: domain, pion anchor, GMOR; baryon candidate
Doc. 190	P35, P40; register R62 (baryon correction, legacy corpus)
Doc. 275	$k^* \approx 3609$, scale recursion, $1/\varphi$ passage
Doc. 230/231/232	T^4/Z_3 Hilbert-space bridge

Chapter 46

A271 · Landauer and Phase-Space Bookkeeping

Abstract. The Landauer bound is developed here not as a statement about *information* but as a statement about *regions in phase space*. A physical system occupies a region; a map that reduces several source regions onto one target region shrinks that volume; Liouville's theorem forbids volume shrinkage in a closed system; hence a bath must absorb the volume, and the price is heat $Q \geq k_B T \ln(W_{\text{before}}/W_{\text{after}})$ — with $\ln 2$ merely the binary special case. Counting proceeds by distinguishability (orthogonality), not by energy; the absolute base discretisation is supplied by h^f. Content neutrality follows immediately: thermodynamics counts regions, it does not read them. Three levels are kept strictly apart — construction (engineer), state transition (clock), description (program) — and from this it follows that a real computer does not erase bits discretely but carries a continuous entropy current (DRAM refresh, qubit decoherence). The epistemic caveat is booked explicitly: not measurable does not mean not present — Landauer holds because of the mechanics (Liouville), not because of ignorance; the irreversibility is statistical, not ontologically absolute. *Contact with the ξ scheme of FFGFT: none* — Landauer accounting lives at the level of statistical mechanics (its own level, its own accounting).

Layer markings. [SETZUNG] declared starting point (German corpus token: [SETZUNG]) — **[B]** derived algebraically/mathematically from declared foundations — **[Q]** source — external primary source or measured value — **[S]** plausibility sketch — **[H]** open research question.

Check script. `a271_landauer.py` (checks 1–10). Every numerical statement in this document is backed there by assertions. Check 7: state count in the colloid region; check 8: analogue case with declared δx.

46.1 The region comes first

Two glasses stand on a table — region A and region B. Next to them, the world ocean. One of the glasses is poured into the sea. The volume of the glass does not vanish — it distributes itself across 1.4 billion cubic kilometres. This image carries the whole document: there is no vanishing information, no annihilated bits — only a reduction of regions that has a measurable price.

[SETZUNG] A physical system occupies a region in phase space. That region has a size — measurable in units of h^f (Section 49.6). It exists independently of how one names, partitions or describes it. That is the physical starting point. Everything else comes afterwards.

[B] Descriptions come from outside. If one decides to partition the region into two subregions and to call them "A" and "B", and the target region "Z", one has introduced a description of the region reduction. If one calls them "0", "1" and "0" instead (the traditional notation), the confusion with logical bits arises — that convention hides the fact that "0" denotes both a source region and the target region, and it suggests an information content where there is none. If one partitions more finely — into 1000 zones — one has introduced ~ 10 logical bits. If one merely assigns a number between 0 and 1, one has introduced an analogue value. Physics enforces none of these decisions. In all cases the region is the same physical region.

The terms "bit", "byte", "qubit", "analogue value" are labels for descriptive decisions — not for physical objects. They partition the region; they do not create it.

46.1.1 The Landauer bit is hardware — not information

[B] This is regularly confused. The clarification in one paragraph: a Landauer bit is a physical region with two stable macrostates — a transistor, a capacitor, a trace held high or low. A bus bit is the same: a physical region with two stable macrostates, embedded in a parallel hardware architecture. A memory cell is the same. The three are structurally identical. Thermodynamics treats all three alike:

$$Q \geq k_B T \ln\left(\frac{W_{A+B}}{W_Z}\right)$$

— regardless of whether the region is called "Landauer bit", "bus bit" or "memory cell".

The hardware is fixed before any computation runs. It does not change because of what one computes with it. A 64-bit bus has 64 physical regions — before the computation, during it, after it. What the computation does with those regions — which maps it executes, which regions are reduced onto which target regions — is a logical description of those physical processes. The description does not exist in the hardware; it is an external interpretation.

The information content of a bit — whether it means "true" or "false", whether it belongs to a correct computational step, whether it is read at all — is a decision at the level of description. Thermodynamics does not know that level. It knows only: region A and region B exist; after the region reduction the target region Z exists; $W_Z \leq W_{A+B}$; the bath absorbs the difference.

46.1.2 Three levels — strictly separated

[B] Level 1 — construction. The engineer decides at chip-design time which region operations exist — bijective or non-bijective. That is cast in silicon, unchangeable during operation, prior to any computation. This decision fixes the thermodynamic cost structure — not the computation, not the program.

[B] Level 2 — state transition during operation. Once the computer runs, state transitions occur clock cycle by clock cycle — uniformly, at the clock frequency, independently of what is being computed. Landauer costs arise only for **non-bijective** region maps — when a component, by its physical construction, reduces several input regions onto one target region Z. That is fixed by the hardware architecture, not by a computational operation. A transistor flipping from region A to region B executes a bijective map — no volume loss, no Landauer cost. What the descriptive level subsequently does with the component's state is thermodynamically irrelevant.

Landauer costs run with the clock frequency, practically constant, as long as the computer is running — because the non-bijective region maps are fixed by the designer and are incurred with every clock cycle. CPU utilisation says little about this: it measures how many clock cycles are used for computations at the descriptive level — but at idle the clock transitions keep running. The thermodynamic difference between high and low utilisation is small as long as no physical parts are actually switched off. Clock gating, power gating — these are hardware decisions at level 1, not program decisions. The hardware determines the cost — not the program.

[B] Level 3 — description. The program, the computation, an AI controller may decide which hardware is used — switch off parts, choose routes, determine that for a given result the full bit width is not needed. This has nothing to do with Landauer. It is resource management at the descriptive level. The thermodynamic costs of the hardware in use are incurred regardless — fixed by construction and switching moment, independently of what the descriptive level does with the results.

46.1.3 How much is inside a physical region?

[B] A colloidal particle in a 1 µm trap physically contains roughly $\mathbf{7.9 \times 10^9}$ **orthogonal quantum states** *(check script, check 7)*. If one coarsely names two zones "left" and "right", one describes that region with 1 logical bit. Partitioned into 1000 zones, one describes the same region with ~ 10 logical bits. The physical region is identical in both cases — the depth of description is the observer's choice.

[B] The cost of erasure attaches to the region, not to the description. The formula reads

$$\boxed{Q \;\geq\; k_B T \ln\!\left(\frac{W_{\text{before}}}{W_{\text{after}}}\right)}$$

W_{before} and W_{after} are physical region sizes in units of h^f. How one names that ratio — "erasing 1 bit" or "reduction by a factor of 2" — changes nothing about the physics.

[Q] The analogue case shows this particularly clearly — and shows its complexity at the same time. How many distinguishable states an analogue signal contains depends on several factors, all of them physical: on the bandwidth B, the temperature T and the resistance R (Johnson–Nyquist noise, $U_{\mathrm{rms}} = \sqrt{4k_BTRB}$ [B20]), on the sensitivity of the measurement apparatus, on the signal-to-noise ratio of the concrete system, and on further external sources of interference. There is no universal resolution limit for analogue signals — it is a system property, not a constant of nature.

What can be said in general: if the resolution limit is set by the thermal noise of the bath itself (the physically deepest case), then the principle already stated in Section 49.6 applies: the same bath that collects the erasure cost limits the gradation of the analogue signal. The region count $W = L/\delta x$ and the erasure cost $k_BT \ln W$ scale with the same temperature quantity — the accounting is self-consistent. How many logical bits one ascribes to the analogue region is independent of this and remains a descriptive decision *(check script, check 8: illustrative example with declared δx)*.

46.1.4 Shannon and Boltzmann are different ledgers

[B] [B18][B19] Shannon describes how much uncertainty a *description* carries: $H = -\sum_i p_i \log_2 p_i$ — a quantity about descriptions. Boltzmann describes how large a *physical region* is: $S = k_B \ln W$ — a quantity about regions. The two are identical only if one maps the description exactly onto the physical microstates and those are uniformly distributed. In every other case — coarse partitioning, unequal occupation, continuous systems — they are different things with different units.

[Q] What Landauer actually said in 1961 [B1]: he wrote explicitly "logically irreversible" — logically, not physically irreversible. He meant a map in which different logical input states lead to the same logical output state. That is a statement about descriptions.

[B] And therefore, taken by itself, it is not a physical condition. What is sufficient for heat production is solely the **non-bijectivity of the region map**: that several source regions fall onto a smaller target region. “Operation” is the wrong word at this point — an operation is a description, and descriptions cost nothing. Logical irreversibility coincides with the region map only under an additional realisation assumption that has to be declared explicitly: that the logical labels are mapped onto disjoint physical regions and that no region is carried along through the transition. If that assumption is violated, logical irreversibility carries nothing — Bennett [B3] executes any logically irreversible function as a sequence of bijective region maps, with no lower bound (Section 49.8). And conversely, every non-bijective region map costs, whether or not it is described as an operation at all. Every physically irreversible process produces heat, regardless of whether it is described as “logically irreversible”; that is exactly the point of Section 49.7.3. The physics beneath the description can be far richer than the description itself.

[B] The error of the standard presentation is therefore this: it treats the descriptive unit — the bit — as if it were a physical minimum quantity. It is not. A physical minimum quantity does exist: h^f. But it has nothing to do with bits. The region comes first; how many bits one ascribes to it is a subsequent decision.

46.2 The one idea from which everything follows

A bit is physical: two **distinguishable** states of a system. Two wells for a particle, two magnetisation directions, two voltage levels. All that matters is that the system can be in state “0” or “1” and that the two can be told apart.

Region reduction: no matter which region — A or B — the system was in before, afterwards it is in the target region Z. The target region Z is freely choosable; all that counts is the ratio W_{A+B}/W_Z.

```
A   ---+
       +--->  Z
B   ---+
```

Two regions are reduced onto one — the target region Z. This map is **not invertible**: whoever knows only the target region cannot reconstruct which source region the system was in. Exactly that — and nothing else — generates the cost (Section 49.4).

46.3 What "phase space" means here

Phase space is the space of all possible microstates (all positions and momenta of all particles). A macrostate such as "region A" is not a point in it but a **region**: many microstates (thermal jitter in the left well — every jitter configuration counts as "0").

[SETZUNG] Boltzmann entropy measures this region:

$$S = k_B \ln W$$

W = phase-space volume of the region. Entropy here is a **measure of the size of the region occupied** — nothing mystical.

```
Phase space before:        Phase space after:

+-------+-------+          +-------+-------+
|       |       |          |       |       |
|  A    |  B    |    ⟶     |  Z    | empty |
|       |       |          |       |       |
+-------+-------+          +-------+-------+
 volume: 2W0 (A+B)          volume: W0 (Z only)
```

[B] Erasure halves the accessible volume:

$$\Delta S_{\text{region}} = -k_B \ln 2$$

Pure accounting over region sizes — no model, no hardware. (Generalisation to N levels and to non-uniform distributions: Section 49.6.)

46.4 Why this is not free: Liouville

The microscopic dynamics (classically Hamiltonian, quantum-mechanically unitary) has one hard property, **Liouville's theorem** [B2]:

> **[B]** The time evolution of a closed system **preserves phase-space volume**. Deforming, displacing, stirring — yes. Shrinking — never.

A refinement: Liouville holds exactly only for the closed system *without* a bath. For the subsystem "bit" alone — once coupled to the bath — it no longer holds. Only the total system bit + bath is closed; for that total the volume is preserved, while the bit subregion may shrink at the expense of the bath region.

Intuitively: the phase-space flow is an **incompressible fluid**. A blot of ink in water can be distorted arbitrarily; its volume remains.

The trap now closes:

- Erasure **demands**: $2W_0 \to W_0$ (halving, Section 3).
- Liouville **forbids**: volume shrinkage in a closed system.

[B] Consequence: a closed system cannot erase. Not an engineering problem — a theorem. The only way out: the system must not be closed. The volume has to go somewhere — into the bath (Section 49.5).

46.5 The bath as volume recipient

If the bit is coupled to a heat bath, Liouville applies to the **total system** bit+bath. If the bit region is to shrink by half (the uniform case: both regions equally large), the bath region must at least **double**. In the general case $\Delta S_{\text{bath}} \geq -\Delta S_{\text{bit}}$ (second law); the generalisation to $\ln(W_{\text{before}}/W_{\text{after}})$ follows in Section 49.6:

$$\Delta S_{\text{bath}} \;\geq\; +k_B \ln 2$$

A bath at temperature T enlarges its region by absorbing energy; the basic relation $dS = \delta Q/T$ translates this into heat:

$$\textbf{[B]} \quad Q \;\geq\; T\Delta S_{\text{bath}} \;=\; k_B T \ln 2$$

This is the Landauer bound [B1][B3]:

> **Liouville (volume is preserved) + accounting (bit region halved) ⇒ bath region doubled ⇒ heat $\geq k_B T \ln 2$ into the bath.**

At 300 K: $k_B T \ln 2 \approx 2.87 \times 10^{-21}$ J per bit *(check script, check 1)*.

The two conditions can be read off directly:

1. **No erasure without a bath** — there is no other volume recipient (Section 49.4).
2. **Equality only quasistatically** — any finite speed generates additional dissipation $W \approx k_B T \ln 2 + B/\tau$ [B4][B5] (Section 49.10).

46.6 Why 2, actually? — The counting base

Binarity is **convention, not physics**. In general:

$$Q \geq k_B T \ln\left(\frac{W_{\text{before}}}{W_{\text{after}}}\right)$$

— a ratio of region sizes. N levels erased onto one: $\ln N$ *(check script, check 4)*. The $\ln 2$ is the computer-science special case.

[B] What counts as a level? Not energy. Energy levels are unsuitable as a counting base: they may be degenerate or have unequal spacings without the count changing. The criterion is **distinguishability**: two states count separately if they can in principle be told apart without error. For **quantum systems** this means orthogonality in Hilbert space; for **classical systems** it means distinguishability within the thermal resolution limit set by the bath — i.e. the separation of the states must exceed the thermal fluctuations. A bit may sit in two degenerate states of equal energy — ideally so, in fact: holding then costs nothing, only forgetting does. Level spacings act only indirectly, through the thermal occupation probabilities. The true currency is therefore the Gibbs form [B6]

$$S = -k_B \sum_i p_i \ln p_i$$

with $\ln N$ as the special case of uniform distribution. Unequally spaced, unequally occupied systems automatically carry less entropy *(check script, checks 5–6)*.

[B] The base discretisation exists — and it is h^f. Classically the phase-space volume carries units; W is defined only up to an arbitrary cell size. This does not disturb the accounting, because only differences enter — the cell cancels. Quantum mechanics fixes the cell absolutely: the number of orthogonal states in a region Γ of a system with f degrees of freedom is

$$N_{\mathrm{max}} = \Gamma/h^f \qquad \text{[B7]}$$

(check script, check 7: colloid in a 1 µm trap, roughly $7.9 \times 10^{9)}$ *orthogonal states — the "2" of the experiment is a coarse choice, not something fundamental*.

46.6.1 Fundamental versus emergent two-valuedness

	Spin-½	**Colloid in a double well**
origin of the 2	fundamental: Hilbert space $\mathbb{C}^2$	emergent: 2 metastable macro-regions
validity	always	only while the barrier $\gg k_B T$ on the process timescale
underneath	nothing — no finer gradation	quasi-continuum of microstates

46.6.2 The quasi-analogue case

A value in a range L with resolution δx carries ln$(L/\delta x)$ levels; region reduction costs $k_B T$ln$(L/\delta x)$ *(check script, check 8)*. δx is not free: the bath noise sets it — the same bath that collects the erasure cost limits the gradation. No analogue trick evades the bound.

46.7 Why the content is irrelevant

At **no point** did the entire calculation ask what meaning region A or B carries, whether the bit belongs to a correct computation, whether it is interpreted as "true" or "false". Only this entered: **how many** distinguishable regions before, **how many** after.

Thermodynamics counts regions; it does not read them. Region A and region B are physically equivalent — physics knows no preferred direction, and the target region Z is freely choosable. What meaning one assigns to them (“true”/“false”, 0/1, high/low) is a decision outside thermodynamics:

> **[B] Region reduction costs the same for every assignment — regardless of how one interprets the regions.** There is no thermodynamic detector of content or of truth.

In the same language: a non-terminating computation dissipates energy without bound and has no truth value while doing so. Energy turnover and semantic validity lie on different levels — phase-space accounting here, representation there. A separation of categories: physical validity (hardware level) versus representational validity (content level).

46.7.1 Hardware and computational operation are different things

A fundamental error of reasoning in the standard presentation of Landauer: the hardware is treated as an image of the computational operation — as if the physical apparatus existed, and were thermodynamically effective, only while a logical operation is being executed. That is not so.

[B] The hardware is not the computational operation. A 64-bit processor is a physical system with a 64-bit state space. That state space exists independently of whether a meaningful operation is running, of whether the result requires 1 bit or 64, of what meaning the result carries, of whether anyone reads the result at all. The hardware is not an abstraction — it is silicon, wiring, fields, capacitors, interacting continuously with their environment.

[B] The computational operation is an abstraction over the hardware. It describes what the programmer or mathematician sees — input, output, function. The hardware does not know that abstraction. It knows only state transitions, charge displacements, heat release.

The consequence for Landauer. Asking “what does erasing a bit cost” takes the abstraction as the starting point and looks for the thermodynamic cost of the logical operation. That is a legitimate question — but it misses the physics of the real system. The real computer produces entropy continuously, independently of the logical abstraction currently running on it.

46.7.2 The same statement from the hardware side

The statement of Section 49.7 — thermodynamics reads no content — can be derived independently of the accounting, directly from the physical structure of the computer.

[B] The computer does not shrink. A 64-bit processor has, before and after every operation, the same number of transistors, the same number of wires, the same number of degrees of freedom. The phase space of the hardware stays constant — 2^{64} possible states before, 2^{64} after. The result may need 1 bit; the apparatus that produced it has stayed exactly as large.

[B] What actually goes into the bath is the correlation. The logical map is many-to-one: many different input states land on the same output. The 63 discarded bits are logically gone, but physically their trace still sits somewhere in the transistor states — only no longer reconstructible from the result. Exactly this non-reconstructibility, the correlation between input and output that is no longer contained in the result, goes as heat into the lattice. Not the silicon, but the forgetting of the correlation.

[B] The act of forgetting is content-neutral. A processor computing $2 + 2 = 4$ and one computing $2 + 2 = 5$ switch the same number of transistors, discard the same bit width and pay the same thermodynamic bill to the bath. Physics does not distinguish the two. A result in region A costs the same as one in region B — physics does not distinguish the regions.

This is not a new claim — it is a derivation from below. Landauer's abstraction "region reduction costs, content is irrelevant" follows from the concrete physics of the computer, without a formula. The accounting in Section 49.7 and the hardware structure here arrive at the same statement by different routes.

46.7.3 The computer always computes

A physical computer does not stop computing while it waits for a result. The clock runs, the transistors switch, the power supply delivers current — continuously, independently of whether "useful" work is happening. An idle processor computes idle loops. A waiting server sends keepalive packets. An operating system polls interrupts. There is no state "the computer is not computing" as long as it is switched on.

This means: the act of forgetting runs **always**. The thermodynamic bill — loss of correlation, heat into the lattice — does not stop, no matter whether the result is needed or not, no matter what meaning it carries.

This sharpens Landauer's formulation into practical irrelevance. He isolates a single discrete act of region reduction — "region reduction costs $k_B T \ln 2$". Reality is a **continuous current** of state transitions, losses of correlation and heat releases, three billion times per second in a modern processor, without pause.

A concrete example: dynamic RAM. A DRAM bit is a capacitor holding charge. But capacitors lose charge through leakage currents — continuously, unstoppably. Therefore every DRAM cell must be **refreshed** every 32–64 milliseconds: the refresh operation reads the state and writes it back. This happens cycle by cycle, automatically, for each of the billions of cells — no matter whether the data are currently needed, no matter what meaning they carry, no matter whether the program is asleep.

This is not a side effect — it is the physical nature of the system. The DRAM refresh is **not** an act of region reduction in Landauer's sense; it is an act of preservation, and its direct costs are primarily engineering costs (leakage currents, recharging). But the reason it has to happen at all is thermodynamic: the capacitor is permanently coupled to the heat bath. The charge — the physical region "bit charged" — is continuously pushed towards the common equilibrium by the thermal coupling. That is a permanent, unceasing interaction with the bath, independently of whether a logical operation is taking place. The refresh is the technical answer to this thermodynamic reality — it renormalises the region back, because the bath permanently blurs it.

But that is not yet the whole point. The refresh **itself** is not a passive reaction — it requires active computational work: a counter must run, the address of the next row to be refreshed must be computed, a timing signal must be generated, the memory bus must be blocked, the sense amplifier must be activated. That is real computational work — with real state transitions, real losses of correlation, a real entropy current into the bath. And for a bank of 8192 rows at a 64 ms refresh interval the memory controller issues roughly 1.3×10^5 refresh commands per second — without interruption, entirely independently of whether the running program is computing anything or asleep.

That is the actual point: it is not the refresh operation itself that is Landauer-relevant in the sense of individual bit erasures, but the fact that **the computer must compute continuously in order to maintain its own state** — refresh counters, clock generators, watchdog timers, operating-system schedulers, memory controllers. This is a permanent self-computation that has nothing to do with the user program. Landauer does not describe the thermodynamics of this self-computation, because he considered individual operations; it is an additional, more fundamental cost category: **the computer pays entropy for existing — not only for computing.**

[B] A DRAM cell that "does nothing" for an hour has nevertheless run through roughly 1.12×10^5 refresh cycles in that hour (at a 32 ms interval; at 64 ms roughly 5.6×10^4), producing entropy continuously. The stored information was the same throughout — but the thermodynamic bill kept running.

That is the physical computer as it really is: not an automaton that discretely erases bits, but a system that continuously fights thermodynamic decay and pays for it. The question is therefore not: what does region reduction cost? But: how large is the continuous entropy current of a physical computer as a function of its clock frequency, its bit width and its logical irreversibility — independently of what it happens to be computing?

That is an open question. Fredkin, Toffoli and Landauer himself thought about it [B3][B13]. The standard presentation still starts from the individual act of bit region reduction, because that is tractable. It is an abstraction that misses the physics of the real computer: the computer is not an automaton that erases individual bits on command — it is a physical system producing entropy continuously, whether one looks or not, whether the result is correct or not.

46.7.4 Quantum computers: decoherence as a continuous entropy current

For the quantum computer the same principle holds as for DRAM — the physical apparatus fights thermodynamic decay continuously — but the physics is more unforgiving.

What decoherence is. A qubit in superposition has a phase relation to its environment. Every smallest interaction — a phonon, a photon, a magnetic-field fluctuation — alters that phase. The qubit loses its quantum nature not because information is annihilated, but because the correlation between qubit and environment is distributed into the bath. Liouville again: the volume is preserved, but it spreads into qubit-plus-environment correlations that nobody controls any more.

The aggravation relative to DRAM. With classical DRAM one can refresh — read the state, write it back; it costs thermodynamically, and it works because 0 and 1 are robust. For a qubit this is impossible in principle: a measurement collapses the superposition. An unknown quantum state can neither be copied (no-cloning theorem [B14]) nor measured without altering it. Refresh in the classical sense does not exist.

What exists instead is **quantum error correction** [B15]. But it is thermodynamically even more expensive than DRAM refresh — it needs 100–1000 physical qubits per logical qubit, continuous syndrome measurements and classical correction operations. All of this runs on, cycle by cycle, for every logical qubit, even when no gate operation is taking place.

[Q] Decoherence time as the DRAM analogue. A superconducting qubit typically has $T_2 \sim 100\,\mu\text{s}$ [B16] — after which the phase information is distributed in the bath. That is roughly 640 times shorter than a DRAM refresh interval of 64 ms, but qualitatively different: DRAM loses information slowly and steadily, the qubit loses it exponentially fast, and the “information” lost is not a bit but a continuous complex amplitude over a high-dimensional Hilbert space (bounded in practice by h^f, Section 49.6).

The entropy current in a quantum computer. For the classical computer the continuous entropy current is determined by clock frequency and bit width. For the quantum computer an additional term appears: the decoherence-induced entropy current, which runs proportional to the number of qubits and inversely proportional to T_2 — independently of whether a gate is being executed. A waiting quantum computer produces entropy more aggressively than a waiting classical computer, because the quantum nature itself fights the environment.

Decoherence is not region reduction in Landauer's sense — it is a loss of quantum correlations. But thermodynamically it pays the same price: the correlation goes into the bath, the bath absorbs the volume. The decisive difference from DRAM: with DRAM the information can be recovered (refresh), with a qubit it cannot (no-cloning theorem [B14]). Decoherence and Landauer erasure are different processes with the same thermodynamic mechanism.

Content-neutral as always. Decoherence does not distinguish between a qubit carrying one quantum state and one carrying another. It destroys both equally fast. The thermodynamic price of quantum computing does not attach to the content — it attaches to the physics of the system.

[H] Open question. For the classical computer the continuous entropy current is in principle reducible by reversible logic (Section 49.8). For the quantum computer this is more fundamentally limited: whether there is a matter-of-principle minimum price for maintaining quantum coherence — independently of the quality of error correction — is an open question of quantum thermodynamics [B17].

46.8 Region operations — volume-preserving or volume-reducing

Every physical change of state is a **map between phase-space regions**. The thermodynamically decisive question is exclusively: does the map preserve the region volume, or does it reduce it?

Whether that map belongs to a "computation", whether it is described as a logical operation, whether anyone calls it meaningful — for thermodynamics this is entirely irrelevant.

[B] Volume-preserving maps — the region is reshaped, not reduced. Liouville applies. No thermodynamic lower bound:

```
A ---> B    (bijective: each input region
onto its own output region)
B ---> A
```

Traditionally such maps are called “reversible gates” (NOT, CNOT, Toffoli) — physically they are **bijective region maps**. Whether they are part of a computation or not is irrelevant.

[B] Volume-reducing maps — several input regions land on the same target region Z. The volume shrinks. The bath must absorb the difference. Thermodynamic lower bound:

```
AA ---+
AB ---+---> Z      4 input regions → 2
output regions
BA ---+             volume shrinks ⇒ the
bath pays
BB -------> Z
```

Traditionally such maps are called “irreversible gates” (AND, OR, ERASE) — physically they are **non-bijective region maps**. Again: whether they are part of a computation or not is irrelevant.

[B] Bennett [B3] showed that any sequence of region maps can in principle be executed as a pure sequence of volume-preserving maps — intermediate states are carried along and released reversibly at the end. The thermodynamic costs do not arise at the intermediate steps but exclusively at the final relinquishing of regions:

> **[B] The lower bound does not attach to the map, but to the relinquishing of regions — to the forgetting.**

46.9 The accounting in one table

Step	Bit region	Bath region	Total (Liouville [B2])
before	$2W_0$	W_{bath}	$2W_0 \cdot W_{\text{bath}}$
reduction onto Z	$W_Z \leq W_0$	?	must stay $\geq 2W_0 W_{\text{bath}}$
hence	W_Z	$\geq 2\,W_{\text{bath}}$	preserved (yes)
price	$\Delta S = -k_B \ln \frac{W_b}{W_Z}$	$\Delta S \geq +k_B \ln \frac{W_b}{W_Z}$	$Q \geq k_B T \ln \frac{W_b}{W_Z}$ into the bath

46.10 The experiments and their limits

The image. Two glasses stand on a table — region A and region B. Next to them, the world ocean. Region reduction means: one of the glasses is poured into the ocean. The volume of the glass does not vanish — it distributes itself across 1.4 billion cubic kilometres. Liouville is satisfied: not a drop is lost.

But: which glass it was — A or B — is distributed so finely in the ocean's noise that no real instrument can ever extract it again. Not because the information were annihilated (it sits in every current, every temperature gradient), but because the resolution limit of every real observer lies ten orders of magnitude below the signal.

And: afterwards, under the same external conditions, the sea cannot offer the same representational capacity for the two glasses. It can no longer display them separately — the target region Z is smaller than $A + B$. That is not ignorance; that is a physical fact about region sizes.

The price of the pouring: $Q \geq k_B T \ln 2$. Not because of ignorance — but because the sea had to absorb the volume, and heat flowed for it.

46.10.1 The orders of magnitude

[Q] *(check script, checks 2–3)* A real CMOS switching event lies at $\sim 10^{-17}$–10^{-16} J — four to five orders of magnitude above Landauer. All bit erasures in the world taken together would have Landauer costs in the watt to kilowatt range (depending on the assumed global erasure rate, declared in the script); in fact data centres consume some tens of gigawatts [B8] — a factor of 10^{7}–10^{10}. In every real machine the bound is invisible; it is a lower bound imposed by natural law, the remainder is engineering loss (CV^2 recharging, leakage currents, data transport).

[S] The world-ocean image. The bit volume that tips into the bath during the region reduction stands in so extreme a disproportion to the bath (1 degree of freedom against 10^{23}) that it lies **below the resolution limit** of every real instrument. The bath absorbs the volume — Liouville is satisfied — but no measuring device can still separate the signal from the bath's noise. Not because the instrument were imperfect, but because the structure of the system itself sets the resolution limit.

46.10.2 Epistemic versus ontological — a necessary caveat

[B] Not measurable does not mean not present. After the region reduction the microstates of the bath all still exist; every particle has a definite position and momentum; the information about which source region (A or B) the system was in still sits in the correlations — Liouville guarantees this explicitly. The bath has absorbed the volume, not annihilated it.

Landauer does **not** hold **because** information is ontologically annihilated. It holds **because** the bath must absorb the volume (Liouville) — that is a statement about mechanics, not about knowledge or perception. A real observer cannot resolve the correlations distributed in the bath, but that is the epistemic side; the thermodynamic side (heat flow) follows from Liouville alone.

[B] What is ontologically true, however: a smaller region can carry fewer states under the same conditions than a larger one. After the region reduction $\{A, B\} \to Z$ we have $W_Z \leq W_{A+B}/2$ — the target region is at most half the size of the source region. Under the same external conditions (same temperature, same pressure, same coupling to the bath) it therefore cannot carry the same number of distinguishable states as before. This is not observer ignorance but a physical fact about region sizes. Whoever wants to represent, after the region reduction and under the same conditions, the same thing as before, needs a different — larger — physical system.

The correlations distributed in the bath are real. They are merely so finely distributed that no observer with finite means, under the same conditions, can reconstruct them. That is the transition from the ontological to the epistemic: the region is smaller (ontological), and its reconstruction would require bath-sized knowledge (epistemic).

[Q] Maxwell's demon [B12] and Bennett's resolution [B3]. A being that knows all 10^{23} microstates of the bath — a Laplace demon — could trace the bit back in thought. For it, erasure would be reversible and would cost nothing. Maxwell's demon shows exactly this: an observer with complete microstate knowledge can apparently reduce entropy at no cost. Bennett's resolution of 1982 [B3]: the demon must **erase its measurement notes** in order to begin a new cycle — and exactly that costs $k_B T \ln 2$. The actual site of entropy production is the erasure of the notes, not the measurement itself. The second law is thereby preserved, but the justification is sharper: not "information is lost", but "region reduction of information produces entropy".

In summary: Landauer is grounded **ontologically** by Liouville. The **irreversibility** is statistical, not ontologically absolute — for a demon with complete knowledge it would be liftable. For every real observer, however, the resolution limit lies so far below the signal that the statistical irreversibility is practically absolute. Both must be said.

46.10.3 The single-particle experiments

Bérut et al. 2012 [B4] (colloid in an optical double well, cycle times 10–40 s), Jun/Gavrilov/Bechhoefer 2014 [B5] (feedback trap, higher precision). Both approach the bound and confirm the $1/\tau$ form of the additional dissipation *(check script, check 9)*. They work in the quasistatic limiting case — the smallest possible bath, extremely slow processes — in order to come anywhere near the bound at all. A real computer erases bits in a nanosecond into the world ocean, in a storm: the stirring dominates everything, the bound lies ten orders of magnitude below the noise.

Three declared points of scepticism (objections to the evidential force of the experiments, not to the principle — the principle rests on Section 49.4):

1. **Extrapolation:** what is confirmed is a limit ($\tau \to \infty$) that no experiment enters; the bound is fitted as an asymptote.
2. **Signal $\approx$ noise:** $k_B T \ln 2$ is of the order of the thermal fluctuation of a single measurement; the result exists only as an ensemble average; some individual cycles "gain" energy.
3. **Success rate:** the protocols erase with ~ 90–95 % reliability; the accounting of failed erasures enters the balance.

[Q] Feedback extension. With measurement and feedback the generalised second law $W_{\text{ext}} \leq -\Delta F + k_B T I$ applies (Sagawa–Ueda [B9]), experimentally confirmed [B10][B11]. In the accounting, information is a resource with an energy exchange rate of $k_B T$ per nat *(check script, check 10)*.

46.11 Connection to open questions (internal)

- **Feedback accounting in fast systems:** computational operations at ~ns timescales lie ten orders of magnitude away from the tested quasistatic regime; there the B/τ term dominates everything. Anyone wishing to measure a Landauer-like residual in fast recurrent systems must take the Sagawa–Ueda term $k_B T I$ [B9] explicitly into the conventional budget before a residual can count as new — otherwise one is measuring feedback thermodynamics, not new physics.
- **Contact with FFGFT: none.** Landauer accounting lives at the level of statistical mechanics; the ξ scheme is untouched (its own level, its own accounting — analogous to P20/P39).

46.12 Summary in five sentences

Region reduction shrinks the phase-space region of the system; Liouville forbids volume shrinkage in closed systems; hence a bath must take over the volume, and the price is heat $\geq k_B T \ln(W_{\text{b}}/W_{\text{a}})$ — the $\ln 2$ only as the binary special case. Counting proceeds by distinguishability (orthogonality), not by energy; the absolute base discretisation is supplied by h^f, everything coarser is emergent counting at a chosen level. Not measurable does not mean not present: the microstates of the bath all still exist, the information about the erased bit sits in correlations — Landauer holds because of the mechanics (Liouville), not because of ignorance; the irreversibility is statistical, not ontologically absolute (Maxwell's demon, Bennett's resolution). The accounting counts regions and reads no content — every region assignment costs the same. In reality the bound is never reached, because any finite speed generates additional dissipation that dominates by orders of magnitude — the bit tips into the world ocean in a storm, and the resolution limit of the bath lies ten orders of magnitude above the Landauer bound.

46.13 Layer-status summary

The following table is a *selection* of the load-bearing statements; further passages are marked in the running text.

Statement	**Status**
Bit $\neq$ physical region (Shannon vs. Boltzmann)	**[B]**
$S = k_B \ln W$ as the starting point	**[SETZUNG]**
Region reduction $\{A, B\} \to Z$ (physics, not description)	**[SETZUNG]**
Region reduction $\Rightarrow \Delta S = -k_B \ln(W_{A+B}/W_Z)$	**[B]**
Liouville's theorem	**[B]**
Non-bijectivity of the region map is sufficient, not logical irreversibility	**[B]**
A closed system cannot erase	**[B]**
Landauer bound $Q \geq k_B T \ln 2$	**[B]**
Numerical value 2.87×10^{-21} J at 300 K	**[B]**
Counting base = orthogonality, not energy	**[B]**
h^f as the base discretisation	**[B]**
Gibbs form as the general entropy	**[SETZUNG]**
Content (true/false) does not enter the calculation	**[B]**
Hardware $\neq$ computational operation	**[B]**
The computer does not shrink	**[B]**
Correlation into the bath (not the silicon)	**[B]**
The act of forgetting is content-neutral	**[B]**
DRAM $\sim 1.12 \times 10^5$ refresh cycles per hour (32 ms)	**[B]**
No-cloning theorem	**[B]**
$T_2 \sim 100$ µs for superconducting qubits	**[Q]**
Minimum price for maintaining coherence	**[H]**
Volume-preserving maps: no lower bound	**[B]**
Volume-reducing maps: the bath pays	**[B]**
Epistemic vs. ontological: correlations remain	**[B]**
Sagawa–Ueda term $k_B T I$	**[Q]**

Balance of the table: 19× **[B]** (algebraic/mathematical from declared foundations) · 2× **[Q]** (external primary source or measured value) · 3× **[SETZUNG]** (declared starting points) · 1× **[H]** (open research question) = 25 entries.

46.14 Use of AI tools

This document was produced with the aid of AI-assisted tools. The questions, the stipulations and all decisions of content are the author's; the tools draft the prose, check the arithmetic and produce the verification scripts. Responsibility for content and for errors rests entirely with the author. The full wording is given in A010.

46.15 Check script

- `python/A_Serie_Skripte/a271_landauer.py` (checks 1–10)

46.16 What remains open

First. Whether there is a matter-of-principle minimum price for maintaining quantum coherence — independently of the quality of error correction — is an open question of quantum thermodynamics [B17].

Second. The continuous entropy current of a physical computer as a function of clock frequency, bit width and logical irreversibility is not settled in the literature.

46.17 Supersedes

This document has no direct counterpart in the older corpus; it is a self-contained document on Landauer accounting and phase-space analysis.

Sources

[B1] R. Landauer, *Irreversibility and Heat Generation in the Computing Process*, IBM J. Res. Dev. **5**, 183 (1961).

[B2] Liouville's theorem: standard result of Hamiltonian mechanics; e.g. Landau/Lifshitz, *Mechanics*, §46; quantum-mechanical counterpart: unitarity of time evolution.

[B3] C. H. Bennett, *The Thermodynamics of Computation — a Review*, Int. J. Theor. Phys. **21**, 905 (1982); id., *Notes on Landauer's principle*, Stud. Hist. Phil. Mod. Phys. **34**, 501 (2003).

[B4] A. Bérut et al., *Experimental verification of Landauer's principle*, Nature **483**, 187 (2012).

[B5] Y. Jun, M. Gavrilov, J. Bechhoefer, *High-Precision Test of Landauer's Principle in a Feedback Trap*, Phys. Rev. Lett. **113**, 190601 (2014).

[B6] Gibbs/Shannon entropy: standard; e.g. E. T. Jaynes, *Information Theory and Statistical Mechanics*, Phys. Rev. **106**, 620 (1957).

[B7] Semiclassical state counting Γ/h^f: standard; e.g. Landau/Lifshitz, *Statistical Physics*, §7.

[B8] Data-centre energy consumption: IEA reports 2024/25, of the order of several hundred TWh per year worldwide ($\approx$ some tens of GW continuous power). The figure serves only for an order-of-magnitude comparison.

[B9] T. Sagawa, M. Ueda, *Second Law of Thermodynamics with Discrete Quantum Feedback Control*, Phys. Rev. Lett. **100**, 080403 (2008); id., *Generalized Jarzynski Equality under Nonequilibrium Feedback Control*, Phys. Rev. Lett. **104**, 090602 (2010).

[B10] S. Toyabe et al., *Experimental demonstration of information-to-energy conversion*, Nature Physics **6**, 988 (2010).

[B11] *(reconstructed)* J. V. Koski et al., *Experimental observation of the role of mutual information in the nonequilibrium dynamics of a Maxwell demon*, Phys. Rev. Lett. **113**, 030601 (2014).

[B12] *(reconstructed)* Maxwell's demon — origin of the thought experiment: J. C. Maxwell, *Theory of Heat*, Longmans (1871); standard presentation in H. S. Leff and A. F. Rex (eds.), *Maxwell's Demon 2*, IOP Publishing (2003).

[B13] *(reconstructed)* E. Fredkin and T. Toffoli, *Conservative Logic*, Int. J. Theor. Phys. **21**, 219 (1982) — reversible gates and the question of the continuous entropy current of physical computers.

[B14] W. K. Wootters and W. H. Zurek, *A single quantum cannot be cloned*, Nature **299**, 802 (1982) — no-cloning theorem.

[B15] P. W. Shor, *Scheme for reducing decoherence in quantum computer memory*, Phys. Rev. A **52**, R2493 (1995); A. M. Steane, *Error correcting codes in quantum theory*, Phys. Rev. Lett. **77**, 793 (1996) — quantum error correction.

[B16] F. Arute et al. (Google AI Quantum), *Quantum supremacy using a programmable superconducting processor*, Nature **574**, 505 (2019) — typical T_2 times of superconducting qubits ~ 100 µs.

[B17] M. Esposito and C. Van den Broeck, *Three faces of the second law: I. Master equation formulation*, Phys. Rev. E **82**, 011143 (2010); P. Kammerlander and J. Anders, *Coherence and measurement in quantum thermodynamics*, Sci. Rep. **6**, 22174 (2016) — quantum thermodynamics and the minimum price of maintaining coherence.

[B18] C. E. Shannon, *A Mathematical Theory of Communication*, Bell System Technical Journal **27**, 379 (1948) — definition of the bit as $\log_2$ of the distinguishable states; explicitly information-theoretic, not thermodynamic.

[B19] E. T. Jaynes, *Information Theory and Statistical Mechanics*, Phys. Rev. **106**, 620 (1957) — connection of Shannon entropy to Boltzmann: holds exactly only for a uniform distribution over physical microstates; not for a coarsely chosen logical resolution.

[B20] *(reconstructed)* Johnson–Nyquist noise: J. B. Johnson, *Thermal Agitation of Electricity in Conductors*, Phys. Rev. **32**, 97 (1928); H. Nyquist, *Thermal Agitation of Electric Charge in Conductors*, Phys. Rev. **32**, 110 (1928).

The entries marked (reconstructed) *were missing from the working note and have been supplied from the citation context; they are to be checked against the working files before publication.*

Chapter 47

A272 · Carrier and Information

Abstract. The familiar claim that "erasing a bit produces at least $kT \ln 2$ of heat" goes beyond what Landauer's 1961 paper proves. Landauer shows something precise: if a physical two-state system in thermal equilibrium is forced into a definite state irrespective of its initial state, at least $kT \ln 2$ must be dissipated. That is a statement about *carrier operations*. The step from there to a universal claim about *information* as such is not taken in Landauer's paper; it is encouraged by a conceptual ambiguity in the word "bit", but it is not a proved consequence of his arguments. This paper separates the two, documents Landauer's own restriction, traces the generalisation through the reception literature, and names the linguistic mechanism — the hypostatisation of an abstraction — that carried it. *Relation to A271:* A272 adds nothing to the physics. A271 does the accounting; A272 does the textual criticism. The dictionary mapping the two vocabularies is given in Section 47.1.1.

Layer markings. [SETZUNG] declared starting point (German corpus token: [SETZUNG]) — **[B]** derived algebraically/mathematically from declared foundations — **[Q]** source — external primary source or measured value — **[S]** plausibility sketch — **[H]** open research question.

47.1 The question at issue

Since Landauer's 1961 paper [B1] it has often been claimed in the literature that every erased bit must produce at least $kT \ln 2$ of heat. The claim has become a fixture of computer science, physics and the philosophy of science.

The present paper holds that this generalisation goes beyond what Landauer's proof shows. Landauer makes a precise statement about physical carrier operations. The jump from there to a universal statement about information as an abstract quantity is not his step — it was taken in the reception, encouraged by a conceptual imprecision that is built into the discipline.

Two delimitations first, so that the reach of this paper is not overestimated.

[B] First: this is not a refutation of Landauer's principle. What is contested here is the *popular short form*, not Landauer's calculation. For the physical operation Landauer considers, his result is correct. What is contested is the jurisdiction of that result over a subject matter — abstract information — about which it makes no statement.

[B] Second: the physical substance is in A271, not here. That the thermodynamic lower bound attaches to the physical region and not to its description is derived there from Liouville. A272 adds nothing to that. What A272 does is different and stands on its own: the textual analysis of Landauer 1961, the documentation of Landauer's own restriction, the reception history of the generalisation, and the critique of the language of digital technology.

47.1.1 Dictionary: A271 and A272

[SETZUNG] The two documents draw the same distinction in different words. So that the corpus does not end up carrying two parallel terminologies for one matter, the mapping is fixed here and is binding:

A271 (accounting)	A272 (textual criticism)	Level
region in phase space	carrier	physical
description	information	abstract
region reduction $\{A, B\} \to Z$	carrier operation RESTORE TO ONE	physical
non-bijectivity of the region map	shrinking of the carrier state space	physical
descriptive decision	interpretive erasure	abstract

[B] The mapping is a translation, not an extension. Whoever reads "carrier" may think "region", and conversely. A statement that holds in one column holds in the other — otherwise there is an error in one of the two documents, not a new matter of fact.

47.2 The core distinction

47.2.1 Definitions

Definition 47.2.1 (Information)**.** Information is an abstract quantity. A bit is a unit of information representing one of two states (0 or 1). Information has no mass, no energy, no entropy and no physical location.

Definition 47.2.2 (Carrier)**.** The carrier of the information is a physical system that represents the information: an electrical circuit, a magnetic spin, a photon or any other physical two-state system. The carrier has energy, entropy and further physical properties.

[SETZUNG] Both definitions are stipulations. They are not proved; they fix what is being talked about in what follows. The entire yield of the paper depends on keeping the separation intact.

47.2.2 Energy attaches to the carrier

Theorem 47.2.3. *No energy can be assigned to a bit of information, only to its carrier.*

[B] Proof. Suppose a bit of information had an intrinsic energy. Then that energy would have to be independent of the physical realisation. But a bit with the value 1 can be represented by different physical systems with different energies: by 5 V in a CMOS circuit, by 1V in another technology, by a magnetic spin with a given exchange energy, by a photon of a given frequency. Since the information content is identical in all cases while the energy varies, the energy cannot be a property of the information. It is a property of the carrier. □

The argument is a multiple-realisability argument and carries exactly as far as multiple realisability reaches — that is, all the way: there is no class of realisations in which the energy would co-vary with the information content.

47.2.3 The bus example

[B] An instructive example is the bit width of a data bus:

- The **bus width** (say 64 bits) is a fixed physical property of the hardware. It determines how many carriers are switched simultaneously.
- The **information** is what is encoded by the patterns. The same bit pattern can mean different things in different contexts.

A 64-bit bus always switches 64 carriers, regardless of whether the information is a number, a letter or a picture element. The lines have a certain capacitance and require energy for recharging. The information does not have that energy.

47.3 Landauer's model — what it shows

47.3.1 The particle in the double well

[Q] Landauer's central model (Fig. 1 in [B1]) is a classical particle in a bistable potential well:

- left minimum: state 0;
- right minimum: state 1;
- the operation RESTORE TO ONE forces the particle into the right well irrespective of its initial state.

This is a classical macroscopic model. It does not capture the full breadth of modern quantum-information concepts, but it remains valid as an analogy for bistable systems — spin-½ systems are formally equivalent. What matters is what the model does *not* do: it shows under which physical conditions dissipation is unavoidable. It does *not* show that those conditions are met whenever a bit is erased in the information-theoretic sense.

What Landauer's model does	What Landauer's model does not do
proof for physical carrier operations	proof for information-theoretic erasure
lower bound for RESTORE TO ONE	statement about abstract bits
validity for a thermalised ensemble	validity for arbitrary computational processes
physical entropy $S = k \ln W$	bridge to Shannon entropy

47.3.2 The entropy calculation

[Q] Landauer computes the entropy change on resetting as

$$\Delta S \;=\; k \ln 2 - k \ln 1 \;=\; k \ln 2. \tag{47.1}$$

This calculation presupposes:

1. **before:** the carrier can take 2 physical states ($W = 2$);
2. **after:** the carrier can take only 1 physical state ($W = 1$).

Both presuppositions are statements about the carrier. Neither is a statement about the information content.

47.4 The objection and its reach

[B] The entropy change in equation (47.1) is a property of the **physical carrier**, not of the information. If a bit is erased *interpretively* — by redefining its meaning, declaring the value invalid, striking the address from a table — then the number of physical states of the carrier does **not** change. It still holds that $W = 2$, hence

$$\Delta S \;=\; k \ln 2 - k \ln 2 \;=\; 0. \tag{47.2}$$

Theorem 47.4.1. *Landauer's principle bites only if the operation physically shrinks the state space of the carrier. A purely interpretive erasure does not shrink it.*

[B] Proof. Let W be the number of physical microstates of the carrier. An interpretive erasure declares one state no longer valid; it is an operation at the level of description and leaves the physical state space unchanged. The entropy of the carrier remains $S = k \ln W$. Only when the carrier itself is forced into a single state ($W' = 1$) does the entropy change to $S' = 0$. That is a physical operation on the carrier, not on the information content. □

47.4.1 What this objection is not directed against

[B] Precision is needed here, otherwise the objection is aimed at an opponent who does not exist. Equation (47.2) does *not* refute Landauer. Landauer nowhere claims that a conceptual redesignation costs heat; nobody in the technical literature claims that. What equation (47.2) shows is solely that the two processes — interpretive erasure and carrier reset — come apart, and that only the second falls under Landauer's bound.

That is not a small thing, but it is also not what it looks like at first glance. The yield is a *finding about jurisdiction*, not a refutation:

> **[B]** Landauer's bound has jurisdiction over carrier operations that shrink the state space. For everything else that ordinary language calls "erasing", it has no jurisdiction — neither confirming nor refuting.

[B] The real question thereby shifts, and it is harder than the original one: *does information-theoretic erasure enforce a state-space-shrinking carrier operation?* Bennett [B2][B8] showed that in general it does not — any logically irreversible function can be executed reversibly if the intermediate states are carried along. Only the final relinquishing of carrier states costs. So the answer for the general case is negative, and it is an answer Landauer could not yet give in 1961.

47.5 Landauer's own restriction

[Q] Landauer formulated a restriction that the later reception mostly passes over:

> *"Our reset operation, in the preceding discussion, was applied to a thermal equilibrium ensemble. In*

> *actuality we would like to know what happens in a particular computing circuit which will work on information which has not yet been thermalized …"* [B1]

[Q] Landauer resolves the question himself immediately afterwards: for a random sequence of ones and zeros he equates the statistical ensemble with the time average and thereby arrives at the same dissipation per reset as for the thermalised ensemble. For non-random data — if, say, all inputs are already ONE — he expressly concedes that no entropy change occurs.

Three things follow:

1. **[Q]** Landauer is aware of the limits of his model and works them out himself. The charge of over-generalisation hits the reception, not him.
2. **[B]** His result $kT \ln 2$ holds strictly for the equilibrium ensemble. For correlated, non-random data, $kT \ln 2$ per carrier is an *upper* bound; the applicable bound is kTH, with H the entropy of the source in nats. This is not a weakening of the principle but its correct statement: the bound follows the actual state count, not the bit width.
3. **[B]** Above all: this extension too remains a statement about *carrier states*. The transition to abstract information is not made here either.

47.6 What Landauer proves — and what he does not

The popular short form reads:

> *"Every erased bit produces $kT \ln 2$ of heat."*

[B] This claim goes beyond Landauer's proof. What is proved is:

> *If a physical two-state system in thermal equilibrium is forced into a definite state irrespective of its initial state, at least $kT \ln 2$ of heat must be released.*

> *This dissipation is a property of the physical carrier operation.*

[Q] The step Landauer's proof does not contain is the bridge from the carrier operation to abstract information. Landauer uses "bit" throughout in a double sense — sometimes for the physical object, sometimes for the unit of information. This ambiguity is present in the original paper and encouraged the later reception, in which a statement about carriers became a statement about information.

[B] That every information-theoretic erasure requires a state-space-shrinking carrier operation is presupposed, not derived — and by Bennett it is in general false.

47.7 The modern thermodynamics of information

[Q] The thermodynamics of information with feedback [B5][B6] gives, for the erasure process, the generalised bound

$$Q_{\text{min}} = kT(\ln 2 - I), \qquad I \in [0, \ \ln 2], \tag{47.3}$$

where I is the mutual information between measuring device and system state, measured in **nats** and therefore dimensionless. Equivalently in the form of Sagawa and Ueda: $W_{\text{ext}} \leq -\Delta F + kTI$.

- **[B]** If the experimenter knows whether the carrier is at 0 or 1 ($I = \ln 2$), the carrier can be reset reversibly — $Q_{\text{min}} = 0$.
- **[B]** If not ($I = 0$), the reset must be irreversible — $Q_{\text{min}} = kT \ln 2$.

[B] The heat therefore arises not *because* information is lost, but because of the lack of knowledge about the carrier state. The information is the label stuck onto the carrier; the price attaches to how far the label actually pins the carrier down.

[B] And the symmetry must be seen at the same time: the knowledge I is itself stored on a carrier. Whoever uses it to reset reversibly has not avoided the price but displaced it — to the later moment at which the carrier of the knowledge is itself reset. That is Bennett's resolution of Maxwell's demon [B8], and it prevents equation (47.3) from being read as a circumvention of the bound.

47.8 The perpetuation in the reception

[Q] The conceptual imprecision would have remained inconsequential had the subsequent literature corrected it. The opposite happened. The majority of publications after 1961 adopt the principle in generalised form without going back to Landauer's own restrictions.

- **[Q] Lairez** [B9] notes that criticism of the generalisation has been available for decades but has had no broad effect.
- **[Q] Norton** [B4] shows that the standard argument rests on an untenable identification of dynamic and static probabilities; **Earman and Norton** [B3] had already worked through the line from Szilárd to Landauer in 1999.
- **[Q] Hemmo and Shenker** [B10] show that in the light of the multiple-computations theorem the principle cannot be universally valid.

[S] The reason for the persistence lies not in a refutation of this criticism but in structural features of scientific communication: the formula $kT\ln 2$ is catchy and easy to cite. The 1961 original is rarely read; what gets cited are secondary sources citing secondary sources. Trust in the authority of a distinguished industrial researcher does the rest. The consequence is that a step never taken in the proof is traded as a "fundamental law of nature" — a description that does not appear in the original paper.

[B] This diagnosis is a plausibility sketch and is marked as such. It names mechanisms that are individually observable but whose combined effect is not quantified. A defensible version would be a citation analysis: the share of papers citing [B1] that also state the ensemble restriction. That analysis is not provided here.

47.9 The imprecision in the language of digital technology

[B] The problem is not confined to Landauer. The whole of digital technology uses a language that systematically states carrier properties as properties of information. The practice is so deeply entrenched that practitioners rarely perceive it as problematic.

Customary expression	**Physical reality (carrier)**	**Problematic implication**
“set a bit”	apply a voltage to a capacitor	information has a state
“erase a bit”	set the carrier to a reference voltage	information can be annihilated
“transmit a bit”	reconstruct a signal on a new line	information is transportable
“copy a bit”	read a voltage, generate it elsewhere	information multiplies itself
“the RAM holds 8 GB”	8 billion physical memory cells	information is a bulk quantity
“bit rate”	voltage changes per second	amount of information per unit time

In each case a property of the carrier is phrased as if it were a property of the information. In practice engineers know that they are handling voltages, currents and charges. In the descriptive language, however, the bit becomes the acting subject: it “flows through the bus”, “is stored”, “is erased”. This language is convenient for many purposes — but it tempts one to apply physical laws directly to the abstraction, as the popular Landauer reception did.

[B] A more precise usage would distinguish:

- **Not:** “the bit has energy.” **But:** “the carrier of this bit has energy.”
- **Not:** “the bit is erased.” **But:** “the carrier is set to a definite state.”
- **Not:** “the bit costs $kT \ln 2$.” **But:** “this carrier operation costs at least $kT \ln 2$ under these conditions.”

The connection to Landauer is direct: he uses “bit” in exactly this double sense. The ambiguity is not his invention; it is built into the discipline and it carried the generalisation.

47.10 Epistemological background: hypostatisation

[S] The mechanism described — an abstraction is treated as a physical quantity and equipped with physical properties — has a name: *hypostatisation* (or reification), the transformation of an abstract concept into a supposedly real, substantial object.

[Q] The bit is an abstraction. Shannon [B11] defined it as a statistical property of signals, not as a physical object. Wiener stated the same point explicitly: information is information, not matter or energy [B12]. That clarity was lost in engineering practice. The bit became an object to which energy, entropy and location could be ascribed.

[S] The mechanism is not new in the history of science. Entropy, the electric field, the gene — all of these abstractions were at certain times treated as substantial things, with consequences for theory. In the case of information the hypostatisation leads to a false physics: if the bit itself had entropy, then erasing it would have to produce heat — independently of the carrier. That conclusion is compelling only for someone who has confused the abstraction with physical reality.

[B] This section is marked as a sketch because it offers an interpretation and provides no demonstration. The historical parallels are analogies; they support the thesis, they do not prove it.

47.11 Conclusion

1. **[B]** Landauer's proof for the physical operation RESTORE TO ONE on a thermal equilibrium ensemble is correct and precise.
2. **[B]** It shows a lower bound for *carrier operations* — not a universal statement about abstract information.
3. **[B]** The transfer to the information-theoretic notion of the bit presupposes that every information-theoretic erasure is bound up with a state-space-shrinking carrier operation.

Landauer does not prove this connection — and by Bennett it does not hold in general.

4. **[Q]** The double sense of “bit” is present in the original paper and encouraged the generalisation in the reception.
5. **[Q]** Landauer himself discusses the limits of his ensemble argument and concedes that non-random input data lower the dissipation below $kT \ln 2$.
6. **[B]** What is not shown here: that Landauer's principle is false. What is shown is that it has jurisdiction over a subject matter other than the one it is usually cited for.

47.12 Outlook

[B] A formulation that reflects the actual proof reads:

> *The physical operation that puts a bistable system in thermal equilibrium into a definite state irrespective of its initial state requires at least $kT \ln 2$ of dissipation. If this system serves as an information carrier and the operation corresponds to erasing a bit, the bound applies to the associated carrier operation — not to information-theoretic erasure as such.*

[H] Open question. Under what conditions does a given computational process enforce a state-space-shrinking carrier operation? Bennett answers the limiting case (in principle never, if everything is carried along); the practically interesting question is quantitative: how much carrier state space must an architecture finally give up at a given memory depth? This is the same open question as the continuous entropy current in A271 — posed from the other side.

47.13 Layer-status summary

Statement	Status
Definition of information (abstract, without energy)	**[SETZUNG]**
Definition of carrier (physical, with energy)	**[SETZUNG]**
Dictionary mapping A271 ↔ A272	**[SETZUNG]**
Proposition 1: energy attaches to the carrier, not to the bit	**[B]**
Bus example: bus width is hardware, not information	**[B]**
Landauer's double-well model and RESTORE TO ONE	**[Q]**
$\Delta S = k \ln 2$ presupposes $W{=}2 \to W{=}1$ at the carrier	**[Q]**
Proposition 2: interpretive erasure does not shrink W	**[B]**
The objection is a finding about jurisdiction, not a refutation	**[B]**
Bennett: logically irreversible functions executable reversibly	**[B]**
Landauer's ensemble restriction (original quotation)	**[Q]**
Landauer's own resolution for random data	**[Q]**
For correlated data the bound is kTH, not $kT \ln 2$	**[B]**
Landauer uses "bit" in a double sense	**[Q]**
Feedback bound $Q_{\text{min}} = kT(\ln 2 - I)$, I in nats	**[Q]**
The knowledge I is itself carrier-bound (Bennett)	**[B]**
Reception findings: Lairez / Norton / Hemmo–Shenker	**[Q]**
Structural reasons for the persistence	**[S]**
Language table: carrier versus information	**[B]**
Hypostatisation as an interpretive frame	**[S]**
Historical parallels (entropy, field, gene)	**[S]**
No demonstration that Landauer's principle is false	**[B]**
Quantitative form of the unavoidable state-space relinquishment	**[H]**

Balance: 10× **[B]** · 8× **[Q]** · 3× **[SETZUNG]** · 3× **[S]** · 1× **[H]** = 25 entries.

47.14 Use of AI tools

This document was produced with the aid of AI-assisted tools. The questions, the stipulations and all decisions of content are the author's; the tools draft the prose, check the arithmetic and produce the verification scripts. Responsibility for content and for errors rests entirely with the author. The full wording is given in A010.

47.15 Check script

This document carries no figures of its own; the numerical statements are held in A271 (`a271_landauer.py`) and A273 (`a273_rechenkugel.py`).

47.16 What remains open

Under what conditions a given computational process enforces a state-space-shrinking carrier operation is quantitatively open; Bennett answers the limiting case, the practically interesting question remains.

47.17 Supersedes

This document has no direct counterpart in the older corpus; it supplements A271 with the textual criticism and the reception history.

Sources

[B1] R. Landauer, *Irreversibility and Heat Generation in the Computing Process*, IBM J. Res. Dev. **5**, 183 (1961).

[B2] C. H. Bennett, *Logical Reversibility of Computation*, IBM J. Res. Dev. **17**, 525 (1973).

[B3] J. Earman and J. D. Norton, *Exorcist XIV: The Wrath of Maxwell's Demon. Part II: From Szilard to Landauer and Beyond*, Stud. Hist. Phil. Mod. Phys. **30**, 1 (1999).

[B4] J. D. Norton, *All Shook Up: Fluctuations, Maxwell's Demon and the Thermodynamics of Computation*, Entropy **15**, 4432 (2013).

[B5] T. Sagawa and M. Ueda, *Nonequilibrium Thermodynamics of Feedback Control*, Phys. Rev. E **85**, 021104 (2012).

[B6] J. M. R. Parrondo, J. M. Horowitz and T. Sagawa, *Thermodynamics of Information*, Nature Physics **11**, 131 (2015).

[B7] J. Bub, *Maxwell's Demon and the Thermodynamics of Computation*, Stud. Hist. Phil. Mod. Phys. **32**, 569 (2001).

[B8] C. H. Bennett, *The Thermodynamics of Computation — a Review*, Int. J. Theor. Phys. **21**, 905 (1982).

[B9] *(incomplete — identifier to be supplied)* D. Lairez, *Landauer's Principle: A Critical Assessment*, arXiv preprint (2024).

[B10] M. Hemmo and O. Shenker, *The Multiple-Computations Theorem and the Physics of Singling Out a Computation*, The Monist **105**, 175 (2022).

[B11] C. E. Shannon, *A Mathematical Theory of Communication*, Bell System Technical Journal **27**, 379 (1948).

[B12] *(reconstructed — page number to be checked)* N. Wiener, *Cybernetics: or Control and Communication in the Animal and the Machine*, MIT Press, Cambridge, MA (1948).

Entries marked incomplete *or* reconstructed *were supplied from the citation context or carry an open placeholder; they are to be checked against the working files before publication.*

Chapter 48

A273 · The Reckoning Bead

Abstract. Computing with tokens — abacus beads, shells, coins — is the limiting case in which Landauer's argument comes apart into its two halves. The *accounting* holds there exactly and is for once visible: the stock from which the tokens come and into which they flow back is a bath one can point at. The *thermal conversion* $\Delta S \to Q = T\Delta S$ does not hold there: an abacus bead costs about 5×10^{15} times the Landauer bound to move, its positional bit carries 4×10^{-23} of its own thermal entropy, and its positional barrier stands at $3.6 \times 10^{15}\, kT$. The token computer therefore falls squarely under the very restriction Landauer himself stated and the reception dropped (A272): the counted states are not a thermal ensemble. From this follows an ordering that runs against intuition: **visibility of the structure and binding force of the number run in opposite directions.** The cruder the carrier, the clearer the accounting and the more irrelevant the bound. The passage back is continuous — shrink the token until its barrier approaches kT and it becomes a Brownian reckoning bead, which is Bérut's colloid.

Layer markings. [SETZUNG] declared starting point (German corpus token: [SETZUNG]) — **[B]** derived algebraically/mathematically from declared foundations — **[Q]** source — external primary source or measured value — **[S]** plausibility sketch — **[H]** open research question.

Check script. `a273_rechenkugel.py` (checks 1–14). Every number in this document is backed there by assertions; the parameter choices are marked in the script as stipulations and can be varied freely.

48.1 The stock as a visible bath

A271 works with the world ocean: a glass is poured in, the volume does not vanish, but nobody gets it back. The image is correct and has one drawback — the bath has to be inferred. Nobody sees the 10^{23} degrees of freedom into which the bit volume tips.

[SETZUNG] With token computation this is different. An abacus, a heap of shells, a till with coins: next to the working area there lies a **stock**. The tokens come from it and flow back into it. The stock is a bath one can point at.

```
 Working area            Stock (bath)
+-----------------+     +-------------------------+
| o o . o . . o . |     | o o o o o o o o o o o o |
| . o o . o o . o |  ---> o o o o o o o o o o o o |
+-----------------+     +-------------------------+
 W = 2^N config.         tokens conserved, arrangement mixed
```

[B] The structure is step for step the same as in A271:

- **Region.** A token in the left position and the same token in the right position are two distinguishable macro-regions.
- **Region reduction.** Clearing the board maps all configurations onto one — all tokens in the stock. Non-bijective, exactly like $\{A, B\} \to Z$.
- **Conservation.** No token vanishes. They spread into an arrangement from which the previous configuration can no longer be read off. That is the Liouville analogue — and here it is literally countable.
- **Content neutrality.** The stock does not know whether a token stood for a debit, for a five, or for nothing. It absorbs it the same way.

[B] Cash with coins and a till is the same figure, and the double meaning of the word "accounting" stops being a metaphor here: the balance of a till and phase-space accounting have the same form. Barter is the case without tokens — there is no stock there and correspondingly no region reduction, only bijective exchange.

48.2 The accounting holds exactly

[B] For N tokens with two positions each, before clearing $W_{\text{before}} = 2^N$ and after $W_{\text{after}} = 1$, hence

$$\frac{\Delta S}{k_B} = \ln \frac{W_{\text{before}}}{W_{\text{after}}} = N \ln 2. \tag{48.1}$$

This is exact and carrier-independent *(check script, check 10)*. It holds for beads, for shells, for transistors, for colloids. Nothing in equation (48.1) demands that the token be small, light, or thermally driven. The count is combinatorics over regions — exactly what A271 identifies as primary.

[B] Up to this point the token computer is a complete model of Landauer accounting, and a better one than any other, because every step can be seen.

48.3 The bound does not hold

[SETZUNG] For the following numbers it is stipulated: an abacus bead of mass $m = 0.5$ g, a displacement $d = 1$ cm, a friction coefficient $\mu = 0.3$, an ambient temperature $T = 300$ K, a molar mass of 100 g/mol and a molar entropy of 50 J/(mol·K) for a solid. All stipulations are marked in the check script and can be changed; the orders of magnitude do not depend on their precise values.

[B] The work per bead displacement. Sliding friction gives $W = \mu m g d = 1.47 \times 10^{-5}$ J *(check script, check 2)*. The Landauer bound at 300 K is $k_B T \ln 2 = 2.87 \times 10^{-21}$ J *(check script, check 1)*. The ratio is

$$\frac{W_{\text{bead}}}{k_B T \ln 2} \approx 5.1 \times 10^{15} \qquad \textit{(check script, check 3)}. \tag{48.2}$$

48.3.1 The scale band

Carrier	Cost per switching event	Multiple of $k_B T \ln 2$
colloid, quasistatic [B4][B5]	$\sim 3 \times 10^{-21}$ J	1–10
CMOS transistor (A271)	10^{-17}–10^{-16} J	$3.5 \times 10^{3} - 3.5 \times 10^{4}$
abacus bead	1.5×10^{-5} J	5.1×10^{15}

[B] Between bead and CMOS lie roughly **eleven decades** *(check script, check 5)*. The ordering is thus the reverse of the obvious conjecture: Landauer binds not at the primitive end but at the small one. The cruder the carrier, the further the bound lies below the actual expenditure.

48.3.2 Why the bead pays nothing

Two numbers account for the finding, and both concern the bead itself, not the accounting.

[B] First: the positional bit is vanishing against the bead's own entropy. A bead of 0.5 g carries, under the stipulations above, an entropy of 0.25 J/K, that is $1.8 \times 10^{22}\, k_B$ *(check script, check 6)*. The positional bit carries $\ln 2 = 0.69\, k_B$. Its share is

$$\frac{\ln 2}{S_{\text{bead}}/k_B} \approx 3.8 \times 10^{-23} \qquad \textit{(check script, check 7)}. \qquad (48.3)$$

Moving the bead changes essentially nothing in the thermally accessible phase space. The region reduction takes place in a coordinate decoupled from the bead's 10^{22} thermal degrees of freedom.

[B] Second: the positional barrier is thermally insurmountable. It stands at $3.6 \times 10^{15}\, k_B T$ *(check script, check 8)*. An Arrhenius lifetime $\propto \exp(3.6 \times 10^{15})$ is not numerically representable. There is no thermal transfer between "left" and "right", and therefore no equilibrium ensemble over those two states.

[B] The bead is thus the extreme case of the row "emergent two-valuedness" in A271: the two is a partition imposed on a quasi-continuum of 10^{22} microstates, and it survives not because it is fundamental but because the barrier is absurdly high.

48.4 The two halves of Landauer's argument

[B] The token computer thereby does something no other model does as cleanly: it separates the two steps of which Landauer's argument consists, and shows that they are independent.

Step	Statement	Presupposition
combinatorics	$\Delta S = k_B \ln(W_\text{b}/W_\text{a})$	regions are distinguishable and countable
conversion	$Q = T\Delta S$	the counted states are thermally occupied

[B] The first step holds exactly for the bead. The second does not hold for it at all. The usual presentation runs the two together in one move and thereby makes the second presupposition invisible. Only when a carrier violates it maximally does it come into view.

> **[B]** Landauer's bound is not the statement "region reduction costs". It is the statement "region reduction *in thermally occupied states* costs". The qualifier is not cosmetic — it carries the whole conversion.

[B] This sharpens A271 at a place where it needs to be more precise. A271 §6 requires, for classical systems, distinguishability *above* the thermal resolution limit. That is the condition for two regions to count as two. It is *not* the condition for their reduction to cost heat — for that one additionally needs the barrier *not* to lie arbitrarily far above $k_B T$. The two conditions point in opposite directions, and between them lies the narrow band in which Landauer binds.

48.5 Landauer's own restriction

[Q] The finding is not a new discovery but the extreme case of a restriction Landauer stated himself in 1961: his reset operation was applied to a thermal equilibrium ensemble; for information not yet thermalised the calculation does not apply directly [B1]. A272 documents the passage and shows that the reception dropped it.

[B] With the abacus the restriction is maximally violated, and in two ways at once:

1. **No thermal ensemble.** The distribution over "left" and "right" does not arise from thermal occupation but from the hand that placed the bead.
2. **Complete knowledge.** Whoever placed the beads knows the configuration. In the feedback bound (A272) this makes $I = \ln 2$ per token and

$$Q_{\min} = k_B T(\ln 2 - I) = 0 \qquad \textit{(check script, check 11)}. \tag{48.4}$$

[B] Thermodynamically, then, clearing a known abacus costs nothing. What it does cost is friction — and friction is an engineering quantity with no lower bound in nature, reducible at will by better bearings.

48.6 The passage back: the Brownian reckoning bead

[B] The passage back into Landauer's domain of validity is continuous and can be computed. Setting the positional barrier equal to $k_B T$ gives, for the critical token mass,

$$\mu m g d = k_B T$$

$$m_{\text{crit}} = \frac{k_B T}{\mu g d} \approx 1.4 \times 10^{-19}\,\text{kg} \quad \textit{(check 9)}.$$

[B] For comparison: a silica colloid of 2 µm diameter weighs 8.4×10^{-15} kg, roughly 6×10^{4} times more *(check script, check 9)*. But it holds its position not by friction, rather by an optical trap, and the height of that barrier is adjustable down to a few $k_B T$. That is exactly what makes it a Brownian reckoning bead — and exactly why Bérut's apparatus [B4] is the only one in which the bound actually binds.

> **[B]** Bérut's colloid is not a model of the bit. It is an abacus bead small enough for the bath to push it.

[B] The series thereby closes: shell, coin, abacus bead, transistor, colloid. It is a scale, not a jump. What changes along the scale is not the accounting — it is the same everywhere — but the ratio of the positional barrier to $k_B T$.

48.7 Two axes, running opposite

[B] From this follows the ordering that gives the document its yield:

Carrier	Visibility of the structure	Binding force of the bound
abacus, shells, coins	complete (bath can be pointed at)	none (10^{15} above)
CMOS	slight (bath is the lattice)	weak (10^{4} above)
colloid in an optical trap	none (bath not separable)	full (factor 1–10)

> **[B]** At the most primitive level the accounting is maximally visible and the bound maximally irrelevant. Visibility of the structure and binding force of the number are two different axes, and they run in opposite directions.

[B] This also explains why intuition regularly misleads in this matter. Anyone wanting to make Landauer plausible reaches for the coarse image — the bead, the drawer, the glass of water — and thereby lands exactly where the bound has ceased to mean anything. The image carries the structure and loses the number.

48.8 Two floors: $\hbar c/L$ and k_BT

[B] The thermal floor k_BT is not the only one. There is a second, quantum-geometric floor: $\hbar c/L$, the energy of a mode of extension L. Both limit how finely a state can be pinned down at all — but they are of different origin, and the larger of the two governs.

[B] The quantum-geometric floor $\hbar c/L$ is the energy of a mode of extension L. It applies to massless or field-like carriers — photon, winding mode — and is temperature-independent. It says how finely a state can be pinned down at all at a given extension.

[B] The thermal floor k_BT applies wherever a carrier couples to a bath. It says how deep a well must be to hold against fluctuations — and it is what decides about Landauer.

[B] The larger of the two governs. This sorts the cases:

- **Massive mechanical carriers.** For bead and colloid, $\hbar c/L$ is not the relevant quantum scale at all; the relevant one would be $\hbar^2/(2mL^2)$, and that amounts to $3 \times 10^{-41}\, k_BT$ for the abacus bead and $2 \times 10^{-22}\, k_BT$ for Bérut's colloid *(check script, check 12)*. The quantum-geometric floor lies so far beneath everything here that only the thermal floor counts — the criterion of this document is the whole story there.

- **Field-like carriers.** There $\hbar c/L$ carries. At 300 K the two floors cross at $L = \hbar c/(k_B T) = 7.6$ µm *(check script, check 13)*; below that length the quantum-geometric floor dominates, above it the thermal one.
- **The qubit** is the case in which the quantum-geometric floor exceeds the thermal one: $hf/k_B T = 24$ at 5 GHz and 10 mK *(check script, check 14)*. That is exactly why it behaves quantum-mechanically at all — and exactly why it lies close enough to the thermal band to need error correction.

[B] Landauer therefore binds where the thermal floor leads *and* the barrier lies within a few $k_B T$. Where the quantum-geometric floor leads, $k_B T \ln 2$ is not the relevant quantity. In that sense, but only in that sense, the bound is a special case: it holds for a class of carriers, not for all of them.

48.9 The token stock as a cost category of its own

[B] One point remains that thermodynamic accounting does not capture and that becomes especially clear with the token computer: **without tokens one cannot compute.** The stock is an exhaustible resource. An abacus with too few beads does not compute more slowly; it does not compute at all.

[B] This is a *material bound*, not a thermodynamic one. It has no lower limit in $k_B T$; it has a lower limit in the number of items. It is related to what A271 §7.3 opens up — the computer pays for existing, not only for computing — but it is not the same thing: there the subject is a running entropy current, here a standing inventory.

[S] With semiconductors this category drops out of view, because the stock is cast in silicon: the regions are laid down in fabrication and not drawn during operation. The inventory is there, so it is not counted. On the abacus it is visible, because it can be used up.

[H] Open question. Whether inventory and current can be carried in a single ledger — token stock and entropy current as two balances of one computer — is open. The question is the same as the one at the end of A272, posed from the material side.

48.10 Placement in the corpus

[B] The three pieces stand to one another as follows:

- **A271** does the accounting: region, region reduction, Liouville, bath, bound. The physical core.
- **A272** does the textual criticism: what Landauer proved in 1961, his own ensemble restriction, the reception history, the language.
- **A273** exhibits the limiting case at which both findings converge and become visible: the token computer satisfies A271's accounting exactly and violates A272's ensemble condition maximally.

[B] The dictionary of A272 §1.1 applies unchanged. In the language of this document: token = carrier = region; stock = bath; clearing the board = carrier operation = region reduction.

[B] Contact with FFGFT: none. Like A271 and A272, this document lives at the level of statistical mechanics; the ξ scheme is untouched.

48.11 Layer-status summary

Statement	Status
The token stock is a bath analogue one can point at	**[SETZUNG]**
Parameters of the abacus bead (mass, distance, friction, T)	**[SETZUNG]**
Clearing the board is a non-bijective region map	**[B]**
Tokens are conserved (Liouville analogue)	**[B]**
$\Delta S/k_B = N \ln 2$, exact and carrier-independent	**[B]**
$k_B T \ln 2 = 2.87 \times 10^{-21}$ J at 300 K	**[B]**

Statement	Status
Bead displacement 1.47×10^{-5} J	**[B]**
Bead $\approx 5.1 \times 10^{15} \times$ Landauer	**[B]**
Scale band colloid / CMOS / bead	**[Q]**
Eleven decades between bead and CMOS	**[B]**
Bead's own entropy $1.8 \times 10^{22}\,k_B$	**[B]**
Positional bit $= 3.8 \times 10^{-23}$ of that entropy	**[B]**
Positional barrier $3.6 \times 10^{15}\,k_B T$	**[B]**
Combinatorics and thermal conversion are independent	**[B]**
The conversion presupposes thermally occupied states	**[B]**
Landauer's ensemble restriction (original passage)	**[Q]**
Known abacus: $I = \ln 2$, hence $Q_{\text{min}} = 0$	**[B]**
Critical token mass 1.4×10^{-19} kg	**[B]**
Silica colloid 2 µm: 8.4×10^{-15} kg	**[B]**
Bérut's colloid as a Brownian reckoning bead	**[B]**
Visibility and binding force run in opposite directions	**[B]**
Token stock as a material bound	**[B]**
Why the category drops out of view with semiconductors	**[S]**
A single ledger for inventory and current	**[H]**

Balance: 17× **[B]** · 2× **[Q]** · 2× **[SETZUNG]** · 1× **[S]** · 1× **[H]** = 23 entries.

48.12 Use of AI tools

This document was produced with the aid of AI-assisted tools. The questions, the stipulations and all decisions of content are the author's; the tools draft the prose, check the arithmetic and produce the verification scripts. Responsibility for content and for errors rests entirely with the author. The full wording is given in A010.

48.13 Check script

- `python/A_Serie_Skripte/a273_rechenkugel.py` (checks 1–14)

48.14 What remains open

Whether inventory and current can be carried in a single ledger — token stock and entropy current as two balances of one computer — is open.

48.15 Supersedes

This document has no direct counterpart in the older corpus; it works out the limiting case for A271 and A272.

Sources

[B1] R. Landauer, *Irreversibility and Heat Generation in the Computing Process*, IBM J. Res. Dev. **5**, 183 (1961).

[B2] Liouville's theorem: standard result of Hamiltonian mechanics; e.g. Landau/Lifshitz, *Mechanics*, §46.

[B3] C. H. Bennett, *The Thermodynamics of Computation — a Review*, Int. J. Theor. Phys. **21**, 905 (1982).

[B4] A. Bérut et al., *Experimental verification of Landauer's principle*, Nature **483**, 187 (2012).

[B5] Y. Jun, M. Gavrilov, J. Bechhoefer, *High-Precision Test of Landauer's Principle in a Feedback Trap*, Phys. Rev. Lett. **113**, 190601 (2014).

[B6] T. Sagawa and M. Ueda, *Nonequilibrium Thermodynamics of Feedback Control*, Phys. Rev. E **85**, 021104 (2012).

[B7] Semiclassical state counting Γ/h^f and the entropy of solids: standard; e.g. Landau/Lifshitz, *Statistical Physics*, §7 and §64.

[B8] CODATA 2018 values; $k_B = 1.380649 \times 10^{-23}$ J/K exactly defined since the SI revision of 2019.

The numerical values in this document come from the check script `a273_rechenkugel.py`; the parameters set there are model choices, not measured values. The orders of magnitude are insensitive to their variation.

Part V

New Extensions

Chapter 49

A280 · Landauer: Phase-Space Bookkeeping

Abstract. The Landauer bound is developed here not as a statement about *information* but as a statement about *regions in phase space*. A physical system occupies a region; a map that reduces several source regions onto one target region shrinks that volume; Liouville's theorem forbids volume shrinkage in a closed system; hence a bath must absorb the volume, and the price is heat $Q \geq k_B T \ln(W_{\text{before}}/W_{\text{after}})$ — with $\ln 2$ merely the binary special case. Counting proceeds by distinguishability (orthogonality), not by energy; the absolute base discretisation is supplied by h^f. Content neutrality follows immediately: thermodynamics counts regions, it does not read them. Three levels are kept strictly apart — construction (engineer), state transition (clock), description (program) — and from this it follows that a real computer does not erase bits discretely but carries a continuous entropy current (DRAM refresh, qubit decoherence). The epistemic caveat is booked explicitly: not measurable does not mean not present — Landauer holds because of the mechanics (Liouville), not because of ignorance; the irreversibility is statistical, not ontologically absolute. *Contact with the ξ scheme of FFGFT: none* — Landauer accounting lives at the level of statistical mechanics (its own level, its own accounting).

Layer markings. **[AXIOM]** declared starting point (German corpus token: [SETZUNG]) — **[B]** derived algebraically/mathematically from declared foundations — **[K]** empirically tested or documented from primary sources — **[S]** plausibility sketch — **[H]** open research question (an extension of the A-series scheme; see the appendix, item 6).

Check script. `landauer_pruefskript.py` (checks 1–10). Every numerical statement in this document is backed there by assertions. Check 7: state count in the colloid region; check 8: analogue case with declared δx.

49.1 The region comes first

Two glasses stand on a table — region A and region B. Next to them, the world ocean. One of the glasses is poured into the sea.

The volume of the glass does not vanish — it distributes itself across 1.4 billion cubic kilometres. This image carries the whole document: there is no vanishing information, no annihilated bits — only a reduction of regions that has a measurable price.

[AXIOM] **A physical system occupies a region in phase space.** That region has a size — measurable in units of h^f (Section 49.6). It exists independently of how one names, partitions or describes it. That is the physical starting point. Everything else comes afterwards.

[B] **Descriptions come from outside.** If one decides to partition the region into two subregions and to call them "A" and "B", and the target region "Z", one has introduced a description of the region reduction. If one calls them "0", "1" and "0" instead (the traditional notation), the confusion with logical bits arises — that convention hides the fact that "0" denotes both a source region and the target region, and it suggests an information content where there is none. If one partitions more finely — into 1000 zones — one has introduced ~ 10 logical bits. If one merely assigns a number between 0 and 1, one has introduced an analogue value. Physics enforces none of these decisions. In all cases the region is the same physical region.

The terms "bit", "byte", "qubit", "analogue value" are labels for descriptive decisions — not for physical objects. They partition the region; they do not create it.

49.1.1 The Landauer bit is hardware — not information

[B] This is regularly confused. The clarification in one paragraph: a Landauer bit is a physical region with two stable macrostates — a transistor, a capacitor, a trace held high or low. A bus bit is the same: a physical region with two stable macrostates, embedded in a parallel hardware architecture. A memory cell is the same. The three are structurally identical. Thermodynamics treats all three alike:

$$Q \;\geq\; k_B T \ln\!\left(\frac{W_{A+B}}{W_Z}\right)$$

— regardless of whether the region is called "Landauer bit", "bus bit" or "memory cell".

The hardware is fixed before any computation runs. It does not change because of what one computes with it. A 64-bit bus has 64 physical regions — before the computation, during it, after it. What the computation does with those regions — which maps it executes, which regions are reduced onto which target regions — is a logical description of those physical processes. The description does not exist in the hardware; it is an external interpretation.

The information content of a bit — whether it means “true” or “false”, whether it belongs to a correct computational step, whether it is read at all — is a decision at the level of description. Thermodynamics does not know that level. It knows only: region A and region B exist; after the region reduction the target region Z exists; $W_Z \leq W_{A+B}$; the bath absorbs the difference.

49.1.2 Three levels — strictly separated

[B] **Level 1 — construction.** The engineer decides at chip-design time which region operations exist — bijective or non-bijective. That is cast in silicon, unchangeable during operation, prior to any computation. This decision fixes the thermodynamic cost structure — not the computation, not the program.

[B] **Level 2 — state transition during operation.** Once the computer runs, state transitions occur clock cycle by clock cycle — uniformly, at the clock frequency, independently of what is being computed. Landauer costs arise only for **non-bijective** region maps — when a component, by its physical construction, reduces several input regions onto one target region Z. That is fixed by the hardware architecture, not by a computational operation. A transistor flipping from region A to region B executes a bijective map — no volume loss, no Landauer cost. What the descriptive level subsequently does with the component’s state is thermodynamically irrelevant.

Landauer costs run with the clock frequency, practically constant, as long as the computer is running — because the non-bijective region maps are fixed by the designer and are incurred with every clock cycle. CPU utilisation says little about this: it measures how many clock cycles are used for computations at the descriptive level — but at idle the clock transitions keep running. The thermodynamic difference between high and low utilisation is small as long as no physical parts are actually switched off. Clock gating, power gating — these are hardware decisions at level 1, not program decisions. The hardware determines the cost — not the program.

[B] Level 3 — description. The program, the computation, an AI controller may decide which hardware is used — switch off parts, choose routes, determine that for a given result the full bit width is not needed. This has nothing to do with Landauer. It is resource management at the descriptive level. The thermodynamic costs of the hardware in use are incurred regardless — fixed by construction and switching moment, independently of what the descriptive level does with the results.

49.1.3 How much is inside a physical region?

[K] A colloidal particle in a 1 µm trap physically contains roughly **7.9×10^9 orthogonal quantum states** *(verification script, check 7)*. If one coarsely names two zones "left" and "right", one describes that region with 1 logical bit. Partitioned into 1000 zones, one describes the same region with ~ 10 logical bits. The physical region is identical in both cases — the depth of description is the observer's choice.

[B] The cost of erasure attaches to the region, not to the description. The formula reads

$$\boxed{Q \geq k_B T \ln\left(\frac{W_{\text{before}}}{W_{\text{after}}}\right)}$$

W_{before} and W_{after} are physical region sizes in units of h^f. How one names that ratio — "erasing 1 bit" or "reduction by a factor of 2" — changes nothing about the physics.

[K] The analogue case shows this particularly clearly — and shows its complexity at the same time. How many distinguishable states an analogue signal contains depends on several factors, all of them physical: on the bandwidth B, the temperature T and the resistance R (Johnson–Nyquist noise, $U_{\text{rms}} = \sqrt{4 k_B T R B}$ [B20]), on the sensitivity of the measurement apparatus, on the signal-to-noise ratio of the concrete system, and on further external sources of interference. There is no universal resolution limit for analogue signals — it is a system property, not a constant of nature.

What can be said in general: if the resolution limit is set by the thermal noise of the bath itself (the physically deepest case), then the principle already stated in Section 49.6 applies: the same bath that collects the erasure cost limits the gradation of the analogue signal. The region count $W = L/\delta x$ and the erasure cost $k_B T \ln W$ scale with the same temperature quantity — the accounting is self-consistent. How many logical bits one ascribes to the analogue region is independent of this and remains a descriptive decision *(verification script, check 8: illustrative example with declared δx)*.

49.1.4 Shannon and Boltzmann are different ledgers

[B] [B18][B19] Shannon describes how much uncertainty a *description* carries: $H = -\sum_i p_i \log_2 p_i$ — a quantity about descriptions. Boltzmann describes how large a *physical region* is: $S = k_B \ln W$ — a quantity about regions. The two are identical only if one maps the description exactly onto the physical microstates and those are uniformly distributed. In every other case — coarse partitioning, unequal occupation, continuous systems — they are different things with different units.

[K] **What Landauer actually said in 1961** [B1]: he wrote explicitly “logically irreversible” — logically, not physically irreversible. He meant a map in which different logical input states lead to the same logical output state. That is a statement about descriptions.

[B] **And therefore, taken by itself, it is not a physical condition.** What is sufficient for heat production is solely the **non-bijectivity of the region map**: that several source regions fall onto a smaller target region. “Operation” is the wrong word at this point — an operation is a description, and descriptions cost nothing. Logical irreversibility coincides with the region map only under an additional realisation assumption that has to be declared explicitly: that the logical labels are mapped onto disjoint physical regions and that no region is carried along through the transition. If that assumption is violated, logical irreversibility carries nothing — Bennett [B3] executes any logically irreversible function as a sequence of bijective region maps, with no lower bound (Section 49.8). And conversely, every non-bijective region map costs, whether or not it is described as an operation at all. Every physically irreversible process produces heat, regardless of whether it is described as "logically irreversible"; that is exactly the point of Section 49.7.3. The physics beneath the description can be far richer than the description itself.

[B] **The error of the standard presentation** is therefore this: it treats the descriptive unit — the bit — as if it were a physical minimum quantity. It is not. A physical minimum quantity does exist: h^f. But it has nothing to do with bits. The region comes first; how many bits one ascribes to it is a subsequent decision.

49.2 The one idea from which everything follows

A bit is physical: two **distinguishable** states of a system. Two wells for a particle, two magnetisation directions, two voltage levels. All that matters is that the system can be in state "0" or "1" and that the two can be told apart.

Region reduction: no matter which region — A or B — the system was in before, afterwards it is in the target region Z. The target region Z is freely choosable; all that counts is the ratio W_{A+B}/W_Z.

```
A   ---+
       +--->   Z
B   ---+
```

Two regions are reduced onto one — the target region Z. This map is **not invertible**: whoever knows only the target region cannot reconstruct which source region the system was in. Exactly that — and nothing else — generates the cost (Section 49.4).

49.3 What "phase space" means here

Phase space is the space of all possible microstates (all positions and momenta of all particles). A macrostate such as "region A" is not a point in it but a **region**: many microstates (thermal jitter in the left well — every jitter configuration counts as "0").

[AXIOM] Boltzmann entropy measures this region:

$$S = k_B \ln W$$

W = phase-space volume of the region. Entropy here is a **measure of the size of the region occupied** — nothing mystical.

```
Phase space before:       Phase space after:
+-------+-------+         +-------+-------+
|       |       |         |       |       |
|   A   |   B   |   ⟶     |   Z   | empty |
|       |       |         |       |       |
+-------+-------+         +-------+-------+
 volume: 2W0 (A+B)         volume: W0 (Z only)
```

[B] Erasure halves the accessible volume:

$$\Delta S_{\text{region}} = -k_B \ln 2$$

Pure accounting over region sizes — no model, no hardware. (Generalisation to N levels and to non-uniform distributions: Section 49.6.)

49.4 Why this is not free: Liouville

The microscopic dynamics (classically Hamiltonian, quantum-mechanically unitary) has one hard property, **Liouville's theorem** [B2]:

> **[B]** The time evolution of a closed system **preserves phase-space volume**. Deforming, displacing, stirring — yes. Shrinking — never.

A refinement: Liouville holds exactly only for the closed system *without* a bath. For the subsystem "bit" alone — once coupled to the bath — it no longer holds. Only the total system bit + bath is closed; for that total the volume is preserved, while the bit subregion may shrink at the expense of the bath region.

Intuitively: the phase-space flow is an **incompressible fluid**. A blot of ink in water can be distorted arbitrarily; its volume remains.

The trap now closes:

- Erasure **demands**: $2W_0 \to W_0$ (halving, Section 3).
- Liouville **forbids**: volume shrinkage in a closed system.

[B] **Consequence: a closed system cannot erase.** Not an engineering problem — a theorem. The only way out: the system must not be closed. The volume has to go somewhere — into the bath (Section 49.5).

49.5 The bath as volume recipient

If the bit is coupled to a heat bath, Liouville applies to the **total system** bit+bath. If the bit region is to shrink by half (the uniform case: both regions equally large), the bath region must at least **double**. In the general case $\Delta S_{\text{bath}} \geq -\Delta S_{\text{bit}}$ (second law); the generalisation to $\ln(W_{\text{before}}/W_{\text{after}})$ follows in Section 49.6:

$$\Delta S_{\text{bath}} \;\geq\; +k_B \ln 2$$

A bath at temperature T enlarges its region by absorbing energy; the basic relation $dS = \delta Q/T$ translates this into heat:

$$\textbf{[B]} \quad Q \;\geq\; T\Delta S_{\text{bath}} \;=\; k_B T \ln 2$$

This is the Landauer bound [B1][B3]:

> **Liouville (volume is preserved) + accounting (bit region halved) ⇒ bath region doubled ⇒ heat $\geq k_B T \ln 2$ into the bath.**

At 300 K: $k_B T \ln 2 \approx 2.87 \times 10^{-21}$ J per bit *(verification script, check 1)*.

The two conditions can be read off directly:

1. **No erasure without a bath** — there is no other volume recipient (Section 49.4).
2. **Equality only quasistatically** — any finite speed generates additional dissipation $W \approx k_B T \ln 2 + B/\tau$ [B4][B5] (Section 49.10).

49.6 Why 2, actually? — The counting base

Binarity is **convention, not physics**. In general:

$$Q \ \geq \ k_B T \ln\left(\frac{W_{\text{before}}}{W_{\text{after}}}\right)$$

— a ratio of region sizes. N levels erased onto one: $\ln N$ *(verification script, check 4)*. The $\ln 2$ is the computer-science special case.

[B] What counts as a level? Not energy. Energy levels are unsuitable as a counting base: they may be degenerate or have unequal spacings without the count changing. The criterion is **distinguishability**: two states count separately if they can in principle be told apart without error. For **quantum systems** this means orthogonality in Hilbert space; for **classical systems** it means distinguishability within the thermal resolution limit set by the bath — i.e. the separation of the states must exceed the thermal fluctuations. A bit may sit in two degenerate states of equal energy — ideally so, in fact: holding then costs nothing, only forgetting does. Level spacings act only indirectly, through the thermal occupation probabilities. The true currency is therefore the Gibbs form [B6]

$$S \ = \ -k_B \sum_i p_i \ln p_i$$

with $\ln N$ as the special case of uniform distribution. Unequally spaced, unequally occupied systems automatically carry less entropy *(verification script, checks 5–6)*.

[B] The base discretisation exists — and it is h^f. Classically the phase-space volume carries units; W is defined only up to an arbitrary cell size. This does not disturb the accounting, because only differences enter — the cell cancels. Quantum mechanics fixes the cell absolutely: the number of orthogonal states in a region Γ of a system with f degrees of freedom is

$$N_{\mathrm{max}} = \Gamma/h^f \qquad \text{[B7]}$$

(verification script, check 7: colloid in a 1 µm trap, roughly 7.9×10^9 *orthogonal states — the “2” of the experiment is a coarse choice, not something fundamental)*.

49.6.1 Fundamental versus emergent two-valuedness

	Spin-½	**Colloid in a double well**
origin of the 2	fundamental: Hilbert space $\mathbb{C}^2$	emergent: 2 metastable macro-regions
validity	always	only while the barrier $\gg k_B T$ on the process timescale
underneath	nothing — no finer gradation	quasi-continuum of microstates

49.6.2 The quasi-analogue case

A value in a range L with resolution δx carries $\ln(L/\delta x)$ levels; region reduction costs $k_B T \ln(L/\delta x)$ *(verification script, check 8)*. δx is not free: the bath noise sets it — the same bath that collects the erasure cost limits the gradation. No analogue trick evades the bound.

49.7 Why the content is irrelevant

At **no point** did the entire calculation ask what meaning region A or B carries, whether the bit belongs to a correct computation, whether it is interpreted as “true” or “false”. Only this entered: **how many** distinguishable regions before, **how many** after.

Thermodynamics counts regions; it does not read them. Region A and region B are physically equivalent — physics knows no preferred direction, and the target region Z is freely choosable. What meaning one assigns to them (“true”/“false”, 0/1, high/low) is a decision outside thermodynamics:

> [B] **Region reduction costs the same for every assignment — regardless of how one interprets the regions.** There is no thermodynamic detector of content or of truth.

In the same language: a non-terminating computation dissipates energy without bound and has no truth value while doing so. Energy turnover and semantic validity lie on different levels — phase-space accounting here, representation there. A separation of categories: physical validity (hardware level) versus representational validity (content level).

49.7.1 Hardware and computational operation are different things

A fundamental error of reasoning in the standard presentation of Landauer: the hardware is treated as an image of the computational operation — as if the physical apparatus existed, and were thermodynamically effective, only while a logical operation is being executed. That is not so.

[B] **The hardware is not the computational operation.** A 64-bit processor is a physical system with a 64-bit state space. That state space exists independently of whether a meaningful operation is running, of whether the result requires 1 bit or 64, of what meaning the result carries, of whether anyone reads the result at all. The hardware is not an abstraction — it is silicon, wiring, fields, capacitors, interacting continuously with their environment.

[B] **The computational operation is an abstraction over the hardware.** It describes what the programmer or mathematician sees — input, output, function. The hardware does not know that abstraction. It knows only state transitions, charge displacements, heat release.

The consequence for Landauer. Asking "what does erasing a bit cost" takes the abstraction as the starting point and looks for the thermodynamic cost of the logical operation. That is a legitimate question — but it misses the physics of the real system. The real computer produces entropy continuously, independently of the logical abstraction currently running on it.

49.7.2 The same statement from the hardware side

The statement of Section 49.7 — thermodynamics reads no content — can be derived independently of the accounting, directly from the physical structure of the computer.

[B] The computer does not shrink. A 64-bit processor has, before and after every operation, the same number of transistors, the same number of wires, the same number of degrees of freedom. The phase space of the hardware stays constant — 2^{64} possible states before, 2^{64} after. The result may need 1 bit; the apparatus that produced it has stayed exactly as large.

[B] What actually goes into the bath is the correlation. The logical map is many-to-one: many different input states land on the same output. The 63 discarded bits are logically gone, but physically their trace still sits somewhere in the transistor states — only no longer reconstructible from the result. Exactly this non-reconstructibility, the correlation between input and output that is no longer contained in the result, goes as heat into the lattice. Not the silicon, but the forgetting of the correlation.

[B] The act of forgetting is content-neutral. A processor computing $2 + 2 = 4$ and one computing $2 + 2 = 5$ switch the same number of transistors, discard the same bit width and pay the same thermodynamic bill to the bath. Physics does not distinguish the two. A result in region A costs the same as one in region B — physics does not distinguish the regions.

This is not a new claim — it is a derivation from below. Landauer's abstraction "region reduction costs, content is irrelevant" follows from the concrete physics of the computer, without a formula. The accounting in Section 49.7 and the hardware structure here arrive at the same statement by different routes.

49.7.3 The computer always computes

A physical computer does not stop computing while it waits for a result. The clock runs, the transistors switch, the power supply delivers current — continuously, independently of whether "useful" work is happening. An idle processor computes idle loops. A waiting server sends keepalive packets. An operating system polls interrupts. There is no state "the computer is not computing" as long as it is switched on.

This means: the act of forgetting runs **always**. The thermodynamic bill — loss of correlation, heat into the lattice — does not stop, no matter whether the result is needed or not, no matter what meaning it carries.

This sharpens Landauer's formulation into practical irrelevance. He isolates a single discrete act of region reduction — "region reduction costs $k_B T \ln 2$". Reality is a **continuous current** of state transitions, losses of correlation and heat releases, three billion times per second in a modern processor, without pause.

A concrete example: dynamic RAM. A DRAM bit is a capacitor holding charge. But capacitors lose charge through leakage currents — continuously, unstoppably. Therefore every DRAM cell must be **refreshed** every 32–64 milliseconds: the refresh operation reads the state and writes it back. This happens cycle by cycle, automatically, for each of the billions of cells — no matter whether the data are currently needed, no matter what meaning they carry, no matter whether the program is asleep.

This is not a side effect — it is the physical nature of the system. The DRAM refresh is **not** an act of region reduction in Landauer's sense; it is an act of preservation, and its direct costs are primarily engineering costs (leakage currents, recharging). But the reason it has to happen at all is thermodynamic: the capacitor is permanently coupled to the heat bath. The charge — the physical region "bit charged" — is continuously pushed towards the common equilibrium by the thermal coupling. That is a permanent, unceasing interaction with the bath, independently of whether a logical operation is taking place. The refresh is the technical answer to this thermodynamic reality — it renormalises the region back, because the bath permanently blurs it.

But that is not yet the whole point. The refresh **itself** is not a passive reaction — it requires active computational work: a counter must run, the address of the next row to be refreshed must be computed, a timing signal must be generated, the memory bus must be blocked, the sense amplifier must be activated. That is real computational work — with real state transitions, real losses of correlation, a real entropy current into the bath. And for a bank of 8192 rows at a 64 ms refresh interval the memory controller issues roughly 1.3×10^5 refresh commands per second — without interruption, entirely independently of whether the running program is computing anything or asleep.

That is the actual point: it is not the refresh operation itself that is Landauer-relevant in the sense of individual bit erasures, but the fact that **the computer must compute continuously in order to maintain its own state** — refresh counters, clock generators, watchdog timers, operating-system schedulers, memory controllers. This is a permanent self-computation that has nothing to do with the user program. Landauer does not describe the thermodynamics of this self-computation, because he considered individual operations; it is an additional, more fundamental cost category: **the computer pays entropy for existing — not only for computing.**

[K] A DRAM cell that “does nothing” for an hour has nevertheless run through roughly 1.12×10^5 refresh cycles in that hour (at a 32 ms interval; at 64 ms roughly 5.6×10^4), producing entropy continuously. The stored information was the same throughout — but the thermodynamic bill kept running.

That is the physical computer as it really is: not an automaton that discretely erases bits, but a system that continuously fights thermodynamic decay and pays for it. The question is therefore not: what does region reduction cost? But: how large is the continuous entropy current of a physical computer as a function of its clock frequency, its bit width and its logical irreversibility — independently of what it happens to be computing?

That is an open question. Fredkin, Toffoli and Landauer himself thought about it [B3][B13]. The standard presentation still starts from the individual act of bit region reduction, because that is tractable. It is an abstraction that misses the physics of the real computer: the computer is not an automaton that erases individual bits on command — it is a physical system producing entropy continuously, whether one looks or not, whether the result is correct or not.

49.7.4 Quantum computers: decoherence as a continuous entropy current

For the quantum computer the same principle holds as for DRAM — the physical apparatus fights thermodynamic decay continuously — but the physics is more unforgiving.

What decoherence is. A qubit in superposition has a phase relation to its environment. Every smallest interaction — a phonon, a photon, a magnetic-field fluctuation — alters that phase. The qubit loses its quantum nature not because information is annihilated, but because the correlation between qubit and environment is distributed into the bath. Liouville again: the volume is preserved, but it spreads into qubit-plus-environment correlations that nobody controls any more.

The aggravation relative to DRAM. With classical DRAM one can refresh — read the state, write it back; it costs thermodynamically, and it works because 0 and 1 are robust. For a qubit this is impossible in principle: a measurement collapses the superposition. An unknown quantum state can neither be copied (no-cloning theorem [B14]) nor measured without altering it. Refresh in the classical sense does not exist.

What exists instead is **quantum error correction** [B15]. But it is thermodynamically even more expensive than DRAM refresh — it needs 100–1000 physical qubits per logical qubit, continuous syndrome measurements and classical correction operations. All of this runs on, cycle by cycle, for every logical qubit, even when no gate operation is taking place.

[K] **Decoherence time as the DRAM analogue.** A superconducting qubit typically has $T_2 \sim 100$ µs [B16] — after which the phase information is distributed in the bath. That is roughly 640 times shorter than a DRAM refresh interval of 64 ms, but qualitatively different: DRAM loses information slowly and steadily, the qubit loses it exponentially fast, and the “information” lost is not a bit but a continuous complex amplitude over a high-dimensional Hilbert space (bounded in practice by h^f, Section 49.6).

The entropy current in a quantum computer. For the classical computer the continuous entropy current is determined by clock frequency and bit width. For the quantum computer an additional term appears: the decoherence-induced entropy current, which runs proportional to the number of qubits and inversely proportional to T_2 — independently of whether a gate is being executed. A waiting quantum computer produces entropy more aggressively than a waiting classical computer, because the quantum nature itself fights the environment.

Decoherence is not region reduction in Landauer's sense — it is a loss of quantum correlations. But thermodynamically it pays the same price: the correlation goes into the bath, the bath absorbs the volume. The decisive difference from DRAM: with DRAM the information can be recovered (refresh), with a qubit it cannot (no-cloning theorem [B14]). Decoherence and Landauer erasure are different processes with the same thermodynamic mechanism.

Content-neutral as always. Decoherence does not distinguish between a qubit carrying one quantum state and one carrying another. It destroys both equally fast. The thermodynamic price of quantum computing does not attach to the content — it attaches to the physics of the system.

[H] **Open question.** For the classical computer the continuous entropy current is in principle reducible by reversible logic (Section 49.8). For the quantum computer this is more fundamentally limited: whether there is a matter-of-principle minimum price for maintaining quantum coherence — independently of the quality of error correction — is an open question of quantum thermodynamics [B17].

49.8 Region operations — volume-preserving or volume-reducing

Every physical change of state is a **map between phase-space regions**. The thermodynamically decisive question is exclusively: does the map preserve the region volume, or does it reduce it?

Whether that map belongs to a "computation", whether it is described as a logical operation, whether anyone calls it meaningful — for thermodynamics this is entirely irrelevant.

[B] **Volume-preserving maps** — the region is reshaped, not reduced. Liouville applies. No thermodynamic lower bound:

```
A ---> B    (bijective: each input region
onto its own output region)
B ---> A
```

Traditionally such maps are called "reversible gates" (NOT, CNOT, Toffoli) — physically they are **bijective region maps**. Whether they are part of a computation or not is irrelevant.

[B] **Volume-reducing maps** — several input regions land on the same target region Z. The volume shrinks. The bath must absorb the difference. Thermodynamic lower bound:

```
AA ---+
AB ---+---> Z      4 input regions → 2
output regions
BA ---+             volume shrinks ⇒ the
bath pays
BB -------> Z
```

Traditionally such maps are called "irreversible gates" (AND, OR, ERASE) — physically they are **non-bijective region maps**. Again: whether they are part of a computation or not is irrelevant.

[B] **Bennett** [B3] showed that any sequence of region maps can in principle be executed as a pure sequence of volume-preserving maps — intermediate states are carried along and released reversibly at the end. The thermodynamic costs do not arise at the intermediate steps but exclusively at the final relinquishing of regions:

> [B] **The lower bound does not attach to the map, but to the relinquishing of regions — to the forgetting.**

49.9 The accounting in one table

Step	**Bit region**	**Bath region**	**Total (Liouville [B2])**
before	$2W_0$	W_{bath}	$2W_0 \cdot W_{\text{bath}}$
reduction onto Z	$W_Z \le W_0$	?	must stay $\ge 2W_0 W_{\text{bath}}$
hence	W_Z	$\ge 2\,W_{\text{bath}}$	preserved (yes)
price	$\Delta S = -k_B \ln \frac{W_b}{W_Z}$	$\Delta S \ge +k_B \ln \frac{W_b}{W_Z}$	$Q \ge k_B T \ln \frac{W_b}{W_Z}$ into the bath

49.10 The experiments and their limits

The image. Two glasses stand on a table — region A and region B. Next to them, the world ocean. Region reduction means: one of the glasses is poured into the ocean. The volume of the glass does not vanish — it distributes itself across 1.4 billion cubic kilometres. Liouville is satisfied: not a drop is lost.

But: which glass it was — A or B — is distributed so finely in the ocean's noise that no real instrument can ever extract it again. Not because the information were annihilated (it sits in every current, every temperature gradient), but because the resolution limit of every real observer lies ten orders of magnitude below the signal.

And: afterwards, under the same external conditions, the sea cannot offer the same representational capacity for the two glasses. It can no longer display them separately — the target region Z is smaller than $A + B$. That is not ignorance; that is a physical fact about region sizes.

The price of the pouring: $Q \geq k_B T \ln 2$. Not because of ignorance — but because the sea had to absorb the volume, and heat flowed for it.

49.10.1 The orders of magnitude

[K] *(verification script, checks 2–3)* A real CMOS switching event lies at $\sim 10^{-17}$–10^{-16} J — four to five orders of magnitude above Landauer. All bit erasures in the world taken together would have Landauer costs in the watt to kilowatt range (depending on the assumed global erasure rate, declared in the script); in fact data centres consume some tens of gigawatts [B8] — a factor of 10^7–10^{10}. In every real machine the bound is invisible; it is a lower bound imposed by natural law, the remainder is engineering loss (CV^2 recharging, leakage currents, data transport).

[K] **The world-ocean image.** The bit volume that tips into the bath during the region reduction stands in so extreme a disproportion to the bath (1 degree of freedom against 10^{23}) that it lies **below the resolution limit** of every real instrument. The bath absorbs the volume — Liouville is satisfied — but no measuring device can still separate the signal from the bath's noise. Not because the instrument were imperfect, but because the structure of the system itself sets the resolution limit.

49.10.2 Epistemic versus ontological — a necessary caveat

[B] Not measurable does not mean not present. After the region reduction the microstates of the bath all still exist; every particle has a definite position and momentum; the information about which source region (A or B) the system was in still sits in the correlations — Liouville guarantees this explicitly. The bath has absorbed the volume, not annihilated it.

Landauer does **not** hold **because** information is ontologically annihilated. It holds **because** the bath must absorb the volume (Liouville) — that is a statement about mechanics, not about knowledge or perception. A real observer cannot resolve the correlations distributed in the bath, but that is the epistemic side; the thermodynamic side (heat flow) follows from Liouville alone.

[B] What is ontologically true, however: a smaller region can carry fewer states under the same conditions than a larger one. After the region reduction $\{A, B\} \to Z$ we have $W_Z \leq W_{A+B}/2$ — the target region is at most half the size of the source region. Under the same external conditions (same temperature, same pressure, same coupling to the bath) it therefore cannot carry the same number of distinguishable states as before. This is not observer ignorance but a physical fact about region sizes. Whoever wants to represent, after the region reduction and under the same conditions, the same thing as before, needs a different — larger — physical system.

The correlations distributed in the bath are real. They are merely so finely distributed that no observer with finite means, under the same conditions, can reconstruct them. That is the transition from the ontological to the epistemic: the region is smaller (ontological), and its reconstruction would require bath-sized knowledge (epistemic).

[K] Maxwell's demon [B12] and Bennett's resolution [B3]. A being that knows all 10^{23} microstates of the bath — a Laplace demon — could trace the bit back in thought. For it, erasure would be reversible and would cost nothing. Maxwell's demon shows exactly this: an observer with complete microstate knowledge can apparently reduce entropy at no cost. Bennett's resolution of 1982 [B3]: the demon must **erase its measurement notes** in order to begin a new cycle — and exactly that costs $k_B T \ln 2$. The actual site of entropy production is the erasure of the notes, not the measurement itself. The second law is thereby preserved, but the justification is sharper: not "information is lost", but "region reduction of information produces entropy".

In summary: Landauer is grounded **ontologically** by Liouville. The **irreversibility** is statistical, not ontologically absolute — for a demon with complete knowledge it would be liftable. For every real observer, however, the resolution limit lies so far below the signal that the statistical irreversibility is practically absolute. Both must be said.

49.10.3 The single-particle experiments

Bérut et al. 2012 [B4] (colloid in an optical double well, cycle times 10–40 s), Jun/Gavrilov/Bechhoefer 2014 [B5] (feedback trap, higher precision). Both approach the bound and confirm the $1/\tau$ form of the additional dissipation *(verification script, check 9)*. They work in the quasistatic limiting case — the smallest possible bath, extremely slow processes — in order to come anywhere near the bound at all. A real computer erases bits in a nanosecond into the world ocean, in a storm: the stirring dominates everything, the bound lies ten orders of magnitude below the noise.

Three declared points of scepticism (objections to the evidential force of the experiments, not to the principle — the principle rests on Section 49.4):

1. **Extrapolation:** what is confirmed is a limit ($\tau \to \infty$) that no experiment enters; the bound is fitted as an asymptote.
2. **Signal $\approx$ noise:** $k_B T \ln 2$ is of the order of the thermal fluctuation of a single measurement; the result exists only as an ensemble average; some individual cycles "gain" energy.
3. **Success rate:** the protocols erase with ~ 90–95 % reliability; the accounting of failed erasures enters the balance.

[K] **Feedback extension.** With measurement and feedback the generalised second law $W_{\text{ext}} \leq -\Delta F + k_B T I$ applies (Sagawa–Ueda [B9]), experimentally confirmed [B10][B11]. In the accounting, information is a resource with an energy exchange rate of $k_B T$ per nat *(verification script, check 10)*.

49.11 Connection to open questions (internal)

- **Feedback accounting in fast systems:** computational operations at $\sim$ns timescales lie ten orders of magnitude away from the tested quasistatic regime; there the B/τ term dominates everything. Anyone wishing to measure a Landauer-like residual in fast recurrent systems must take the Sagawa–Ueda term $k_B T I$ [B9] explicitly into the conventional budget before a residual can count as new — otherwise one is measuring feedback thermodynamics, not new physics.
- **Contact with FFGFT: none.** Landauer accounting lives at the level of statistical mechanics; the ξ scheme is untouched (its own level, its own accounting — analogous to P20/P39).

49.12 Summary in five sentences

Region reduction shrinks the phase-space region of the system; Liouville forbids volume shrinkage in closed systems; hence a bath must take over the volume, and the price is heat $\geq k_B T \ln(W_\text{b}/W_\text{a})$ — the $\ln 2$ only as the binary special case. Counting proceeds by distinguishability (orthogonality), not by energy; the absolute base discretisation is supplied by h^f, everything coarser is emergent counting at a chosen level. Not measurable does not mean not present: the microstates of the bath all still exist, the information about the erased bit sits in correlations — Landauer holds because of the mechanics (Liouville), not because of ignorance; the irreversibility is statistical, not ontologically absolute (Maxwell's demon, Bennett's resolution). The accounting counts regions and reads no content — every region assignment costs the same. In reality the bound is never reached, because any finite speed generates additional dissipation that dominates by orders of magnitude — the bit tips into the world ocean in a storm, and the resolution limit of the bath lies ten orders of magnitude above the Landauer bound.

49.13 Layer-status summary

The following table is a *selection* of the load-bearing statements; further passages are marked in the running text.

Statement	Status
Bit $\neq$ physical region (Shannon vs. Boltzmann)	[B]
$S = k_B \ln W$ as the starting point	[AXIOM]
Region reduction $\{A, B\} \to Z$ (physics, not description)	[AXIOM]
Region reduction $\Rightarrow \Delta S = -k_B \ln(W_{A+B}/W_Z)$	[B]
Liouville's theorem	[B]
Non-bijectivity of the region map is sufficient, not logical irreversibility	[B]
A closed system cannot erase	[B]
Landauer bound $Q \geq k_B T \ln 2$	[B]
Numerical value 2.87×10^{-21} J at 300 K	[K]
Counting base = orthogonality, not energy	[B]
h^f as the base discretisation	[B]
Gibbs form as the general entropy	[AXIOM]
Content (true/false) does not enter the calculation	[B]
Hardware $\neq$ computational operation	[B]
The computer does not shrink	[B]
Correlation into the bath (not the silicon)	[B]
The act of forgetting is content-neutral	[B]
DRAM $\sim 1.12 \times 10^5$ refresh cycles per hour (32 ms)	[K]
No-cloning theorem	[B]
$T_2 \sim 100$ µs for superconducting qubits	[K]
Minimum price for maintaining coherence	[H]
Volume-preserving maps: no lower bound	[B]
Volume-reducing maps: the bath pays	[B]
Epistemic vs. ontological: correlations remain	[B]
Sagawa–Ueda term $k_B T I$	[K]

Balance of the table: 17× **[B]** (algebraic/mathematical from declared foundations) · 4× **[K]** (empirically tested or primary sources) · 3× **[AXIOM]** (declared starting points) · 1× **[H]** (open research question) = 25 entries.

Appendix: Corrections relative to the working note

This A-series edition follows the working note of 24 July 2026 completely in content. The following points were changed and are declared openly here so that the changes remain checkable.

1. **State count in the colloid region unified.** The note gave 7.9×10^9 in one place and $\approx 10^5$ orthogonal states in another, both citing check 7. Here 7.9×10^9 is used throughout. *To be checked:* which value the check script actually contains.

2. **Decoherence time versus refresh interval.** The note gave $T_2 \sim 100$ µs as "seven times shorter" than 64 ms; the correct factor is roughly 640. Corrected.

3. **DRAM rates separated.** The note mixed $\sim 15{,}000$ refresh operations per second with $\sim 112{,}000$ cycles per hour and cell; the two together are inconsistent by a factor of 10^3. Separated here: per cell roughly 1.12×10^5 cycles per hour at a 32 ms interval (at 64 ms roughly 5.6×10^4); the controller issues roughly 1.3×10^5 refresh commands per second and bank at 8192 rows and 64 ms.

4. **Layer balance corrected.** The note counted 15× [B] and 8× [K]; the table in fact contains 16× [B] and 4× [K] across 24 entries (with the additional entry from item 11, now 17× [B] across 25 entries). Recounted and corrected; at the same time declared as a *selection*, since further passages are marked in the running text.

5. **Source list completed.** [B11], [B12], [B13] and [B20] were cited in the text but not listed; entry [B17] inadvertently contained the texts of [B13] (Fredkin/Toffoli) and [B12] (Maxwell's demon). All four added, [B17] cleaned up. The reconstructed entries are marked as such in the source list and are to be checked against the working files before publication.

6. **The marker [H].** The A-series scheme knows [SETZUNG], [K], [B] and [S]. The note additionally uses [H] for "open research question". Retained here and declared in the legend

— *open* whether [H] is taken into the scheme or the passage is moved to [S].

7. **Order.** The note placed section 6c before 6b. Brought into logical order here (hardware $\neq$ computational operation, then the hardware side, then continuous operation, then quantum computers) and carried as subsections of Section 49.7.
8. **Figures.** The box and arrow diagrams of the note used Unicode drawing characters; here they are set in pure ASCII so that the build stays free of missing glyphs. Content unchanged.
9. **Missing paragraph head.** The paragraph on orders of magnitude (CMOS, data centres) began in the note without a head; here it is given as the subsection "The orders of magnitude" with **[K]** .
10. **Language.** Individual grammatical errors of the German source were corrected silently.
11. **"Operation" replaced by "region map" (Landauer paragraph).** The note wrote that logical irreversibility is "a sufficient condition for heat production in a certain class of *operations*". This contradicts the document's own programme — an operation is a description, and descriptions cost nothing. What is sufficient is solely the non-bijectivity of the *region map*. At the same time the claim is stated more sharply: logical irreversibility coincides with the region map only under a declared realisation assumption (disjoint regions, no region carried along); without it, it carries nothing, because Bennett executes any logically irreversible function bijectively. The paragraph was moreover marked **[K]** throughout, although only the Landauer quotation is a primary-source statement; the derived half now carries **[B]** . A corresponding entry has been added to the layer-status table.

Not changed: all physical statements, the layer assignments of the individual passages, the formulae, the line of argument, and the finding "contact with FFGFT: none".

Sources

[B1] R. Landauer, *Irreversibility and Heat Generation in the Computing Process*, IBM J. Res. Dev. **5**, 183 (1961).

[B2] Liouville's theorem: standard result of Hamiltonian mechanics; e.g. Landau/Lifshitz, *Mechanics*, §46; quantum-mechanical counterpart: unitarity of time evolution.

[B3] C. H. Bennett, *The Thermodynamics of Computation — a Review*, Int. J. Theor. Phys. **21**, 905 (1982); id., *Notes on Landauer's principle*, Stud. Hist. Phil. Mod. Phys. **34**, 501 (2003).

[B4] A. Bérut et al., *Experimental verification of Landauer's principle*, Nature **483**, 187 (2012).

[B5] Y. Jun, M. Gavrilov, J. Bechhoefer, *High-Precision Test of Landauer's Principle in a Feedback Trap*, Phys. Rev. Lett. **113**, 190601 (2014).

[B6] Gibbs/Shannon entropy: standard; e.g. E. T. Jaynes, *Information Theory and Statistical Mechanics*, Phys. Rev. **106**, 620 (1957).

[B7] Semiclassical state counting Γ/h^f: standard; e.g. Landau/Lifshitz, *Statistical Physics*, §7.

[B8] Data-centre energy consumption: IEA reports 2024/25, of the order of several hundred TWh per year worldwide ($\approx$ some tens of GW continuous power). The figure serves only for an order-of-magnitude comparison.

[B9] T. Sagawa, M. Ueda, *Second Law of Thermodynamics with Discrete Quantum Feedback Control*, Phys. Rev. Lett. **100**, 080403 (2008); id., *Generalized Jarzynski Equality under Nonequilibrium Feedback Control*, Phys. Rev. Lett. **104**, 090602 (2010).

[B10] S. Toyabe et al., *Experimental demonstration of information-to-energy conversion*, Nature Physics **6**, 988 (2010).

[B11] *(reconstructed)* J. V. Koski et al., *Experimental observation of the role of mutual information in the nonequilibrium dynamics of a Maxwell demon*, Phys. Rev. Lett. **113**, 030601 (2014).

[B12] *(reconstructed)* Maxwell's demon — origin of the thought experiment: J. C. Maxwell, *Theory of Heat*, Longmans (1871); standard presentation in H. S. Leff and A. F. Rex (eds.), *Maxwell's Demon 2*, IOP Publishing (2003).

[B13] *(reconstructed)* E. Fredkin and T. Toffoli, *Conservative Logic*, Int. J. Theor. Phys. **21**, 219 (1982) — reversible gates and the question of the continuous entropy current of physical computers.

[B14] W. K. Wootters and W. H. Zurek, *A single quantum cannot be cloned*, Nature **299**, 802 (1982) — no-cloning theorem.

[B15] P. W. Shor, *Scheme for reducing decoherence in quantum computer memory*, Phys. Rev. A **52**, R2493 (1995); A. M. Steane, *Error correcting codes in quantum theory*, Phys. Rev. Lett. **77**, 793 (1996) — quantum error correction.

[B16] F. Arute et al. (Google AI Quantum), *Quantum supremacy using a programmable superconducting processor*, Nature **574**, 505 (2019) — typical T_2 times of superconducting qubits ~ 100 µs.

[B17] M. Esposito and C. Van den Broeck, *Three faces of the second law: I. Master equation formulation*, Phys. Rev. E **82**, 011143 (2010); P. Kammerlander and J. Anders, *Coherence and measurement in quantum thermodynamics*, Sci. Rep. **6**, 22174 (2016) — quantum thermodynamics and the minimum price of maintaining coherence.

[B18] C. E. Shannon, *A Mathematical Theory of Communication*, Bell System Technical Journal **27**, 379 (1948) — definition of the bit as $\log_2$ of the distinguishable states; explicitly information-theoretic, not thermodynamic.

[B19] E. T. Jaynes, *Information Theory and Statistical Mechanics*, Phys. Rev. **106**, 620 (1957) — connection of Shannon entropy to Boltzmann: holds exactly only for a uniform distribution over physical microstates; not for a coarsely chosen logical resolution.

[B20] *(reconstructed)* Johnson–Nyquist noise: J. B. Johnson, *Thermal Agitation of Electricity in Conductors*, Phys. Rev. **32**, 97 (1928); H. Nyquist, *Thermal Agitation of Electric Charge in Conductors*, Phys. Rev. **32**, 110 (1928).

The entries marked (reconstructed) *were missing from the working note and have been supplied from the citation context; they are to be checked against the working files before publication.*

Chapter 50

A283 · Thermodynamics in FFGFT

Abstract. Thermodynamics in FFGFT is anchored on five levels – not as a postulate, but as a consequence of geometry. This document consolidates the distributed individual results, organizes them into a coherent argumentative chain, and identifies where the levels interlock and where they remain separate. A concluding section acknowledges that both framework descriptions – expanding space and fractal refinement – are models that interpret the same observations in structurally opposing ways.

Layer markers. **[SET]** declared starting point — **[B]** algebraically/-mathematically derived from declared foundations — **[K]** numerically calculated from corpus constants — **[Q]** external primary source — **[S]** plausibility sketch — **[H]** open research question.

50.1 Why a comprehensive document

The thermodynamic content of FFGFT has been developed across several documents: Doc. 157 addresses the arrow of time, A271–A282 cover Landauer accounting, and Doc. 312/313 treat cosmological thermodynamics. Each document is self-contained; none provides an overview of how the parts connect.

[B] Three questions remain unanswered without a comprehensive document: (1) Where do the levels touch, and where do they remain separate? (2) Which statements depend on the ξ-scheme, and which are independent of it? (3) What is the epistemic status of each statement? This document answers these questions and contains no new substantive claims.

50.2 Level 1 – The arrow of time is geometric (Doc. 157)

50.2.1 The standard problem

All fundamental equations of physics are time-reversal invariant. The second law ($\Delta S \geq 0$) is statistical in nature and shifts the problem to the unexplained low initial entropy (Penrose: $\sim 10^{-10^{123}}$).

50.2.2 The FFGFT answer

[B] Three geometric arguments interlock (Doc. 157):

50.2.2.1 Argument 1: Algebraic asymmetry

$\tilde{T} \cdot m = 1$ enforces $\tilde{T} > 0$ because mass is definitively positive. Time reversal would require $m \to -m$ – physically impossible.

50.2.2.2 Argument 2: Topological winding direction

The vacuum field winds with $\dot{\theta} = m = 1/\tilde{T}$ along the T^4. This chirality is topologically protected.

50.2.2.3 Argument 3: Fractal dimensional asymmetry

$D_f = 3 - \xi_0 \approx 2.9999$ breaks exact mirror symmetry; path integrals for forward and backward directions differ by $O(\xi_0)$.

50.2.3 Consequence

[B] The logic reverses: $\Delta S \geq 0$ is in FFGFT a *consequence* of geometry, not a postulate. The "Past Hypothesis" becomes unnecessary – the low initial entropy is geometrically enforced, not an initial condition.

50.3 Level 2 – Landauer as phase-space accounting (A271)

50.3.1 The phase-space formulation

[SET] A physical system occupies a region in phase space of size W (measured in h^f). The region exists independently of how one labels it.

[B] A mapping that reduces multiple input regions $\{A, B\}$ to a single target region Z reduces the volume: $\Delta S_{\text{system}} = -k_B \ln(W_{A+B}/W_Z) \leq 0$. Liouville's theorem forbids volume shrinkage in a closed system. A heat bath must absorb the volume:

$$\boxed{Q \geq k_B T \ln\left(\frac{W_{\text{before}}}{W_{\text{after}}}\right)}$$

with $\ln 2$ as the binary special case.

50.3.2 Three separations

[B] (1) *Bit ≠ physical region.* Shannon and Boltzmann entropy are identical only under uniform distribution over physical microstates. (2) *Content neutrality.* Thermodynamics counts regions, does not read them. Bit "true" and bit "false" cost identically. (3) *Three levels separated.* Construction / state transition / description. Landauer costs occur at levels 1 and 2, not at 3.

50.4 Level 3 – Phase-space accounting universal (A280–A282)

[B] A280: The region comes first. A glass into the ocean – the volume does not disappear, it distributes over 1.4×10^9 km^3. The price: the bath must absorb the volume, heat flows. The counting basis is distinguishability (orthogonality), not energy; absolute fundamental discretization: h^f.

[B] A281: Landauer's own restriction – logical irreversibility is a description property, not a physical one. Sufficient for heat is solely the non-bijectivity of the region mapping. Bennett's result: Every logically irreversible function can be executed as a sequence of bijective mappings; costs stick to *forgetting*, not to the operation.

[B] A282: The token computer as a limiting case. Accounting holds exactly, Landauer's bound does *not* apply there (engineering costs dominate). The limit to the Brownian computing sphere continuously connects to Landauer's domain of validity.

[H] Open question: Minimal coherence cost for quantum coherence independent of error correction (A271, § 7.4).

50.5 Level 4 – Cosmological thermodynamics (Doc. 312/313)

50.5.1 Thermal history as geometric condition

[K] Doc. 313: Three phases on the observable half-cycle ($\tau_c/2 \approx 46$ Gyr):

Phase	Range	θ/π	Character
Radiation domination	$z > 3000$	> 0.90	$m \to m_{\text{min}}$, clocks slowest
Transition	$z \approx 3000$–1090	0.90–0.979	Recombination, scattering surface
Matter domination	$z < 3000$	< 0.90	$m(\theta)$ grows, structure formation

50.5.2 The five-step chain

[K] Complete derivation of Ω_m^* from ξ_0 alone:

	Step	Result
(1)	$H_0 = (\pi/2)\, c\, \xi_0^{10}/\bar{\lambda}_e$	$h = 0.6682$
(2)	$k_B T_0 = (16/9)\, \xi_0$ (Doc. 061)	$T_0 = 2.751$ K
(3)	$\Omega_r \propto T_0^4/H_0^2$	9.643×10^{-5}
(4)	$K_{\text{frak}} = 1 - 100\, \xi_0$ (A040)	0.98667
(5)	$\int_0^\infty dz/E(z) = \pi/K_{\text{frak}}$	$\boxed{\Omega_m^* = 0.3136}$

Planck: $\Omega_m = 0.315 \pm 0.007$ ($-0.19\,\sigma$). No step contains a parameter fitted to Ω_m. Radiation ($\Omega_r \propto T_0^4/H_0^2$) is the link between the particle and cosmic sectors.

50.5.3 Overdetermination and open core

[K] T_0 is determined via two independent paths: Particle sector (Path 1: $k_B T_0 = (16/9)\xi_0$) and cosmic sector (Path 2: antipode condition). Both yield Ω_m agreeing to $0.1\,\%$ – the same overdetermination pattern as α in A130.

[K] Open (D(ii)): Coarse-grained entropy on the return half-cycle. Poincaré discrepancy: $\sim 10^{10^{104}}$ dynamical times versus $\tau_c \approx 92$ Gyr. Periodicity is a boundary condition, not a consequence of dynamics. Three cases (A050/Doc. 295): Case A ($d = 0$, 1 revolution), Case B (ξ frozen, 75 revolutions), Case C (ξ runs, logarithmic spiral – removes entropy requirement, but costs τ_c).

50.5.4 Falsification points

[K] Low-ℓ cutoff: $\ell_{\text{min}} = 2\pi D_{\text{LSS}}/L^* = 3.08$ – hits the quadrupole suppression unexplained since COBE (1992). Primary falsification test: $m_\tau = 1776.97$ MeV (Belle-II).

50.6 Level 5 – What separates the levels

[B] Landauer accounting (Levels 2 and 3) does not touch the ξ-scheme. It holds in any system with Hamiltonian or unitary dynamics – independent of the torus geometry. Explicitly logged in A271: "Contact with the FFGFT ξ-scheme: none."

The arrow of time (Level 1), by contrast, is directly tied to $D_f = 3 - \xi_0$ and $\tilde{T} \cdot m = 1$ – FFGFT-specific. Cosmological thermodynamics (Level 4) depends entirely on ξ_0, H_0, and the torus geometry.

Level	**Document**	**ξ-dependence**	**Status**
Arrow of time	Doc. 157	direct	[B]/[S]
Landauer	A271	none	[B]
Universal accounting	A280–A282	none	[B]
Cosmol. thermodynamics	Doc. 312/313	direct	[K]
Information physics	Doc. 243–301	indirect	[S]/[B]

50.7 Argument chain: from geometry to heat

[B] The five levels form a chain from FFGFT geometry to classical thermodynamics: (1) ξ_0 and T^4 fix the geometry. (2) $\tilde{T} \cdot m = 1$ and the torus winding define geometrically a direction of time. (3) In this directed time, there is more phase-space volume in the forward direction – $\Delta S \geq 0$ as geometric consequence. (4) Non-bijective phase-space mappings (Landauer) cost heat – independent of ξ_0, in the same phase-space picture. (5) The thermal history (Ω_m^*, Λ^*) follows from ξ_0 directly.

Liouville's theorem and the second law are in this chain *consequences*, not postulates.

50.8 Information thermodynamics in the IPI context (Doc. 243–301)

[S] A further group of documents connects FFGFT thermodynamics to the information physics community (IPI context): Doc. 251 (comparison with Vopson's information mass hypothesis), Doc. 247 (Landauer's principle in the context of information states), Doc. 290 (information question magnitude/phase), Doc. 301 (information reversal test). These documents are plausibility sketches or partial algebraic derivations; they open connections to current debates, without contributing new core results to FFGFT thermodynamics.

50.9 Phases, radiation, and temperature in the time cycle (Doc. 313 in detail)

50.9.1 Gradient on a circle

[B] The observable history shows a clear thermal sequence: radiation domination, transition, matter domination. This gives the impression of a directed process with a beginning. The FFGFT answer (Doc. 313): The gradient is not an evolution *from* a beginning, but a map *along* the time cycle – like climate zones along a meridian circle.

50.9.2 Three phases on the half-cycle

[K] The observable history fills a half-cycle ($\tau_c/2 \approx 46$ Gyr). Cycle positions:

		θ/π	
Event	z	smooth	fractal
Matter-radiation equality	~ 3400	0.93	0.92
Scattering surface (recombination)	1090	0.992	**0.979**
Particle horizon	z_{max}	1.012	**0.9986**

50.9.2.1 Phase 1 — Radiation domination (near the antipode)

In the mass-run reading: masses $m(\theta)$ are minimal, intrinsic times $\tilde{T} = 1/m$ maximal – clocks run slowest. The endpoint slope of the mass profile at the antipode:

$$\left|\frac{dm}{d\theta}\right|_{\text{antipode}} = \sqrt{\Omega_r} \approx 0.0096$$

No free parameter – the observed radiation density appears here as a *smoothness condition* of the periodic closure.

50.9.2.2 Phase 2 — Matter-radiation transition ($z \approx 3000$ to 1090)

With the fractal path correction $K_{\text{frak}} = 1 - 100\xi_0$ (A040, independently derived), the particle horizon sits at $0.14\,\%$ from the antipode – improvement by factor 9 over the smooth calculation ($1.2\,\%$), without fitted parameter.

The path correction affects path/structure, shortens path/-path. Invariance test: CMB peak positions remain unchanged (test passed without having been designed for it).

50.9.2.3 Phase 3 — Matter domination (present epoch)

Energy density $\propto (1+z)^3$. In the mass-run reading, masses grow with decreasing z; clocks run faster; structure formation unfolds.

50.9.3 Temperature as a structural quantity

[K] In the standard model: $T(z) = T_0(1+z)$ – adiabatic back-extrapolation. In FFGFT: T_0 comes from the atomic scale (Doc. 061):

$k_B T_0 = \frac{16}{9}\xi_0 \approx 2.370\times10^{-4}$ eV (measurement: 2.349×10^{-4} eV, $+0.92\,\%$)

The CMB temperature is a *structural quantity* of the ξ-scheme, not the endpoint of a cooling history.

[K] Range limit (Doc. 313): Equilibria at other locations of the time cycle are *not* derivable from today by forward-extrapolation. Unaffected: the Ω_m^*-chain (runs via ξ_0, not via adiabatic back-calculation).

50.9.4 Overdetermination of T_0

[K] Two independent paths: Path 1 (particle sector): $k_B T_0 = (16/9)\xi_0$, no cosmological input. Path 2 (cosmic sector): antipode condition, no particle physics input. Both yield Ω_m agreeing to $0.1\,\%$ (0.3136 vs. 0.3139) – the same pattern as α in A130. Leverage: $dT_0/T_0 \big/ d\Omega_m/\Omega_m \approx -12$.

50.9.5 Lower bound and Hawking

[K] Energy quantum of the time cycle (Doc. 312): $E_{\text{min}} = \hbar H_0 = 2.28\times10^{-52}$ J. Finite $z_{\text{max}} = m_e/m_{\text{min}} - 1 = 3.59\times10^{38}$. Singularity excluded: masses cannot run to zero.

[K] A KMS rule connects Gibbons–Hawking, Unruh, and Hawking: $T = \hbar/(k_B\tau)$ with τ as local time period. The compact time direction yields T_{GH} without horizon. The family ladder meets the cosmos at $r_s = R_H/2$ (exactly 0.500). Hawking evaporation for $M_\odot$: 56 orders of magnitude too slow for τ_c – sharpens the entropy obstacle D(ii).

50.9.6 Low-ℓ cutoff

[K] On a torus of side length L^*, modes with wavelength $> L^*$ are missing:

$$\ell_{\text{min}} = \frac{2\pi D_{\text{LSS}}}{L^*} = 3.08$$

No fitted parameter. Hits the quadrupole suppression ($\ell = 2$, factor $\sim 1/6$), an unexplained finding in ΛCDM since COBE (1992). Order statement [K|P20]; exact mode summation on the torus remains pending.

50.9.7 Robust and non-robust statements

[K] Three classes (Doc. 313):

Class	Examples	Robustness
Counts	$\lvert\mathrm{Aut}(D4)\rvert = 1152$, kissing number 24	exact
Ratios on same level	α, m_μ/m_e, Koide Q	to 10^{-10}
With scale bridge	G, H_0, absolute masses	only with anchor
Number and duration quantities	$\eta = n_B/n_\gamma$, window durations	not transferable

BBN observables (Y_p, D/H, ^{7}Li): not a pure ratio on the same level – the same holds for ΛCDM.

50.10 Mass formation, fractal refinement, and the appearance of expansion

50.10.1 What is missing at the antipode: metric sharpness

[K] All physical lengths that depend on mass scale with $1/m$: Compton wavelength $\bar{\lambda} = \hbar/(mc)$, Bohr radius $a_0 \propto 1/m_e$, intrinsic time $\tilde{T} = 1/m$. At the antipode ($m \to m_{\text{min}}$), all run to their maximum. Without localizable massive objects, *metric sharpness* is missing.

[H] What metric structure precisely means at $m \to m_{\text{min}}$ has not yet been elaborated in the corpus.

50.10.2 Photons: no freezing

[B] Photons carry no rest mass – $\tilde{T} \cdot m = 1$ does not apply to them directly. A photon at the antipode does not freeze; it propagates at c. CMB photons lose energy geometrically on their way to us (A265):

$$\frac{dE}{dx} = -\xi E \;\Rightarrow\; 1 + z = e^{\xi x}$$

Achromatic: all wavelengths stretched equally. From our perspective, everything runs uniformly. Looking back is looking at a location on the time cycle, not looking into a past.

50.10.3 Expansion as fractal refinement inward

[B] In the standard model: space expands outward, scale factor $a(t)$ grows. In FFGFT, there is no expanding space. With increasing $m(\theta)$, all characteristic lengths ($\propto 1/m$) become smaller. Matter becomes more structured, more finely resolved. This is **refinement inward**, not growth outward.

[B] $K_{\text{frak}} = 1 - 100\xi_0$ (A040): dimension deficit $D_f = 3 - \xi_0$, locally $\sim 0.003\,\%$, cumulative over 17 orders of magnitude ~ 100.

[B] Recursion $\xi_{n+1} = \xi_n(1 - 100\xi_n)$ (A050), fixed point: zero. Rolled up: winding closes after 75 revolutions (τ_c-cycle). Unrolled: logarithmic spiral $\rho = 75\,e^{\beta/2\pi}$ – what the standard model calls expansion.

[K] Both descriptions are observationally equivalent (A265). Rulers and clocks change in the same ratio; no local observer notices the scaling. What we see as expansion is the difference between our present masses and the redshifted light from the antipode.

[S] Image: Mandelbrot zoom. Zooming inward reveals ever finer structures, no outward growth. Antipode = unzoomed view (minimal differentiation, pure radiation). Present position = highly zoomed view (sharp atomic and stellar structures). The standard model turns the image around – a coordinate choice, not a physical statement.

50.11 Minimal length and fractal refinement: an apparent contradiction

[S] At first glance, there appears to be a tension: On the one hand, FFGFT speaks of a *minimal length* at the antipode ($\bar{\lambda} =$

$\hbar/(m_{\text{min}}c)$), on the other hand, of a *fractal refinement inward* along the time cycle. Is this a contradiction?

The answer is **no** – the key lies in the *reference-frame logic* of the theory.

50.11.1 What does "minimal length" mean?

The "minimal length" in FFGFT is *not* an absolute Planck length, but a **mass-dependent quantity**:

$$\bar{\lambda} = \frac{\hbar}{mc}, \qquad a_0 \propto \frac{1}{m_e}, \qquad \tilde{T} = \frac{1}{m}.$$

At the antipode ($m \to m_{\text{min}}$), these lengths become *maximal*. There, massive structures are least sharply resolved – a regime of *minimal metric sharpness*, dominated by radiation. The "minimal length" is thus actually the *largest characteristic length* of the system.

50.11.2 What does "fractal refinement inward" mean?

With increasing mass $m(\theta)$ along the time cycle, all characteristic lengths become *shorter*:

$$\bar{\lambda} \propto \frac{1}{m(\theta)}, \qquad a_0 \propto \frac{1}{m_e(\theta)}.$$

This means: matter becomes *more differentiated*, *more structured*, *more finely resolved* – like a **zoom inward**. Space itself does not expand; the *scale of objects* changes.

50.11.3 Why is this not a contradiction?

The apparent tension dissolves once one distinguishes two *different perspectives*:

Perspective	Meaning in FFGFT
Minimal length (at the antipode)	Maximal Compton wavelength, minimal structuring, radiation-dominated regime
Fractal refinement inward	Increasing differentiation, growing masses, shorter wavelengths, present matter domination

The **direction** is unambiguous:

$$\text{Antipode } (m_{\text{min}},\ \bar{\lambda}_{\text{max}}) \longrightarrow \text{ present position } (m_0,\ \bar{\lambda}_0)$$

Here, *structural resolution increases* (fractal inward), while *Compton lengths decrease*. This is not a contradiction, but a **complementary description** of the same process.

50.11.4 Comparison with the standard model

The standard model says: "The space expands, wavelengths are stretched."

FFGFT says: "Masses grow, wavelengths shrink, structure becomes finer."

Both descriptions are **observationally equivalent** (A265), because *clocks and rulers change together* – no local observer can measure the scaling.

50.11.5 Conclusion

The concepts of "minimal length" and "fractal refinement inward" are *not* a contradiction, but two sides of the same geometric coin:

- **Length perspective:** from large (antipode) to small (today)
- **Structure perspective:** from coarse (antipode) to fine (today)

Both follow from the same mass profile $m(\theta)$ and are consistently connected in FFGFT.

[S] This clarification makes visible the internal consistency of FFGFT geometry and resolves a common point of misunderstanding.

50.12 Open points

[H] Three questions remain open: (1) Minimal coherence cost for quantum coherence independent of error correction (A271, § 7.4). (2) Continuous entropy flow as a function of clock cycle, bit width, and logical irreversibility (A271, § 8). (3) Metric structure in the limit $m \to m_{\text{min}}$ – whether a regime of pure field geometry without localizable particle modes exists.

50.13 Layer status summary

Statement	Status
$\tilde{T} > 0$ enforces geometric arrow of time	**[B]**
Torus winding: topologically protected direction of time	**[B]**
$D_f = 3 - \xi_0$ breaks time-reversal symmetry	**[B]**
$\Delta S \geq 0$ as geometric consequence	**[B]**
Landauer: $Q \geq k_B T \ln 2$ from Liouville	**[B]**
Content neutrality of thermodynamics	**[B]**
Bennett's reversible logic	**[B]**
$\Omega_m^* = 0.3136$ from ξ_0-chain	**[K]**
$\Lambda^* = 1/R_H^2$ geometric carrier	**[K]**
Overdetermination T_0: two sectors to $0.1\,\%$	**[K]**
Singularity excluded by $\hbar H_0$	**[K]**
$\ell_{\text{min}} = 3.08$ (Low-ℓ cutoff)	**[K|P20]**
Slowest clocks at the antipode	**[K]**
Compton wavelength, Bohr radius $\propto 1/m$	**[B]**
Photons achromatically redshifted: $1 + z = e^{\xi x}$	**[B]**
Expansion = fractal refinement inward	**[B]**
Recursion: fixed point zero, spiral in unrolled case	**[B]**
Observational equivalence with ΛCDM	**[K]**
Landauer does not touch ξ-scheme	**[B]**
Minimal length = maximal Compton wavelength	**[S]**
Fractal refinement = increasing structure	**[S]**
No contradiction – two sides of same geometry	**[S]**
Coarse-grained closure D(ii): open	open
$m = 0$ structurally excluded: no empty state (A060)	**[B]**
Double cutoff: $L_0 \Rightarrow E_{\text{max}}$, $L^* \Rightarrow E_{\text{min}}$	**[K]**

Statement	Status
$m_{\text{min}} = \hbar H_0/c^2$ from IR cutoff (Doc. 312/313)	**[K]**
Local field modes at thinnest point resolvable?	**[H]**
Past Hypothesis as geometric necessity	**[S]**
Mandelbrot zoom as image	**[S]**

Balance: 14× **[B]** · 8× **[K]** · 6× **[S]** · 2× **[H]** · 1× open.

50.14 Concluding remark: Two opposing models

[S] An honest acknowledgment belongs at the end.

For us – as observers with our present masses and clocks – our perspective naturally counts. We think in light paths. We measure redshifts. We experience time as running forward. This is real in the sense that it is what we measure, and our instruments are calibrated to it.

The standard model (ΛCDM) takes exactly this perspective as its starting point: expanding space, Big Bang as beginning, redshift as space stretching, CMB as snapshot of an early epoch. It is a model that agrees excellently with observations.

FFGFT proposes a different coordinate choice: no expanding space, no Big Bang as an event, redshift as geometric energy loss, CMB as a view of a location on the time cycle. This description also agrees with the standard tests (A265, cosmological degeneracy).

[B] What distinguishes the two models:

- ΛCDM generalizes from the observer perspective *outward*: expanding space.
- FFGFT generalizes from geometry *inward*: compact torus, fractal refinement.

Both are models. Both leave direct observation behind and add an interpretative layer.

[S] What remains: two structurally opposing ways of describing the same thing. In one, everything expands outward;

in the other, everything refines inward. In one, time has a beginning; in the other, it is a closed cycle. In one, the CMB is a window into the past; in the other, a window onto a location.

This is not a flaw – it is an open question, formulated precisely enough to be decidable. The falsification tests (m_τ, $\ell_{\min}$, Ω_m^*-chain) will show whether the geometric inward perspective of FFGFT carries more than the expanding outward perspective – or less.

50.15 References

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A040 Pascher, J., *Fractal correction*. A-series, 2026.

A050 Pascher, J., *Recursion*. A-series, 2026.

A265 Pascher, J., *Redshift as static geometry effect*. A-series, 2026.

A271 Pascher, J., *Landauer and phase-space accounting*. A-series, July 2026.

A280–A282 Pascher, J., *Phase-space accounting, carrier and information, The computing sphere*. A-series, July 2026.

50.16 Use of AI tools

This document was created with the assistance of AI-based tools. Questions, foundational decisions, and substantive choices originate with the author; the tools assist in formulation and organization. Responsibility for content and errors lies entirely with the author. Full wording in A010.

Chapter 51

A284 · Measurement, Projection and the Origin of Probabilism

Abstract. This document closes a gap that A070, A090 and A160 each only hint at: the closed argument for *why* measurement in FFGFT affects the whole system, and *why* evaluation is probabilistic despite deterministic field equations. The three levels are carried separately and brought together at the end in the context of the instantaneity question of Bell correlations.

Core theses: **(1)** Measurement is a geometric Type-II projection (A090) — it acts on the global state on T^4, not on a local subsystem. **(2)** Probabilism does not arise from fundamental randomness but from the information loss of the projection: the Born rule is the statistics of Type-II loss. **(3)** What appears as instantaneous action between two measurement points A and B is the statistical signature of a global mode, read out at two projection sites of the same system — no process runs from A to B.

Layer markers. [STIPULATION] declared starting point — **[B]** algebraically/mathematically derived — **[K]** numerically computed from corpus constants — **[S]** plausibility sketch — **[H]** open research question.

51.1 What this is about

[STIPULATION] In standard quantum mechanics the measurement process is the most conceptually unsatisfying element: a unified, deterministic evolution of the wave function breaks down at measurement, a probabilistic outcome appears, and two distant particles seem to know about each other instantaneously. Three decades after Bell violation, the question of what physically happens remains open.

[B] FFGFT already has the elements of an answer distributed across other documents: A090 (projection chain, Type I/II/III), A070 (measurement as pole transition, Born rule from phase ignorance), A160 (Bell correlations as topological connection), A030 (winding classes without location). What is missing is the closed chain of argument. This document draws it.

51.2 Light rays as observer abstraction — the EM field as carrier

51.2.1 What a light ray actually is

[S] A light ray is not a physical quantity. It is a *geometric abstraction* distilled from the EM field under specific conditions — namely when the wavelength λ is small compared to all relevant geometric dimensions (apertures, objects, observer distance). This condition is *ray optics as a limiting case of wave optics* — mathematically: $\lambda \to 0$ in the wave equation yields the eikonal equation, from which rays follow as normal trajectories of phase surfaces.

[B] The EM field is the carrier. It exists and evolves according to Maxwell's equations, entirely independent of observers. The light ray appears only when an observer or detector arrangement makes a position measurement: the detector responds to the field at its location and *constructs* backwards a ray as the geometric line connecting source and detector. The ray is the result of this construction — not its precursor.

[B] This is not new — it is fully present in classical wave optics. What FFGFT adds: field evolution on T^4 has no locations in the standard sense. A photon has no trajectory; it has a mode structure on the torus. What we observe as a "light path" is the statistical accumulation of detection events in a spatial region — a projection statistic, not a physical trajectory (A090, Type-II projection).

51.2.2 Why it is so hard to let go of the light-path picture

[S] The light-path picture sits deep for four reasons:

(1) Evolutionary. The human brain is optimised for causality through spatial transfer: stone flies, hits, acts. "Cause before effect, carrier connects, distance separates" — these heuristics were survival-relevant for 10^5 years. They sit deeper than any theory.

(2) Linguistic. Every natural language has subjects performing predicates: "Light goes from A to B." The idea that a field exists and an observer reacts locally — without "something" travelling from A to B — is not grammatically natural. Even physically correct sentences sound like negations, not positive statements.

(3) Historical. Newton thought in corpuscles — light particles on trajectories. Maxwell solved this: fields instead of trajectories. But then photons arrived, and everyday physics regressed to the trajectory picture: "The photon goes through the double slit." Wave optics became a calculation method, the particle trajectory became the intuition. This regression is deeply embedded in physics education.

(4) Visibility only via detours. Light rays become visible only through intermediaries: dust, smoke, droplets — scattering centres that locally absorb and re-emit the field. What we see in a searchlight beam is millions of scattering events, each at a different location. The "beam" is the statistical line of these events. Without scatterers the EM field is invisible — physically real, but not an object of observation.

51.2.3 Why do lenses work then?

[B] The lens question is the most precise test of the thesis. If light rays are abstractions and the EM field is the carrier — why does a lens focus?

The answer lies in two layers:

Layer 1 — Wave optics (complete description): The EM field encounters the glass medium of the lens. Inside the glass the phase velocity is $v = c/n$, less than in vacuum. The wave front — a surface of equal phase — slows down more in the thicker part of the lens than in the thinner part. After passing through, the wave front is no longer flat but curved: it converges. The field behind develops toward a focal-point structure — a region of maximum field intensity. No ray is needed: Maxwell's equations in the inhomogeneous medium describe everything completely.

Layer 2 — Ray optics (limiting-case description): In the limit $\lambda \ll f$ (wavelength ≪ focal length), Snell's law holds: $n_1 \sin\theta_1 = n_2 \sin\theta_2$. Rays are now the normal trajectories of wave fronts — they describe *where the field concentrates*, not where a particle comes from. The lens does not deflect bodies; it reshapes phase surfaces.

[B] The decisive point: the lens acts on the field, not on rays. The ray is a useful geometric description of the result — but the field is the cause. An observer behind the lens sees the image because the field has high intensity at the detector location. The observer *constructs* from this a ray path.

[S] In the FFGFT reading: the photon is a field mode on T^4. It has no trajectory. The lens modifies the local metric of the EM field — it changes the phase structure of the mode. Behind the lens the probability of detecting the photon at the focal point is maximal. The focal point is not a station on a trajectory but a maximum of the projection probability. That lenses "work" is a statement about the distribution of Type-II projection events in space — not about trajectories.

51.2.4 The connection to the Bell question

[B] The connection to the instantaneity question is direct: whoever asks "how can the measurement at A instantaneously affect B?" is thinking in light paths. The light path is the implicit assumption that every effect has a spatial carrier that moves from A to B.

But if the EM field is the carrier — and the field exists globally on T^4 — then there is no "from A to B". The field is already everywhere. The measurement at A is a local interaction with the global field. The correlation at B is the response of the same global field at a different projection point. No carrier needs to travel.

[S] The lens example makes this tangible: when light passes through a lens and arrives at the focal point, no one asks "which ray caused the focal point?" That is the wrong question. The field has the phase structure that leads to this focal point. Likewise: when an entangled photon pair is measured, FFGFT does not ask "which signal ran from A to B?" The field has the mode structure that leads to these correlations.

51.3 What measurement is in FFGFT

51.3.1 Measurement is a Type-II projection

[B] In the projection chain (A090) there are three qualitatively distinct operations: Type I (duality decompactification, nearly reversible), Type II (geometric projection, lossy, irreversible), Type III (representational translation, bijective).

[B] A measurement is a Type-II operation. It collapses a dimension of the state space — it projects the global state on T^4 onto a lower-dimensional subspace. The kernel of this projection $\ker(\pi) = \mathbb{R}$ contains information no longer present in the image (A090). This lost information is not gone — it is deposited in the measuring apparatus, which thereby becomes part of the total system.

[B] Three consequences follow immediately:

1. **Measurement acts globally.** The T^4 is boundary-free ($\partial T^4 = \emptyset$, A030) and carries global eigenmodes without a distinguished location. A projection at one projection point changes the global mode content — there is no local measurement that leaves the rest of the system untouched.
2. **The apparatus becomes part of the system.** What was previously system state + apparatus state is afterwards a joint state. The boundary is not a physical boundary but a model boundary (interface definition).
3. **Repetition changes the system.** Every subsequent measurement encounters a different global state than the first. The torus "knows" about the first measurement — not through a signal, but because its global mode content has changed.

51.3.2 The geometric reading of collapse

[S] In standard QM the wave function collapses onto an eigenstate at measurement. In the FFGFT reading (A070) this is a deterministic motion in mode space: the (z, r, θ) point is driven to the nearest stable extreme ($z = +1$ or $z = -1$). This is not an instantaneous action at a distance but a local geometric motion at the projection point. The "collapse" is the Type-II projection step itself.

[B] What is lost in the process is the phase θ — the azimuthal information of the mode. It is no longer accessible in the image after measurement, but it is not destroyed: it is encoded in the measuring apparatus (Type-II loss).

51.4 Where probabilism comes from

51.4.1 Three possible explanations

There are three structurally distinct answers to the question of why quantum measurements yield probabilistic outcomes:

Standard reading (ontological randomness): $|\psi|^2$ is an ontological probability. The outcome is fundamentally random, not determined by hidden variables (Bell, Kochen-Specker).

Hidden-variable reading (local): The outcome is determined by hidden local parameters. This reading is experimentally excluded by Bell.

FFGFT reading (projection loss): The outcome is determined by the deterministic field equations on T^4, but the phase θ is not accessible to the observer. Stochasticity arises at the measuring apparatus (the Type-II projection), not in the natural process.

51.4.2 Why the FFGFT reading does not violate Bell's theorem

[B] Bell excludes *local* hidden variables on the projected $\mathbb{R}^3$. The phase θ on T^4 is not a local hidden variable in the sense of the theorem — it is a global feature of the torus mode, not a parameter sitting at a spatial location. Bell's proof assumes a factorisation of the hidden-variable density across the spatial separation. This factorisation does not hold on T^4 because locations on T^4 are defined by their winding classes, not by spatial coordinates (A030).

[B] The quantitative form: the Born rule

$$P_{\mathsf{FFGFT}}(z=+1)=\frac{1+z}{2}=|\alpha|^2=P_{\mathsf{QM}}(0)$$

follows from the uniform distribution over $\theta \in [0, 2\pi)$ at large statistics (A070). Probabilism is the ensemble average over all phase states — not fundamental randomness, but averaging over what the embedded observer cannot see.

51.4.3 Born rule as projection statistics

[S] The structural thesis: the Born rule $P = |\psi|^2$ is the statistics of Type-II information loss. In a projection $\pi : \mathbb{C}^n \to \mathbb{C}^{n-1}$ the information of the lost dimension distributes over the ensemble of repeated measurements. The quadratic form arises because the norm in Hilbert space is Euclidean and the projection acts isometrically on the subspace.

[H] The quantitative connection between Type-II loss and the exact form of the Born rule ($|\psi|^2$ rather than $|\psi|$ or $|\psi|^3$) is not worked out. That is the genuine open point — not whether but precisely how the projection statistics enforces the quadratic form.

51.5 Three levels of non-locality

[B] The instantaneity question in Bell tests has three levels that must be kept strictly separate. All three are consistently resolvable in the FFGFT framework — in different ways.

51.5.1 Level 1 — Torus picture: topological neighbourhood

[STIPULATION] (A160) Bell correlations on T^4 are a topological phenomenon. What appears in the projected $\mathbb{R}^3$ as action at a distance are points connected by closed paths in the compact picture — topological neighbours.

[B] For terrestrial separations $d \ll L^*$: both measurement points A and B lie on the same torus sheet. The T^4 with the closure scale $L^* = 4\bar{\lambda}_e/\xi_0^{10} \approx 1.3 \times 10^{26}$ m is practically flat on terrestrial scales ($d \sim 10^6$ m). *But:* the topological connection runs not via the Euclidean distance but via the winding class. An entangled pair shares the same winding class — that is the meaning of the topological connection, independent of the projected separation.

51.5.2 Level 2 — Field picture: global eigenmode

[B] (A030) Winding classes and eigenmodes have no location. An eigenmode is global — it is a property of the entire torus, not of a region. An entangled pair is not a pair of separate objects with a connection but *one* object with two projection sites (A160).

[B] It follows immediately: nothing propagates from A to B. The correlation is identity, not transmission. Signal-freedom is not an additional condition that must be imposed — it follows from the construction.

51.5.3 Level 3 — Measurement process: global projection

[S] This is the deepest level, not explicitly stated in the previous documents.

The measurement at A is not a local disturbance that subsequently propagates to B. It is a Type-II projection of the *global* state on T^4. This projection changes the global mode content — the joint state of the total system to which both projection sites A and B belong. What appears at B as an "instantaneous effect" is not an effect coming from A, but the changed boundary condition of the global state from which B reads off.

[S] Put differently: measurement at A reads the global mode and modifies it in doing so. The next measurement at B encounters an already modified global mode — not because a signal travelled from A to B, but because both are part of the same global system.

51.5.4 Why evaluation remains probabilistic in principle

[B] All three levels converge on the same point. Evaluation is probabilistic because:

1. Every single measurement is a Type-II projection (loss of the phase θ).
2. The global mode is modified after every measurement — the starting state is never exactly the same a second time.
3. The ensemble over many measurements delivers the phase averaging from which the Born rule follows.

[B] A Bell correlation cannot be read from a single measurement. An ensemble is always required. The ensemble is the statistics of the projection — and the projection is global. Evaluation is therefore probabilistic in principle, not for convenience, but because the single measurement process systematically hides the information.

51.6 Field travel time, not path travel time — the measurable delay

51.6.1 The crucial distinction

[B] There are two fundamentally different ways in which a delay $\Delta t = d/c$ can arise:

Path travel time: An object or signal carrier moves from point A to point B. There is *something* that traverses the path. The detector at B registers this something when it arrives. This is the intuitive picture underlying the concept of the light ray.

Field travel time: The field already exists everywhere. At point A something happens — a measurement, a disturbance, an interaction — and this *change in field state* spreads as a wave front. The detector at B does not register an arriving object but a changed field configuration at its location.

[B] The mathematical expression is identical for both cases — $\delta(|\mathbf{x} - \mathbf{x}'| - c(t - t'))$ — but the physical interpretation is fundamentally different. In the Green's function (Doc. 131):

$$G(\mathbf{x}, \mathbf{x}', t - t') = \theta(t - t') \cdot \frac{\delta(|\mathbf{x} - \mathbf{x}'| - c(t - t'))}{4\pi|\mathbf{x} - \mathbf{x}'|}$$

$\mathbf{x}'$ is the *source point of the disturbance* and $\mathbf{x}$ the *observation point*. The δ function says: the change reaches the observation point exactly when $|\mathbf{x} - \mathbf{x}'| = c(t - t')$. What propagates is the *wave front of the field change* — not a carrier.

51.6.2 What propagates in a Bell measurement

[S] At the moment of measurement at detector A the following happens in field terms:

1. The global EM field (the entangled state) interacts with the measuring apparatus at location A. This is a local interaction — not a global event.
2. Through this interaction the field state at A changes: phase, amplitude, mode occupation are modified. This is a *local field change*.
3. This local field change spreads as a retarded wave — with the Green's function $G(\mathbf{x}, \mathbf{x}_A, t - t_A)$. It reaches every point $\mathbf{x}$ at time $t_A + |\mathbf{x} - \mathbf{x}_A|/c$.
4. Detector B at location $\mathbf{x}_B$ registers the changed field configuration at time $t_A + d_{AB}/c$ — not because a signal travelled from A to B, but because the field change has reached it.

[B] This is field travel time, not path travel time. The distinction is physically real:

- With path travel time: B must *wait* until the signal arrives. Before that, B *knows nothing*.
- With field travel time: B interacts with the global field the whole time. Before, B responds to the undisturbed field state. After, B responds to the field state modified by the A measurement. The transition occurs when the wave front arrives.

[B] No information has been transmitted from A to B. The modified field configuration at B is not a message from A — it is the natural evolution of the global field after the A interaction. Since B did not know the unmodified field state anyway (Type-II projection, phase θ unknown), B cannot extract information about the A measurement from the modified field state either — except by comparison with A via a classical channel.

51.6.3 The ξ delay as a field phase offset

[K] (A165) The FFGFT prediction:

$$\Delta t_{\xi} = \xi \cdot \frac{d}{c} \approx 445\,\text{ns} \quad (d = 1000\,\text{km})$$

[S] This delay is *not* a path travel time and *not* an additional signal transmission. It is a *phase offset in the field wave front* arising from the fractal torus geometry.

Precisely: the global field on T^4 is not completely flat — on terrestrial scales there is a residue of compactness (A090). This residue generates a geometric phase shift of the field modes. When the wave front of the field change spreads out from A, it carries this phase offset — not because it takes a detour, but because the field geometry on the torus deviates slightly from the Euclidean.

[S] This means: what detector B measures is the field amplitude with a phase offset $\xi \cdot d/c$ relative to what B would measure without this geometric effect. The 445 ns is the time over which this phase offset shifts the correlation statistics — not the travel time of a signal.

[B] Whether this effect is measurable depends on whether one can measure the correlation statistics with a time resolution $\ll 445$ ns relative to a common time reference Q. Measuring the path travel time (signal A→B) would be conceptually wrong. Measuring the field travel time (change in correlation statistics relative to $t_Q + d_B/c$) is the right question.

51.6.4 Why it has never been measured

[K] *Preliminary: what has actually been proved and what has not.* The four generations of Bell experiments proved one thing: the correlation statistics violates the classical bound ($S > 2$).

What they did *not* prove and could not prove: that the correlation arises *instantaneously*. Instantaneity is not an experimental observation but an *interpretation* from the standard QM formalism, in which collapse is by construction timeless. None of the four generations posed the timing question — time resolution was either too coarse (Aspect: μs), or the experiment had no common source-point timestamp (Micius). The statement "Bell correlations are instantaneous" is correct as a *statement of the QM formalism*, but it is *not experimentally verified*.

[K] *Comparison with the corpus Bell documents:* A160 (topological connection, CHSH, ξ residue) and A165 (hardware validation, 445-ns prediction) treat Bell tests extensively. Doc. 131 (apparent instantaneity) shows that $T \cdot E = 1$ is a *local constraint*, not a dynamic action at a distance, and that the field equations have finite propagation velocity $v \leq c$. None of these documents states explicitly: *the instantaneity of the correlation has never been directly measured, only inferred from the standard QM interpretation.* That is the gap this section closes.

[K] *Classification of Doc. 023 (Bell tests in the T0 context, legacy corpus):* Doc. 023 — now superseded by A160 — describes a ξ modification of the Bell correlation:

$$E^{\mathrm{T0}}(a, b) = -\cos(a - b) \cdot (1 - \xi \cdot f(n, l, j))$$

This gives $\mathrm{CHSH}^{\mathrm{T0}} \approx 2.827$ instead of $2\sqrt{2} = 2.828$ — a reduction of $\Delta \approx 10^{-4}$, consistent with A165 (ΔCHSH $\approx \xi/(2\pi) \approx 2.1\times10^{-5}$; the numerical difference is explained by the different approach via $f(n, l, j)$ factors). The principle is the same in both: FFGFT modifies the correlation by $O(\xi)$. Additionally, Doc. 023 derives an extended formula from ML divergence at $\Delta\theta > \pi$ — a simulation-artefact inference that was never incorporated into the main corpus and remains [S].

Crucially: Doc. 023 treats exclusively the *angular dependence* of the correlation (whether $S > 2$). The timing question — *when* does the correlation arise? — is not posed there either. This confirms the statement of this section: all Bell documents of the corpus, including Doc. 023, have not posed the instantaneity question. A284 adds this dimension — without contradiction to the CHSH corrections.

The complete survey of the legacy documents yields five content groups:

[K] Group 1 — Consistent with main corpus (Docs. 023, A160, A165): CHSH correction $O(\xi)$ *below* the QM bound: $E^{\mathrm{T0}}(a,b) = -\cos(a-b)\cdot(1-\xi\cdot f(n,l,j))$, giving $S^{\mathrm{T0}} \approx 2.827 < 2\sqrt{2}$. Prediction: ΔCHSH $\approx 10^{-4}$, measurable. Anchored in main corpus (A160, A165) and remains valid.

[S→?] Group 2 — Contradiction with main corpus (Doc. 074): Doc. 074 claims T0 *exceeds* the Tsirelson bound $2\sqrt{2}$ by 133 ppm: $|S_{\mathrm{T0}}| = 2.389133$. It simultaneously uses superdeterminism (violation of the measurement freedom assumption) as the explanatory strategy: apparatus and quantum system are correlated via spatially extended energy fields. This directly contradicts A160: *"What does not change: CHSH and the Tsirelson bound."* Two alternatives: (a) The 133-ppm prediction is a computational artefact (Doc. 074 sets $\varepsilon_{T0,\mathrm{direct}} = \xi$ without gravitational suppression) and is to be discarded. (b) It is a genuine alternative to A160 — then both are simultaneously falsifiable in the same experiments. Doc. 074 is not incorporated in the main corpus; the superdeterminism framework is not congruent with the topological thesis of A160.

[S→contradiction] Group 3 — Distance-dependent damping (Doc. 023a): $\rho^{\mathrm{T0}} = \rho_0 \cdot \exp\left(-\xi\cdot\Delta d/D_f\right)$ — entanglement decreases with distance. This is in *direct contradiction* with A160 (winding classes have no location, no distance; entanglement is identity, not transmission). Two structurally distinct frameworks: damping model (Doc. 023a) vs. topological identity model (A160). The main corpus (A160) is authoritative; Doc. 023a is superseded.

[S] Group 4 — Energy-dependent timing difference (Doc. 055): For entangled photons with $\omega_1 \neq \omega_2$:

$$\Delta t_{\mathrm{corr}} = \frac{G\langle m\rangle}{r}\left|\frac{1}{\omega_1} - \frac{1}{\omega_2}\right|$$

Timing statement about *differences within the pair*, not about instantaneity between A and B. Mechanism (Gm/r, gravitational) differs from $\xi\cdot d/c$ (A165, geometric-fractal). Possible bridge: if $G\langle m\rangle/r$ is reducible to ξ via $\xi = 2\sqrt{G}\cdot m$, both predictions would be compatible. This reduction is not worked out in the corpus — open bridge [H].

[Q] Group 5 — External evidence for finite formation time (Doc. 142, Jiang et al., PRL 2024): Jiang, W.-C. et al., *Time*

Delays as Attosecond Probe of Interelectronic Coherence and Entanglement, Phys. Rev. Lett. 133, 163201 (2024): quantum entanglement does not arise instantaneously but builds up over ~ 232 attoseconds (theoretical prediction, no laboratory experiment). The only peer-reviewed source in the corpus establishing directly: entanglement formation has a finite time structure. Directly supports A284 — in a different system (helium under EUV laser), but structurally identical.

[K] Overall assessment:

1. Group 1 (Docs. 023/A160/A165): valid, consistent, main corpus.
2. Group 2 (Doc. 074): Tsirelson excess and superdeterminism — contradiction with A160; to be clarified whether artefact or alternative.
3. Group 3 (Doc. 023a): distance-dependent damping — contradiction with A160; main corpus is authoritative.
4. Group 4 (Doc. 055): energy-dependent timing difference — compatible, bridge to ξ still open [H].
5. Group 5 (Doc. 142): external evidence for finite formation time — supports A284.

Previous Bell experiments fall into four generations. None of them posed the question that would be decisive for the FFGFT prediction.

Generation 1: Demonstrating Bell violation. Aspect, Dalibard and Roger (1982) [1] demonstrated Bell violation with time-varying analysers for the first time. The goal was to show $S > 2$. Timing information was used only to identify coincidences (windows $\sim \mu$s), not to resolve the correlation structure in time.

Generation 2: Closing the locality loophole. Weihs, Jennewein, Simon, Weinfurter and Zeilinger (1998) [2] closed the locality loophole via fast random choice of measurement setting ($\tau_{\text{decision}} \ll d/c$). Time resolution ($\sim 1$ ns) was sufficient for that purpose, but the experiment measured A against B — not relative to the source point Q.

Generation 3: Loophole-free Bell tests. Three independent groups (Hensen et al. 2015 [3]; Giustina et al. 2015 [4]; Shalm et al. 2015 [5]) simultaneously closed locality and detection loopholes. Time resolution: ~ 1 ns, separation $d \sim 1$ km. For

$d = 1$ km FFGFT would predict $\Delta t = \xi \cdot d/c \approx 0.44$ ns — just below the time resolution of these experiments. But again the question was: does S violate the classical bound? Not: is there a delay in the correlation structure?

Generation 4: Cosmic separations. Yin et al. (2017) [6] demonstrated Bell violation with the Micius satellite over ~ 1200 km. For this distance FFGFT would predict $\Delta t \approx 530$ ns — well above the detectability threshold of atomic clocks (~ 10 ns). But Micius had *no* precise source-point reference Q. Timestamps were used for coincidence windows, not for an analysis of the causal propagation structure. Moreover Micius did not have three synchronised clocks (Q, A, B) with a precise common time base.

The structural problem. All four generations measure the same thing: whether the correlation statistics over many events violates the classical bound. That is an ensemble question. The FFGFT prediction of the 445 ns delay is instead a question about the *temporal structure* of the field modification per individual event — statistically extracted, but conceptually distinct.

An FFGFT test would require:

1. A precisely time-stamped source point Q (pair generation time to $\ll 1$ ns accuracy).
2. Three synchronised atomic clocks: Q, A, B (common time base to ~ 1 ns over $d \sim 1000$ km).
3. A statistical analysis extracting the delay Δt of the correlation change at B relative to $t_Q + d_B/c$ — not relative to t_A.
4. Sufficient event count to extract $\Delta t \approx 445$ ns from the noise (shot noise $\sim 1/\sqrt{N}$).

No one has performed this experiment because the question — is there a ξ-scaled delay in the propagation of the field modification? — makes no sense in the standard QM framework: in the standard reading nothing propagates; collapse is instantaneous. The question is only meaningful under a theory that assumes a causal field structure behind the correlations.

Experiment	d	Δt_{FFGFT}	Resolution	Question
Aspect 1982	~ 10 m	~ 0 ps	μs	Bell violation
Weihs 1998	400 m	0.18 ns	~ 1 ns	Locality loophole
Hensen 2015	1.3 km	0.58 ns	~ 1 ns	Loophole-free
Micius 2017	1200 km	530 ns	10 ns	Cosmic distance
FFGFT test	1000 km	**445** ns	≤ 10 ns	ξ **delay?**

51.7 Open question: are A and B the same field point?

51.7.1 The question

[H] In the projected $\mathbb{R}^3$ two detectors A and B are separated by $d = 1000$ km. In the torus picture:

$$\mathbf{x}_A \equiv \mathbf{x}_B \pmod{L^*}$$

Since $L^* = 4\bar{\lambda}_e/\xi_0^{10} \approx 1.3\times10^{26}$ m, the terrestrial separation $d \sim 10^6$ m is negligible: $d/L^* \approx 10^{-20}$. In the torus picture A and B lie in the same fundamental domain — practically at the same location.

[H] The deeper question is not metric but *field-ontological*: are A and B two different points for the entangled field — or the *same* field point, read out at two projection sites?

The corpus has the elements of the question (A030: winding classes have no location; A090: identification $t \sim t + R_m$; A160: entangled pair is one object with two projection images) but does not state the question explicitly and gives no answer. This is a genuine gap, not a knowledge gap.

51.7.2 Three structurally distinct possibilities

[H] There are three answers that are not equivalent:

Possibility 1 — two points, topologically connected [S]: A and B are different points in the torus picture, but connected by the same winding class. The entangled pair is a mode that encompasses both points. The field modification after the A measurement propagates as a wave front to B. This implies: the ξ delay $\Delta t = \xi \cdot d/c \approx 445$ ns is measurable (§ 51.2).

Possibility 2 — identical point in the fundamental domain [H]: Since $d \ll L^*$, A and B are practically the same point in the fundamental domain of the torus: $\mathbf{x}_A \approx \mathbf{x}_B$. The entangled mode is fully local within the fundamental domain — there is no separation between them in the torus picture. The correlation is instantaneous in the torus picture; the 445 ns is an artefact of projection.

Possibility 3 — identical point in mode space [H]: The relevant "location variable" is not $\mathbf{x} \in \mathbb{R}^3$ but the winding class $(n_1, n_2, n_3, n_4) \in \mathbb{Z}^4$. Two entangled photons share the same winding class — they are *identical* in mode space. Spatial separation is entirely a projection artefact. Field travel time is then also a concept of the projected space, not of the compact field.

Possibility	Status	Prediction
Two points, topologically connected	[S]	$\Delta t = \xi \cdot d/c$ measurable
Identical in fundamental domain	[H]	445 ns is projection artefact
Identical in mode space	[H]	Field travel time is projection concept

51.7.3 What the corpus already implies

[B] A030 (winding classes without location) and A090 (identification $t \sim t + R_m$) together provide a structure in which "location" for field modes is not a primary concept. This speaks for Possibility 2 or 3 rather than 1.

[B] A160 says the entangled pair is "one object with two projection images". If it is *one* object, it has *one* field configuration in the torus picture — not two at different locations. The two projection images only arise through the Type-II projection (A090) onto $\mathbb{R}^3$. This speaks for Possibility 3.

[S] If Possibility 3 is correct, this has a strong consequence: the question "why is the correlation instantaneous?" is then not a question about propagation speed — it is a question about why we take the projection of one object at two sites to be two measurements. The answer is not physics but epistemology: the projection image deceives us about the number of objects.

51.7.4 Consequence for the 445-ns prediction

[H] The prediction $\Delta t = \xi \cdot d/c$ from A165 assumes Possibility 1. It is correct if A and B are different field points in the torus picture, between which a field modification can propagate.

If instead Possibility 2 or 3 holds, the prediction shifts conceptually: the effect then does not exist as field travel time between A and B, but as a phase offset in the projection itself — measurable as a timing asymmetry in the correlation statistics, but with a different origin.

[H] This question cannot currently be decided within the FFGFT corpus. It requires a precise formulation of what "location in the field" means for a mode structure on T^4 — a bridge between the topological structure of A030 and the measurement process of A090 that has not yet been built.

Until clarified, A165 remains valid as a prediction — under the explicit assumption of Possibility 1. The assumption is to be declared, not concealed.

51.8 The most plausible explanation — a judgement

[S] This section contains a judgement by the author. It is not a proved statement of the corpus but a reasoned assessment based on the analysis in § 51.2 through § 51.7.

51.8.1 Group 2 (Doc. 074): Artefact, not alternative

A160 states explicitly as a [STIPULATION] falsification condition: *"If a CHSH value is measured that exceeds the Tsirelson bound, the theory is refuted at this point."* The Tsirelson bound $2\sqrt{2}$ is an algebraic proof, not an experimental result. Exceeding it would mean refuting standard quantum mechanics itself.

Doc. 074 achieves the 133-ppm excess by dropping the gravitational suppression factor ($G\langle m\rangle/r \approx 10^{-30}$) and inserting ξ directly. This is an unjustified parameter choice, not a physical argument — Doc. 074 itself calls it an "Alternative Interpretation".

[S] Judgement: Computational artefact. Discarded.

51.8.2 Group 3 (Doc. 023a): Experimentally refuted

A160 establishes **[B]** algebraically: the $\mathbb{C}^3$ tensor factor is invisible to correlation quantities — the observable acts as identity there (A070). Distance-dependent damping would have measurable consequences: entanglement would decrease with separation. Micius 2017 shows Bell violation over 1200 km at the same level as laboratory experiments. This is experimentally excluded.

[S] Judgement: Experimentally refuted. Discarded.

51.8.3 Group 4 (Doc. 055): Compatible, but second order

In natural units $\xi = 2\sqrt{G} \cdot m$, hence $G = \xi^2/(4m^2)$. The gravitational term of Doc. 055 is therefore $\propto \xi^2$ — second order. The predicted time difference is $\sim 10^{-30}$ s, experimentally inaccessible.

[S] Judgement: Consistent with the ξ scheme at second order, many orders of magnitude below A165. Compatible, but in the noise.

51.8.4 Group 5 (Doc. 142, Jiang PRL 2024): Structurally strongest argument

The Jiang work is independent of the FFGFT framework. It shows from the time-dependent Schrödinger equation for helium that entanglement has a finite build-up time. This is not an FFGFT-specific result but a general finding of time-dependent quantum mechanics.

[S] Judgement: Quantum mechanics itself excludes strict instantaneity when the time-dependent evolution is taken seriously. Standard QM introduces instantaneity through the stationary eigenstate basis, not through the time-dependent dynamics. This is structurally identical to the field travel-time thesis of A284 — in a different system, but with the same mechanism.

51.8.5 The overall judgement

[S] The most plausible explanation for Bell correlations in FFGFT is **Group 1 and Group 5 combined**:

- The correlation does *not* arise instantaneously (Jiang, [Q]) — it has a finite build-up time in the field geometry.
- In the compact torus picture this is no action at a distance, because A and B are projection sites of the same global field (A160, [B]).
- The CHSH correction lies at $O(\xi) \approx 10^{-4}$ below the Tsirelson bound (A165, [K]).
- Distance-dependent damping (Group 3) is experimentally excluded; Tsirelson excess (Group 2) is a computational artefact.

[H] What remains open: whether A and B are the same field point in the torus picture (§ 51.7). This question decides whether the 445-ns delay (A165) has a measurable consequence — or is itself a projection artefact, because there is no field separation between A and B.

[S] This is the honest remainder. The qualitative answer is clear: no action at a distance, no instantaneity, no distance dependence. The quantitative prediction (445 ns) depends on the unresolved question of whether field points on T^4 carry a well-defined separation at all.

51.9 Open questions

[H] Three genuine open points:

1. **Born rule from Type-II loss:** The quantitative derivation of the quadratic form $P = |\psi|^2$ from projection loss is not worked out. Why square and not absolute value or higher power? The isometric property of the Hilbert-space projection is the candidate (A090), but the proof is missing.
2. **Path assignment for entanglement:** Which closed paths on T^4 correspond to which concrete entangled states (A160, [S]). Without this assignment the topological thesis remains a structural statement, not an operational criterion.
3. **Global mode and local measurement:** How precisely does a Type-II projection at one projection point modify the global mode content? The qualitative mechanism is clear (Type-II

loss); the quantitative field equation for the global back-action is not worked out.

51.10 Layer-status summary

Statement	Status
Measurement is Type-II projection (A090)	**[B]**
$\partial T^4 = \emptyset$: boundary-free, global eigenmodes without location	**[B]**
Measuring apparatus becomes part of the total system	**[B]**
Winding classes have no location (A030)	**[B]**
Born rule: $P = (1+z)/2 = \lvert\alpha\rvert^2$ from phase averaging	**[B]**
Bell: no local hidden variables on $\mathbb{R}^3$	**[B]**
FFGFT: θ is global feature, not local parameter	**[B]**
Entangled pair: one object, two projection sites (A160)	**[B]**
Signal-freedom: consequence of construction, not extra condition	**[B]**
Spatial delay $\Delta t = \xi \cdot d/c$ (A165)	**[K]**
Measurement at A modifies global mode content	**[S]**
Born rule as statistics of Type-II loss	**[S]**
ξ delay as field modification, not signal	**[S]**
Born rule quadratic form from projection	**[H]**
Path assignment entangled state $\leftrightarrow$ T^4 path	**[H]**
Quantitative field equation for global back-action	**[H]**

Balance: 9× **[B]** · 1× **[K]** · 3× **[S]** · 3× **[H]**.

51.11 Source references

A030 Pascher, J., *Compact Geometry*. A-series, 2026. Winding classes without location, $\partial T^4 = \emptyset$.

A070 Pascher, J., *Hilbert-Space Translation*. A-series, 2026. Measurement as pole transition, Born rule from phase ignorance, bijectivity.

A090 Pascher, J., *Projection Chain*. A-series, 2026. Type-I/II/III operations, reversibility, information loss.

A160 Pascher, J., *Quantum Mechanics and Bell Correlations*. A-series, 2026. Topological connection, CHSH, ξ residue.

A165 Pascher, J., *Bell Hardware and Quantum Computing*. A-series, 2026. $\Delta t = \xi \cdot d/c$, IBM Heron measurements.

Doc. 131 Pascher, J., *Apparent Instantaneity*. Legacy corpus, 2025. Green's function, causality, time scales.

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[3] Hensen, B. et al. (2015). *Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres*. Nature, 526, 682–686.

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51.12 Use of AI tools

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